Class 10

Mathematics

(Solutions)

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1. Real Numbers

Solutions

LEVEL-I

1. Correct option: (B)

Given that $a = 2^2 \times 3^3 \times 5^4, b = 2^3 \times 3^2 \times 5$ then HCF of (a, b) =

It is known that HCF (a, b) is calculated by the product of the smallest power of the each common prime factor in the numbers

In this case common prime factors are 2,3 and 5

Their smallest powers are $2^2, 3^2, 5^1$ respectively and their product yields 180

The correct Answer is B

2. Correct option: (C)

Given that HCF $(2^3 \times 3^2 \times 5), (2^2 \times 3^3 \times 5^2), (2^4 \times 3 \times 5^3 \times 7))$ =

It is known that HCF between the numbers is calculated by the product of the smallest power of the each common prime factor in the numbers

In this case common prime factors are 2,3 and 5

Their smallest powers are $2^2, 3, 5^1$ respectively and their product yields 60

The correct Answer is C

3. Correct option: (D)

LCM of $(2^3 \times 3 \times 5)(2^4 \times 5 \times 7)$ is

It is known that HCF between the numbers is calculated by the product of the greatest power of the each prime factor in the numbers

In this case greatest powers of each prime factor are 2^4,3, 5, and 7 and the product yields 1680

The correct Answer is D

4. Correct option: (D)

Given that HCF of the two numbers is 27 and their LCM is 162

Let one number be 'x' and its value is given as 54

Let the Other number be 'y' and its value to be calculated

If one of the numbers is 54, what is the other number?

It is known that the product of two numbers will be equal to product of their HCF and LCM

This implies that $x \times y = 27 \times 162$

Substitute the value of 'x' in above mathematical expression yields

$54 \times y = 27 \times 162$

$$\Rightarrow y = \frac{27 \times 162}{54} = 81$$

Therefore, the value of the other number is 81

The correct Answer is D

5. Correct option: (D)

The simplest form of $\frac{1095}{1168}$ is

To find the simplest form of a fraction, both numerator and denominator of the fraction is divided denominator by the greatest common factor that divides them completely

The factors of 1095 are 1, 3, 5, 15, 73, 219, 365, 1095.

Similarly, the factors of 1168: 1, 2, 4, 8, 16, 73, 146, 292, 584, 1168.

The Highest common factor between them is 73

By dividing the Numerator and Denominator of the given fraction with '73' implies

$$\frac{109\cancel{3}}{11\cancel{6}8}=\frac{15}{16}$$

The correct Answer is D

6. Correct option: (D)

A number, when divided by 143, leaves 31 as the remainder. What will be the remainder when the same number is divided by 13

Let the given number be expressed in Euclid division Lemma using the given conditions implies that $143n+31$

Given that the same number is multiplied by 13, if it is divided by 13 then as 143 is completely divided by 13 but while dividing 31 it leaves reminder 5

Hence the answer is 5.

The correct Answer is D

7. Correct option: (D)

A rational number is the one which can be represented in the form of P/Q where P and Q are integers and Q ≠ 0.

An irrational number cannot be written in the form of simple fractions.

a) $\frac{22}{7}=3.142857142857142857142857142857$ this is a recurring decimal value 3.142857 and hence this is a rational number

b) 3.1416 since this is terminating fraction this can be easily represented in the form of $\frac{p}{q}, q\neq 0$ and hence this is also a rational number

c) $3.\overline{1416}$ 3.1416141614161416 as this also a recurring decimal this is also a rational number

d) 3.141141114…. since this is an non terminating decimal and hence this is an irrational number

The correct Answer is D

8. Correct option: (C)

Which of the following rational number is expressible as a terminating decimal

$$\frac{124}{165}=0.7515151=0.7\overline{51}$$

$$\frac{131}{30}=4.3666..=4.3\overline{6}$$

$$\frac{2027}{625}=3.2432$$

$$\frac{1625}{462}=3.51731601731160=3.51\overline{731160}$$

From this it is clear that fraction in option 'C' is expressed as a terminating decimal

9. Correct option: (B)

The Decimal expansion of the rational number $\frac{37}{2^2\times 5}$ will terminate after $\frac{37}{2^2\times 5}=1.85$

This is a rational number terminated after two decimal places

The correct Answer is B

10. Correct option: (C)

As $\sqrt{2}$ is an irrational number $\frac{1}{\sqrt{2}}$ is also an irrational number

The correct Answer is C

11. Correct option: (D)

$(-1)^{2001}+(-1)^{2002}+.....(-1)^{2010}$

$\Rightarrow(-1)^{2001}\left\{1+(-1)^{1}+....(-1)^{9}\right\}$

$\Rightarrow(-1)^{2001}[5-5]=0$

The correct Answer is D

12. Correct option: (B)

The product of the cube of $\frac{-2}{3}$ and the square of $\frac{-4}{5}$ is

$\left(\frac{-2}{3}\right)^{3}\times\left(\frac{-4}{5}\right)^{2}=\frac{-8}{27}\times\frac{16}{25}=\frac{-128}{625}$

The correct Answer is B

13. Correct option: (C)

$\left[(3^{2}-2^{2})\div\left(\frac{1}{5}\right)^{2}\right]=\left[5\div\left(\frac{1}{5}\right)^{2}\right]=\frac{5}{\frac{1}{25}}=125$

The correct Answer is C

14. Correct option: (B)

The exponential form of 3125 is=

$3125=5\times625=5\times5\times125=5^{2}\times5^{3}=5^{5}$

The correct Answer is B

15. Correct option: (A)

$5^{n-2}\times3^{2n-3}=135$

$\Rightarrow\frac{5^{n}}{25}\times\frac{\left(3^{2}\right)^{n}}{27}=135$

$\Rightarrow5^{n}\times9^{n}=135\times25\times27$

$\Rightarrow(45)^{n}=(45)^{4}\Rightarrow n=4$

The correct Answer is A

16. Correct option: (D)

$\frac{10\times5^{n+1}+25\times5^{n}}{3\times5^{n+2}+10\times5^{n+1}}=\frac{2\times5\times5^{n}\times5+5^{n}\times5^{2}}{3\times5^{n}\times5^{2}+5\times2\times5\times5^{n}}=\frac{5^{n}\times5^{2}(2+1)}{5^{n}\times5^{2}(3+2)}=\frac{3}{5}$

The correct Answer is D

17. Correct option: (B)

$\log_{10}^{8}=\log_{10}^{2^{3}}=3\left(\log_{10}^{2}\right)=3(0.3010)=0.9030$

The correct Answer is B

18. Correct option: (C)

$\log_{30}^{3}=a,\log_{30}^{5}=b,\log_{30}^{32}=?$

$\Rightarrow\log_{30}^{32}=\log_{30}^{2^{5}}=5\log_{30}^{2}=5\log_{30}^{\frac{30}{15}}=5\left(\log_{30}^{30}-\log_{30}^{15}\right)$

$\Rightarrow5\left(\log_{30}^{30}-\log_{30}^{5\times3}\right)=5\left(\log_{30}^{30}-\log_{30}^{5}-\log_{30}^{3}\right)=5(1-b-a)$

The correct Answer is C

19. Correct option: (C)

$log_{\sqrt{2}}\left[log_2^{\left\{log_4^{(x-15)}\right\}}\right] = 0, x = ?$

$\Rightarrow \left[log_2^{\left\{log_4^{(x-15)}\right\}}\right] = 1 \Rightarrow \left\{log_4^{(x-15)}\right\} = 2 \Rightarrow x - 15 = 16 \Rightarrow x = 31$

The correct Answer is C

20. Correct option: (C)

$(3.7)^a = (0.37)^b = 1000, \frac{1}{a} - \frac{1}{b} = ?$

$\Rightarrow (3.7)^a = 10^3$

$\Rightarrow a\, log_{10}^{3.7} = 3 \Rightarrow log_{10}^{3.7} = \frac{3}{a} \Rightarrow \frac{3}{a} = 0.5682$

$\Rightarrow (0.37)^b = 1000$

$\Rightarrow a\, log_{10}^{0.37} = 3 \Rightarrow log_{10}^{0.37} = \frac{3}{b} \Rightarrow \frac{3}{b} = -0.4317$

$\Rightarrow \frac{3}{a} - \frac{3}{b} = 0.999 \approx 1 \Rightarrow \frac{1}{a} - \frac{1}{b} = \frac{1}{3}$

The correct Answer is C

LEVEL – II

1. Correct option: (B)

The number of natural numbers up to 150 that are divisible by 7 is

If 150 is divided by 7 then reminder left will be '3'. If '3' is subtracted from 150 implies 147 will get.

Dividing 147 with 7, 21 will yielded as a quotient

Therefore, there are 21natural numbers are divisible by 7 upto 150

2. Correct option: (A)

The number of numbers between 11 and 111 that are multiples of both 2 and 5 is

The number who has both the 2 and 5 as multiples is 10. There are 10 multiples between 11 and 111, and hence there are 10 numbers of numbers between 11 and 111

3. Correct option: (B)

Unit digit in the product of first 50 odd natural numbers is

$1,3,5,7...99$, these are first 50 odd natural numbers

Product of these numbers implies

$1\times3\times5\times7....\times99$

$\Rightarrow 3\times5\times7....\times99 = 5(3\times7....\times99) = 5(oddnumber)$

If an odd number is multiplied with 5, the unit digit of the product is always '5'

4. Correct option: (D)

If LCM of three numbers is 12, numbers are in the ratio1:2:3, then the numbers are

If the numbers are in the ratio of 1:2:3, let 'x' be a common multiple of the number

Then the numbers will be x, 2x and 3x respectively and the LCM of these numbers will be 6x

Here $6x = 12 \Rightarrow 2$

Therefore the numbers will be 2,4 and 6 respectively

5. Correct option: (B)

$\log_e^{x-1} + \log_e^{x} + \log_e^{x+0} = 0$

$\log_e^{(x-1)(x)(x)} = 0$

$\Rightarrow (x-1)(x)(x) = e^0$

$\Rightarrow x^3 - x - 1 = 0$

The correct Answer is B

6. Correct option: (A,B,C)

The following rational numbers lie between $\frac{-1}{2}, \frac{1}{2}$

$a) \frac{3}{10} = 0.3$

$b) 0$

$c) \frac{21}{100} = 0.21$

$d) \frac{2}{3} = 0.666666\ldots..$

as $\frac{-1}{2}, \frac{1}{2}$ includes $-0.5 to 0.5$ on number line 0,3,0 and 0.21 lies in that interval

The correct Answer is A,B,C

7. Correct option: (A, B)

The terminating decimals among the following is/ are

$a) \frac{5}{32} = 0.15625$

$b) \frac{27}{80} = 0.3375$

$c) \frac{11}{24} = 0.458333\ldots$

$d) \frac{5}{12} = 0.4166666\ldots..$

Terminating decimal implies when dividing with its denominator the numerator must be divided with reminder 0

The correct Answer is A, B

8. Correct option: (A, D)

Solving the Value of $0.\overline{2}$

$0.\overline{2} = 0.22222....$

$let x = 0.22222....$

$10x = 2.2222$

$10x - x = 2.2222 - 0.2222$

$9x = 2$

$x = \frac{2}{9}$

Solving the Value of $0.\overline{3}$

$0.\overline{3} = 0.3333$

$let x = 0.3333$

$10x = 3.3333$

$x = \frac{3}{9}$

$x = \frac{1}{3}$

Solving the Value of $0.\bar{4}$

$0.\bar{4} = 0.44444444.....$

$let x = 0.44444444.....$

$10x = 4.44444444.....$

$9x = 4 \Rightarrow x = \frac{4}{9}$

Solving the Value of $0.\bar{9}$

$0.\bar{9} = 0.9999$

$x = 0.9999$

$10\,\mathrm{x} = 9.9999$

$9\,\mathrm{x} = 9$

$\mathrm{x} = 1$

Solving the Value of $0.\overline{39}$

$let x = 0.\overline{39} = 0.39393939.....$

$100x = 39.39393939.....$

$99x = 39 \Rightarrow x = \frac{39}{99} = \frac{13}{33}$

Summing all the values together implies

$\frac{2}{9} + \frac{1}{3} + 1 + \frac{4}{9} + \frac{13}{33}$

$\frac{237}{99} = \frac{79}{33} = 2\frac{13}{33}$

The correct Answer is A, D

9. Correct option: (C, D)

If $\log_a^x = b$ for permissible values of a and z, then the identify the statement which can be correct?

a) If a and b are two irrational numbers, then x can be rational

Assume two irrational numbers for a & b as below

$let a = \sqrt{2}, b = \sqrt{3}$

$log_a^x = b \Rightarrow x = a^b \Rightarrow x = \left(\sqrt{2}\right)^{\sqrt{3}}$

The value of 'x' is again an irrational number hence this statement is false

b) If a is rational and b is irrational numbers, then x can be rational

$let a = 2, \mathrm{b} = \sqrt{3}$

$\log_a^x = b \Rightarrow x = a^b \Rightarrow x = \left(2\right)^{\sqrt{3}}$

The value of 'x' is again an irrational number hence this statement is false

c) If a is irrational and b rational numbers, then x can be rational

$let b = 2, \mathrm{a} = \sqrt{3}$

$\log_a^x = b \Rightarrow x = a^b \Rightarrow x = \left(\sqrt{3}\right)^2 = 3$

The value of 'x' is a rational number hence this statement is True

d) If a is rational and b is rational numbers, then x can be rational

$let a = 2, b = 3$

$\log_a^x = b \Rightarrow x = a^b \Rightarrow x = (2)^3 = 8$

The value of 'x' is a rational number hence this statement is True

The correct Answer is C, D

10. Correct option: (A)

$\log^{(x+y)} = \log^{(\frac{3x-3y}{2})}$

$2x + 2y = 3x - 3y$

$2y + 3y = 3x - 2x$

$x = 5y$

$\frac{x}{y} = 5$

$\log \frac{x}{y} = \log 5$

$\log x - \log y = \log 5$

The Correct Answer is A

11. Correct option: (C)

$(-1)^3 \times (a^2)^3 \times (5a)^2$

$(-1)^3 \times (a^6) \times 5^2 \times a^2$

$-1 \times a^6 \times 5^2 \times a^2$

$-25a^8 = -5^2 a^8$

The Correct Answer is C

12. Correct option: (C)

$\{[(\frac{-1}{5})^{-2}]^2\}^1$

$\{[(-5^{-1})^{-2}]^2\}^1$

$\{[(-5)^2]^2\}^1$

$= -5^4$

The Correct Answer is C

13. Correct option: (B)

$8^{-2} \times 5^{-3} \times 2^4$

$5^3 \times 2^{-6} \times 2^4$

$5^3 \times 2^{-2} = \frac{5^3}{2^2} = \frac{125}{4}$

The Correct Answer is B

14. Correct option: (A)

$\frac{5^m}{5^{-3}} = 5^5$

$5^m = 5^5 \times 5^{-3}$

$5^m = 5^2$

$m = 2$

The Correct Answer is A

15. Correct Matching is: a -Q, b – R, c – S, d – P

(a) $2^{a+1} \times 5^a = 200$, then a=?

$2^a \times 2^1 \times 5^a = 200$

$2^a \times 5^a = 100$

$(10)^a = 100$

$a = 2$

'a' is matching with 'Q'

(b)

$$\frac{\frac{2x}{2} \times 4^{2x} \times 4^2}{\frac{8x}{8}} = 64$$

$$\frac{2^x \times 4^{2x} \times 2 \times 8}{8^x} = 64$$

$$\frac{2^x \times (2^2)^{2x}}{8^x} = 4$$

$$\frac{2^x \times 2^{4x}}{(2^3)^x} = 4$$

$$\frac{2^{5x}}{2^{3x}} = 4$$

$$2^{2x} = 2^2$$

$$2x = 2$$

$$x = 1$$

'b' is matching with 'R'

(c)

$$\frac{2^1 + 2^0 + 2^{-1}}{2^{-2} + 2^{-3} + 2^{-4}}$$

$$\frac{2 + 1 + \frac{1}{2}}{\frac{1}{4} + \frac{1}{8} + \frac{1}{16}} = \frac{\frac{7}{2}}{\frac{4+2+1}{16}} = \frac{\frac{7}{2}}{\frac{7}{16}} = \frac{16}{2} = 8$$

'c' is matching with 'S'

(d)) $5^{-5} \times 5^{2n+1} = 5^{13} \div 5^5$ Then value of n=?

$$\frac{5^{-5+2n+1}}{\frac{5^{13}}{5^5}}$$

$$\frac{5^{2n-4}}{5^3}.5^5 = 1$$

$$5^{2n-4+5} = 5^{13}$$

$$2n + 1 = 13$$

$$2n = 12$$

$$n = 6$$

'd' is matching with 'P'

16. Correct Answer is: 2

$\log_2^{\log_3^{(8+\log_3^{\log_2^{x^2+4}})}} = 1, x = ?$

$\Rightarrow \log_3^{(8+\log_3^{\log_2^{x^2+4}})} = 2 \Rightarrow 8 + \log_3^{\log_2^{x^2+4}} = 9$

$\Rightarrow \log_3^{\log_2^{x^2+4}} = 1 \Rightarrow \log_2^{x^2+4} = 3 \Rightarrow x^2 + 4 = 8 \Rightarrow x = \pm 2$

But the value of 'x' is +2 as given that x is a positive number

17. Correct Answer is:2

$2^{2x-1} = \frac{1}{8^{x-3}}$

$2^{x-1} = 8^{-(x-3)}$

$2^{x-1} = 2^{-3(x-3)}$

$2x - 1 = -3x + 9$

$2x + 3x = 9 + 1$

$5x = 10$

$x = 2.$

18. Correct Answer is: $(\frac{4}{3})^{-40}$

$\frac{p}{q} = \frac{\frac{4}{3} \times \frac{4}{3}}{(\frac{3}{4})^2}$

$\frac{p}{q} = \frac{\frac{4}{3} \times \frac{4}{3}}{\frac{3}{4} \times \frac{3}{4}}$

$\frac{p}{q} = \frac{4}{3} \times \frac{4}{3} \times \frac{4}{3} \times \frac{4}{3} = (\frac{4}{3})^4$

$(\frac{p}{q})^{-10} = ((\frac{4}{3})^4)^{-10} = (\frac{4}{3})^{-40}$

19. Correct Answer is: 81

$\frac{3^{n+5} - 6.3^{n+2}}{3^{n-1} \times 7}$

$\frac{3^n.3^5 - 6.(3^n \times 3^2)}{\frac{3^n}{3^3} \times 7}$

$\frac{3^n(3^5 - 6 \times 3^2)}{3^n \times 7} \times 3$

$\frac{3(3^5 - 6 \times 3^2)}{7}$

$\frac{3(243 - 54)}{7}$

$3 \times 27 = 81.$

20. Correct Answer is: $\frac{5}{3}$

$3^{x-5} = \frac{1}{9^x}$

$\frac{3^x}{3^5} = 9^{-x}$

$\frac{3^x}{3^5} = 3^{2x}$

$3^{x-5} = 3^{-2x}$

$x - 5 = -2x$

$x + 2x = 5$

$x = \frac{5}{3}$

STUDENT PRACTICE QUESTIONS

1. Correct option: (C)

Given that the product of the two numbers is 1600

HCF is 5.

The LCM of the numbers is to be calculated, let the LCM of the two numbers be 'L'

It is known that the product of two numbers will be equal to product of their HCF and LCM

This implies that $P = LCM \times HCF$, here P is product of the given two numbers

Substitute known values in above mathematical expression yields

$P = LCM \times HCF$

$$\Rightarrow LCM = L = \frac{HCF}{P} = \frac{1600}{5} = 320$$

The correct Answer is C

2. Correct option: (C)

Euclid's division lemma stated that for any positive integers a and b. There exist unique integers q and r such that $a = bq + r$, where r must satisfy

Given that both a & b are positive numbers

For the given equation $a = bq + r$ to satisfy the condition $a > 0, b > 0$

The value of r should satisfy below two conditions

$r < b$

$0 \le r$

Combining above two implies that $0 \le r < b$

The correct Answer is C

3. Correct option: (B)

The number 3.24636363… is a non-terminating repeating number, this number is a rational number

The correct Answer is B

4. Correct option: (B)

The number 1.732 is clearly a terminating decimal and hence this can be represented in form pf $\frac{p}{q}, q \neq 0$ and hence this is a rational number

The correct Answer is B

5. Correct option: (B)

$$(-2)^5 \times 3^2 = -32 \times 9 = -288$$

The correct Answer is B

6. Correct option: (D)

$$\frac{3^5 \times 10^5 \times 25}{5^7 \times 6^5} = \frac{3^5 \times (5 \times 2)^5 \times 25}{5^7 \times 6^5} = \frac{(3^5 \times 2)^5 \times 5^7}{5^7 \times 6^5} = 1$$

The correct Answer is D

7. Correct option: (C)

$\log_{\sqrt{8}}^{b} = 3\frac{1}{3}$

$\Rightarrow \log_{\sqrt{8}}^{b} = \frac{10}{3} = \log_{2^{\frac{3}{2}}}^{b} = \frac{10}{3}$

$\Rightarrow \log_{2^{\frac{3}{2}}}^{b} = \frac{2}{3}\log_2^b = \frac{10}{3}\left(\because \log_{a^m}^{b} = \frac{1}{m}\log_2^b\right)$

$\Rightarrow \log_2^b = 5 \Rightarrow b = 2^5 = 32$

The correct Answer is C

8. Correct option: (A)

$\frac{1}{\log_x^{xy}} + \frac{1}{\log_y^{xy}}$

$= \frac{1}{\log_x^x + \log_x^y} + \frac{1}{\log_y^x + \log_y^y}$

$= \frac{1}{1 + \log_x^y} + \frac{1}{1 + \log_y^x}$

$= \frac{1 + \log_y^x + 1 + \log_x^y}{\left(1 + \log_x^y\right)\left(1 + \log_y^x\right)} = \frac{2 + \log_y^x + \log_x^y}{1 + \log_y^x + \log_y^x \log_x^y + \log_x^y} = \frac{2 + \log_y^x + \log_x^y}{2 + \log_y^x + \log_x^y} = 1$

The correct Answer is A

9. Correct option: (C)

$\log_4^{\left(\log_3^x\right)} = \frac{1}{2}$

$\Rightarrow \log_3^x = 4^{\frac{1}{2}} \Rightarrow \log_3^x = 2 \Rightarrow x = 3^2 = 9$

The correct Answer is C

10. Correct option: (D)

$\frac{1}{\log_x^{10}} = \frac{1}{\log_a^{10^{-2}}}$

$\frac{1}{\frac{\log 10}{\log x}} = \frac{1}{\frac{\log 10^{-2}}{\log a}} \Rightarrow \frac{\log x}{\log 10} = \frac{\log a}{\log 10^{-2}} \Rightarrow \frac{\log x}{\log 10} = \frac{\log a}{(-2)\log 10}$

$\Rightarrow -2\log x = \log a$

$\Rightarrow \log x = \frac{-1}{2}\log a \Rightarrow x = a^{\frac{-1}{2}}$

The correct Answer is D

11. Correct option: (B)

LCM of two numbers is 105, HCF of that two numbers is 1, the difference of numbers is 8, then the sum of numbers is

Let the two numbers be 'a' and 'b'

LCM is 105 and HCF is 1

Product of two numbers is equals to product of LCM and HCF

$ab = 105, a - b = 8, a + b = ?$

$\Rightarrow (a+b)^2 = (a-b)^2 + 4ab = 64 + 420 = 484$

$\Rightarrow a + b = 22$

12. Correct option: (C)

$x^{\frac{1}{3}} + y^{\frac{1}{3}} + z^{\frac{1}{3}} = 0, log\left(\frac{x+y+z}{3}\right) = ?$

$let x^{\frac{1}{3}} = a, y^{\frac{1}{3}} = b, z^{\frac{1}{3}} = c \Rightarrow a + b + c = 0$

$a + b = -c$

$(a+b)^3 = (-c)^3 \Rightarrow a^3 + b^3 + 3ab(a+b) = -c^3$

$\Rightarrow a^3 + b^3 + c^3 = -3ab(a+b) = -3ab(-c) = 3abc$

$\therefore x + y + z = 3abc$

$\Rightarrow log\left(\frac{x+y+z}{3}\right) = log\left(\frac{3abc}{3}\right) = log(abc) = log\, a + log\, b + log\, c$

$\Rightarrow \frac{1}{3} log\, x + \frac{1}{3} log\, y + \frac{1}{3} log\, z = \frac{1}{3}(logx + logy + logz())$

The correct Answer is C

13. Correct option: (C)

$x^{log_x^{(x+3)^2}} = 16$

$(x+3)^2 = 16$

$x^2 + 6x + 9 - 16 = 0$

$x^2 + 6x - 7 = 0$

$x^2 + 7x - x - 7 = 0$

$x(x+7) - 1(x+7) = 0$

$(x-1)(x+7) = 0$

$x = 1, x = -7$

As there are two values for 'x', there are two solutions

The Correct Answer is C

14. Correct option: (C)

$3 \cdot x^{\log_5^2} + 9^{\log_5^x} = 64$

$\Rightarrow 3 \cdot x^{\log_5^2} + 3^{2^{\log_5^x}} = 64 (\because a^{\log_c^b} = c^{\log_c^a})$

$\Rightarrow 3 \cdot x^{\log_5^2} + 3^{x^{\log_5^2}} = 64, put x^{\log_5^2} = a$

$\Rightarrow 3 \cdot a + 3^a = 64$

15. Correct option: (B)

$2^{\log_6^{(-4x)}} = \log_4^{2401}$

$2^{\log_6^{(-4x)}} = 4$

$2^{\log_6^{(-4x)}} = 2^2$

$log_6^{-4x} = 2$

$-4x = 6^2$

$4x = -36$

$x = -9$

As there is one value for 'x' there is only one solution for this expression

The Correct Answer is B

16. Correct Matching is: a – s b - t c – q d – p

(1)

$$\log_{0.5}^{0.0625} = \log_{\frac{5}{10}}^{\frac{625}{10000}} = \log_{\frac{5}{10}}^{(\frac{5}{10})^4} = 4$$

So, answer is a perfect square,

so '1' is matching with 's'

(2)

$$\log_2^8 = \log_2^{2^3} = 3.\log_2^2 = 3$$

'3' is a prime number and hence '2' is matching with 't'

(3)

$$\log_4^3 . \log_5^4 . \log_5^3$$

$$\frac{\log 3}{\log 4} . \frac{\log 4}{\log 5} . \frac{\log 5}{\log 3}$$

$$= 1$$

So answer is odd number and hence '3' is matching with 'q'

$$\log_3^{\log_2^{512}} = \log_3^{\log_2^{2^9}} = \log_3^{9^{\log_2^2}} = \log_3^9 = 2$$

So answer is odd number and hence '4' is matching with 'p'

17. Correct Answer is: 2

$$3^{x-1} = 27$$

$$4^{y+2} = 256$$

$$3^{x-1} = 3^3$$

$$4^{y+2} = 4^4$$

$$x - 1 = 3$$

$$y + 2 = 4$$

$$x = 4, y = 2$$

$$\frac{x}{y} = \frac{4}{2} = 2$$

2. Polynomials and Pair of Linear Equations in Two Variables

Solutions

FACTOR, REMAINDER, THEOREMS

LEVEL – I

1. Correct option: (C)

Polynomial is defined as an expression consists of coefficients, variables and exponentials (positive), and its standard form is $p(x) = a_n x^n + a_{n-1} x^{n-1} + ... + a_0$

From the given options only Option 'C' can be realized in above form

The Correct option is C.

2. Correct option: (C)

A trinomial defined as a specific type of polynomial that consists of three terms. Each term in a trinomial is typically separated by addition or subtraction. Trinomials can involve variables raised to different powers and coefficients. Here's a general form of a trinomial: $ax^n + bx^m + c$. Where a, b, c are real numbers. m, n are non negative integers representing exponents of variables of x.

The Correct option is C.

3. Correct option: (B)

A bi-quadratic polynomial is a polynomial of degree 4. It is also known as a bi-quadratic polynomial. This type of polynomial has terms that can be written in the form of $ax^4 + bx^3 + cx^2 + dx + e$. Where a, b, c, d, and e are coefficients and x is the variable.

The Correct option is B.

4. Correct option: (C):

Given equation is $x^3 - ax^2 + 6x - a = p(x)$.

By using remainder theorem, we can conclude that if polynomial p(x) is divide with x-a, then remainder will be equal to p(a).

P (a)

$= a^3 - a.a^2 + 6a - a$

$= a^3 - a^3 + 6a - a$

$= \cancel{a}^3 - \cancel{a}^3 + 6a - a$

$= 5a$

5a is the remainder.

The Correct option is C

5. Correct option: (D)

Given $\dfrac{1 + x + x^2 + x^3 + ----+ x^{2019}}{x - 1}$

According to remainder theorem,

$p(a) = p(1) = 1 + 1^2 + 1^3 + ------ + 1^{2019}$

$p(a) = 2020$

The Correct option is D

6. **Correct option: (B)**

Given $x^4-3x^4-7x^2+27x-18$ is divided by x+3.

$$
\begin{array}{l}
x+3\overline{)\,x^4-3x^3-7x^2+27x-18\;\;\;

$27p+41=p+15$

$26p=-26$

$p=-1$

The Correct option is B.

9. **Correct option: (C)**

R_1 is remainder when $p_1(x)=4x^3+3x^2-12x-5$ divided by $x-1$.

R_2 Is remainder when $p_2(x)=2x^3+ax^2-6x+2$ divided by $x+2$

Also $3R_1+R_2+28=0$

According to remainder theorem,

$R_1=p_1(1)=4(1)^3+3(1)^2-12(1)-5$

$p_1(1)=4+3-12-5$

$p_1(1)=-10$

$R_2=p_2(-2)=2(-2)^3+a(-2)^2-16(-2)+2$

$P_2(-2)=-16+a(4)+12+2$

$P_2(-2)=4a-2$

Substituting these values in the given condition implies

$3R_1+R_2+28=0$

$3(-10)+(4a-2)+28=0$

$-30+4a-2+28=0$

$4a-4=0$

$a=1$

The Correct option is C.

10. **Correct option: (D)**

Factorize the given polynomial implies

$$2\sqrt{2}x^2+5x+\sqrt{2}=0$$

$$2\sqrt{2}x^2+4x+x+\sqrt{2}=0$$

$$2\sqrt{2}x(x+\sqrt{2})+1(x+\sqrt{2})=0$$

$$(2\sqrt{2}x+1)(x+\sqrt{2})=0$$

The Correct option is D.

11. **Correct option: (D)**

According to the remainder theorem, if x-a is factor $p(a)=0$ here a=3

$p(3)=k^2(9)-3k-2=0$

$p(3)=9k^2-3k-2=0$

$p(3)=9k^2-6k+3k-2$

$p(3)=3k(3k-2)+1(3k-2)=0$

$p(3)=(3k+1)(3k-2)=0$

$k=\frac{-1}{3},k=\frac{2}{3}$

The Correct option is D.

12. **Correct option: (A)**

Given x^2-1 is factor of $p(x)=ax^4+bx^3+cx^2+dx+e$ and asked to find $a+c+e$

$$x^2-1=(x+1)(x-1)$$

$$x^2-1=(x+1)(x-1)$$
$$R_1=P_1(1)=a(1)^4+b(1)^3+c(1)^2+d(1)^1+e$$
$$R_1=P_1(1)=a+b+c+d+e$$
$$R_2=P_1(-1)=a(-1)^4+b(-1)^3+c(-1)^2+d(-1)^1+e$$
$$R_2=P_1(-1)=a-b+c-d+e$$
$$R_1=R_2=0$$

$R_1=R_2=0$ As $(x+1),(x-1)$ are factors of p(x).

As 1 and -1 are factors of polynomial p(x) from remainder theorem

$$p(1)=0\Rightarrow a+b+c+d+e=0$$

Similarly

$$p(-1)=0=a-b+c-d+e=0$$
$$\Rightarrow a+c+e=b+d$$

Also $R_1=R_2$

$$(a+c+e)+(b+d)=0$$
$$\Rightarrow (b+d)+(b+d)=0\Rightarrow b+d=0$$

This implies,

$$a+c+e=b+d$$
$$\Rightarrow a+c+e=0$$

The Correct option is A.

13. **Correct option: (C)**

Given that polynomial $p(x)=kx^2+2k^2x+m^3$ is divided by $x+k$.

As per remainder theorem

$$p(-k)=k(-k)^2+2k^2(-k)+m^3$$
$$p(-k)=k(-k)^2+2k^2(-k)+m^3$$
$$p(-k)=k^3-2k^2+m^3$$
$$p(-k)=m^3-k^3=0\Rightarrow m=k$$

Asked to find the value of k^2+mk+m^2

$$\therefore k^2+mk+m^2=m^2+m^2+m^2=3m^2.$$

The Correct option is none of above.

14. **Correct option: (B)**

Given $p_1(x)=6x^2+mx+4, p_2(x)=nx^2-11x+3$ are divided by $2x-1$

$$2x-1=0\Rightarrow x=\frac{1}{2}$$

According to remainder theorem,

$$p_1\left(\frac{1}{2}\right)=0=6\left(\frac{1}{2}\right)^2+m\left(\frac{1}{2}\right)-4=0$$
$$\Rightarrow 6\left(\frac{1}{4}\right)+\frac{m}{2}-4=0$$
$$=\frac{3}{2}+\frac{m}{2}-4=0\Rightarrow\frac{m+3}{2}-4=0$$
$$=m+3=8\Rightarrow m=5$$

In the similar manner

$$p_2\left(\frac{1}{2}\right)=0$$

$$\mathrm{n}\left(\frac{1}{2}\right)^2-11\left(\frac{1}{2}\right)+3=0$$

$$\Rightarrow \frac{n}{4}-\frac{11}{2}+\frac{3}{1}=0 \Rightarrow n-22+12=0$$

$$\therefore n=10$$

$$\therefore n^2-m^2+m-n=10^2-5^2+5-10$$

$$\Rightarrow 100-25+5-10=70$$

The Correct option is B.

15. **Correct option: (B)**

Given the polynomial $\sqrt{3}x^2+27x+5\sqrt{3}$ and asked to find Sum and products of zeros of a polynomial

Sum of zeros $=\frac{-b}{a}=\frac{-27}{\sqrt{3}}=-9\sqrt{3}$

Product of zeros$=\frac{c}{a}=\frac{5\sqrt{3}}{\sqrt{3}}=5$

The Correct option is B.

16. **Correct option: (D)**

Given that α,β is zeros of polynomial $4x^2-9=0$ and asked to find the product of zeros

Product of Zeros$=\frac{c}{a}=\frac{-9}{4}$

The Correct option is D.

17. **Correct option: (A)**

Given that $\frac{1}{4},-1$ are roots of a polynomial

Then the polynomial is $(x-\frac{1}{4})(\mathrm{x}+1)$

$$=(x-\frac{1}{4})(\mathrm{x}+1)$$

$$=\mathrm{x}^2+x-\frac{x}{4}-\frac{1}{4}$$

$$\Rightarrow x^2+\frac{3x}{4}-\frac{1}{4}=4x^2+3x-1$$

The Correct option is A.

18. **Correct option: (A)**

Zeros of a quadratic polynomial are 7, 8.

$$(\mathrm{x}-7)(\mathrm{x}-8)$$

$$\mathrm{x}^2-8x-7x+56$$

$$x^2-15x+56$$

Therefore polynomial consists of above zeros is $x^2-15x+56$

The Correct option is A.

19. **Correct option: (C)**

Given α, β are zeros of $x^2 + 4x + 3$.

$$\therefore \alpha + \beta = -4$$
$$\alpha\beta = 3$$
$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$
$$\alpha^2 + \beta^2 = 16 - 2(3)$$
$$\alpha^2 + \beta^2 = 16 - 6 = 10$$

The Correct option is C.

20. **Correct option: (B)**

: Given α, β are the zeros of $x^2 - x - 6$

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta}$$
$$\alpha + \beta = \frac{-b}{a} = 1$$
$$\alpha\beta = -6$$
$$\therefore \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{1}{-6} = \frac{-1}{6}$$

The Correct option is B.

LEVEL – II

1. Correct option: (B)

$$\frac{x^{100} - 3x^2 + x^6}{x^2}$$
$$x^{98} - 3 + x^4$$

Degree=98

The Correct option is B.

2. Correct option: (B)

Bionomical means a polynomial that is sum of two terms.

The Correct option is B.

3. Correct option: (D)

Zero polynomial means there are zero or no terms in that polynomial and hence degree of this polynomial is undefined.

The Correct option is D

4. Correct option: (C)

Given $p(x) = x^3 - 3x^2 + x + 2$ is divided by $g(x)$ with quotient x-2 and remainder -2x+4.

$$p(x) = g(x).quotient + remainder$$
$$p(x) = g(x).quotient + remainder$$
$$g(x) = \frac{p(x) - remainder}{quotient}$$
$$g(x) = \frac{x^3 - 3x^2 + x + 2 + 2x - 4}{x - 2}$$
$$g(x) = \frac{x^3 - 3x^2 + 3x - 2}{x - 2}$$

$$x-2\overline{)x^3-3x^2+3x-2}\quad\Big|\quad x^2-x+1$$

x^3-2x^2

$-x^2+3x-2$

$-x^2+2x$

$x-2$

$x-2$

0

Therefore, g(x) = x^2-x+1.

The Correct option is C.

5. **Correct option: (B)**

Given $p(x)=8x^4+12x^3+16x^2+8x+10$ and divided by 4x+2,

According to remainder theorem $4x+2=0 \Rightarrow x=\frac{-2}{4}=\frac{-1}{2}$

P ($\frac{-1}{2}$) = remainder.

$$p(\frac{-1}{2})=8(\frac{-1}{2})^4+12(\frac{-1}{2})^3+16(\frac{-1}{2})^2+8(\frac{-1}{2})+10$$

$$p(\frac{-1}{2})=\frac{\cancel{8}^{1}}{\cancel{16}_{2}}-\frac{\cancel{12}^{3}}{\cancel{8}_{2}}+\frac{\cancel{16}^{2}}{\cancel{4}_{1}}-\frac{\cancel{8}^{4}}{\cancel{2}_{1}}$$

$$p(\frac{-1}{2})=\frac{1}{2}-\frac{3}{2}+4-4+10$$

$$p(\frac{-1}{2})=-1+10=9$$

The Correct option is B.

6. **Correct option: (A)**

Given polynomial $p(x)=3x^4+6x^3-2x^2-10x-5$ with $\frac{\sqrt{5}}{3},-\frac{\sqrt{5}}{3}$ as zeros

So the factor of above polynomial is $\left(x+\frac{\sqrt{5}}{3}\right)\left(x-\frac{\sqrt{5}}{3}\right)=x^2-\frac{5}{3}$

If $p(x)$ is divided with above expression remaining zeros can be extracted.

$$x^2-\frac{5}{3}\overline{)3x^4+6x^3-2x^2-10x-5}\quad\Big|\quad 3x^2+6x+3$$

$3x^4-5x^2$

$6x^3+3x^2-10x-5$

$6x^3-10x$

$3x^2-5$

$3x^2-5$

0

Therefore, quotient is $3x^2+6x+3$

$$\Rightarrow 3x^2+6x+3$$
$$=3x^2+3x+3x+3$$
$$=3(x+1)(x+1)$$

This implies the other zeros are -1

The Correct option is A.

7. **Correct option: (B)**

Common factors of $3x^2-x-10, 2x^2-x-6$

$$3x^2-x-10$$
$$3x(x-2)+5(x-2)\ ,$$
$$(3x+5)(x-2)$$
$$2x^2-x-6$$
$$2x(x-2)+3(x-2)$$
$$(2x+3)(x-2)$$

$(x-2)$ is common factor between these two polynomials .

The Correct option is B

8. **Correct option: (A)**

Given that x^2+x-1 is factor of polynomial $p(x)=x^4+px^3+qx^2-1$ then the value of p,q is

As x^2+x-1 is the factor of p(x) dividing it with x^2+x-1 yields remainder zero

$$x^2+x-1\overline{)\,x^4+px^3+qx^2-1}\ \Big|\ x^2+(p-1)x+(q-p+2)$$
$$x^4+x^3-x^2$$
$$(p-1)x^3+(q+1)x^2-1$$
$$(p-1)x^3+(p-1)x^2-(p-1)x$$
$$(q+1-p+1)x^2+(p-1)x-1$$
$$(q+1-p+1)x^2+(q+1-p+1)x-(q+1-p+1)$$
$$(2p-3-q)x+(q-p+1)$$

By equating remainder to zero implies $(2p-3-q)x+(q-p+1)$

$$2p-3-q=0, q-p+1=0$$

This implies

$$2p-q=3---(1)$$
$$p-q=1----(2)$$

Solving above pair of equation by subtracting equation 2 from equation 1 implies that

$$p=2, q=1$$

The Correct option is. A

9. **Correct option: (A)**

Given that zeros of a polynomial is $\frac{1}{2}, \frac{3}{2}$

Quadratic equation having above zeros is

$$\left(x-\frac{1}{2}\right)\left(x-\frac{3}{2}\right)$$

$$\Rightarrow\left(\frac{2x-1}{2}\right)\left(\frac{2x-3}{2}\right)=\frac{1}{4}\left(4x^2-6x-2x+3\right)$$

$$=\frac{1}{4}\left(4x^2-8x+3\right)$$

The Correct option is A

10. **Correct option: (A)**

Given that $\alpha+\beta=\frac{1}{\sqrt{2}},\alpha\beta=\sqrt{2}$

Then the equation for quadratic polynomial is

$$x^2-(\alpha+\beta)x+\alpha\beta=0$$

$$\Rightarrow x^2-\left(\frac{1}{\sqrt{2}}\right)x+(\sqrt{2})=0$$

$$\Rightarrow\sqrt{2}x^2-x+2=0$$

The Correct option is A

11. **Correct option: (B)**

Real roots of $2x^{99}+3x^{98}+2x^{97}+3x^{96}+------+2x+3=0$ is

$$\Rightarrow(2x+3)\left(x^{98}+x^{97}+-----+1\right)=0$$

Here $\left(x^{98}+x^{97}+-----+1\right)$ Cannot be zero.

$$\therefore(2x+3)=0$$

$$x=-\frac{3}{2}$$

The Correct option is B

12. **Correct option: (A, B, C)**

Given $x-y$ is a factor of $p(x)=x^n-y^n$. Then 'n' is a natural number.

Let n=4, $\frac{x^4-y^4}{x-y}=\frac{\left(x^2+y^2\right)\left(x^2-y^2\right)}{x-y}$

Therefore, n is even number.

Let n=3, $\frac{x^3-y^3}{x-y}=\frac{x-y\left(x^2+xy+y^2\right)}{x-y}$

Therefore, divides when 'n' is odd also.

So, $x-y$ is a factor of x^n-y^n when 'n' is even and odd.

'n' is natural number to become $x-y$ factor of x^n-y^n.

Correct answer is A, B, C.

13. **Correct option: (B)**

Given $\alpha+\beta=14,\alpha\beta=-1$

Quadratic polynomial equation with sum of zeros and product of zeros is

$$x^2-(\alpha+\beta)x+\alpha\beta=0$$

$$x^2-\frac{1}{4}x-1=0$$

$$4x^2-x-4=0$$

The Correct option is B.

14. **Correct option: (B)**

Given $p_1(x) = ax^3 + 3x^2 - 3$, $p_2(x) = 2x^3 - 5x + a$ are divided by x-4.

Given that $R_1 = R_2$

From remainder theorem $p_1(4) = p_2(4) = 0$ as x-4 is factor of both the polynomials

$$p_1(4) = a(4)^3 + 3(4)^2 - 3$$

$$p_2(4) = 2(4)^3 - 5(4) + a$$

As $R_1 = R_2$

$$\Rightarrow 64a + 45 = 128 - 20 + a$$
$$\Rightarrow 64a - a = 128 - 20 - 45$$
$$\Rightarrow 63a = 63$$
$$\Rightarrow a = 1$$

The Correct option is B

15. **Correct option: (A)**

Given that $R_1 + R_2 = 0$ and from above problem the values of R_1, R_2 can be considered

Therefore

$$64a + 45 + 108 + a = 0$$
$$65a + 153 = 0$$
$$a = \frac{-153}{65}$$

The Correct option is A

16. **Correct option: (C)**

Given that $2R_1 - R_2 = 0$

And from above considerations $R_1 = 64a + 45, R_2 = 108 + a$

$2R_1 = 128a + 90$

$R_2 = 108 + a$

By using the given condition $2R_1 - R_2 = 0$

$$128a + 90 - 108 - a = 0$$
$$127a - 18 = 0$$
$$a = \frac{18}{127}$$

The Correct option is C

17. Correct Matching is: a-S, b-R, c-R, d-S

(a) If x^{11} is divisible by $x+1$ then remainder is

$$x+1\overline{)\begin{array}{r} x^{10}+x^9+x^8+x^7+x^6+x^5+x^4+x^3+x^2+x^1+1 \\ \hline x^{11} \\ x^{11}+x^{10} \\ \hline x^{10} \\ x^{10}+x^9 \\ \hline x^9 \\ x^9+x^8 \\ \hline x^8 \\ x^8+x^7 \\ \hline x^7 \\ x^7+x^6 \\ \hline x^6 \\ x^6+x^5 \\ \hline x^5 \\ x^5+x^4 \\ \hline x^4 \\ x^4+x^3 \\ \hline x^3 \\ x^3+x^2 \\ \hline x^2 \\ x^2+x \\ \hline x \\ x+1 \\ \hline -1 \end{array}}$$

Therefore remainder is -1

Therefore 'a' is mapped with S.

(b) For a polynomial f(x), $f(-3)=0$

According to remainder theorem $x=-3$ is zero of that polynomial f(x)

$\therefore x+3$ is factor of that polynomial

Therefore 'b' is matched with R.

(c) One of the factors of x^2+x-6

$$\begin{aligned} & x^2+x-6 \\ &= x^2+3x-2x-6 \\ &= x(x+3)-2(x+3) \\ &= (x+3)(x-2) \end{aligned}$$

Therefore 'c' is matched with R.

(d) If $x-2$ is factor of polynomial $p(x)=x^2+3px-2p$.

According to remainder theorem $p(2)=0$

$$p(2)=2^2+3p(2)-2p=0$$
$$p(2)=4+6p-2p=0$$

Therefore $$p(2)=4p+4=0$$
$$4p=-4$$
$$p=-1$$

Therefore 'd' is matched with S.

18. Correct Answer is: 0

Given that $1+x+x^2=0$, then $\left(ax^2+bx+c\right)+\left(a+bx^2+cx\right)+\left(ax+b+cx^2\right)=?$

$$\left(ax^2+bx+c\right)+\left(a+bx^2+cx\right)+\left(ax+b+cx^2\right)$$
$$\Rightarrow\left(ax^2+bx^2+cx^2\right)+\left(ax+bx+cx\right)+\left(a+b+c\right)$$
$$\Rightarrow x^2\left(a+b+c\right)+x\left(a+b+c\right)+\left(a+b+c\right)$$
$$\Rightarrow\left(a+b+c\right)\left(1+x+x^2\right)=0$$

19. Correct Answer is: 3

Given $8m^2-6mn$ is divided by 2m and quotient 4m-kn

$$\therefore 8m^2-6mn=\left(4m-kn\right)2m$$
$$\Rightarrow 8m^2-6mn=8m^2+2kmn$$
$$\Rightarrow k=3$$

20. Correct Answer is: 1568

Given α,β,γ are roots of cubic equation $x^3-13x^2+56x+17$ then $\alpha\beta+\gamma\beta+\alpha\gamma=\frac{x}{28}$

Then x=?

From the given cubic equation the value of $\alpha\beta+\gamma\beta+\alpha\gamma$ is 56

$$\therefore 56=\frac{x}{28}\Rightarrow x=28\times 56=1568$$

STUDENT PRACTICE QUESTIONS

1. Correct option: (A)

Polynomial is defined as an expression consists of coefficients, variables and exponentials (positive), and its standard form is $p(x)=a_nx^n+a_{n-1}x^{n-1}+.....+a_0$

From the given options only Option 'A' can be realized in above form

The Correct option is A.

2. Correct option: (C)

The degree of a polynomial is explained as the highest power of the variable present in any of its terms.

In the given polynomial $x-x^5+1$ highest power is 5.Therefore, the degree of the given polynomial is 5.

The Correct option is C

3. Correct option: (B)

Given polynomial is $\frac{\sqrt{2}x^5-x^{10}+1}{x}$.

The degree of a polynomial is explained as the highest power of the variable present in any of its terms. Here, 10 is the degree of polynomial.

The Correct option is B.

4. Correct option: (B)

The number of terms in a polynomial is determined by how many separate expressions are added or subtracted within the polynomial.

In the polynomial $4x^2+5x+3$, there are three separate expressions: $4x^2$, 5x, and 3.

Each of these expressions is a term. Therefore, the polynomial $4x^2+5x+3$ contains 3 terms.

The Correct option is B.

5. Correct option: (C)

Given polynomial equation is $4+x^3-x+2x^2$. It can also be written as x^3+2x^2-x+4.

It is asked to find the value at x=-2,

$\Rightarrow (-2)^3+2(-2)^2-(-2)+4$

$\Rightarrow -8+8+2+4$

$\Rightarrow 6$

The Correct option is C.

6. Correct option: (B):

Given polynomial is $3x^2-1$. Which equal to zero. It can be written as

$3x^2-1=0$

$3x^2=1$

$x^2=\frac{1}{3}$

$x=\pm\frac{1}{\sqrt{3}}$

Therefore, zeros of above polynomial is at $x=+\frac{1}{\sqrt{3}},-\frac{1}{\sqrt{3}}$.

The Correct option is B.

7. Correct option: (A)

If $7+3x$ divides $3x^2+7x$, By using remainder theorem, we can conclude that if polynomial p(x) is divide with x-a, then remainder will be equal to p(a).

Here x= $\frac{-7}{3}$

$\Rightarrow p\left(\frac{-7}{3}\right)=3\left(\frac{-7}{3}\right)^2+7(\frac{-7}{3})$

$\therefore p\left(\frac{-7}{3}\right)=3\left(\frac{49}{9}\right)-\frac{49}{3}=0$

The Correct option is A

8. Correct option: (A)

Given $x^{17}+x^{17}+x^{13}+x^7+x^5+x^3$ is divided by x^2+1.

as $x^2+1=0$

$x^2+1=0$

$x^2=-1$

$x=\pm i$

$$p(\mathrm{i}) = (\mathrm{i})^{19} + (\mathrm{i})^{17} + (\mathrm{i})^{13} + (\mathrm{i})^{7} + (\mathrm{i})^{5} + (\mathrm{i})^{3}$$
$$p(\mathrm{i}) = (\mathrm{i}^2)^9.\mathrm{i} + (\mathrm{i}^2)^8.\mathrm{i} + (\mathrm{i}^2)^6.\mathrm{i} + (\mathrm{i}^2)^3.\mathrm{i} + (\mathrm{i}^2)^2.\mathrm{i} + (\mathrm{i}^2)\mathrm{i}$$
$$p(\mathrm{i}) = -i + i + i - i + i - i$$
$$p(\mathrm{i}) = 0$$

The Correct option is A.

9. **Correct option: (B)**

Given $3x^3 + x^2 + 2x + 5$ is divided by $1 + 2x + x^2$ Then remainder

$$\begin{array}{r} 3x-5 \\ x^2+2x+1 \overline{)\ 3x^3 + x^2 + 2x + 5} \\ 3x^3 + 6x^2 + 3x \\ \hline -5x^2 - x + 5 \\ -5x^2 - 10x - 5 \\ \hline 9x + 10 \end{array}$$

Therefore, the remainder is $9x + 10$

The Correct option is B.

10. **Correct option: (C)**

$p(\mathrm{x}) = 2\mathrm{x}^2 + 3\mathrm{x} + 1$ is divided by $x + 2$

According to remainder theorem,

$p(-2) = 2(-2)^2 + 3(-2) + 1$

$p(-2) = 2(4) - 6 + 1$

$p(-2) = 9 - 6 = 3$

The Correct option is C.

11. **Correct option: (B)**

Given $p(\mathrm{x}) = \mathrm{x}^4 - 5\mathrm{x} + 6$ divided by $2 - x^2$ then remainder and quotient is

$$\begin{array}{r} -x^2-2 \\ 2\text{-}x^2 \overline{)\ x^4 - 5x + 6} \\ x^4 - 2x^2 \\ \hline -2x^2 - 5x + 6 \\ -2x^2 - 4 \\ \hline -5x + 10 \end{array}$$

Therefore, quotient is $-x^2 - 2$ and remainder is $-5x + 10$

The Correct option is B.

12. **Correct option: (A)**

Given polynomial is $3x^3 + ax^2 + 3x + 5$ and $a - 2x + x^2 + 4x^3$ have same remainder divided by $x - 2$.

$$p_1(\mathrm{x}) = 3x^3 + ax^2 + 3x + 5,\ p_2(\mathrm{x}) = a - 2x + x^2 + 4x^3$$
$$x - 2 \Rightarrow x = 2$$

$$p_1(x) = 3x^3 + ax^2 + 3x + 5$$
$$p_1(2) = 3(2)^3 + a(2)^2 + 3(2) + 5$$
$$\Rightarrow p_1(2) = 4a + 35$$
$$p_2(x) = a - 2x + x^2 + 4x^3$$
$$p_2(2) = a - 2(2) + (2)^2 + 4(2)^3$$
$$\Rightarrow p_2(2) = a - 2(2) + (2)^2 + 4 = a + 32$$

As both leave the same remainders and hence $p_1(2) = p_2(2)$

$4a + 35 = a + 32$

$3a = 32 - 35$

$3a = -3 \Rightarrow a = -1$

The Correct option is A

13. **Correct option: (C)**

According to remainder theorem,

$$R_1 = p(1) = (+1)^3 - a(+1)^2 - 13(+1) + b$$
$$R_1 = p(1) = 1 - a - 13 + b = -a - 12 + b$$
$$R_2 = p(-3) = (-3)^3 - a(-3)^2 - 13(-3) + b$$
$$R_2 = p(-3) = -27 - 9a + 39 + b = -9a + 12 + b$$

Since both are factors

$$p(\mathrm{x}) = \mathrm{R}_1 = R_2$$
$$\Rightarrow -a - 12 + b = -9a + 12 + b$$
$$\Rightarrow -a + 9a = 2a$$
$$\Rightarrow a = 3$$

Now, it's known that $p_1(+1) = 0$

$$(1)^3 - 3(1)^2 - 13(1) + \mathrm{b} = 0$$
$$1 - 3 - 13 + \mathrm{b} = 0$$
$$\mathrm{b} = 15$$
$$\therefore \mathrm{a} + \mathrm{b} = 3 + 15 = 18.$$

The Correct option is C.

14. **Correct option: (B)**

Given $\mathrm{x}^4 + 2\mathrm{x}^3 + 3\mathrm{x}^2 - nx + 3$ may divisible by '3' this implies from remainder theorem

$$R_1 = p(3) = (3)^4 - 2(3)^3 + 3(3)^2 - n(3) + 3 = 0$$
$$R_1 = p(3) = 81 - 54 + 27 - 3n + 3 = 0$$
$$R_1 = p(3) = 81 - 27 + 3 - 3n = 0$$
$$R_1 = p(3) = 57 - 3n = 0 \Rightarrow n = 19$$

The Correct option is B.

15. **Correct option: (C)**

Given $p(\mathrm{x}) = \mathrm{kx}^3 + 9\,\mathrm{x}^2 + 4\,\mathrm{x} - 8$ is divided by $x + 3$ and remainder $R_1 = -20$

According to remainder theorem $R_1 = -20$ when $p(-3)$

$p(\mathrm{x}) = \mathrm{kx}^3 + 9\,\mathrm{x}^2 + 4\,\mathrm{x} - 8$

$p(-3) = \mathrm{k}(-3)^3 + 9(-3)^2 + 4(-3) - 8 = -20$

$p(-3) = -27k$

$27k = 81 - 12 - 8 + 20$

$27k = 81$

$k = 3$

The Correct option is C.

16. Correct option: (D)

Given that α, β is zeros of polynomial $x^2 - 2x - 8 = 0$ and asked to find the product of zeros

Product of Zeros of a polynomial = $\frac{c}{a} = \frac{-8}{1} = -8$

The Correct option is D.

17. Correct option: (A)

Given $p(x) = x^3 - 4x^2 + 5x - 2$ and $\alpha, \beta, \gamma = 0$

Sum of zeros $\alpha + \beta + \gamma = 0$

$$\alpha + \beta + \gamma = \frac{-b}{a}$$

$$\alpha + \beta + \gamma = -(\frac{-4}{1})$$

$$\alpha + \beta + \gamma = 4$$

Sum of products of zeros $\alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}$

$$\alpha\beta + \beta\gamma + \gamma\alpha = \frac{5}{1} = 5$$

The Correct option is A.

18. Correct option: (C)

Given zeros of a cubic polynomial is 1, 2,-3.

Polynomial consists of above zeros is $(x-1)(x-2)(x+3)$

$$= x^3 + 3x^2 - 3x^2 - 9x + 2x + 6$$

$$= x^3 - 7x + 6$$

The Correct option is C.

19. Correct option: (D)

Zeros of a quadratic polynomial is $\sqrt{2}, 0$

Polynomial consists of above zeros is

$$(x - \sqrt{2})(x - 0)$$

$$x^2 - \sqrt{2}x$$

The Correct option is D.

20. Correct option: (C)

Given the zeros of a cubic polynomial 3, –1, $\frac{-1}{3}$.

Polynomial consists of above zeros is $(x-3)(x+1)(x+\frac{1}{3})$

$$= (x-3)(x+1)(x+\frac{1}{3})$$

$$= (x^2 - 2x - 3)(x+\frac{1}{3})$$

$$= x^3 + \frac{1}{3}x^2 - 2x^2 - \frac{2}{3}x - 3x - 1$$

$$= x^3 + x^2(\frac{1}{3} - 2) - x(\frac{2}{3} + 3)$$

$$\Rightarrow x^3 - \frac{5}{3}x^2 - \frac{11}{3}x - 1 = 3x^3 - 5x^2 - 11x - 3$$

The Correct option is C.

21. **Correct option: (C)**

Given α, β are zeros of $x^2 + 8x + 6$.

$$(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$$
$$(\alpha - \beta)^2 = 64 - 24$$
$$(\alpha - \beta)^2 = 40$$
$$(\alpha - \beta) = \sqrt{40}$$

The Correct option is C.

22. **Correct option: (C)**

Given α, β are zeros of $2x^2 - 4x + 5$,

$$\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$$
$$\alpha + \beta = \frac{-b}{a} = \frac{4}{2} = 2$$
$$\alpha\beta = \frac{5}{2}$$
$$\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$$
$$\alpha^3 + \beta^3 = (2)^3 - \frac{-15}{2}(2)$$
$$\alpha^3 + \beta^3 = 8 - 15 = -7$$

The Correct option is C.

23. **Correct option: (D)**

Given α, β are zeroes of $x^2 - \sqrt{2}x$.

$$\alpha + \beta = \sqrt{2}$$
$$\alpha\beta = 0$$
$$(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$$
$$(\alpha - \beta)^2 = (\sqrt{2})^2 - 0$$
$$(\alpha - \beta) = \sqrt{2}$$
$$\therefore \alpha^2 - \beta^2 = (\alpha + \beta)(\alpha - \beta)$$
$$\alpha^2 - \beta^2 = (\sqrt{2})(\sqrt{2})$$
$$\alpha^2 - \beta^2 = 2$$

The Correct option is D.

24. **Correct option: (C)**

Given α, β are zeroes of $x^2 - 11x + 30$.

$\alpha^3 - \beta^3 = (\alpha - \beta)^3 + 3\alpha\beta(\alpha - \beta)$

$\alpha + \beta = 11$

$\alpha\beta = 30$

$(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$

$(\alpha - \beta)^2 = 121 - 120 = 1$

$\alpha - \beta = 1$

$\therefore \alpha^3 - \beta^3 = (1)^3 + 3(30)(1)$

$\alpha^3 - \beta^3 = 91$

The Correct option is C.

25. **Correct option: (B)**

Given α, β are the zeros of x^2+2x+1

$$\alpha+\beta=-2$$
$$\alpha\beta=1$$
$$\frac{1}{\alpha^2}+\frac{1}{\beta^2}=\frac{\alpha^2+\beta^2}{\alpha\beta}$$
$$\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta$$
$$\alpha^2+\beta^2=4-2=2$$
$$\frac{\alpha^2+\beta^2}{\alpha\beta}=\frac{2}{1}=2$$

The Correct option is B.

26. Correct option: (C)

Cubic polynomial is a polynomial of degree 3.

The Correct option is C.

27. **Correct option: (B)**

Given that $(x+2)$ is factor of $p(x)=x^2+ax+2b, \text{a}+\text{b}=4$.

According to remainder theorem $\text{p}(-2)=0$ as $(x+2)$ is factor of $p(x)$

$$\Rightarrow \text{p}(-2)=(-2)^2+a(-2)+2b=0$$
$$\Rightarrow \text{p}(-2)=-2a+2b+4=0$$
$$2a-2b=4$$
$$a+b=4$$

$a-b=2$

$a+b=4$

Solving above two equations implies that $2a=6, a=3, b=1$

The Correct option is B.

28. **Correct option: (A)**

It is asked to find $(x+2)$ is root of which given polynomial

According to remainder theorem $p(-2)=0$

Let's consider polynomial in Option A

$p(x)=x^5-4x^4-3x^3+8x^2-14x+12$

$\therefore p(-2)=(-2)^5-4(-2)^4-3(-2)^3+8(-2)^2-14(-2)+12=0$

The Correct option is A.

29. **Correct option: (A)**

Remainder when x^4-y^4 divided with $x-y$

$$\frac{x^4-y^4}{x-y}$$
$$=\frac{(x^2-y^2)(x^2+y^2)}{x-y}$$
$$=\frac{(x-y)(x+y)(x^2+y^2)}{x-y}=(x+y)(x^2+y^2)$$

Since x^4-y^4 is completely divided with $x-y$ remainder will be zero.

The Correct option is A

30. **Correct option: (B)**

Given that sum of zeros $\Rightarrow \alpha+\beta+\gamma=2$

The product of its zeroes taken two at a time $\Rightarrow \alpha\beta+\beta\gamma+\gamma\alpha=-5$

The product of its zeroes $\Rightarrow \alpha\beta\gamma=-20$

Then the equation of cubic polynomial is given as

$$x^3-(\alpha+\beta+\gamma)x^2+(\alpha\beta+\beta\gamma+\gamma\alpha)x-\alpha\beta\gamma=0$$
$$\Rightarrow x^3-(2)x^2+(-5)x-(-20)=0$$
$$\Rightarrow x^3-2x^2-5x+20=0$$

The Correct option is B

31. **Correct option: (A)**

Given that zeros of a polynomial are $\frac{1}{\sqrt{2}},\sqrt{2}$

Then the equation of the polynomial having two zeros $\frac{1}{\sqrt{2}},\sqrt{2}$ is

$$\left(x-\frac{1}{\sqrt{2}}\right)\left(x-\sqrt{2}\right)$$
$$\Rightarrow x^2-\frac{1}{\sqrt{2}}x-\sqrt{2}x+1$$
$$\Rightarrow x^2-\left(\frac{1}{\sqrt{2}}+\sqrt{2}\right)x+1$$
$$\Rightarrow \sqrt{2}x^2-3x+\sqrt{2}$$

The Correct option is A

32. **Correct option: (D)**

Given $\alpha+\beta=3$, $\alpha\beta=2$.

$$\left(1+\frac{\alpha}{\beta}\right)\left(1+\frac{\beta}{\alpha}\right)$$
$$\left(\frac{\beta+\alpha}{\beta}\right)\left(\frac{\alpha+\beta}{\alpha}\right)$$
$$\left(\frac{(\alpha+\beta)^2}{\alpha\beta}\right)=\frac{9}{2}$$

The Correct option is D

33. **Correct option: (B)**

Given polynomial is $11x^2-88x^3+14x^4$ a Bi-quadratic equation as highest degree of this polynomial is 4

The Correct option is B

34. **Correct option: (B)**

Given $2x^{3m+2n}+4x^{5m+2n}+2x^{m+n}$ is a polynomial with $m=2,n=1$.

After substituting the values of m and n then polynomial $p(x)$ will become

$$p(x)=2x^{3(2)+2(1)}+4x^{5(2)+2(1)}+2x^{(2)+(1)}$$
$$p(x)=2x^8+4x^{12}+2x^3$$

Since highest power of x is equal to 12.Degree of p(x) is equal to 12.

The Correct option is B

35. **Correct option: (A, B)**

Given that $x-2, x+3$ both are factors of $p(x) = x^3 + ax^2 + bx - 12$.

$$(x-2)(x+3) = x^2 - 5x - 6$$

To get the value of a & b p(x) should divide by applying remainder theorem.

$$p(+2) = p(-3) = 0$$

$$p(+2) = p(-3) = 0$$

$$p(+2) = 2^3 + a(2^2) + b(2) - 12$$

$$p(+2) = 8 + 4a + 2b - 12 = 0$$

$$p(+2) = 4a + 2b = 4$$

$$p(-3) = (-3)^3 + a(-3)^2 + b(-3) - 12$$

$$p(-3) = -27 + 9a - 3b - 12 = 0$$

$$p(-3) = 9a - 3b = 39$$

$$3a - b = 13 \rightarrow \times 2$$

$$4a + 2b = 4 \rightarrow \times 1$$

$$_____$$

$$6a - 2b = 26$$

$$4a + 2b = 4$$

$$_____$$

$$10a = 30$$

$$a = 3$$

$$4(3) + 2b = 4$$

$$2b = 4 - 12$$

$$b = -4$$

Correct answer is A, B.

36.**Correct option: (A, B, C)**

Given factors of polynomial $p(x) = x^3 + 6x^2 + 11x + 6$.

As this is polynomial of degree '3'.

There are 3 roots and products of roots is equal to 6 and hence possible factors are $\pm 1, \pm 2, \pm 3, \pm 6$.

Let x=1, then p(1) should be zero.

$$p(1) = 1^3 + 6(1)^2 + 11(1) + 6$$

$$p(1) = 1 + 6 + 4 \neq 0$$

$$p(-1) = (-1)^3 + 6(-1)^2 + 11(-1) + 6$$

$$p(-1) = -1 + 6 - 11 - 6 = 0$$

Therefore x+1 is the factor.

$$
\begin{array}{r}
x^2+5x+6 \\
x+1\overline{)\,x^3+6x^2+11x+6} \\
x^3+x^2 \\
\hline
5x^2+11x+6 \\
5x^2+5x \\
\hline
6x+6 \\
6x+6 \\
\hline
0 \\
\hline
\end{array}
$$

Therefore the quotient x^2+5x+6 is another factor.

Factorizing it implies

$$
\begin{aligned}
& x^2+5x+6 \\
&= x^2+3x+2x+6 \\
&= x(x+3)+2(x+3) \\
&= (x+2)(x+3)
\end{aligned}
$$

$$\therefore x^3+6x^2+11x+6=(x+3)(x+2)(x+1)$$

The Correct option is A, B, C.

37. **Correct option: (A, B)**

Given $\sqrt{2}, -\sqrt{2}$ are zeros of $p(x) = 2x^4 - 3x^3 - 3x^2 + 6x - 2$

The other zeros of p(x) can be determined by dividing $p(x)$ with $(x-\sqrt{2})(x+\sqrt{2})$

$$
\begin{array}{r}
2x^2-3x+1 \\
x^2-2\overline{)\,2x^4-3x^3-3x^2+6x-2} \\
2x^4-4x^2 \\
\hline
-3x^3+x^2+6x-2 \\
-3x^3+6x \\
\hline
x^2-2 \\
x^2-2 \\
\hline
0 \\
\hline
\end{array}
$$

Therefore, quotient is $2x^2 - 3x + 1$, factorizing this will gives another pair of zeros

$$
\begin{aligned}
& 2x^2-3x+1 \\
&= 2x^2-2x-x+1 \\
&= 2x(x-1)-1(x-1) \\
&= (2x-1)(x-1)
\end{aligned}
$$

$x = \frac{1}{2}, 1$ are the remaining zeros

The Correct option is A, B.

38. **Correct option: (A, B)**

α, β are the zeros of polynomial $p(x) = x^2 - 20x - 30$

Then, sum of zeros is $\frac{-b}{a}$ this implies $\frac{-b}{a} = \frac{-(-20)}{1}$

Therefore $\frac{-b}{a} = \alpha + \beta = 20$

The Correct option is A, B.

39. **Correct option: (A)**

Given $f(x) = x^5 - x^4 + x^3 - x^2 + x - 1$.

Here $a_0 = 1, a_1 = -1, a_2 = 1, a_3 = -1, a_4 = +1, a_5 = 1$

$$\therefore a_0 + a_1 + a_2 + a_3 + a_4 + a_5 = 1 - 1 + 1 - 1 + 1 - 1 = 0$$

It is known that sum of all coefficients in an polynomial is zero then $(x-1)$ is factor of that polynomial

Therefore $(x-1)$ is a factor.

The Correct option is A

40. **Correct option: (B)**

Given $g(x) = x^3 + x^2 + x + 1$

Here

$$a_0 = 1, a_1 = 2, a_2 = 1, a_3 = 1$$

It is clearly evident that $a_0 + a_2 = a_1 + a_3$

It is known that sum of all odd and even coefficients of a polynomial is equal then $(x+1)$ is factor of that polynomial

The Correct option is B.

41. **Correct option: (D)**

Given $f(x) = x^3 - 7x + 6$ it can be written as $f(x) = x^3 + 0x^2 - 7x + 6$

From the polynomial the values of coefficients extracted as $a_0 = 1, a_1 = 0, a_2 = -7, a_3 = 6$

It is clear that $a_0 + a_2 = a_1 + a_3$

By substituting

$$\Rightarrow 1 + (-7) = 0 + 6$$
$$\Rightarrow 6 = 6$$

It is known that sum of all odd and even coefficients of a polynomial is equal then $(x+1)$ is factor of that polynomial

The Correct option is D.

42. **Correct option: (B)**

Given that $(3x-1)^7 = a_7x^7 + a_6x^6 + ...a_0$ then $a_0 + a_1 + a_2 + a_3 + a_4 + a_5 + a_6 + a_7$ is

$$(3x-1)^7 = a_7x^7 + a_6x^6 + ...a_0$$

Substituting the value of x=1 implies

$$(3-1)^7 = a_7 + a_6 + ...a_0$$

Therefore $a_0 + a_1 + a_2 + a_3 + a_4 + a_5 + a_6 + a_7 = 128$

The Correct option is B.

43. **Correct option: (C)**

Given that $\frac{1}{2},\frac{1}{3},\frac{1}{5}$ are zeros of given cubic polynomial $ax^3+bx^2+cx+d=0$

$$\alpha=\frac{1}{2},\beta=\frac{1}{3},\gamma=\frac{1}{5}$$

Therefore $\alpha+\beta+\gamma=\frac{1}{2}+\frac{1}{3}+\frac{1}{5}$

$$\alpha+\beta+\gamma=\frac{15+10+6}{30}=\frac{31}{30}$$

The Correct option is C.

44. **Correct option: (B)**

As mentioned in earlier problem $\alpha=\frac{1}{2},\beta=\frac{1}{3},\gamma=\frac{1}{5}$

$$\alpha\beta+\beta\gamma+\alpha\gamma=\left(\frac{1}{2}\right)\left(\frac{1}{3}\right)+\left(\frac{1}{3}\right)\left(\frac{1}{5}\right)+\left(\frac{1}{5}\right)\left(\frac{1}{2}\right)$$

$$\alpha\beta+\beta\gamma+\alpha\gamma=\frac{1}{6}+\frac{1}{15}+\frac{1}{10}$$

$$\alpha\beta+\beta\gamma+\alpha\gamma=\frac{5+2+3}{30}$$

$$\alpha\beta+\beta\gamma+\alpha\gamma=\frac{1}{3}$$

The Correct option is B.

45. **Correct option: (A)**

As mentioned earlier $\alpha=\frac{1}{2},\beta=\frac{1}{3},\gamma=\frac{1}{5}$

$$\alpha\beta\gamma=\left(\frac{1}{2}\right)\left(\frac{1}{3}\right)\left(\frac{1}{5}\right)$$

$$\alpha\beta\gamma=\frac{1}{30}$$

The Correct option is A.

46. **Correct Matching is: a-T, b-P, c-QS, d-R**

(a) Given '2' is root of polynomial $p(x)=x^4+px+26=0$

From remainder theorem if '2' is root of polynomial p(x) then p(2)=0

$$\therefore p(2)=2^4+p(2)+26=0$$

$$p(2)=2p+30=0$$

$$2p+30=0$$

$$p=-15$$

Therefore 'a' is matched with T.

(b) Given '5' is root of polynomial $p(5)=x^2-(k-1)x+10=0$

From remainder theorem if '5' is root of p(x) then p(5)=0

$$p(5)=5^2-(k-1)5+10=0$$

$$p(5)=25-5k+5+10=0$$

$$40-5k=0$$

$$k=8$$

Therefore 'b' is matched with P.

(c) Given zeros of polynomial $4x^2-9$

$$x^2=\frac{9}{4}$$

$$x=\pm\frac{3}{2}$$

Therefore 'c' is matched with Q, S.

(d) Given $(x-a)(x-b)(x-c)(x-d)$

It is clear that highest power of 'x' is equal to 4 hence the degree is 4.
Therefore 'd' is matched with R

47. Correct Matching is: a-R, b-S, c-T, d-P

48. Correct Answer is: 0

Given polynomial $p(x)=x^5-x^4+x^3-x^2+x-1$ is divided by $x-1$then remainder is

Here $a_0=1, a_1=-1, a_2=1, a_3=-1, a_4=1, a_5=-1$

$$a_0+a_1+a_2+a_3+a_4+a_5=0$$

Therefore $x-1$ is factor of p(x).

If any polynomial is divided by its factor then remainder will be zero.

49. Correct Answer is: 2

Given $x-2$ is a factor of $p_1(x)=x^2+cx+d$, $p_2(x)=x^2+ax+b$

According to remainder theorem $p_1(2)=p_2(2)=0$

$$p_1(2)=2^2+c(2)+d=0$$

$$p_2(2)=2^2+a(2)+b=0$$

$$4+2c+a=4+2a+b$$

$$b-d=2c-2a$$

$$b-d=2(c-a)$$

$$\frac{b-d}{(c-a)}=2$$

50. Correct Answer is: 3

Given $x-2$ and $x+3$ both are factors of $p(x)=x^3+ax^2+bx-12$

According to remainder theorem if $x-2$ and $x+3$ are factors of p(x) then $p(2)=p(-3)=0$

$$p(x)=x^3+ax^2+bx-12$$

$$p(2)=8+4a+2b-12=0$$

$$\Rightarrow 2a+b=2$$

$$p(-3)=-27+9a-3b-12=0$$

$$\Rightarrow 9a-3b=39$$

$$\Rightarrow 3a-b=13$$

Solving the above equation pair by adding them together implies that a=3

51. Correct Answer is: 4

Given that $x-1, x+1, x-2$ are factors of $x^4+(p-3)x^3-(3p-5)x^2+(2p-9)x+6$ then p=?

According to remainder theorem $p(1)=p(-1)=p(2)=0$

Let $p(-1)=0$

Therefore $p(-1)=0$ implies that

$$(-1)^4+(p-3)(-1)^3-(3p-5)(-1)^2+(2p-9)(-1)+6=0$$

$$\Rightarrow 24-6p=0 \Rightarrow p=4$$

52. Correct Answer is: 1

Sum of roots of $2x^2-179x+325=0$ is $\frac{179}{b}, b=?$

Sum of roots of a polynomial is given as

$$\frac{-b}{a}=\frac{179}{1}$$

It is given that $\frac{179}{b}$ is equal to sum of roots which implies $\frac{-b}{a}=\frac{179}{1}$

Therefore

$$\frac{179}{b}=179\Rightarrow b=1$$

53. Correct Answer is: 5

Given that α,β are roots of $x^2-15x+3=0$ then $\frac{1}{\alpha}+\frac{1}{\beta}$

From the given equation

$$\alpha+\beta=15$$
$$\alpha\beta=3$$

Therefore

$$\frac{1}{\alpha}+\frac{1}{\beta}=\frac{\alpha+\beta}{\alpha\beta}=\frac{15}{3}=5$$

Pair of Linear Equations

Solutions

LEVEL-I

1. Correct option: (B)

The given equation is

$$6-3(2x-3)=5(6-x)-4x$$
$$\Rightarrow 6-6x+9=30-5x-4x$$
$$\Rightarrow -6x+9x=30-15\Rightarrow x=5$$

2. Correct option: (B)

$$\frac{2x+3}{3x+4}=\frac{2x-3}{3x-2}$$
$$\Rightarrow (2x+3)(3x-2)=(2x-3)(3x+4)$$
$$\Rightarrow 6x^2+5x-6=6x^2-x-12$$
$$\Rightarrow 6x=-6\Rightarrow x=-1$$

3. Correct option: (D)

$$2+\frac{2x-3}{2x+3}=\frac{3x+4}{x+2}$$
$$\Rightarrow \frac{4x+6+2x-3}{2x+3}=\frac{3x+4}{x+2}$$
$$\Rightarrow \frac{6x+3}{2x+3}=\frac{3x+4}{x+2}\Rightarrow (6x+3)(x+2)=(3x+4)(2x+3)$$
$$\Rightarrow 6x^2+15x+6=6x^2+17x+12$$
$$\Rightarrow -2x=6\Rightarrow x=-3$$

4. Correct option: (A)

$$x+\frac{1}{x}=5$$

Squaring on both sides implies

$$x^2+\frac{1}{x^2}+2(x)\left(\frac{1}{x}\right)=25$$

$$\Rightarrow x^2+\frac{1}{x^2}=25-2\Rightarrow x^2+\frac{1}{x^2}=23$$

5. Correct option: (C)

$$\frac{x}{2}-\frac{1}{5}=\frac{x}{3}+\frac{1}{4}$$

Simplifying the above expression implies

$$\frac{x}{2}-\frac{x}{3}=\frac{1}{5}+\frac{1}{4}$$

$$\Rightarrow \frac{x}{6}=\frac{9}{20}\Rightarrow x=\frac{27}{10}$$

6. Correct option: (A)

Sol. Let the number of students be 'x'

If each student was given by two chocolates implies total number of chocolates are with students will be equal to 2x

After distributing accordingly number of chocolates left with teacher is 7

This entire hypothesis can be represented in the form of equation as follows

$$2x+7=45$$

$$\Rightarrow 2x=38$$

$$\Rightarrow x=19$$

7. Correct option: (A)

Sol. Let the two numbers 'x' and 'y'

Given that their sum and sum of their squares is 15 and 113 respectively

This can be represented in the form of equation as follows

$$x+y=15$$

$$x^2+y^2=113$$

Considering 1st equation and squaring it on both sides implies

$$(x+y)^2=15^2$$

$$\Rightarrow x^2+y^2+2xy=225$$

$$\Rightarrow 113+2xy=225$$

$$\Rightarrow 2xy=112\Rightarrow xy=56$$

$$\therefore (x-y)^2=(x+y)^2-4xy=225-224=1$$

$$\Rightarrow x-y=1$$

Now we have an equation pair with only 'x' and 'y'

$$x+y=15$$

$$x-y=1$$

Adding both of them yields the value of 'x' as 8

Substitute the value of 'x' in any of the equation pair yields the value of 'y=7'

8. Correct option: (B)

Let the sides of perpendicular of a right angle be 3x and 4 respectively

The perimeter of the right angle triangle is given by 96cm. Formula for the perimeter of a right angle triangle is given below

$$p = a + b + \sqrt{a^2 + b^2}$$

Here a, b are the perpendicular sides of the right angle triangles respectively.

Substitute the values of perpendicular sides of right angle triangle in above formula implies

$$p = a + b + \sqrt{a^2 + b^2}$$
$$\Rightarrow 96 = 3x + 4x + \sqrt{(3x)^2 + (4x)^2}$$
$$\Rightarrow x = 8cm$$

One perpendicular side of a right angle triangle will be 3x=24cm

Another perpendicular side of a right angle triangle will be 4x=32cm

If two perpendicular sides of the right angle triangle is known the other side can be calculated by using Pythagarous theorem

According to it $d^2 = \sqrt{p_1^2 + p_2^2}$

Here d is diagonal and p_1, p_2 are perpendicular sides of right angle triangle respectively

$$d^2 = \sqrt{24^2 + 32^2} \Rightarrow d = 40cm$$

The largest side of the triangle is 40cm

9. Correct option: (D)

$$x + y = 3$$
$$xy = 2$$

$$(x - y)^2 = (x + y)^2 - 4xy$$
$$\Rightarrow (x - y)^2 = (3)^2 - 4(2) = 1$$
$$\Rightarrow x - y = 1$$

Solving $x + y, x - y$ together implies

$$x + y = 3$$
$$x - y = 1$$

Adding both the equations implies $x = 2$ this yield $y = 1$

$$\Rightarrow x^3 - y^3 = 2^3 - 1^3 = 7$$

10. Correct option: (A)

$$4x + 3y = 14$$
$$9x - 5y = 55$$

To solve this linear equation pair it is essential to find the values of x & y.

In order to do that either of 'x' or 'y' term should be eliminated.

Let's eliminate y term, multiply 1st equation with '5' and multiply second equation with '3' implies

$$20x + 15y = 70$$
$$27x - 15y = 165$$

Adding above two equations implies

$$47x = 235$$

$$\Rightarrow x = \frac{235}{47} = 5$$

Now substitute value of 'x' in any of the equation will yields the value of 'y'

$$4(5) + 3y = 14$$

$$\Rightarrow 3y = 14 - 20$$

$$\Rightarrow y = \frac{-6}{3} = -2$$

Solution of given equation pair is (5,-2)

11. **Correct option: (A)**

$$3^{x+y} = 27$$

$$3^{x-y} = 243$$

$$3^{x+y} = 27 = 3^3$$

$$\Rightarrow x + y = 3 (\because if a^m = a^n \Rightarrow m = n)$$

$$3^{x-y} = 243 = 3^5$$

$$\Rightarrow x - y = 5 (\because if a^m = a^n \Rightarrow m = n)$$

Adding both the equations implies

$$2x = 8 \Rightarrow x = 4$$

12. **Correct option: (B)**

$$\frac{1}{2x} - \frac{1}{y} = -1$$

$$\frac{1}{x} + \frac{1}{2y} = 8.$$

After simplifying above two equations implies

$$y - 2x = -2xy$$

$$2y + x = 16xy$$

In order to solve above equation pair multiply 1st equation with '1' and 2nd equation with '2' and add the resultant equations implies

$$5y = 30xy$$

$$\Rightarrow 6x = 1$$

$$\therefore x = \frac{1}{6}$$

Substitute the value of 'x' in any of the above equation, implies

$$\Rightarrow y - 2\left(\frac{1}{6}\right) = -2\left(\frac{1}{6}\right)y$$

$$\Rightarrow y + \frac{y}{3} = \frac{1}{3}$$

$$\Rightarrow \frac{4y}{3} = \frac{1}{3} \Rightarrow y = \frac{1}{4}$$

13. **Correct option: (B)**

Let 'x' and 'y' be the two numbers and according to given data

$$x+y=40$$
$$x-y=6$$

Adding both equations implies

$$2x=46$$
$$x=\frac{46}{2}=23$$

Substituting value of 'x' in any of above equations implies $23+y=40\Rightarrow y=17$

14. **Correct option: (A)**

Let the fraction be $\frac{x}{y}$

It is given that both the numerator and the denominator are decreased by one, it is Equal to $\frac{1}{2}$ this implies the resultant fraction is equals to $\frac{x-1}{y-1}=\frac{1}{2}$

Also it is given that if the numerator is increased by 2 it is equal to $\frac{x+2}{y}=\frac{2}{3}$

Simplifying these two equations implies that

$$2x-2=y-1, 3x+6=2y$$
$$2x-y=1\rightarrow\times 2$$
$$3x-2y=-6\rightarrow\times 1$$
$$4x-2y=2$$
$$-3x+2y=+6$$
$$x=8$$
$$y=15$$

Therefore the resultant fraction $\frac{x}{y}=\frac{8}{15}$

15. **Correct option: (D)**

The sum of three angles in a triangle is 180 degrees $x+3x+y=180^0$

Given the relation between three angles is $3y-5x=30^\circ$

Solving these two together implies that

$$4x+y=180^0\rightarrow\times 5$$
$$-5x+3y=30^0\rightarrow\times 4$$
$$20x+5y=900$$
$$-20x+12y=120$$
$$17y=1020^0$$
$$y=60^0$$
$$4x+60^0=120^0$$
$$4x=120^0$$
$$x=30^0$$
$$3x=90^0$$
$$30^0, 90^0, 60^0$$

Since one angle of triangle is 90 degrees and hence it is an right angled triangle

16. **Correct option: (B)**

Given the ratio of ages between 'A' and 'B' is $A:B=5:3$

Let the Age be 'x'

After 6 years their respective ages with ratio is $5x+6:3x+6=7:5$

$$\frac{5x+6}{3x+6}=\frac{7}{5}$$

$$25x+30=21x+42$$

$$4x=42-30$$

$$4x=12, x=3$$

$$\therefore A=15, B=9$$

17. **Correct option: (A)**

5 pencils and 7 pens together cost Rs. 50 this implies that $5p_e+7p=50$

7 pencils and 5 pens together cost Rs.46 this implies that $7p_e+5p=46$

The sum of costs of 1 pencil and 1 pen is $p+p_e$

Solving these two together implies that

$$5p_e+7p=50 \rightarrow \times 5$$

$$7p_e+5p=46 \rightarrow \times 7$$

$$25p_e+35p=250$$

$$-49p_e-35p=-322$$

$$-24p_e=-172$$

$$p_e=3$$

$$5(3)+7p=50$$

$$7p=35, p=5$$

$$p+p_e=8$$

18. **Correct option: (D)**

$$2x-y=5 \rightarrow \times 2$$

$$3x+2y=11 \rightarrow \times 1$$

$$4x-2y=10$$

$$3x+2y=11$$

$$7x=21$$

$$x=3$$

$$6-y=5 \Rightarrow y=1$$

$$2x+3y \Rightarrow 2(3)+3(1)=9$$

19. **Correct option: (C)**

The condition for pair of equations is coincident implies $\frac{a_1}{a_2}=\frac{b_1}{b_2}=\frac{c_1}{c_2}$

Given pair of equations

$$3x+4y+2=0$$

$$9x+12y+k=0$$

$$\frac{\cancel{3}}{\cancel{4}}=\frac{\cancel{4}}{\cancel{12}}=\frac{2}{k}$$

$$\frac{1}{3}=\frac{2}{k} \Rightarrow k=6$$

20. **Correct option: (134)**

Let length and Breadth be l, b respectively and the Area be A=l∗b.

Given that the area of a rectangle gets reduced by 80 sq. units if its length is reduced by 5 units and the breadth increased by 2 units, this implies that

$$(l-5)(b+2)=lb-80$$

Simplifying implies that

$$lb+2l-5b-10=lb-80$$

$$2l-5b=-70$$

If we increase the length by 10 units and decrease the breadth by 5 units, the area will be increased by 50 sq. units this implies that

$$(l+10)(b-5)=lb+50$$

Simplifying implies that

$$lb-5l+10b-50=lb+50$$

$$-5l+10b=100$$

$$-5l+10b=100 \rightarrow \times 1$$

$$2l-5b=-70 \rightarrow \times 2$$

$$-5l+10b=100$$

$$4l-10b=-140$$

$$-l=-40$$

$$l=40$$

$$2(40)-5b=-70$$

$$80+70=5b$$

$$b=30$$

$$2(l+b)=140$$

$$2(35+37)=134$$

LEVEL - II

1. Correct option: (C)

Given

$$\frac{4p+9q}{p}=\frac{5q}{p-q}$$

Cross- multiply

$$(4p+9q)(p-q)=(5q)(p)$$

$$4p^2-4pq+9pq-9q^2=5pq$$

$$4p^2+5pq-5pq=9q^2$$

$$4p^2=9q^2$$

$$\frac{p^2}{q^2}=\frac{9}{4}$$

$$\frac{p}{q}=\sqrt{\frac{9}{4}}$$

$$\frac{p}{q}=\frac{3}{2}$$

Hence, the value of $\frac{p}{q}$ is $\frac{3}{2}$.

2. Correct option: (B)

Given that x, y and $y-\frac{1}{x}$ are not equal to zero

So

$$\frac{x-\frac{1}{y}}{y-\frac{1}{x}}=\frac{\frac{xy-1}{y}}{\frac{xy-1}{x}}$$

$$\frac{\mathrm{x}-\frac{1}{\mathrm{y}}}{\mathrm{y}-\frac{1}{\mathrm{x}}}=\frac{x}{y}$$

Hence, the value of $\frac{\mathrm{x}-\frac{1}{\mathrm{y}}}{\mathrm{y}-\frac{1}{\mathrm{x}}}$ is $\frac{x}{y}$.

3. Correct option: (B)

Let x be the number of coins of 25 paise and y be the number of coins of 50 paise.

Given that there are 47 coins in total. So

$$x+y=47$$

Also worth of coins is Rs. 14.50 = 1450 paise

Then

$$25x\ +50y=1450$$

Consider

$$x+y=47$$

$$y=47-x$$

Put value of y in equation $25\mathrm{x}\ +50\mathrm{y}=1450$

$$25\mathrm{x}\ +50\mathrm{y}=1450$$

$$25x+50(47-x)=1450$$

$$25x+2350-50x=1450$$

$$2350-1450=25x$$

$$25x=900$$

$$x=\frac{900}{25}$$

$$x=36$$

Then y is

$$y=47-x$$

$$y=47-36$$

$$y=11$$

Hence, there are 36 coins of 25 paise and 11 coins of 50 paise.

4. Correct option: (C)

Given

$$10x+3y=75$$

$$6x-5y=11$$

Solve equations to get value of x and y

For this consider

$$6x - 5y = 11$$
$$x = \frac{11 + 5y}{6}$$

Put the value of x in equation $10x + 3y = 75$

$$10x + 3y = 75$$
$$10\left(\frac{11 + 5y}{6}\right) + 3y = 75$$
$$\frac{110 + 50y}{6} + 3y = 75$$
$$\frac{110 + 50y + 18y}{6} = 75$$
$$110 + 68y = 450$$
$$68y = 450 - 110$$
$$68y = 340$$
$$y = \frac{340}{68}$$
$$y = 5$$

Then value of x is

$$x = \frac{11 + 5y}{6}$$
$$x = \frac{11 + 5(5)}{6}$$
$$x = \frac{11 + 25}{6}$$
$$x = \frac{36}{6}$$
$$x = 6$$

Now
$$x + y = 6 + 5$$
$$x + y = 11$$

Hence, value of x + y is 11.

5. Correct option: (C)

Let the numerator of fraction be N

Given that denominator is 1 more than the numerator. So, denominator is N+1

Also, if 1 is subtracted from the numerator and denominator, the fraction becomes equal to $1/2$

$$\frac{N - 1}{(N + 1) - 1} = \frac{1}{2}$$
$$\frac{N - 1}{N} = \frac{1}{2}$$
$$2(N - 1) = N$$
$$2N - 2 = N$$
$$2N - N = 2$$
$$N = 2$$

Numerator is 2

Then denominator is 2 + 1 = 3

Hence, the fraction is $\frac{2}{3}$

6. Correct option: (B)

Let the number of goats be x and the number of hens be y.

Goats have 4 legs and hens have 2 legs, which are $4x + 2y$

Given that legs are 10 more than twice the number of heads, which can be formulated as

$$4x + 2y = 2(x + y) + 12$$
$$4x + 2y = 2x + 2y + 12$$
$$2x = 12$$
$$x = 6$$

Hence, the number of goats is 6.

7. Correct option: (B)

Given that (1, 4) is the point of intersection of the lines $2x + by = 6$ and $3y = 8 + ax$

Therefore, the value x=1 and y=4 satisfy the set of equations

$$2x + by = 6$$
$$3y = 8 + ax$$

So

$$2(1) + b(4) = 6$$
$$2 + 4b = 6$$
$$4b = 4$$
$$b = 1$$

and

$$3y = 8 + ax$$
$$3(4) = 8 + a(1)$$
$$12 = 8 + a$$
$$a = 4$$

Now

$$a - b = 4 - 1 = 3$$

Hence, value of a - b is 3.

8. Correct option: (D)

Given (a + b, a - b) is the solution of equation $3x + 2y = 20$ and $4x - 5y = 42$

So put value of x as (a + b) and y as (a - b) in above equation

$$3x + 2y = 20$$
$$3(a + b) + 2(a - b) = 20$$
$$3a + 3b + 2a - 2b = 20$$
$$5a + b = 20$$

And

$$4x - 5y = 42$$
$$4(a + b) - 5(a - b) = 42$$
$$4a + 4b - 5a + 5b = 42$$
$$9b - a = 42$$
$$a = 9b - 42$$

Put the value of a in equation 5a + b = 20

$$5a + b = 20$$

$$5(9b - 42) + b = 20$$

$$45b - 210 + b = 20$$

$$46b = 210 + 20$$

$$46b = 230$$

$$b = 5$$

Hence, the value of b is 5

9. Correct option: (B)

Let N be numerator and D be denominator

If 1 is added to the numerator and 1 is subtracted from denominator, the fraction becomes equal to 1

$$\frac{N+1}{D-1} = 1$$

$$N + 1 = D - 1$$

$$N = D - 2$$

Also, if 1 is added to denominator, then fraction becomes equal to $1/2$

$$\frac{N}{D+1} = \frac{1}{2}$$

$$2N = D + 1$$

Put the value of N in above equation

$$2N = D + 1$$

$$2(D - 2) = D + 1$$

$$2D - 4 = D + 1$$

$$2D - D = 4 + 1$$

$$D = 5$$

Then numerator is

$$2N = D + 1$$

$$2N = 5 + 1$$

$$2N = 6$$

$$N = 3$$

The fraction is $3/8$

Hence, sum of numerator and denominator is 3+5=8

10. Correct option: (A)

Given $kx + 2y = 5$

Or $kx + 2y - 5 = 0$

And $3x + y = 1$

Or $3x + y - 1 = 0$

Comparing with general form of equation

$$a_1x + b_1y + c_1 = 0$$

$$a_1 = k, b_1 = 2, c_1 = -5$$

And $a_2x + b_2y + c_2 = 0$

$$a_2 = 3, b_2 = 1, c_2 = -1$$

Now for the system of equation to have zero solution

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

$$\frac{k}{3} = \frac{2}{1} \neq \frac{-5}{-1}$$

Clearly

$$\frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

Then the given system of equation will have no solution, if

$$\frac{\mathrm{a}_1}{\mathrm{a}_2} = \frac{\mathrm{b}_1}{\mathrm{b}_2}$$

$$\frac{\mathrm{k}}{3} = \frac{2}{1}$$

$$k = 6$$

Hence, the given system of equations will have zero solution, if k=6

11. Correct option: (D)

Given

$$\frac{ax}{b} - \frac{by}{a} = a + b$$

or

$$a^2x - b^2y = ab(a + b)$$

or

$$\mathrm{a}^2\mathrm{x} - \mathrm{b}^2\mathrm{y} - \mathrm{ab}(\mathrm{a} + \mathrm{b}) = 0$$

And

$$ax - by = 2ab$$

$$\mathrm{ax} - \mathrm{by} - 2\mathrm{ab} = 0$$

Comparing with general form of equation

$$\mathrm{a}_1\mathrm{x} + \mathrm{b}_1\mathrm{y} + \mathrm{c}_1 = 0$$

$$\mathrm{a}_1 = a^2, \mathrm{b}_1 = -b^2, \mathrm{c}_1 = -\mathrm{ab}(\mathrm{a} + \mathrm{b})$$

And

$$\mathrm{a}_2\mathrm{x} + \mathrm{b}_2\mathrm{y} + \mathrm{c}_2 = 0$$

$$\mathrm{a}_2 = \mathrm{a}, \mathrm{b}_2 = -\mathrm{b}, \mathrm{c}_2 = -2\mathrm{ab}$$

Formula for cross- multiplication is

$$\frac{x}{b_1c_2 - b_2c_1} = \frac{y}{c_1a_2 - c_2a_1} = \frac{1}{a_1b_2 - a_2b_1}$$

$$\frac{\mathrm{x}}{(-b^2)(-2\mathrm{ab}) - (-\mathrm{b})[-ab(a+b)]} = \frac{\mathrm{y}}{[-ab(a+b)](\mathrm{a}) - (-2\mathrm{ab})(a^2)} = \frac{1}{a^2(-\mathrm{b}) - (\mathrm{a})(-b^2)}$$

$$\frac{x}{2ab^3 - a^2b^2 - ab^3} = \frac{y}{-a^3b - a^2b^2 + 2a^3b} = \frac{1}{-a^2b + ab^2}$$

$$\frac{\mathrm{x}}{\mathrm{ab}^3 - \mathrm{a}^2\mathrm{b}^2} = \frac{\mathrm{y}}{\mathrm{a}^3\mathrm{b} - \mathrm{a}^2\mathrm{b}^2} = \frac{1}{\mathrm{ab}^2 - a^2b}$$

$$\frac{x}{ab^2(b-a)} = \frac{y}{-a^2b(b-a)} = \frac{1}{ab(b-a)}$$

$$\frac{x}{ab^2(b-a)} = \frac{1}{ab(b-a)}$$

$$x = b$$

$$\frac{\mathrm{y}}{-\mathrm{a}^2\mathrm{b}(\mathrm{b}-\mathrm{a})} = \frac{1}{\mathrm{ab}(\mathrm{b}-\mathrm{a})}$$

$$y = -a$$

Hence, the value of x are b and y are -a

12. Correct option: (B)

Given $3 < x < 6$

$2 < y < 4$

Multiply 2 in equation $3 < x < 6$ and 3 in equation $2 < y < 4$

$$6 < 2x < 12$$

$$6 < 2y < 12$$

Then $6 < \frac{2x}{2y} < 12$

$$3 < \frac{x}{y} < 6$$

Hence, the greatest value of $\frac{x}{y}$ is 6

13. Correct option: (A, C)

Given

A work can be finished in 18 days

A and B can do a work in 10 days

$$\frac{10}{A} + \frac{10}{B} = 1$$

$$10B + 10A = AB$$

Now, A alone can finish a work in 12 days and B alone can finish a work in 6 days

$$\frac{12}{A} + \frac{6}{B} = 1$$

$$12B + 6A = AB$$

Comparing equation $10\mathrm{B} + 10\mathrm{A} = \mathrm{AB}$

$$12\mathrm{B} + 6\mathrm{A} = \mathrm{AB}$$

$$12\mathrm{B} + 6\mathrm{A} = 10\mathrm{B} + 10\mathrm{A}$$

$$12B - 10B = 10A - 6A$$

$$2B = 4A$$

$$B = 2A$$

Put value of B in equation

$$\frac{10}{\mathrm{A}} + \frac{10}{\mathrm{B}} = 1$$

$$\frac{10}{\mathrm{A}} + \frac{10}{2\mathrm{A}} = 1$$

$$\frac{20 + 10}{2\mathrm{A}} = 1$$

$$\frac{30}{2A} = 1$$
$$2A = 30$$
$$A = 15$$

Then value of B is
$$B = 2A$$
$$B = 2(15)$$
$$B = 30$$

Hence, the time required for A is 15 days and time required for B is 30 days.

14. Correct option: (A)

Given
$$\frac{1}{2(2x+3y)} + \frac{12}{7(3x-2y)} = \frac{1}{2}$$
$$\frac{7}{2x+3y} + \frac{4}{3x-2y} = 2$$

Let
$$\frac{1}{(2x+3y)} = X$$

And
$$\frac{1}{3x-2y} = Y$$

Then
$$\frac{1}{2}X + \frac{12}{7}Y = \frac{1}{2}$$
$$7X + 24Y = 7$$

and
$$7X + 4Y = 2$$

Subtracting above equation, we get
$$7X - 7X + 24Y - 4Y = 7 - 2$$
$$20Y = 5$$
$$Y = \frac{1}{4}$$

Putting value of Y in equation
$$7X + 4Y = 2$$
$$7X + 4\left(\frac{1}{4}\right) = 2$$
$$7X + 1 = 2$$
$$7X = 1$$
$$X = \frac{1}{7}$$

Now
$$\frac{1}{(2x+3y)} = X$$

$$\frac{1}{(2x+3y)} = \frac{1}{7}$$
$$2x + 3y = 7$$

and

$$\frac{1}{3x-2y} = Y$$
$$\frac{1}{3x-2y} = \frac{1}{4}$$
$$3x - 2y = 4$$

Solve above equation to get value of x and y

$2x + 3y = 7$..... (i)

$3x - 2y = 4$..... (ii)

Multiply equation (i) by 2 and equation (ii) by 3

$$4x + 6y = 14$$
$$9x - 6y = 12$$

Adding above equations

$$4x + 9x + 6y - 6y = 14 + 12$$
$$13x = 26$$
$$x = 2$$

Then y is

$$9(2) - 6y = 12$$
$$18 - 6y = 12$$
$$18 - 12 = 6y$$
$$6y = 6$$
$$y = 1$$

Hence value of x is 2 and value of y is 1.

15. Correct option: (C)

x = 2

y = 1

x + y = 2 +1 =3

Hence, value of x + y is 3

16. Correct option: (B)

x = 2

y = 1

x – y = 2 – 1 =1

Hence, the value of x – y is 1

17. Correct Matching is: a-P, b- R, c- S d- Q

A) $2x + 3y + 4 = 0;\ 5x + 6y + 9 = 0$	P) $\left[-1, \frac{-2}{3}\right]$
B) $x + 3y = 1;\ 2x + 6y = 2$	T) $\left(1, \frac{2}{3}\right)$
C) The solution set for inconsistent system of equations	S) Null set

D) $y = mx;\ m \neq 0$ passes through Q) (0, 0)

Explanation:

A) Given

$$2x + 3y + 4 = 0$$
$$5x + 6y + 9 = 0$$

Multiply 2 in equation $2x + 3y + 4 = 0$

$$2(2x + 3y + 4) = 0$$
$$4x + 6y + 8 = 0$$

Subtract this equation by $5x + 6y + 9 = 0$

$$5x + 6y + 9 - (4x + 6y + 8) = 0$$
$$5x - 4x + 6y - 6y + 9 - 8 = 0$$
$$x + 1 = 0$$
$$x = -1$$

Put value of x in equation

$$2x + 3y + 4 = 0$$
$$2(-1) + 3y + 4 = 0$$
$$-2 + 3y + 4 = 0$$
$$3y + 2 = 0$$
$$3y = -2$$
$$y = \frac{-2}{3}$$

Hence the value of x is -1 and y is $\frac{-2}{3}$.

B) Given

$$x + 3y = 1$$
$$2x + 6y = 2$$

or

$$x + 3y = 1$$

x = 0

$$0 + 3y = 1$$
$$y = \frac{1}{3}$$

y = 0

$$x + 0 = 1$$
$$x = 1$$

C) The solution set for inconsistent system of equations is null set.

D) $y = mx;\ m \neq 0$ passes through (0 ,0).

18. Correct Answer is: 4

Given $6x + 3y = 7xy$

$$3x + 9y = 11xy$$

Solve the equation to get value of x and y

Divide both sides of equation by xy

$$\frac{6}{y}+\frac{3}{x}=7$$

$$\frac{3}{y}+\frac{9}{x}=11$$

Multiply above equation by 2

$$\frac{6}{y}+\frac{18}{x}=22$$

Subtract this equation by $\frac{6}{y}+\frac{3}{x}=7$

$$\frac{6}{y}+\frac{18}{x}-\left(\frac{6}{y}+\frac{3}{x}\right)=22-7$$

$$\frac{6}{y}+\frac{18}{x}-\frac{6}{y}-\frac{3}{x}=15$$

$$\frac{18-3}{x}=15$$

$$15x=15$$

$$x=1$$

Put the value of x in $\frac{3}{y}+\frac{9}{x}=11$

$$\frac{3}{y}+\frac{9}{1}=11$$

$$\frac{3}{y}+9=11$$

$$\frac{3}{y}=2$$

$$y=\frac{3}{2}$$

Thus, value of x+2y is

$$x+2y=1+2\left(\frac{3}{2}\right)$$

$$x+2y=1+3$$

$$x+2y=4$$

Hence, the value of x+2y is 4

19. Correct Answer is: 6

Let the number be x

Given that three times of a number is increased by 6 which gives results as 24

$$3x+6=24$$

$$3x=16$$

$$x=6$$

Hence, the number is 6.

20. Correct Answer is: 2

Let the two-digit number be 10x + y

Reverse of two-digit number is 10y + x

Given the sum of two-digit number and reverse of it is 66

$$10x + y + 10y + x = 66$$
$$11x + 11y = 66$$
$$x + y = 6$$

Also given that difference of digit is 2 $x - y = 2$

Add equation $x + y = 6$

And $x - y = 2$

$$x + y + x - y = 8$$
$$2x = 8$$
$$x = 4$$

then value of y is $x - y = 2$

$$4 - y = 2$$
$$y = 4 - 2$$
$$y = 2$$

Hence, two digits number are 42 and 24.

Therefore, there are two such numbers

STUDENT PRACTICE QUESTIONS

1. Correct option: (D)

$$0.16(5x-2) = 0.4x + 7$$
$$\Rightarrow 0.8x - 0.32 = 0.4x + 7$$
$$\Rightarrow 0.4x = 7.32 \Rightarrow x = \frac{7.32}{0.4} = 18.3$$

2. Correct option: (A)

$$\frac{2x+1}{3x-2} = \frac{9}{10}$$
$$\Rightarrow 10(2x+1) = 9(3x-2)$$
$$\Rightarrow 20x + 10 = 27x - 18$$
$$\Rightarrow 7x = 28 \Rightarrow x = \frac{28}{7} = 4$$

3. Correct option: (A)

$$x - \left(\frac{2x+8}{3}\right) = \frac{1}{4}\left(x - \frac{2-x}{6}\right) - 3$$
$$\Rightarrow \frac{3x - 2x - 8}{3} = \frac{6x - 2 + x}{24} - 3$$
$$\Rightarrow \frac{x-8}{3} = \frac{7x - 2 - 72}{24} \Rightarrow 8x - 64 = 7x - 74$$
$$\Rightarrow x = 10$$

4. Correct option: (A)

$$x + \frac{1}{x} = 3$$

Squaring on both sides implies

$$x^2+\frac{1}{x^2}+2(x)\left(\frac{1}{x}\right)=9$$

$$\Rightarrow x^2+\frac{1}{x^2}=9-2\Rightarrow x^2+\frac{1}{x^2}=7$$

Again Squaring on both sides on $x^2+\frac{1}{x^2}=7$ implies

$$x^4+\frac{1}{x^4}+2\left(x^2\right)\left(\frac{1}{x^2}\right)=49$$

$$\Rightarrow x^4+\frac{1}{x^4}=49-2\Rightarrow x^4+\frac{1}{x^4}=47$$

Again Squaring on both sides on $x^4+\frac{1}{x^4}=47$ implies

$$x^8+\frac{1}{x^8}+2\left(x^4\right)\left(\frac{1}{x^4}\right)=2209$$

$$\Rightarrow x^8+\frac{1}{x^8}=2209-2\Rightarrow x^8+\frac{1}{x^8}=2207$$

5. Correct option: (B).

$$\frac{x}{4}+\frac{x}{6}=x-7$$

$$\frac{3x+2x}{12}=x-7$$

$$\Rightarrow\frac{5x}{12}=x-7\Rightarrow 5x=12x-84\Rightarrow 7x=84$$

$$\Rightarrow x=\frac{84}{7}=12$$

6. Correct option: (C)

Sol. Let the two numbers be 3x and 8x respectively

Given that sum of those two numbers is 165

$$\Rightarrow 3x+8x=165$$

$$\Rightarrow 11x=165$$

$$\Rightarrow x=15$$

Therefore the numbers pair is $3\times15=45, 8\times15=120$ respectively

7. Correct option: (D)

Sol. Let the two consecutive even numbers be 'x' and 'x+2' respectively
Given that two-fifth of the smaller exceeds two Eleventh of the large by 4
This can be represented in the form of equation as follows

$$\frac{2x}{5}-\frac{2x+4}{11}=4$$

$$\Rightarrow\frac{22x-10x-20}{55}=4\Rightarrow 12x-20=220$$

$$\Rightarrow 12x=240\Rightarrow x=20$$

Therefore another number is 'x+2=22'.

8. **Correct option: (B)**

$$2y+\frac{7}{x}=3$$
$$\Rightarrow 2xy+7=3x$$
$$y-\frac{3}{x}=8$$
$$\Rightarrow xy-3=8x$$

Adding both the equations implies

$$11x-4=3xy$$
$$\Rightarrow y=\frac{11x-4}{3x}$$

Substitute the value of 'x' in any of the above equation pair implies that

$$2\left(\frac{11x-4}{3x}\right)+\frac{7}{x}=3$$
$$\Rightarrow 22x-8+21=9x$$
$$\Rightarrow 22x-9x=-13$$
$$\Rightarrow x=-1$$

Substituting the value of 'x' in value of 'y' implies

$$y=\frac{11x-4}{3x}$$
$$y=\frac{11(-1)-4}{3(-1)}=5$$

9. **Correct option: (D)**.

$$\frac{x}{a}+\frac{y}{b}=2$$
$$ax-by=a^2-b^2$$
$$\Rightarrow \frac{x}{a}+\frac{y}{b}-2=0$$
$$\Rightarrow bx+ay-2ab=0$$
$$\Rightarrow bx+ay=2ab$$
$$ax-by=a^2-b^2$$
$$bx+ay=2ab$$

In order to solve above equation pair multiply equation 1 with 'a' and equation 2 with 'b' implies

$$(ax-by)a=(a^2-b^2)a$$
$$(bx+ay)b=2ab^2$$

Simplifying further implies that

$$a^2x-aby=(a^2-b^2)a$$
$$b^2x+aby=2ab^2$$

Adding both the equations implies that

$$\left(a^2+b^2\right)x=\left(a^2-b^2\right)a+2ab^2$$
$$\Rightarrow x=\frac{\left(a^2-b^2\right)a+2ab^2}{\left(a^2+b^2\right)}=\frac{a\left(a^2+b^2\right)}{\left(a^2+b^2\right)}=a$$

10. Correct option: (A)

Cost of 5 pens and 7 notebooks is Rs. 82 this implies $5P+7N=82$

Also the cost of 4 pens and 4 notebooks is Rs 52 $4P+4N=52$

Asked to find the cost of 2 pens and 3 notebooks implies that $2P+3N$

$$5P+7N=82$$
$$4P+4N=52$$
$$P+N=13 \rightarrow \times 7$$
$$5P+7N=82 \rightarrow \times 1$$
$$7P+7N=91$$
$$-5P+-7N=-8$$
$$2P=9$$
$$P=4.5$$
$$N=8.5$$
$$2P+3N=2(4.5)+3(8.5)$$
$$9+25.5$$
$$34.5$$

11. Correct option: (B, C)

Given $3x+4y=25$

And $5x-6y=-9$

Solve the equation to get value of x and y

Consider $3x+4y=25$

$$x=\frac{25-4y}{3}$$

Put the value of x in equation $5x-6y=-9$

$$5x-6y=-9$$
$$5\left(\frac{25-4y}{3}\right)-6y=-9$$
$$\frac{125-20y}{3}-6y=-9$$
$$\frac{125-20y-18y}{3}=-9$$
$$125-38y=-27$$
$$125+27=38y$$
$$38y=152$$
$$y=\frac{152}{38}$$
$$y=4$$

Then value of x is $x=\frac{25-4y}{3}$

$$x=\frac{25-4(4)}{3}$$
$$x=\frac{25-16}{3}$$
$$x=\frac{9}{3}$$
$$x=3$$

Hence the value of x is 3 and value of y is 4

Also, 4y=16

12. Correct option: (A,B)

Given

$\sqrt{3}x - \sqrt{2}y = \sqrt{3}$.....(i)

$\sqrt{5}x + \sqrt{3}y = \sqrt{2}$..... (ii)

Solve the equations to get value of x and y

Multiply equation (i) by $\sqrt{3}$ and equation (ii) by $\sqrt{2}$

$$3x - \sqrt{6}y = 3$$
$$\sqrt{10}x + \sqrt{6}y = 2$$

Adding above equation

$$3x - \sqrt{6}y + \sqrt{10}x + \sqrt{6}y = 5$$
$$x\left(\sqrt{10} + 3\right) = 5$$
$$x = \frac{5}{\sqrt{10} + 3}$$
$$\mathrm{x} = \frac{5}{\sqrt{10} + 3} \times \frac{\sqrt{10} - 3}{\sqrt{10} - 3}$$
$$x = \frac{5\left(\sqrt{10} - 3\right)}{10 - 3\sqrt{10} + 3\sqrt{10} - 9}$$
$$x = 5\left(\sqrt{10} - 3\right)$$

Put value of x in equation

$$\sqrt{3}\mathrm{x} - \sqrt{2}\mathrm{y} = \sqrt{3}$$
$$\sqrt{3}\left[5\left(\sqrt{10} - 3\right)\right] - \sqrt{2}\mathrm{y} = \sqrt{3}$$
$$5\sqrt{30} - 15\sqrt{3} - \sqrt{2}y = \sqrt{3}$$
$$\sqrt{2}y = 5\sqrt{30} - 15\sqrt{3} - \sqrt{3}$$
$$y = \frac{5\sqrt{30} - 16\sqrt{3}}{\sqrt{2}}$$
$$y = \frac{5\sqrt{30}}{\sqrt{2}} - \frac{16\sqrt{3}}{\sqrt{2}}$$
$$y = 5\sqrt{15} - 8\sqrt{6}$$

Hence, the value of x is $5\left(\sqrt{10} - 3\right)$ and y is $5\sqrt{15} - 8\sqrt{6}$.

13. Correct option: (B)

Given $2x - ky + 3 = 0$ and $4x + 6y - 5 = 0$ are parallel lines

Comparing with general form of equation

$$a_1x + b_1y + c_1 = 0$$
$$a_1 = 2, b_1 = -k, c_1 = 3$$
$$a_2x + b_2y + c_2 = 0$$
$$a_2 = 4, b_2 = 6, c_2 = -5$$

Condition for parallel lines is

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

$$\frac{2}{4} = \frac{-k}{6} \neq \frac{3}{-5}$$

Taking

$$\frac{2}{4} = \frac{-k}{6}$$
$$-4k = 12$$
$$k = -3$$

Hence, the value of k is -3.

14. Correct option: (D)

Given that lines $3x + y = 5$ and $4x + (k-3)y + 7 = 0$ are intersecting

Comparing with general form of equation

$$a_1x + b_1y + c_1 = 0$$
$$a_1 = 3, b_1 = 1, c_1 = -5$$
$$a_2x + b_2y + c_2 = 0$$
$$a_2 = 4, b_2 = (k-3), c_2 = 7$$

Condition for intersecting lines is

$$\frac{a_1}{a_2} = \frac{b_1}{b_2}$$
$$\frac{3}{4} = \frac{1}{k-3}$$
$$3k - 9 = 4$$
$$3k = 13$$
$$k = \frac{13}{3}$$

Hence, the value of k is $13/3$.

15. Correct Matching is: a-R, b-S, c-Q, d-T

A) $\sqrt{3}x + \sqrt{5}y = \sqrt{3}; \sqrt{5}x + \sqrt{3}y = \sqrt{3}$ R) $\frac{\sqrt{15}-3}{2}, \frac{\sqrt{15}-3}{2}$

B) $2^x + 3^y = 17; 2^{x+2} - 3^{y+1} = 5$ S) (2, 3)

C) $\frac{2}{x} + \frac{3}{y} = 13; \frac{5}{x} - \frac{4}{y} = -2 \; where \; x \neq 0, y \neq 0$ Q) $\frac{1}{2}, \frac{1}{3}$

D) $\frac{5}{x-1} + \frac{1}{y-2} = 2; \frac{6}{x-1} - \frac{3}{y-2} = 1$ T) (4, 5)

Explanation

A) Given

$$\sqrt{3}x + \sqrt{5}y = \sqrt{3}$$
$$\sqrt{5}x + \sqrt{3}y = \sqrt{3}$$

Multiply above equations by $\sqrt{3}$

$$3x + \sqrt{15}y = 3$$
$$\sqrt{15}x + 3y = 3$$

Consider

$$\sqrt{15}x + 3y = 3$$
$$y = \frac{3 - \sqrt{15}x}{3}$$

Put the value of y in equation

$$3x + \sqrt{15}y = 3$$
$$3x + \sqrt{15}\left(\frac{3 - \sqrt{15}x}{3}\right) = 3$$
$$3x + \frac{3\sqrt{15} - 15x}{3} = 3$$
$$9x + 3\sqrt{15} - 15x = 9$$
$$3\sqrt{15} - 6x = 9$$
$$6x = 3\sqrt{15} - 9$$
$$2x = \sqrt{15} - 3$$
$$x = \frac{\sqrt{15} - 3}{2}$$

Then value of y is

$$y = \frac{3 - \sqrt{15}x}{3}$$
$$y = \frac{3 - \sqrt{15}\left(\frac{\sqrt{15} - 3}{2}\right)}{3}$$
$$y = \frac{3 - \left(\frac{15 - 3\sqrt{15}}{2}\right)}{3}$$
$$y = \frac{6 - 15 + 3\sqrt{15}}{6}$$
$$y = \frac{3\sqrt{15} - 9}{6}$$
$$y = \frac{\sqrt{15} - 3}{2}$$

Hence the value of x and y are $\frac{\sqrt{15}-3}{2}$, $\frac{\sqrt{15}-3}{2}$

B) Given

$$2^x + 3^y = 17$$
$$2^{x+2} - 3^{y+1} = 5$$

Let $2^x = a$ and $3^y = b$

$$a + b = 17$$
$$2^x 2^2 - 3^y 3^1 = 5$$
$$4a - 3b = 5$$

Solve the equations to get value of a and b

$$a + b = 17$$
$$4a - 3b = 5$$

Consider

$$a + b = 17$$
$$a = 17 - b$$

Put the value of a in equation

$$4a - 3b = 5$$
$$4(17 - b) - 3b = 5$$
$$68 - 4b - 3b = 5$$
$$7b = 68 - 5$$
$$7b = 63$$
$$b = 9$$

Value of a is

$$a = 17 - b$$
$$a = 17 - 9$$
$$a = 8$$

Now $2^x = a$

$$2^x = 8$$
$$2^x = 2^3$$
$$x = 3$$

and $3^y = b$

$$3^y = 9$$
$$3^y = 3^2$$
$$y = 2$$

Hence, the value of x is 3 and value of y is 2

C) Given

$$\frac{2}{x} + \frac{3}{y} = 13$$
$$\frac{5}{x} - \frac{4}{y} = -2$$

Let $\frac{1}{x} = a$ and $\frac{1}{y} = b$

$2a + 3b = 13$…… (i)

$5a - 4b = -2$…… (ii)

Multiply equation (i) by 4 and (ii) by 3

$$8a + 12b = 52$$
$$15a - 12b = -6$$

Add above equations

$$8a + 15a + 12b - 12b = 52 - 6$$
$$23a = 46$$
$$a = 2$$

Put the value of a in equation

$$2a + 3b = 13$$
$$2(2) + 3b = 13$$
$$3b = 13 - 4$$
$$3b = 9$$

$$b = 3$$

Now $\frac{1}{x} = a$

$$\frac{1}{x} = 2$$

$$x = \frac{1}{2}$$

and $\frac{1}{y} = b$

$$\frac{1}{y} = b$$

$$y = \frac{1}{3}$$

D) Given

$$\frac{5}{x-1} + \frac{1}{y-2} = 2$$

$$\frac{6}{x-1} - \frac{3}{y-2} = 1$$

Let $\frac{1}{x-1} = a$ and $\frac{1}{y-2} = b$

$5a + b = 2$…… (i)

$6a - 3b = 1$…… (ii)

Multiply equation (i) by 3

$$15a + 3b = 6$$

Add this equation by equation (ii)

$$15a + 6a + 3b - 3b = 6 + 1$$

$$21a = 7$$

$$a = \frac{1}{3}$$

Put value of x in equation $6a - 3b = 1$

$$6\left(\frac{1}{3}\right) - 3b = 1$$

$$2 - 3b = 1$$

$$2 - 1 = 3b$$

$$3b = 1$$

$$b = \frac{1}{3}$$

Now $\frac{1}{x-1} = a$

$$\frac{1}{x-1} = a$$

$$\frac{1}{x-1} = \frac{1}{3}$$

$$3 = x - 1$$

$$x = 3 + 1$$

$$x = 4$$

and $\frac{1}{y-2} = b$

$$\frac{1}{y-2} = b$$
$$\frac{1}{y-2} = \frac{1}{3}$$
$$3 = y - 2$$
$$y = 5$$

Hence, the value of x is 4 and value of y is 5

16. Correct Answer is: 2

Given

$$x + y = 1$$
$$kx + y = 2$$

Comparing with general form of equation

$$a_1x + b_1y + c_1 = 0$$
$$a_1 = 1, b_1 = 1, c_1 = -1$$
$$a_2x + b_2y + c_2 = 0$$
$$a_2 = k, b_2 = 1, c_2 = -2$$

Condition for no solution is

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$
$$\frac{1}{k} = \frac{1}{1} \neq \frac{-1}{-2}$$

Consider

$$\frac{1}{k} = \frac{1}{1}$$
$$k = 1$$

Then value of k + 1 is

$$k + 1 = 1 + 1$$
$$k + 1 = 2$$

Hence, value of k + 1 is given by 2

17. Correct Answer is: 3

Let the number of bees be x and number of flowers be y

Given that if one bee sits on each flower, then one bee will be left and if two bees sit on each flower, one flower will be left.

Then $x - y = 1$

$$y - \frac{x}{2} = 1$$

Solve equations to get value of x and y

Consider

$$x - y = 1$$
$$x = y + 1$$

Put value of x in equation

$$y - \frac{x}{2} = 1$$

$$y - \frac{y+1}{2} = 1$$
$$\frac{2y-(y+1)}{2} = 1$$
$$2y - y - 1 = 2$$
$$y = 3$$

Then value of x is

$$x = y + 1$$
$$x = 3 + 1$$
$$x = 4$$

Hence, the number of bees is 4 and number of flowers is 3.

18. Correct Answer is: 0

Given

$$\frac{5x-3}{4} + \frac{y+1}{5} = 2x + y$$
$$\frac{x+11}{6} - \frac{3y-4x}{4} = 2x + 3$$

Solve the equations to get value of x and y

Consider

$$\frac{x+11}{6} - \frac{3y-4x}{4} = 2x + 3$$
$$\frac{4(x+11)-6(3y-4x)}{24} = 2x + 3$$
$$\frac{4x+44-18y+24x}{24} = 2x + 3$$
$$28x - 18y + 44 = 24(2x + 3)$$
$$28x - 18y = 48x + 72 - 44$$
$$-20x - 18y = 28$$
$$20x = -28 - 18y$$
$$x = \frac{-28-18y}{20}$$

Put the value of x in equation

$$\frac{5x-3}{4} + \frac{y+1}{5} = 2x + y$$
$$\frac{5\left(\frac{-28-18y}{20}\right)-3}{4} + \frac{y+1}{5} = 2\left(\frac{-28-18y}{20}\right) + y$$
$$\frac{\left(\frac{-28-18y}{4}\right)-3}{4} + \frac{y+1}{5} = \left(\frac{-28-18y}{10}\right) + y$$
$$\frac{-28-18y-12}{16} + \frac{y+1}{5} = \left(\frac{-28-18y+10y}{10}\right)$$
$$\frac{-40-18y}{16} + \frac{y+1}{5} = \frac{-28-8y}{10}$$
$$\frac{5(-40-18y)+16(y+1)}{80} = \frac{-28-8y}{10}$$

$$\frac{-240-90y+16y+16}{8}=-28-8y$$
$$-224-74y=-224-8y$$
$$0=74y-8y$$
$$0=66y$$
$$y=0$$

Hence, value of y is 0

19. Correct Answer is: 2

Given $2x+3y=34$

$$1+\frac{x}{y}=\frac{13}{8}$$

Solve the equations to get value of x and y

Consider $2x+3y=34$

$$x=\frac{34-3y}{2}$$

Put the value of x in equation

$$1+\frac{x}{y}=\frac{13}{8}$$
$$1+\frac{\left(\frac{34-3y}{2}\right)}{y}=\frac{13}{8}$$
$$1+\frac{34-3y}{2y}=\frac{13}{8}$$
$$\frac{2y+34-3y}{2y}=\frac{13}{8}$$
$$8(34-y)=13(2y)$$
$$272-8y=26y$$
$$34y=272$$
$$y=8$$

Then value of x is $x=\frac{34-3y}{2}$

$$x=\frac{34-3(8)}{2}$$
$$x=\frac{34-24}{2}$$
$$x=\frac{10}{2}$$
$$x=5$$

Thus, value of 2x – y is $2x-y=2(5)-8$

$$2x-y=10-8$$
$$2x-y=2$$

Hence, value of 2x – y is 2.

20. Correct Answer is:

Given $x+\frac{4}{y}=4$

$$y+\frac{6}{x}=6$$

Solve the equation to get value of x and y

Consider $x+\frac{4}{y}=4$

$$xy+4=4y$$
$$4y-xy=4$$
$$y(4-x)=4$$
$$y=\frac{4}{4-x}$$

Substitute value of y in equation $y+\frac{6}{x}=6$

$$\frac{4}{4-x}+\frac{6}{x}=6$$
$$\frac{4x+6(4-x)}{x(4-x)}=6$$
$$4x+24-6x=6x(4-x)$$
$$24-2x=24x-6x^2$$
$$6x^2-24x-2x+24=0$$
$$6x^2-26x+24=0$$
$$3x^2-13x+12=0$$
$$3x^2-9x-4x+12=0$$
$$3x(x-3)-4(x-3)=0$$
$$(x-3)(3x-4)=0$$
$$x=3, x=\frac{4}{3}$$

Value of y $y=\frac{4}{4-x}$

When x =3 $y=\frac{4}{4-3}$

$$y=4$$

when x = 4/3 $y=\frac{4}{4-\left(\frac{4}{3}\right)}$

$$y=\frac{12}{12-4}$$
$$y=\frac{12}{8}$$
$$y=\frac{3}{2}$$

Now, x + y is x = 3 and y =4

$$x+y=3+4=7$$

And x = 4/3 and y =3/2

$$x+y=\frac{4}{3}+\frac{3}{2}$$
$$x+y=\frac{8+9}{6}$$
$$x+y=\frac{17}{6}$$

3. Quadratic Equations

Solutions

LEVEL-I

1. Correct Option: (C)

Side of the smaller equilateral triangle = x cm

$\therefore$ Perimeter of the smaller equilateral triangle = $3x$ cm

The difference between the perimeters of the two equilateral triangles is given as 9 cm.

$\therefore$Perimeter of the bigger equilateral triangle = $(9 + 3x)$ cm

$\Rightarrow$ Side of the bigger equilateral triangle = $\frac{9+3x}{3} cm = (x + 3)$ cm

Area of smaller equilateral triangle + Area of bigger equilateral triangle = $\frac{29\sqrt{3}}{4}$ cm^2

$\therefore \frac{\sqrt{3}}{4}x^2 + \frac{\sqrt{3}}{4}(x+3)^2 = \frac{29\sqrt{3}}{4}$

$\Rightarrow x^2 + (x + 3)^2 = 29$

$\Rightarrow x^2 + x^2 + 6x + 9 - 29 = 0$

$\Rightarrow 2x^2 + 6x - 20 = 0$

$\Rightarrow x^2 + 3x - 10 = 0$

Thus, the equation $x^2 + 3x - 10 = 0$ represents the given information.

2. Correct Option: (A)

The given pair of equations is $x = 2$ and $y = x^2 - 3$

On solving these equations, it is obtained that $x = 2$ and $y = 1$

Thus, the point of intersection of the given line and curve is (2, 1).

Abscissa of the point of intersection = 2

Ordinate of the point of intersection = 1

The quadratic equation whose roots are 2 and 1 is given by the equation:$(x - 2)(x - 1)=0$

$(x - 2)(x - 1) = 0$

$\Rightarrow x^2 - 3x + 2 = 0$

Thus, the required quadratic equation is $x^2 - 3x + 2 = 0$

3. Correct Option: (B)

$x^2-3y = 0$

$\Rightarrow x^2 = 3y$ ...i

$6y-3x-5=0$ ...(*ii*)

From equation (*i*) and (*ii*), we get

$2x^2-3x-5=0$

$\Rightarrow 2x^2+2x-5x-5=0$

$\Rightarrow 2x(x+1)-5(x+1) = 0$

$\Rightarrow (x +1)(2x -5) = 0$

$\Rightarrow x = -1$ or $x = 5/2$

Therefore, for $x \geq 0$, we get $x=5/2$

4. Correct Option: (C)

Let α, β, and γ be the roots of the given equation such that $\alpha = -\beta$.

Then, $\alpha + \beta + \gamma = 2 \Rightarrow \gamma = 2$

Since γ is a root of the given equation, it will satisfy the equation.

$\Rightarrow 8 - 8 + 2\mu - \lambda = 0$

$\Rightarrow 2\mu = \lambda$

5. Correct Option: (B)

$a = x^2 - 2x$ and $5^a + 7^a = 468$

$5^a + 7^a = 125 + 343 \Rightarrow 5^a + 7^a = 5^3 + 7^3 \Rightarrow a = 3$

$x^2 - 2x = 3 \Rightarrow x^2 - 2x - 3 = 0$

$\Rightarrow x^2 - 3x + x - 3 = 0 \Rightarrow (x - 3)(x + 1) = 0$

$\Rightarrow x = 3, x = -1$

Value of $(x + 1)^2$ are $(4)^2, 0^2 \Rightarrow 16, 0$

6. Correct Option: (B)

Consider the equation $80x^2 + 52x - 84 = 0$.

$80x^2 + 52x - 84 = 0$

$\Rightarrow 4(20x^2 + 13x - 21) = 0$

$\Rightarrow 20x^2 + 13x - 21 = 0$

The middle term i.e., $13x$ can be split as

$13x = 28x - 15x$ [$28 \times (-15) = -420 = 20 \times (-21)$]

$$\therefore 20x^2 + 13x - 21 = 0$$
$$\Rightarrow 20x^2 + 28x - 15x - 21 = 0$$
$$\Rightarrow 4x(5x + 7) - 3(5x + 7) = 0$$
$$\Rightarrow (5x + 7)(4x - 3) = 0$$

On comparing this equation with the equation $(ax + 7)(4x + b) = 0$, we obtain

$a = 5$ and $b = -3$

$\therefore a + b = 5 + (-3) = 2$

Thus, the value of $a + b$ is 2.

7. Correct Option: (D)

Consider the equation $60x^2 + 50x - 210 = 0$.

$60x^2 + 50x - 210 = 0$

$\Rightarrow 10(6x^2 + 5x - 21) = 0$

The middle term i.e., $5x$ can be split as

$5x = 14x - 9x$ [$14 \times (-9) = -126 = 6 \times (-21)$]

$$\therefore 10\left[6x^2 + 5x - 21\right] = 0$$
$$\Rightarrow 10\left[6x^2 + 14x - 9x - 21\right] = 0$$
$$\Rightarrow 10\left[2x(3x + 7) - 3(3x + 7)\right] = 0$$
$$\Rightarrow 10(3x + 7)(2x - 3) = 0$$

On comparing this equation with the equation $10(ax + b)(cx + 7) = 0$, we obtain

$a = 2$, $b = -3$, and $c = 3$

It can be observed that

$ab + bc + ca = 2(-3) + (-3)(3) + 3(2) = -6 - 9 + 6 = -9$

8. Correct Option: (C)

$(x - k)^2 - (2x - 3)^2$ is divisible by $(x - 2)$

$$\Rightarrow (2 - k)^2 - (4 - 3)^2 = 0$$
$$\Rightarrow 4 + k^2 - 4k - 1 = 0$$
$$\Rightarrow k^2 - 4k + 3 = 0$$
$$\Rightarrow (k - 3)(k - 1) = 0$$
$$\Rightarrow k = 3, 1$$

The correct answer is C.

9. Correct Option: (A)

$y^2 + \frac{1}{2}y - 1 = 0$

$\Rightarrow y^2 + \frac{1}{2}y = 1$

Adding $\left(\frac{1}{4}\right)^2$ i.e. $\frac{1}{16}$ on both sides, we have

$\Rightarrow y^2 + \frac{1}{2}y + \frac{1}{16} = 1 + \frac{1}{16}$

or $\left(y + \frac{1}{4}\right)^2 = \frac{17}{16}$

$\Rightarrow y + \frac{1}{4} = \pm\sqrt{\frac{17}{16}} = \pm\frac{\sqrt{17}}{4}$

$\Rightarrow y = -\frac{1}{4} \pm \frac{\sqrt{17}}{4} = \frac{-1 \pm \sqrt{17}}{4}$

So, the solutions are $\frac{-1+\sqrt{17}}{4}, \frac{-1-\sqrt{17}}{4}$

10. Correct Option: (B)

$81x^2 - 126x - 16 = 0$

$\Rightarrow (9x)^2 - 2\times(9x)\times 7 + (7)^2 - (7)^2 - 16 = 0$

$\Rightarrow (9x-7)^2 - 49 - 16 = 0$

$\Rightarrow (9x-7)^2 - 65 = 0$

$\Rightarrow (9)^2\left(x - \frac{7}{9}\right)^2 - 65 = 0$

$\Rightarrow 81\left(x - \frac{7}{9}\right)^2 - 65 = 0$

On dividing this equation by 81, we obtain

$\left(x - \frac{7}{9}\right)^2 - \frac{65}{81} = 0$

Thus, the equation $81x^2 - 126x - 16 = 0$ can be written as $\left(x - \frac{7}{9}\right)^2 - \frac{65}{81} = 0$.

11. Correct Option: (B)

Consider the equation $5x^2 - 7x - 3 = 0$.

On dividing this equation by 5, we obtain

$\Rightarrow x^2 - \frac{7}{5}x - \frac{3}{5} = 0$

$\Rightarrow x^2 - 2\times\frac{7}{10}\times x + \left(\frac{7}{10}\right)^2 - \left(\frac{7}{10}\right)^2 - \frac{3}{5} = 0$

$\Rightarrow \left(x - \frac{7}{10}\right)^2 - \frac{49}{100} - \frac{3}{5} = 0$

$\Rightarrow \left(x - \frac{7}{10}\right)^2 - \frac{49}{100} - \frac{60}{100} = 0$

$\Rightarrow \left(x - \frac{7}{10}\right)^2 - \frac{109}{100} = 0$

$\Rightarrow \left(x - \frac{7}{10}\right)^2 = \frac{109}{100}$

On comparing this equation with the equation $\left(x-\frac{b}{c}\right)^2=\frac{109}{100}$, we obtain

$b = 7$ and $c = 10$

$\therefore c^2 - b^2 = 10^2 - 7^2 = 100 - 49 = 51$

Thus, the value of the expression $c^2 - b^2$ is 51.

12. Correct Option: (C)

The given equation is $3x^2 + 5x + 7 = 0$.

On dividing this equation by 3, we obtain

$$x^2+\frac{5}{3}x+\frac{7}{3}=0$$

This equation can also be written as

$$x^2+2\times x\times\frac{5}{6}+\left(\frac{5}{6}\right)^2-\left(\frac{5}{6}\right)^2+\frac{7}{3}=0$$

$$\Rightarrow\left(x+\frac{5}{6}\right)^2-\frac{25}{36}+\frac{7}{3}=0$$

$$\Rightarrow\left(x+\frac{5}{6}\right)^2+\frac{84-25}{36}=0$$

$$\Rightarrow\left(x+\frac{5}{6}\right)^2+\frac{59}{36}=0$$

Thus, $3x^2 + 5x + 7 = 0$ can also be written as $\left(x+\frac{5}{6}\right)^2+\frac{59}{36}=0$.

13. Correct Option: (A)

Comparing the given equation

$x^2-2ax+(a^2-b^2)=0$ with the standard quadratic equation

$ax^2+bx+c=0$

We have $a = 1$

$b = -2a$

And $c=a^2-b^2$

Now Discriminant $D=b^2-4ac$

$$=(-2a)^2-4(1)(a^2-b^2)=4a^2-4a^2+4b^2$$

$$\Rightarrow D=(2b)^2$$

$$\Rightarrow \sqrt{D}=2b$$

Using the quadratic formula, we get

$$x=\frac{-b\pm\sqrt{D}}{2a}=\frac{2a\pm 2b}{2\times 1}=a\pm b$$

Hence $x=a+b$ or $x=a-b$.

14. Correct Option: (D)

It is known that the solution of a quadratic equation $ax^2 + bx + c = 0$ is given by $x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$.

The given quadratic equation is $4x^2 + 12x - 8 = 0$.

On comparing this equation with the general form of quadratic equation i.e., $ax^2 + bx + c = 0$, we obtain $a = 4$, $b = 12$, and $c = -8$.

Hence, the solution of the quadratic equation $4x^2 + 12x - 8 = 0$ is given by

$$x = \frac{-12 \pm \sqrt{12^2 - 4\times 4\times(-8)}}{2\times 4}$$
$$= \frac{-12 \pm \sqrt{144+128}}{8}$$
$$= \frac{-12 \pm \sqrt{272}}{8}$$
$$= \frac{-12 \pm 4\sqrt{17}}{8}$$
$$= \frac{-3 \pm \sqrt{17}}{2}$$

On comparing the expression $x = \frac{-3 \pm \sqrt{17}}{2}$ with the given expression i.e., $x = \frac{a \pm \sqrt{17}}{b}$, we obtain $a = -3$ and $b = 2$.

Hence, $b - a = 2 - (-3) = 2 + 3 = 5$

The correct answer is B.

15. Correct Option: (B

$$x = \sqrt{10+\sqrt{10+\sqrt{10....}}} \qquad ...(1)$$
$$\Rightarrow x = \sqrt{10+x}$$

$\Rightarrow x^2 = 10 + x$ [On squaring both sides]

$\Rightarrow x^2 - x - 10 = 0$

The roots of the quadratic equation $ax^2 + bx + c = 0$ are given by the formula: $\frac{-b \pm \sqrt{b^2-4ac}}{2a}$

$$\therefore x = \frac{1 \pm \sqrt{(-1)^2 - 4(1)(-10)}}{2(1)}$$
$$\Rightarrow x = \frac{1 \pm \sqrt{1+40}}{2}$$
$$\Rightarrow x = \frac{1 \pm \sqrt{41}}{2}$$
$$\Rightarrow x = \frac{1 \pm 6.4}{2} \qquad \left[\sqrt{41} \approx 6.4\right]$$
$$\Rightarrow x = 3.7 \text{ or } -2.7$$

Since x cannot be negative, $x = 3.7$

Thus, the approximate value of $\sqrt{10+\sqrt{10}+\sqrt{10...}}$ is 3.7.

16. Correct Option: (D)

$$x = 1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1+....}}}$$

$$\Rightarrow x = 1 + \frac{1}{x}$$
$$\Rightarrow x = \frac{x+1}{x}$$
$$\Rightarrow x^2 - x - 1 = 0$$
$$\Rightarrow x = \frac{1 \pm \sqrt{5}}{2}$$

17. Correct Option: (D)

If the discriminant of a quadratic equation is greater than zero, then its roots are real and distinct.

If the discriminant of a quadratic equation is zero, then its roots are real and equal.

If the discriminant of a quadratic equation is less than zero, then its roots are imaginary.

I. Consider the equation $3x^2 - 4\sqrt{3}x + 4 = 0$.

Comparing this equation with the general form of a quadratic equation $ax^2 + bx + c = 0$, we obtain

$a = 3$, $b = -4\sqrt{3}$, and $c = 4$

$\therefore$ Discriminant (D) $= b^2 - 4ac$

$= \left(-4\sqrt{3}\right)^2 - 4 \times 3 \times 4$

$= 48 - 48$

$= 0$

Since the discriminant is 0, the equation $3x^2 - 4\sqrt{3}x + 4 = 0$ has equal roots.

Thus, statement I is correct.

II. Consider the equation $2x^2 - 5x + 4 = 0$.

Comparing this equation with the general form of a quadratic equation $ax^2 + bx + c = 0$, we obtain

$a = 2$, $b = -5$, and $c = 4$

$\therefore$ Discriminant (D) $= b^2 - 4ac$

$= (-5)^2 - 4 \times 2 \times 4$

$= 25 - 32$

$= -7$

Since the discriminant is less than 0, the equation $2x^2 - 5x + 4 = 0$ has imaginary roots.

Thus, statement II is correct.

Thus, statements I and II are correct.

18. Correct Option: (B)

If the discriminant of a quadratic equation is greater than zero, then its roots are real and distinct.

If the discriminant of a quadratic equation is zero, then its roots are real and equal.

If the discriminant of a quadratic equation is less than zero, then its roots are imaginary.

Consider the equation $8x^2 + 22x - 21 = 0$.

On comparing this equation with the general form of quadratic equation i.e., $ax^2 + bx + c = 0$, we obtain $a = 8$, $b = 22$, and $c = -21$.

$\therefore$ Discriminant (D) $= b^2 - 4ac = (22)^2 - 4 \times 8 \times (-21) = 484 + 672 = 1156$

Since the discriminant is greater than 0, equation $8x^2 + 22x - 21 = 0$ has real and distinct roots.

19. Correct Option: (A)

Since – 4 is a root of $x^2 + px - 4 = 0$ we have

$(-4)^2 + p(-4) - 4 = 0 \quad \Rightarrow \quad 16 - 4p - 4 = 0$

$\Rightarrow \quad 4p = 12 \quad \Rightarrow \quad p = 3 \qquad \ldots(i)$

Putting p = 3 in equation $x^2 + px + q = 0$, we have

$x^2 + 3x + q = 0$

Equation will have equal roots if D = 0 i.e. $b^2 - 4ac = 0$

$\Rightarrow \quad p^2 - 4q = 0 \quad \Rightarrow \quad (3)^2 - 4q = 0$ [Using (i)]

$\Rightarrow$ 9 – 4q = 0 $\Rightarrow$ $q = \frac{9}{4}$.

Hence, p = 3 and $q = \frac{9}{4}$.

20. Correct Option: (B)

Since – 5 is a root of the equation $2x^2 + px - 15 = 0$ Therefore,

$2(-5)^2 - 5p - 15 = 0$

$\Rightarrow \quad 50 - 5p - 15 = 0$

$\Rightarrow \quad 5p = 35$

$\Rightarrow \quad p = 7$

Putting $p = 7$ in $p(x^2 + x) + k = 0,$ we get

$$7x^2 + 7x + k = 0$$

This equation will have equal roots, if

Discriminant = 0

$\Rightarrow$ 49 – 4 × 7 × k = 0

$\Rightarrow \quad k = \frac{49}{28} \Rightarrow k = \frac{7}{4}$

LEVEL-II

1. Correct Option: (A)

x – y = 4 $\Rightarrow$ x – y = ± 4

xy = 192

If we take x – y = 4

then y = x – 4

so x(x – 4) = 192.

2. Correct Option: (A)

Since $2 + i\sqrt{3}$ is a root, therefore $2 - i\sqrt{3}$ will be other root. Now sum of the roots = 4 = – p and product of roots = 7 = q. Hence (p, q) = (–4, 7).

3. Correct Option: (B)

$$\frac{x-1}{x-2} + \frac{x-3}{x-4} = \frac{10}{3}$$

$$\frac{(x-1)^2(x-4) + (x-3)(x-2)}{(x-2)(x-4)} = \frac{10}{3}$$

$$\frac{x^2 - 5x + 4 + x^2 - 5x + 6}{x^2 - 6x + 8} = \frac{10}{3}$$

$3(2x^2 - 10x + 10) = 101x^2 - 6x + 8)$

$4x^2 - 15x + 25 = 0$

$2x^2 - 10x - 5x + 25 = 0$

$2x^2 - 10x - 5x + 25 = 0$

$2x(x - 5) - 5(x - 5) = 0$

$(x - 5)\ (2x - 5) = 0$

$x = 5, 5/2.$

4. Correct Option: (D)

$x^2 + x + 1$

$D = (1)^2 - 4 \times 1 \times 1$

$= -3$ $D < 0$ then the roots are imaginary.

5. Correct Option: (A)

The given equation is $x^2 - 4x + k = 0$

Here, $a = 1$, $b = -4$ and $c = k$

$\therefore \quad D = (-4)^2 - 4 \times 1 \times k = 16 - 4k$

The given equation will have real and distinct roots, if

$D > 0 \Rightarrow 16 - 4k > 0 \Rightarrow 16 > 4k \Rightarrow 4k < 16 \Rightarrow k < 16/4 = 4$

Hence, the given equation will have distinct roots, if $k < 4$.

6. Correct Option: (B)

7. Correct Option: (D)

If $x = 3$ is a solution of $ax^2 + 2x + 3 = 0$,

$a(3)^2 + 2(3) + 3 = 0 \Rightarrow 9a + 9 = 0 \Rightarrow a = -1$.

8. Correct Option: (B)

Let the sides of the rectangle be x and y.

Given, perimeter = 82 m $\Rightarrow 2(x + y) = 82 \Rightarrow x + y = 41$

$\Rightarrow \quad y = 41 - x$ .....(i)

Given also, area = 400 m^2 $\Rightarrow xy = 400$ .....(ii)

Substituting the value of y from (i) in (ii),

$x(41 - x) = 400 \quad \Rightarrow \quad 41x - x^2 = 400$

$\Rightarrow \quad x^2 - 41x + 400 = 0 \Rightarrow (x - 25)(x - 16) = 0$

$\Rightarrow \quad x = 25$ or $x = 16$

when $x = 25$, $y = 41 - 25 = 16$ when $x = 16$, $y = 41 - 16 = 25$

Therefore, the sides of the rectangle are 16 m and 25 m.

$\therefore$ length of its diagonal $= \sqrt{25^2 + 16^2} = \sqrt{625 + 256} = \sqrt{881}$m.

9. Correct Option: (A)

Let α be a root of equation, then $\frac{1}{\alpha}$ be a root of second equation, therefore

$a\alpha^2 + b\alpha + c = 0$

and $a'\frac{1}{\alpha^2} + b'\frac{1}{\alpha} + c' = 0$

or $c'\alpha^2 + b'\alpha + a' = 0$

On solving eqs. (i) and (ii)

$$\frac{\alpha^2}{ba' - b'c} = \frac{\alpha}{cc' - aa'} = \frac{1}{ab' - bc'}$$

$$\Rightarrow \quad (cc' - aa')^2 = (ba' - cb')(ab' - bc').$$

10. Correct Option: (A)

We have, $x = \sqrt{1 + \sqrt{1 + \sqrt{1 + ...\infty}}}$

$\Rightarrow \quad x = \sqrt{1 + x}$

$\Rightarrow \quad x^2 = 1 + x \Rightarrow x^2 - x - 1 = 0$

$\Rightarrow \quad x = \frac{1 \pm \sqrt{1+4}}{2} = \frac{1 \pm \sqrt{5}}{2}$

As $x > 0$ we take only $x = \frac{1+\sqrt{5}}{2}$.

11. Correct Option: (BC)

Given equation is $3^{x-1} + 5^{x-1} = 152$

$\Rightarrow \quad 3x^{x-1} + 5^{x-1} = 27 + 125$

$\Rightarrow \quad 3x^{x-1} + 5^{x-1} = 3^3 + 5^3$

So, $x - 1 = 3$ obviously $\Rightarrow$ $x = 4$ is the only solution of the given equation.

12. Correct Option: (AB)

Given in equation can be written as

$(ax - b)^2 + (bx - c)^2 + (cx - d)^2 \leq 0$

$\Rightarrow \quad ax - b = 0, \; bx - c = 0, \; cx - d = 0$

$(\because a, b, c, d, x \in R$ and sum of squares of reals cannot be negative) .

$\Rightarrow \quad x = \frac{b}{a} = \frac{c}{b} = \frac{d}{c}$

$\Rightarrow$ a, b, c, d are in G.P.

Note that $\frac{b}{a} = \frac{d}{c} \Leftrightarrow ad - bc = 0$.

13. Correct Option: (AC)

$x^{2/3} + x^{1/3} - 2 = 0 \Rightarrow (x^{1/3})^2 + 1(x^{1/3}) - 2 = 0$

Let $a = x^{1/3}$, then $a^2 + a - 2 = 0$

Hence $x = 1, -8$ (by $a = x^{1/3}$) .

14. Correct Option: (A)

15. Correct Option: (B)

16. Correct Matching is: A - r, B - p, C - qr, D – qrt

(A) $x^2 - |x| - 2 = 0$

$\Rightarrow \quad |x|^2 - |x| - 2 = 0$

$\Rightarrow \quad |x| = 2, -1$ but $|x| \neq -1$

$\therefore \quad |x| = 2 \Rightarrow x = \pm 2$

$\therefore$ The equation $x^2 - |x| = 2 = 0$ has 2 real roots.

(B) $x^2 + x + 1 = x^2 + x + \frac{1}{4} + \left(1 - \frac{1}{4}\right) = \left(x + \frac{1}{2}\right)^2 + \frac{3}{4} \geq \frac{3}{4}$

(C) $4x^2 - 11x + 6 = 0$

$\Rightarrow \quad x = \frac{11 \pm \sqrt{121 - 96}}{8} = \frac{11 \pm 5}{8} = 2, \frac{3}{4}$

(D) Let $f(x) = x^3 - 7x + 6 = 0$

As $f(1) = 1 - 7 + 6 = 0$

$\Rightarrow$ 1 is a root of $x^3 - 7x + 6 = 0$

$\Rightarrow$ x – 1 is a factor of f(x).

$\therefore$ f(x) can be written as (x – 1) $\{x^2 + x - 6\} = 0$ (By long division)

$\Rightarrow$ x = 1 or $x^2 + x - 0$

$\Rightarrow \quad x = \frac{-1 \pm \sqrt{1+24}}{2} = \frac{-1 \pm 5}{2}$.

17. Correct Matching is: A - s, B - p, C - q, D - r

18. Correct Answer is: 0

19. Correct Answer is: 3

(a) The given equation is

$x^2 - 4x + 2 = 0$

Here a = 1, b = –4 and c = 2

$\therefore$ $D = b^2 - 4ac = (-4)^2 - 4 \times 1 \times 2 = 16 - 8 = 8$.

min value $= -\frac{D}{4a} = \frac{0}{4} = 0$.

20. Correct Answer is: 9

STUDENT PRACTICE QUESTIONS

1. Correct Option: (C)

Let x be the breadth of the rectangle.

$\Rightarrow$ The length of the rectangle = $x + 5$

Area of the rectangle = 50 cm^2

$\Rightarrow x(x + 5) = 50$

$\Rightarrow x^2 + 5x = 50$

$\Rightarrow x^2 + 5x - 50 = 0$

2. Correct Option: (C)

An equation is called quadratic if the highest power of the variable in it is 2.

The given equation is $(x + 2)^2 = 3x$.

$\Rightarrow x^2 + 4x + 4 = 3x$

$\Rightarrow x^2 + x + 4 = 0$

Since the highest power of the variable x is 2, this equation is quadratic in nature.

Hence, the equation given in alternative C is quadratic in nature.

3. Correct Option: (B)

Let the breadth of the rectangle be x cm

Then, length of the rectangle = $(2x - 3)$ cm

Area of the original rectangle = $[(2x - 3) \times x]$ cm^2 = $(2x^2 - 3x)$ cm^2

Then, the length is decreased by 4 cm and the breadth is increased by 2 cm

Length of the new rectangle = $(2x - 3)$ cm − 4 cm = $(2x - 7)$ cm

Breadth of the new rectangle = $(x + 2)$ cm

Area of the new rectangle = $[(2x - 7)(x + 2)]$ cm^2 = $(2x^2 + 4x - 7x - 14)$ cm^2 = $(2x^2 - 3x - 14)$ cm^2

According to the given information:

$2x^2 - 3x - 14 = 336$

$\Rightarrow 2x^2 - 3x = 336 + 14$

$\Rightarrow 2x^2 - 3x = 350$

Thus, the area of the original rectangle is $(2x^2 - 3x)$ cm^2 = 350 cm^2.

4. Correct Option: (C)

$4x^2 + 17 = 97$

$\Rightarrow 4x^2 = 97 - 17$

$\Rightarrow 4x^2 = 80$

$\Rightarrow x^2 = \frac{80}{4} = 20$

$\Rightarrow x = \sqrt{20} = 2\sqrt{5}$

Now, $\sqrt{5} = 2.236 \cong 2.2$

$\Rightarrow x = 2 \times 2.2 = 4.4$

5. Correct Option: (C)

$2x^2 + 4x - 3 = 0$

$\alpha + \beta = -4/2$

$\alpha \times \beta = -3/2$

$\Rightarrow 4 [(\alpha + \beta)^2 - 2\alpha\beta] = 4 [(-4/2)^2 - 2(-3/2)] = 4 [4 + 3] = 28$

6. Correct Option: (B)

Let h be the root of the equation $y^2 - y + k = 0$ and let $2h$ be the root of the equation $y^2 - 3y + 2k = 0$.

Then, $h^2 - h + k = 0$ and $4h^2 - 6h + 2k = 0$

$\Rightarrow (h^2/-k) = (h/-k) = 1/2$ (Solving the two equations by cross multiplication)

$\Rightarrow h^2 = -2h \Rightarrow h = 0, -2$

7. Correct Option: (B)

The given equation is: $x^2 - mx - n = 2$

Since m is a root of the given equation, m satisfies the equation $x^2 - mx - n = 2$

$\therefore (m)^2 - m(m) - n = 2$

$\Rightarrow m^2 - m^2 - n = 2$

$\Rightarrow n = -2$

Thus, the value of n is -2.

8. Correct Option: (C)

Let the length of the shorter perpendicular side of the triangle be x m.

Then, length of the other perpendicular side = $(x + 1)$ m

Length of the hypotenuse = 29 m

By applying Pythagoras theorem in the given triangle, we obtain

$x^2 + (x + 1)^2 = 29^2$

$\Rightarrow x^2 + x^2 + 2x + 1 = 841$

$\Rightarrow 2x^2 + 2x - 840 = 0$

$\Rightarrow x^2 + x - 420 = 0$

$\Rightarrow x^2 + 21x - 20x - 420 = 0$

$\Rightarrow x(x + 21) - 20(x + 21) = 0$

$\Rightarrow (x + 21)(x - 20) = 0$

$\Rightarrow x = -21$ or $x = 20$

Since the length of the side cannot be negative, $x = 20$

Therefore, the sides of the given triangle are 20 m, 21 m, and 29 m, out of which, the lengths of the perpendicular sides are 20 m and 21 m.

$\therefore$ Area of the triangle = $\frac{1}{2} \times Base \times Height = \frac{1}{2} \times 21 \times 20 m^2 = 210 m^2$

9. Correct Option: (B)

$3^{2x^2 - 7x + 7} = 9$

$\Rightarrow 2x^2 - 7x + 7 = 2$

$\Rightarrow 2x(x - 1) - 5(x - 1) = 0$

$\Rightarrow x = 1, 5/2$

Hence, the number of the real roots is two.

10. Correct Option: (B)

The equation is $x^2 - 8x + 16 = 0$

$\Rightarrow x^2 - 4x - 4x + 16 = 0$

$\Rightarrow (x-4)(x-4) = 0$

$\Rightarrow x = 4, 4$

11. Correct Option: (C)

Let $y = \sqrt{6+\sqrt{6\sqrt{6+...}}}$

Then, $y = \sqrt{6+y} \Rightarrow y^2 = (6+y)$ (squaring both the sides)

$\Rightarrow y^2 - y - 6 = 0$

$\Rightarrow (y + 2)(y - 3) = 0 \Rightarrow y = -2, 3$

The negative value of y is discarded.

$\therefore y = 3$

12. Correct Option: (B)

We have,

$4x^2 + 4bx - (a^2 - b^2) = 0$

$\Rightarrow x^2 + bx - \left(\frac{a^2 - b^2}{4}\right) = 0$ [Dividing the equation by 4]

$\Rightarrow x^2 + 2\left(\frac{b}{2}\right)x = \frac{a^2 - b^2}{4}$

Adding $\left(\frac{b}{2}\right)^2$ on both sides we have

$\Rightarrow x^2 + 2\left(\frac{b}{2}\right)x + \left(\frac{b}{2}\right)^2 = \frac{a^2 - b^2}{4} + \left(\frac{b}{2}\right)^2$

$= \frac{a^2 - b^2}{4} + \frac{b^2}{4}$

$\Rightarrow \left(x + \frac{b}{2}\right)^2 = \frac{a^2}{4}$

$\Rightarrow x + \frac{b}{2} = \pm\frac{a}{2}$ [taking square root of both sides]

$\Rightarrow x = -\frac{b}{2} \pm \frac{a}{2}$

$\Rightarrow x = \frac{-b-a}{2}, \frac{-b+a}{2}$

Hence, the roots are $\frac{-b-a}{2}$ and $\frac{-b+a}{2}$

13. Correct Option: (C)

We have,

$x^2 - (\sqrt{3}+1)x + \sqrt{3} = 0$

$\Rightarrow x^2 - (\sqrt{3}+1)x = -\sqrt{3}$

Adding $\left(\frac{\sqrt{3}+1}{2}\right)^2$ on both sides we have

$\Rightarrow x^2 - 2\left(\frac{\sqrt{3}+1}{2}\right)x + \left(\frac{\sqrt{3}+1}{2}\right)^2 = -\sqrt{3} + \left(\frac{\sqrt{3}+1}{2}\right)^2$

$\Rightarrow \left(x - \frac{\sqrt{3}+1}{2}\right)^2 = -\sqrt{3} + \frac{3+1+2\sqrt{3}}{4}$

$= \frac{-4\sqrt{3}+3+1+2\sqrt{3}}{4} = \frac{3+1-2\sqrt{3}}{4}$

$\Rightarrow \left(x - \frac{\sqrt{3}+1}{2}\right)^2 = \left(\frac{\sqrt{3}-1}{2}\right)^2$

$\Rightarrow x - \frac{\sqrt{3}+1}{2} = \pm\frac{\sqrt{3}-1}{2}$

$\Rightarrow x = \frac{\sqrt{3}-1}{2} + \frac{\sqrt{3}+1}{2} = \frac{2\sqrt{3}}{2} = \sqrt{3}$

or $x - \frac{\sqrt{3}+1}{2} = -\frac{\sqrt{3}-1}{2}$

$x = -\left(\frac{\sqrt{3}-1}{2}\right) + \frac{\sqrt{3}+1}{2} = \frac{2}{2} = 1$

$\Rightarrow x = \sqrt{3}, 1$

Hence, the roots are $\sqrt{3}$ and 1.

14. Correct Option: (C)

$x^2 + 8x - 4 = 0$

$\Rightarrow x^2 + 8x = 4$ (adding 4 both sides)

$\Rightarrow x^2 + 8x + 16 = 20$

$\Rightarrow (x + 4)^2 = 20$

$\Rightarrow x + 4 = \sqrt{20}$

$\Rightarrow x = -4 \pm 2\sqrt{5}$

15. Correct Option: (A)

$x^2 + 4x - 5 = 0$

$\Rightarrow (x)^2 + 2.(x).2 + (2)^2 - (2)^2 - 5 = 0$

$\Rightarrow (x+2)^2 - 4 - 5 = 0$

$\Rightarrow (x+2)^2 = 9$

$\Rightarrow (x+2)^2 = (3)^2$

$\Rightarrow x + 2 = \pm 3$

Now, $x + 2 = 3$ or $x + 2 = -3$

$\Rightarrow x = 3 - 2$ $\Rightarrow x = -3 - 2$

$\therefore x = 1$ $\therefore x = -5$

Hence, $x = 1, -5$

16. Correct Option: (B)

$5x^2 - 6x - 2 = 0$

$\Rightarrow 25x^2 - 30x - 10 = 0$ [Multiplying both sides by 5, we get]

$\Rightarrow (5x)^2 - 2(5x).3 + (3)^2 - 19 = 0$

$\Rightarrow \quad (5x-3)^2 - 19 = 0$

$\Rightarrow \quad 5x - 3 = \pm\sqrt{19}$ [Taking square root both sides]

$\Rightarrow \quad 5x = 3 \pm \sqrt{19}$

$\Rightarrow \quad x = \frac{3 \pm \sqrt{19}}{5}$

The roots are $\frac{3+\sqrt{19}}{5}$ and $\frac{3-\sqrt{19}}{5}$

17. Correct Option: (C)

The given equation is $2x^2 - 7x + 11 = 0$.

On dividing this equation by 2, we obtain

$x^2 - \frac{7}{2}x + \frac{11}{2} = 0$

This equation can also be written as

$x^2 - 2 \times x \times \frac{7}{4} + \left(\frac{7}{4}\right)^2 - \left(\frac{7}{4}\right)^2 + \frac{11}{2} = 0$

$\Rightarrow \left(x - \frac{7}{4}\right)^2 - \frac{49}{16} + \frac{11}{2} = 0$

$\Rightarrow \left(x - \frac{7}{4}\right)^2 + \frac{88-49}{16} = 0$

$\Rightarrow \left(x - \frac{7}{4}\right)^2 + \frac{39}{16} = 0$

.

On comparing this equation with the equation $(x+a)^2 + \frac{39}{16} = 0$, we obtain $a = -\frac{7}{4}$

18. Correct Option: (B)

$3x^2 - 2x = 8$

$\Rightarrow 3x^2 - 2x - 8 = 0$

$\Rightarrow x^2 - \frac{2}{3}x - \frac{8}{3} = 0$

$\Rightarrow x^2 - 2x\frac{1}{3} - \frac{8}{3} = 0$

$\Rightarrow x^2 - 2x\frac{1}{3} + \frac{1}{9} - \frac{1}{9} - \frac{8}{3} = 0$

$\Rightarrow x^2 - 2x\frac{1}{3} + \left(\frac{1}{3}\right)^2 = \frac{1}{9} + \frac{8}{3}$

$\Rightarrow \left(x - \frac{1}{3}\right)^2 = \frac{1+24}{9}$

$\Rightarrow \left(x - \frac{1}{3}\right)^2 = \frac{25}{9}$

$\therefore$ The value of $\left(x - \frac{1}{3}\right)^2$ is $\frac{25}{9}$.

19. Correct Option: (B)

We have, $p^2x^2 + (p^2 - q^2)x - q^2 = 0$

Comparing the equation with $ax^2 + bx + c = 0$, we have

$a = p^2$, $b = p^2 - q^2$ and $c = -q^2$

$\therefore \quad D = b^2 - 4ac = (p^2 - q^2)^2 - 4 \times p^2 \times -q^2 = (p^2 - q^2)^2 + 4p^2q^2$

$= p^4 + q^4 - 2p^2q^2 + 4p^2q^2 = (p^2 + q^2)^2 > 0$

[$\because$ square of any real number is non-negative]

So, the given equation has real roots given by

$$\alpha = \frac{-b + \sqrt{D}}{2a} = \frac{-(p^2 - q^2) + (p^2 + q^2)}{2p^2} = \frac{2q^2}{2p^2} = \frac{q^2}{p^2}$$

and, $\beta = \frac{-b - \sqrt{D}}{2a} = \frac{-(p^2 - q^2) - (p^2 + q^2)}{2p^2} = -1$

Hence the roots are $\frac{q^2}{p^2}$ and -1

20. Correct Option: (C)

We have, $9x^2 - 9(a+b)x + (2a^2 + 5ab + 2b^2) = 0$

Comparing this equation with $ax^2 + bx + c = 0$, we have

$a = 9,\ b = -9\ (a+b)$ and $c = 2a^2 + 5ab + 2b^2$

$\therefore\ D = b^2 - 4ac$

$\Rightarrow D = 81(a+b)^2 - 36(2a^2 + 5ab + 2b^2)$

$\Rightarrow D = 81\left(a^2 + b^2 + 2ab\right) - 72a^2 - 180ab - 72b^2$

$\Rightarrow D = 81a^2 + 81b^2 + 162ab - 72a^2 - 72b^2 - 180ab$

$\Rightarrow D = 9a^2 + 9b^2 - 18ab$

$\Rightarrow D = 9(a-b)^2 > 0$ [$\because$ square of any real number is non-negative]

So, the roots of the given equation are real and are given by

$$\alpha = \frac{-b + \sqrt{D}}{2a} = \frac{9(a+b) + 3(a-b)}{18} = \frac{12a + 6b}{18} = \frac{2a + b}{3}$$

and, $\beta = \frac{-b - \sqrt{D}}{2a} = \frac{9(a+b) - 3(a-b)}{18} = \frac{6a + 12b}{18} = \frac{a + 2b}{3}$

$\therefore$ Solution is $x = \frac{2a+b}{3}$ and $\frac{a+2b}{3}$

21. Correct Option: (D)

$a^2b^2x^2 - (4b^4 - 3a^4)x - 12a^2b^2 = 0$

Comparing the given equation with $ax^2 + bx + c = 0$,

We have $a = a^2b^2, b = -(4b^4 - 3a^4), c = -12a^2b^2$

Discriminant $D = b^2 - 4ac$

$D = (3a^4 - 4b^4)^2 - 4a^2b^2(-12a^2b^2)$

$= 9a^8 + 16b^8 - 24a^4b^4 + 48a^4b^4$

$= 9a^8 + 16b^8 + 24a^4b^4$

$= (3a^4)^2 + (4b^4)^2 + 2 \times 3a^4 \times 4b^4$

$= (3a^4 + 4b^4)^2 \geq 0$ [$\because$ square of any real number is non-negative]

Hence, the given equation has real roots given by

$$x = \frac{4b^4 - 3a^4 \pm \sqrt{(3a^4 + 4b^4)^2}}{2a^2b^2} \qquad \left[\because x = \frac{-b \pm \sqrt{D}}{2a}\right]$$

The roots are

$$\frac{4b^4 - 3a^4 + 3a^4 + 4b^4}{2a^2b^2} \text{ and } \frac{4b^4 - 3a^4 - 3a^4 - 4b^4}{2a^2b^2}$$

or $\frac{8b^4}{2a^2b^2}$ and $\frac{-6a^4}{2a^2b^2}$

or $\frac{4b^2}{a^2}$ and $-\frac{3a^2}{b^2}$

Hence, $x = \frac{4b^2}{a^2}, -\frac{3a^2}{b^2}$ are the required solutions.

22. Correct Option: (D)

The given equation is $4x^2 - 12x + 7 = 0$

Comparing it with general form of quadratic equation $ax^2 + bx + c = 0$

$a = 4$, $b = -12$, $c = 7$

Thus, the solution is:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2 \times a}$$

$$= \frac{-(-12) \pm \sqrt{(-12)^2 - 4 \times 4 \times 7}}{2 \times 4}$$

$$= \frac{12 \pm \sqrt{144 - 112}}{8}$$

$$= \frac{12 \pm \sqrt{32}}{8}$$

$$= \frac{12 \pm 4\sqrt{2}}{8} \left[\text{Since, } 32 = 4^2 \times 2\right]$$

$$= \frac{4\left(3 \pm \sqrt{2}\right)}{8}$$

$$= \frac{3 \pm \sqrt{2}}{2}$$

Therefore, the required solution is $\left\{\frac{3+\sqrt{2}}{2}, \frac{3-\sqrt{2}}{2}\right\}$

The correct answer is D.

23. Correct Option: (A)

It is known that for a quadratic equation $ax^2 + bx + c = 0$, the solution is given by $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

The given quadratic equation is $x^2 - 2\sqrt{3}x + 3 = 0$.

Comparing this equation with the general form of a quadratic equation $ax^2 + bx + c = 0$, we obtain

$a = 1$, $b = -2\sqrt{3}$, and $c = 3$

Hence, the solution of the quadratic equation $x^2 - 2\sqrt{3}x + 3 = 0$ is given by

$$x = \frac{-\left(-2\sqrt{3}\right) \pm \sqrt{\left(-2\sqrt{3}\right)^2 - 4 \times 1 \times 3}}{2}$$

$$= \frac{2\sqrt{3} \pm \sqrt{12 - 12}}{2}$$

$$= \frac{2\sqrt{3} \pm \sqrt{0}}{2}$$

Comparing the expression $\frac{2\sqrt{3} \pm \sqrt{0}}{2}$ with the given expression $x = \frac{2\sqrt{3} \pm \sqrt{k}}{2}$, the value of k is 0.

24. Correct Option: (D)

An equation has equal roots only if the determinant of the equation is zero.

$\therefore b^2 - 4ac = 0$

$\Rightarrow (2k + 4)^2 - 4(4 - k)(8k + 1) = 0$

$\Rightarrow 4[(k + 2)^2 - (4 - k)(8k + 1)] = 0$

$\Rightarrow 9k^2 - 27k = 0$

$\Rightarrow k = 0, k = 3$

25. Correct Option: (C)

If the discriminant of a quadratic equation is greater than zero, then its roots are real and distinct.

Quadratic equation $ax^2 + bx + c = 0$ will have real and distinct roots if discriminant = $b^2 - 4ac > 0$.

Consider the condition: 'a and c bear opposite signs'

It is clear that if a and c bear opposite signs, then $ac < 0$.

$\therefore -4ac > 0$

Also, we know that $b^2 > 0$ for any value of b.

$\therefore b^2 - 4ac > 0$

Thus, if a and c bear opposite signs, then the quadratic equation $ax^2 + bx + c = 0$ will always have real and distinct roots.

26. Correct Option: (D)

Since –4 is a root of the equation $x^2 + px - 4 = 0$

$\therefore$ $(-4)^2 + p \times (-4) - 4 = 0$ [$\because$ A root always satisfies the equation]

$\Rightarrow$ $16 - 4p - 4 = 0$

$\Rightarrow$ $4p = 12$ $p = 3$

The equation $x^2 + px + k = 0$ has equal roots

$\therefore$ Discriminant = 0

$\Rightarrow$ $p^2 - 4k = 0$ [$\because$ a = 1, b = p and c = k]

$\Rightarrow$ $9 - 4k = 0$ [$\because$ p = 3]

$\Rightarrow$ $k = 9/4$.

27. Correct Option: (B)

Let the digits at tens and units place of the number be x and y respectively. Then,

Number = $10x + y$

It is given that

Number = 4 × Sum of the digits. Also, Number = 3 × Product of digits

$\Rightarrow$ $10x + y = 4(x + y)$ and $10x + y = 3xy$

$\Rightarrow$ $6x - 3y = 0$ and $10x + y = 3xy$

$\Rightarrow$ $y = 2x$ and $10x + y = 3xy$

$\Rightarrow$ $10x + 2x = 3x \times 2x$ [On eliminating y]

$\Rightarrow$ $6x^2 - 12x = 0$ $6x(x - 2) = 0$

$\Rightarrow$ $x = 0$, or $x = 2$

Since the given number is a two digit number. So, its tens digit cannot be zero.

$\therefore$ $x = 2$

$\Rightarrow$ $y = 2 \times 2 = 4$ [$\because$ $y = 2x$]

Hence, required number = $10x + y = 10 \times 2 + 4 = 24$.

28. Correct Option: (A)

Suppose, one year ago, son's age be x years.

Then, man's age one year ago = 8x years.

$\therefore$ Present age of son = (x + 1) years and, present age of man = (8x + 1) years.

$\therefore$ $8x + 1 = (x + 1)^2$ [Given]

$\Rightarrow$ $x^2 - 6x = 0$

$\Rightarrow$ x = 0 or, x = 6

$\Rightarrow$ x = 6 [$\because$ Son's age cannot be 0)

So, Present age of son = (x + 1) years = 7 years.

and, Present age of man = (8x + 1) years = 49 years.

29. Correct Option: (B)

We can write the given equation as $\frac{p}{2x} = \frac{(a+b)\ x\ +\ c\ (b-a)}{x^2 - c^2}$

or $p(x^2 - c^2) = 2\ (a + b)\ x^2 - 2c\ (a - b)\ x$

or $(2a + 2b - p)\ x^2 - 2c\ (a - b)\ x + pc^2 = 0$

For this equation to have equal roots

$c^2(a-b)^2 - pc^2\ (2a + 2b - p) = 0$

$\Rightarrow$ $(a - b)^2 - 2p(a + b) + p^2 = 0$ [$\because$ c^2 0]

$\Rightarrow$ $[p - (a + b)]^2 = (a + b)^2 - (a - b)^2 = 4ab$

$\Rightarrow$ $p - (a + b) = \pm\ 2\ \sqrt{ab}$

$\Rightarrow$ $p = a + b \pm 2\ \sqrt{ab} = (\sqrt{a} \pm\sqrt{b})^2$

Hence option (b) is correct.

30. Correct Option: (A)

Let the total number of swans = x.

Given that 7/2 times the square root of the total number of swans are on the bank.

$\therefore$ number of swans on the bank $= \frac{7}{2}\sqrt{x}$

$\therefore$ $\frac{7}{2}\sqrt{x} + 2 = x \Rightarrow \frac{7}{2}\sqrt{x} = x - 2 \Rightarrow \frac{49}{4}x = x^2 + 4 - 4x$

$\Rightarrow$ $49x = 4x^2 + 16 - 16x \Rightarrow 4x^2 - 65x + 16 = 0$

$\Rightarrow$ $(4x - 1)\ (x - 16) = 0 \Rightarrow x = \frac{1}{4}$ or $x = 16$

Since x cannot be a fraction, x = 16.

31. Correct Option: (BC)

Here, $\sin\theta + \cos\theta = -\frac{b}{a}$ .....(i)

and $\sin\theta + \cos\theta = \frac{c}{a}$ .....(ii)

Squarring (i), we get

$1 + 2\sin\theta\cos\theta = \frac{b^2}{a^2}$

$\Rightarrow$ $1 + \frac{2c}{a} = \frac{b^2}{a^2} \Rightarrow a^2 + 2ac = b^2$ .....(iii)

$\Rightarrow$ $a^2 + 2ac + c^2 = b^2 + c^2$

$\Rightarrow$ $(a + c)^2 = b^2 + c^2$

Also, from (ii), $a^2 - b^2 + 2ac = 0$.

32. Correct Option: (CD)

Given equations is

$$\frac{\lambda}{2x}=\frac{a}{x+c}+\frac{b}{x-c}$$

or $\lambda\left(x^2-c^2\right)=\left\{a\left(x-c\right)+b\left(x+c\right)\right\}2x$

or $\lambda x^2-\lambda c^2=\left(ax+bx\right)2x+\left(bc-ac\right)2x$

or $x^2\left[2a+2b-\lambda\right]+2x\left(b-a\right)c+\lambda c^2=0$

As this equation has equal roots, therefore, disc. = 0

$\Rightarrow \left[2\left(b-a\right)c\right]^2-4\lambda c^2\left[2a+2b-\lambda\right]=0$

$\Rightarrow \left(b-a\right)^2-\lambda\left(2a+2b-\lambda\right)=0$

$$\Rightarrow \lambda=\frac{2\left(a+b\right)\pm\sqrt{4\left(a+b\right)^2-4\left(b-a\right)^2}}{2}$$

$$=a+b\pm\sqrt{4ab}=a+b\pm2\sqrt{ab}$$

$$=\left(\sqrt{a}\pm\sqrt{b}\right)^2.$$

33. Correct Matching is: A - s, B - p, C - q, D – r

34. Correct Answer is: 4

The equation $(|x|-4)(|x|-3)=0$

$\Rightarrow |x|=4\Rightarrow x=\pm4,\ |x|=3\Rightarrow x=\pm3.$

35. Correct Answer is: 5

Let D be the discriminate of the given equation

We have a = k + 4, b = k + 1 and c = 1

$\therefore$ $D = b^2 - 4ac$

$\Rightarrow$ $D = (k + 1)^2 - 4(k + 4)$

$\Rightarrow$ $D = k^2 - 2k - 15 = (k - 5)(k + 3)$

If the roots of the given equation are real, then

$D = 0 \Rightarrow (k - 5)(k + 3) = 0 \Rightarrow k = 5, -3.$

4. Progressions

Solutions

LEVEL – I

1. Correct option: (B)

To find the number of terms in an arithmetic progression (A.P.), we can use the formula:

$$[n = \frac{a_n - a_1}{d} + 1]$$

In our A.P., the first term a_1 is 20, the last term a_n is 100, and the common difference d is 25 - 20 = 5.

Now, plug these values into the formula:

$$[n = \frac{100 - 20}{5} + 1]$$

So, there are 17 terms in the A.P.

Then the correct option is B)

2. Correct option: (A)

Given that: $a_{11} = 80, a_{16} = 40$

Formula for nth term

$$a_n = a_1 + (n-1)d$$

Then

$$80 = a_1 + 10d$$
$$40 = a_1 + 15d$$
$$=> -40 = 5d => d = -8$$
$$=> a_1 = 80 - 10d = 160$$

For a_{31}

$$= 160 + (30)d = 160 - 240 = -80$$

Then the correct option is A)

3. Correct option: (C)

Given that: $a_1 = a_2 = 2$

Our A.P start from the second term as

$$m_1 = 2, m_{n-1} = a_n, d = -1$$

For $a_5 = m_4$

$$= 2 + (3)(-1) = -1$$

Then the correct option is C)

4. Correct option: (D)

Given: 5th term is 5 times the 8th term, which can be written as:

$$5(a_5) = 8(a_8)$$

This leads to:

$$3a + 36d = 0$$

Now, for the 13th term

$$a + 12d = 0$$

Therefore, $a_{13} = 0$ (Option D).

5. Correct option: (B)

Given that

$$\frac{1}{a}, \frac{1}{b}, \frac{1}{c} \; are \; in \; A.P => \frac{1}{a} + \frac{1}{c} = \frac{2}{b}$$

Then we have to find the value of

$$\left(\frac{1}{a} + \frac{1}{b} - \frac{1}{c}\right)\left(\frac{1}{b} + \frac{1}{c} - \frac{1}{a}\right)$$

Putting the value from equation 1

$$\left(\frac{1}{b}+\left(\frac{1}{a}-\frac{1}{c}\right)\right)\left(\frac{1}{b}-\left(\frac{1}{a}-\frac{1}{c}\right)\right)=\left(\frac{1}{b}\right)^2-\left(\frac{1}{a}-\frac{1}{c}\right)^2$$

Then

$$\left(\frac{1}{b}\right)^2-\left(\frac{1}{a}-\frac{1}{c}\right)^2=\left(\frac{1}{b}\right)^2-\left[\left(\frac{1}{a}+\frac{1}{c}\right)^2-\frac{4}{ac}\right]$$
$$=\left(\frac{1}{b}\right)^2-\left[\left(\frac{2}{b}\right)^2-\frac{4}{ac}\right]=\frac{4}{ac}-\frac{3}{b^2}$$

Then the correct option is B)

6. Correct option: (D)

To find the number of two-digit numbers divisible by 6, we can set up an arithmetic progression (A.P.).

Now, we need to find the last term in this progression that is still a two-digit number. The largest two-digit number is 99.

In this case: $a = 12\ d = 6$

Now, we want to find n when $a_n$the largest term is less than or equal to 99:

$$a_n \leq 99$$
$$12 + (n-1) \times 6 \leq 99$$
$$6(n-1) \leq 87$$
$$(n-1) \leq \frac{87}{6}$$
$$n \leq \frac{87}{6} + 1$$
$$n \leq 15.5$$

Since n must be a whole number, the largest n that still give we a two-digit number divisible by 6 is 15.

So, there are 15 two-digit numbers divisible by 6.

The answer is D) 15.

7. Correct option: (C)

The sum of an arithmetic progression (A.P.) can be calculated using the formula:

$$S_n = \left(\frac{n}{2}\right)[2a + (n-1)d] = \frac{n}{2}(a_1 + a_n)$$

Given:

First term (a) = a

Second term (b) = b

Last term (c) = c

The common difference (d) can be calculated as:

$$d = (b - a)$$

For n:

$$c = a + (n-1)d$$
$$n = \frac{c-a}{b-a} + 1 = \frac{c+b-2a}{b-a}$$

Then sum is

$$= \left(\frac{n}{2}\right)[a+c] = \frac{(b+c-2a)(a+c)}{2(b-a)}$$

Sum of given A.P in terms of first second and last term as

$$\frac{(b+c-2a)(a+c)}{2(b-a)}$$

Then the correct option is C)

8. Correct option: (C)

The sum of an arithmetic progression (A.P.) can be calculated using the formula:

$$S_n = \left(\frac{n}{2}\right)[2a + (n-1)d]$$

Given series is

$$\sqrt{2} + \sqrt{8} + \sqrt{18} + \sqrt{32} + \cdots$$

We can write the given series in terms of $\sqrt{2}$ as

$$\sqrt{2} + 2\sqrt{2} + 3\sqrt{2} + 4\sqrt{2} \ldots = \sqrt{2}(1 + 2 + 3 + 4 \ldots)$$

We know that sum of first n natural number

$$\sqrt{2}(sum\ of\ n\ natural\ numbers) = \sqrt{2}\left(\frac{n(n+1)}{2}\right) = \frac{n(n+1)}{\sqrt{2}}$$

Therefore the sum of given series is

$$\frac{n(n+1)}{\sqrt{2}}$$

Then the correct option is C)

9. Correct option: (C)

1. Find the first multiple of 7 greater than or equal to 100.

$100 \div 7 = 14$ with a remainder of 2.

So, the first multiple of 7 is $7 \times 15 = 105$.

2. Find the last multiple of 7 less than or equal to 500.

$500 \div 7 = 71$ with a remainder of 3.

So, the last multiple of 7 is $7 \times 71 = 497$.

3. Calculate the number of multiples of 7 in this range.

$$\frac{Last\ multiple\ of\ 7 - First\ multiple\ of\ 7}{7} + 1 = \frac{497 - 105}{7} + 1 = \frac{399}{7} = 57$$

4. Now, find the number of multiples of 21 in this range.

$$\frac{Last\ multiple\ of\ 21 - First\ multiple\ of\ 21}{21} + 1 = \frac{483 - 105}{21} + 1 = \frac{378}{21} + 1 = 18 + 1 = 19$$

5. Subtract the number of multiples of 21 from the number of multiples of 7.

$$57 - 19 = 38$$

So, there are 38 numbers between 100 and 500 that are divisible by 7 but not by 21.

Therefore, the correct answer is option C) 38.

10. Correct option: (B)

In the given expression for the arithmetic mean (AM) between two numbers a and b:

$$AM = \frac{a^n + b^n}{a^{n-1} + b^{n-1}}$$

We can indeed compare it to the standard formula for the AM between two numbers:

$$AM = \frac{a + b}{2}$$

Now, let's compare the denominators: $a^{n-1} + b^{n-1} = 2$

Only comparable value

$$a^{n-1} = b^{n-1} = 1 \Rightarrow n - 1 = 0 \Rightarrow n = 1$$

So, the value of n is 1 for the given expression to be equivalent to the standard arithmetic mean formula.

Then the correct option is B)

11. Correct option: (A)

Given series $1.2.3 + 2.3.4 + \cdots ..$

We can find the nth term as

$$n(n+1)(n+2) = n^3 + 3n^2 + 2n$$

Then the sum of series

$$S_n = \sum_{n=1}^{n}(n^3 + 3n^2 + 2n) = \sum_{n=1}^{n}(n^3) + \sum_{n=1}^{n}(3n^2) + \sum_{n=1}^{n}(2n)$$

We know that

$$\sum_{n=1}^{n}(n^3) = \left[\frac{n(n+1)}{2}\right]^2$$

$$3\sum_{n=1}^{n}(n^2) = 3\left[\frac{n(n+1)(2n+1)}{6}\right]$$

$$2\sum_{n=1}^{n}(n) = 2\left[\frac{n(n+1)}{2}\right]$$

Combine the all terms as

$$\left[\frac{n(n+1)}{2}\right]^2 + 3\left[\frac{n(n+1)(2n+1)}{6}\right] + 2\left[\frac{n(n+1)}{2}\right]$$

$$= \frac{n(n+1)(n^2+5n+6)}{4} = \frac{n(n+1)(n+2)(n+3)}{4}$$

Then the correct option is A)

12. Correct option: (D)

Given series

$$2, 2\sqrt{2}, 4, \ldots 128$$

The given series is G.P

First, let's find the common ratio (r):

$$Common\ ratio\ (r) = \frac{2nd\ term}{1st\ term} = \frac{2\sqrt{2}}{2} = \sqrt{2}$$

Now, we know that the nth term of a G.P. is given by the formula:

$$T_n = a \times r^{n-1}$$

For $n, T_n = 128$,

$$128 = 2 \times \left(\sqrt{2}\right)^{n-1}$$

Now, simplify the equation:

$$64 = 2^{\frac{n-1}{2}} = 2^6$$

Now, we can equate the exponents:

$$6 = \frac{n-1}{2} => n-1 = 12 => n = 13$$

So, there are 13 terms in the G.P. $2, 2\sqrt{2}, 4, \ldots, 128$.

Then the correct option is D)

13. Correct option: (C)

Given that G.P have

$$T_{m+n} = p, T_{m-n} = q$$

We have to find the *mth* term of given G.P

From above we can write

$$T_{m+n} = p = ar^{m+n-1}, T_{m-n} = q = ar^{m-n-1}$$

Multiply both of equation as

$$pq = a^2r^{m+n-1+m-n-1} = a^2r^{2m-2} = (ar^{m-1})^2$$

Then

$$(pq)^{\frac{1}{2}} = ar^{m-1} = T_m$$

Therefore the *mth* term of the given G.P is

$$T_m = (pq)^{\frac{1}{2}} = \sqrt{pq}$$

Then the correct option is C)

14. Correct option: (B)

If x, y, and z are in an arithmetic progression (A.P.), then the corresponding terms $e^{-x}, e^{-y}, and\ e^{-z}$ will be in a geometric progression (G.P.).

To prove it we can write as

$$x = a, y = a + d, y = a + 2d$$

Where $a = 1^{st}\ term$, $d =$ common difference for A.P.

Then

$$e^{-x} = e^{-a}, e^{-y} = e^{-a-d} = e^{-a}e^{-d}, e^{-z} = e^{-a-2d} = e^{-a}e^{-2d}$$

From above equation we can write that terms $e^{-x}, e^{-y}, and\ e^{-z}$ will be in a geometric progression (G.P.) with common ratio $e^{-d} = e^{y-x}$

and first terms e^{-x}

Then the correct option is B)

15. Correct option: (B)

Let's denote the common ratio of the geometric progression (G.P.) as 'r'.

The first term (a) is given as 1.

Given that the sum of the third and fifth terms is 90, we can write the equation as:

$$ar^2 + ar^4 = 90$$
$$r^2 + r^4 = 90$$
$$(r^2 + 10)(r^2 - 9) = 0$$

Then the factors

$$r^2 = -10$$
$$r^2 = 9$$

For the first equation, $r^2 = -10$, but since we're dealing with real numbers in a geometric progression, we discard the negative solution.

For the second equation, $r^2 = 9$, which means:

$$r = \sqrt{9}$$
$$r = \pm 3$$

So, the common ratio 'r' is ± 3.

Then the correct option is B)

16. Correct option: (D)

In a geometric progression (G.P.), the terms are related by powers of the common ratio 'r'.

Let's denote the first term of the G.P. as 'a', and the common ratio as 'r'.

Now, we are given the

$$a_1 = 3$$
$$a_n = 96$$
$$S_n = 189$$

Using the formula for the nth term, we can write these equations as:

$$a = 3$$
$$a \times r^{n-1} = 96 => r^{n-1} = 32$$

For sum of G.P

$$S_n = 189 = \frac{a(r^n - 1)}{r - 1} => 63 = \frac{r^n - 1}{r - 1} = \frac{r\left(r^{n-1} - \frac{1}{r}\right)}{r - 1}$$

Putting the values from equation

$$63 = \frac{r\left(32 - \frac{1}{r}\right)}{r - 1}$$

Cross multiplication

$$63r - 63 = 32r - 1 => 31r = 62 => r = 2$$

Then the value of n

$$2^{n-1} = 32 => n - 1 = 5 => n = 6$$

Then the correct option is D)

17. Correct option: (B)

Given series

$$3 + 3x + 3x^2 + \cdots + \infty = \frac{45}{8}$$

$$(1 + x + x^2 + \cdots + \infty) = \frac{15}{8}$$

The given series is G.P with x as common ratio

This is a sum of infinite G.P

$$S_\infty = \frac{a}{1-r} = \frac{15}{8}$$

$$8 = 15 - 15r \Rightarrow r = \frac{7}{15}$$

So, the value of $x = \frac{7}{15}$

Then the correct option is B)

18. Correct option: (C)

Given series

$$x^{\frac{1}{2}}.x^{\frac{1}{4}}.x^{\frac{1}{8}}.\ldots.\infty$$

The exponent of the given series is in G.P

Given series can be written as

$$x^{\frac{1}{2}+\frac{1}{4}+\frac{1}{8}\ldots\infty}$$

So we have to find the sum of given G.P

$$\left(\frac{1}{2}+\frac{1}{4}+\frac{1}{8}\ldots+\infty\right) = \frac{\left(\frac{1}{2}\right)}{\frac{1}{2}} = 1$$

Then the value of given series

$$x^{\frac{1}{2}}.x^{\frac{1}{4}}.x^{\frac{1}{8}}.\ldots.\infty = x^1 = x$$

Therefore the correct option is C)

19. Correct option: (A)

To find the geometric mean between two numbers, we can use the formula:

$$GM = \sqrt{ab}$$

In this case, a = -9 and b = -16, so:

$$GM = \sqrt{-9 \times -16} = \sqrt{144} = 12$$

So, the geometric mean between -9 and -16 is 12.

Therefore the correct option is A)

20. Correct option: (B)

In a geometric progression (G.P.), the terms are related by powers of the common ratio 'r'. Let's denote the first term of the G.P. as 'a', and the common ratio as 'r'.

Given series

$$2, a, 2^3, b, 2^5$$

From the above we can write

$$\frac{a}{2} = \frac{2^3}{a} \Rightarrow a^2 = 2^4 \Rightarrow a = 2^2$$

$$\frac{b}{2^3} = \frac{2^5}{b} \Rightarrow b^2 = 2^8 \Rightarrow b = 2^4$$

Therefore

$$ab = 2^6 = 64$$

Then the correct option is B)

LEVEL – II

1. Correct option: (A)

In an arithmetic progression (A.P.), the nth term can be calculated using the formula:

$$T_n = a + (n - 1) \times d$$

Here, the first term (a) is 7, and the common difference (d) is 4.

$$139 = 7 + (n - 1) \times 4$$

$$n = 34$$

So, 139 is the 34th term in the arithmetic progression.

Therefore, the answer is A) 34.

2. Correct option: (A)

In an arithmetic progression (A.P.), the nth term can be calculated using the formula:

$$T_n = a + (n - 1) \times d$$

Here, the first term (a) is $2n - 5$, and the common difference (d) is-3.

$$T_{p+2} = 2n - 5 + (p + 2 - 1) \times (-3)$$

$$= 2n - 3p - 8$$

So, $(p + 2)^{th}$ term of A.P is $2n - 3p - 8$

Therefore, the answer is A).

3. Correct option: (C)

The first common number is 11 in both of the A.P.

Then the next common term is obtained by adding L.C.M of the common differences.

$$D = L.C.M(4, 5) = 20$$

Then the 10th common term will be 10th term of the A.P with first term as 11 and common difference is 20.

$$T_{10} = 11 + (9)20 = 191$$

Therefore, the answer is C).

4. Correct option: (C)

Given that A.P have

$$a_1, a_2, a_3, a_4 \ldots$$

$$a_1 + a_5 + a_{10} + a_{15} + a_{20} + a_{25} = 225$$

We can write that

$$a + a + 4d + a + 9d + a + 14d + a + 19d + a + 23d = 225$$

$$6a + 69d = 225$$

We have to find the value of

$$a_1 + a_2 + a_3 \ldots + a_{23} + a_{24} = 24a + 276d = 4(6a + 69d) = 4 \times 225$$

$$= 900$$

Therefore, the answer is C).

5. Correct option: (A)

Given that

$$p, q, r\ (A.P), x, y, z(G.P)$$

Then

$$x^{q-r}.y^{r-p}.z^{p-q}$$

From above we can write

$$x = a, y = ae, z = ae^2$$

Putting the values

$$a^{q-r}.(ae)^{r-p}.(ae^2)^{p-q} = a^{q-r+r-p+p-q}e^{r-p+2p-2q} = e^{r-p+2p-2q}$$

$$e^{r-p+2p-2q} = e^{r-q-(q-p)} = e^0$$

As p, q, r are in A.P.

Then

$$x^{q-r}.y^{r-p}.z^{p-q} = 1$$

Therefore, the answer is A).

6. Correct option: (A)

1. First, let's suppose the four geometric means (G.M.'s) as $a_1, a_2, a_3, and\ a_4$. The series with these terms will be in a geometric progression (G.P.).

$$a_n = a \times r^{n-1}$$

We are given the first term 2 and the last term as 486. We want to find the common ratio (r) first.

$$2 \times r^5 = 486 => r^5 = 243 => r = 3$$

Now that we have the common ratio (r = 3), we can calculate the missing terms:

$$a_1 = 2r = 6$$
$$a_2 = 2r^2 = 18$$
$$a_3 = 2r^3 = 54$$
$$a_4 = 2r^4 = 162$$

So, the four geometric means between 2 and 486 are 6, 18, 54, and 162.

Therefore, the answer is A).

7. Correct option: (B)

Given that

$$a + ar + ar^2 \ldots . + \infty = 20, a^2 + a^2r^2 + a^2r^4 \ldots + \infty = 100$$

So we can use the sum of infinite G.P

$$\frac{a}{1-r} = 20, \frac{a^2}{1-r^2} = 100$$

We can write

$$\left(\frac{a^2}{(1-r)^2}\right) = 400$$

Divide both of the equation

$$\frac{\left(\frac{a^2}{1-r^2}\right)}{\left(\frac{a^2}{(1-r)^2}\right)} = \frac{(1-r)^2}{1-r^2} = \frac{1-r}{1+r} = \frac{1}{4}$$

$$4 - 4r = 1 + r => \frac{3}{5} = r$$

Then 5 times the common ratio is 3

Therefore, the answer is B).

8. Correct option: (D)

Given that: $16.2, 5.4, 1.8 \ldots$ are in A.P.

First common difference is -10.8, second =-3.6

We can see that common difference are in G.P

Then the next two terms are

$$1.8 + \left(-\frac{3.6}{3}\right) = 0.6, 0.6 + \left(-\frac{1.2}{3}\right) = 0.2$$

Then sum of first 5 terms of the series is

$$16.2 + 5.4 + 1.8 + 0.6 + 0.2 = 24.2$$

Therefore, the answer is D).

9. Correct option: (D)

We have to find the value of $\sum_{n=1}^{\infty}\left(\frac{a}{1+a}\right)^n$

We can say that above is the sum of an infinite G.P with common ratio $\frac{a}{a+1}$ and first term as $\frac{a}{a+1}$.

$$\sum_{n=1}^{\infty}\left(\frac{a}{1+a}\right)^n = \frac{\left(\frac{a}{1+a}\right)}{1-\frac{a}{1+a}} = a$$

Therefore, the answer is D).

10. Correct option: (A)

We have to find the value of

$$\sum_{i=1}^{n}\sum_{j=1}^{i}\sum_{k=1}^{j}1$$

Solving the innermost expression as

$$\sum_{k=1}^{j}1 = j$$

Putting the value of expression and solving the next sum

$$\sum_{j=1}^{i}\sum_{k=1}^{j}1 = \sum_{j=1}^{i} j = \frac{i(i+1)}{2}$$

Putting the value of expression and solving the next sum

$$\sum_{i=1}^{n}\sum_{j=1}^{i}\sum_{k=1}^{j}1 = \sum_{i=1}^{n}\frac{i(i+1)}{2} = \frac{1}{2}\sum_{i=1}^{n} i^2 + i = \frac{1}{2}\left(\frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2}\right)$$

$$= \frac{n(n+1)}{4}\left[\frac{2n+1+3}{3}\right] = \frac{n(n+1)(n+2)}{6}$$

Therefore, the answer is A).

11. Correct option: (B)

Given:

$$a + (a + d) + (a + 2d) = 9$$
$$a^2 + (a + d)^2 + (a + 2d)^2 = 35$$

Solving first equation

$$a + d = 3$$

Solving second equation with first one

$$a^2 + (a + d)^2 + (a + 2d)^2 = 35$$
$$a^2 + (3)^2 + (3 + d)^2 = 35$$
$$a^2 + d^2 + 2d + 4d + 9 + 9 = 35$$
$$9 + 4d + 9 + 9 = 35 => 4d = 8 => d = 2$$

Then $a = 1$

nth term of the A.P

$$S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}[2 + (n-1) \times 2] = \frac{n}{2}[2n] = n^2$$

Then the sum of n terms of the given series is n^2

Therefore, the answer is B).

12. Correct option: (A, B, C, D)

Given:

$$\frac{3k + 4 + k + 12}{2} = 3k - 7 => 2k + 8 = 3k - 7$$

Solving equation $k = 15$

Then $2k + 5 = 30 + 5 = 35$

Therefore, the answer is A, B, C, D).

13. Correct option: (A, D)

Given that $\frac{a+b}{2} = 20, \sqrt{ab} = 16$

Solving for a and b

$$(a - b)^2 = (a + b)^2 - 4ab = 1600 - 4 \times 256 = 576$$

$$a - b = \pm 24$$

Take $a - b = 24$
Then $a = 32, b = 8$
Take $a - b = -24$
Then $a = 8, b = 32$
Therefore, the answer is A, D).

14. Correct option: (B)

Given that series $5, 2, -1, \ldots$
is an A.P. with first term (a) = 5 and common difference (d) = -3
Solving for the sum of n term

$$S_n = \frac{n}{2}\big(2 \times 5 + (n-1)(-3)\big) = \frac{n}{2}(13 - 3n)$$

Therefore, the answer is B).

15. Correct option: (A)

Finding the value of a in terms of S_n , d and n.
We know that

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

$$2S_n = 2an + n^2d - nd \Rightarrow \frac{2S_n - n^2d + nd}{2n} = a$$

Therefore, the answer is A).

16. Correct option: (B)

Given that $S_n = 400, n = 16, a = 5$
We have to find the value of d as

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

$$400 = 8(10 + 15d) \Rightarrow 50 - 10 = 15d$$

$$d = \frac{40}{15} = \frac{8}{3}$$

Therefore, the answer is B).

17. Correct Matching is: a-Q; b-P, c-S, d-R

a) The geometric mean between 25 and 16

$$G.M = \sqrt{ab} = \sqrt{25 \times 16} = 20$$

a) → (Q)

b) If x, 2x + 2 and 3x + 3 are the first three terms of G.P., then the fourth term is

$$r = \frac{2x+2}{x} = \frac{3x+3}{2x+2}$$

$$4(x+1)^2 = 4x^2 + 4 + 8x = 3x^2 + 3x$$

$$x^2 + 5x + 4 = 0 = (x+4)(x+1)$$

$$x = -1\ (not\ defined)$$

Then $x = -4$ only solution for x
Then $a_4 = \frac{3}{2} \times a_3 = -\frac{27}{2}$

b) →(P)

c) sum of the series $a - (a+d) + (a+2d) \ldots . upto\ (2n+1) terms$
Take second and third term

$$-(a+d) + (a+2d) = d$$

Combining the terms in pair we get

$$a + (d + d + _(n\ terms)) = a + nd$$

c) →(S)

d) If sum of the series $2 + 5 + 8 + 11 + \cdots$ upto n terms = 60100 , then n

Given series is A.P then sum of series

$$60100 = \frac{n}{2}(4 + (n-1)3) = \frac{n(n+1)}{2}$$

$$120200 = n(3n+1) => 200 \times 601 = n(3n+1) => n = 200$$

d) →(R)

18. Correct Answer is: (1)

Given that $S_n = 255, T_n = 128$

Also the given series is G.P

$$128 = a2^{n-1} => 256 = a2^n$$

We know that

$$S_n = 255 = \frac{a(2^n - 1)}{1} => ar^n - a = 256 - a = 255$$

$$a = 1$$

So value of a will be 1.

19. Correct Answer is: (5)

Given that $a = 3 + \sqrt{2}, b = 7 - \sqrt{2}$

Then the A.M

$$\frac{a+b}{2} = \frac{3 + \sqrt{2} + 7 - \sqrt{2}}{2} = \frac{10}{2} = 5$$

So the A.M of the given values is 5.

20. Correct Answer is: (7)

Given that two A.P s have the same common difference d and

$$a_1 = -1, b_1 = -8$$

We have to find the difference between 4th term of both the A.P

We can write

$$a_4 = -1 + 3d, b_4 = -8 + 3d$$

Then

$$a_4 - b_4 = \big(-1 + 3d - (-8 + 3d)\big) = 7$$

So the difference between 4th terms of both A.P is 7.

STUDENT PRACTICE QUESTIONS

1. Correct option: (C)

Given: 7th term = 34

13th term = 64

We can find the common difference (d) using the 7th term:

$$34 = a + 6d$$

$$64 = a + 12d$$

$$=> 30 = 6d => d = 5$$

From d we can conclude

$$34 = a + 30 => a = 4$$

Then

$$a_{18} = 4 + 17 \times 5 = 89$$

Therefore, the correct option is C)

2. Correct option: (C)

In an arithmetic progression (A.P.), the nth term T_n can be calculated using the following formula: $T_n = a + (n-1) \times d$

In this case, we are given the terms -17, -21, and -25, which are consecutive terms in the A.P. We can find the common difference (d) by subtracting any two consecutive terms:

$$Common\ Difference\ (d) = (-21) - (-17) = -21 + 17 = -4$$

Now that we have the common difference, we can find the nth term by plugging it into the formula:

$$T_n = (-17) + (n - 1) \times (-4)$$
$$T_n = -17 - 4n + 4$$
$$T_n = -13 - 4n$$

So, the nth term of the A.P. is -13 - 4n. Therefore, the correct option is C)

3. Correct option: (D)

In an arithmetic progression (A.P.), the nth term $T_n$can be calculated using the following formula: $T_n = a + (n - 1) \times d$

Then

$$T_p = a + (p-1)d = q$$
$$T_q = a + (q-1)d = p$$
$$q - p = (p-q)d \Rightarrow d = -1$$

To find the value of a we put d in p_{th} term

$$q = a + 1 - p \Rightarrow a = -1 + (p+q)$$
$$T_r = (p+q) - 1 + (r-1)(-1) = p + q - r$$

Then the correct option is D)

4. Correct option: (C)

In an arithmetic progression (A.P.) with common difference "p," the difference between consecutive terms is always "p." So, for the sequence a, b, c, d, e:

Now, let's express each term in terms of "a" and "p":

b = a + p

c = a + 2p

d = a + 3p

e = a + 4p

Now, we want to find the value of a - 4b + 6c - 4d + e:

a - 4b + 6c - 4d + e = a - 4(a + p) + 6(a + 2p) - 4(a + 3p) + (a + 4p)

Now, simplify the expression:

= a - 4a – 4p + 6a + 12p - 4a – 12p + a + 4p

Now, combine like terms:

a - 4a + 6a - 4a + a - 4a – 4p + 12p – 12p + 4p = 0

So, the value of a - 4b + 6c - 4d + e is 0.

The answer is C) 0.

5. Correct option: (D)

To find the common difference (d) in an arithmetic progression (A.P.) when we know the sum of the first n terms (S_n), the number of terms (n), and the first term (a), we can use the formula for the sum of an A.P.:

$$S_n = \left(\frac{n}{2}\right)[2a + (n-1)d]$$
$$200 = \left(\frac{10}{2}\right)[2(5) + (10-1)d]$$
$$200 = (5)[10 + 9d]$$
$$40 = 10 + 9d$$
$$40 - 10 = 10 - 10 + 9d$$
$$30 = 9d$$
$$\frac{30}{9} = \frac{9d}{9}$$

$$d = \frac{10}{3}$$

So, the common difference (d) is $\frac{10}{3}$.

Then the correct option is D)

6. Correct option: (C)

Given series

$$\frac{1}{1.2} + \frac{1}{2.3} + \cdots \ldots$$

We can find the nth term as

$$\frac{1}{n(n+1)}$$

We can write our nth term as

$$\frac{(n+1)-n}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$$

Then (n-1)th term

$$= \frac{1}{n-1} - \frac{1}{n}$$

Then the sum of series

$$S_n = \sum_{n=1}^{n} \left(\frac{1}{n} - \frac{1}{n+1}\right) = 1 - \frac{1}{n+1} = \frac{n}{n+1}$$

Therefore, the sum of given series is $\frac{n}{n+1}$

Then the correct option is C)

7. Correct option: (C)

In a geometric progression (G.P.), the terms are related by powers of the common ratio 'r'.

Let's denote the first term of the G.P. as 'a', and the common ratio as 'r'.

Now, we are given the 4th, 7th, and 10th terms as p, q, and r, respectively. So, we have:

$$T_4 = p$$
$$T_7 = q$$
$$T_{10} = r$$

Using the formula for the nth term, we can write these equations as:

$$a \times r^3 = p$$
$$a \times r^6 = q$$
$$a \times r^9 = r$$

Now, let's divide the second equation by the first equation:

$$\frac{a \times r^6}{a \times r^3} = \frac{q}{p}$$
$$r^3 = \frac{q}{p}$$

Similarly, divide the third equation by the second equation:

$$\frac{a \times r^9}{a \times r^6} = \frac{r}{q}$$
$$r^3 = \frac{r}{q}$$

Compare both the result $r^3 = \frac{r}{q} = \frac{q}{p} => q^2 = pq$

So, the relation between p, q, and r is:

$q^2 = pq$

Then the correct option is C)

8. Correct option: (D)

Given series $9 + 99 + 999 + \cdots$

$$(10 - 1) + (100 - 1) + (1000 - 1) + \cdots$$

$$10 + 100 + 1000 + \cdots + (-n)$$

The given series is G.P with 10 as common ratio and sum of the series will be sum of G.P - n

For sum of G.P $S_n = \frac{a(r^n-1)}{r-1} = \frac{10}{9}(10^n - 1)$

So, sum of series $\frac{10}{9}(10^n - 1) - n = \frac{1}{9}(10^{n+1} - 10 - 9n)$

Then the correct option is D)

9. Correct option: (A)

To find the geometric mean between two numbers, we can use the formula:

$$GM = \sqrt{ab}$$

In this case, $a = 2\frac{1}{4}$ and $b = \frac{4}{9}$, so:

$$GM = \sqrt{\frac{4}{9} \times \frac{9}{4}} = \sqrt{1} = 1$$

Now two other numbers

$$gM = \sqrt{\frac{4}{9} \times 1} = \frac{2}{3}, Gm = \sqrt{\frac{9}{4} \times 1} = \frac{3}{2}$$

These are 3 geometric mean between $\frac{9}{4}$ and $\frac{4}{9}$.

Therefore the correct option is A)

10. Correct option: (D)

In a geometric progression (G.P.), the terms are related by powers of the common ratio 'r'. Let's denote the first term of the G.P. as 'a', and the common ratio as 'r'.

Now, we are given the 4th, 9th

$$T_4 = 54 = ar^3$$

$$T_9 = 13122 = ar^8$$

Now, let's divide the second equation by the first equation:

$$\frac{a \times r^8}{a \times r^3} = r^5 = 243$$

$$r^5 = 3^5 => r = 3$$

To find the first term

$$54 = a \times 3^3 => a = 2$$

Then the nth term of the given G.P

$$T_n = a \times r^{n-1} = 2 \times 3^{n-1}$$

Then the correct option is D)

21. Correct option: (A, C)

In a geometric progression (G.P.), the sum of an infinite series can be calculated using the formula:

$$Sum\ (S) = \frac{a}{1 - r}$$

Given that

$$a + ar = \frac{5}{3}$$

$$\frac{a}{1 - r} = 3$$

Dividing equation 2 by equation 1

$$\left(\frac{\frac{a}{1-r}}{a(1+r)}\right) = \frac{1}{1-r^2} = \frac{9}{5}$$

$$5 = 9 - 9r^2 => 9r^2 = 4 => r^2 = \frac{4}{9} => r = \pm\frac{2}{3}$$

Then the first term $a\left(1 \pm \frac{2}{3}\right) = \frac{5}{3} => a = 5{,}1$

Therefore, the answer is A, C).

22. Correct option: (B, C)

$$Sum\,(S) = \left(\frac{n}{2}\right) \times [2a + (n-1)d]$$

In our series $a = 54, d = -3$

Now, plug these values into the formula and solve for n:

$$513 = \left(\frac{n}{2}\right) \times [2 \times 54 + (n-1) \times (-3)]$$

$$513 = \left(\frac{n}{2}\right) \times [111 - 3n\,]$$

Now, simplify the equation:

$$1026 = n \times (111 - 3n)$$

$$3n^2 - 111n + 1026 = 0$$

$$n^2 - 37n + 342 = 0$$

Now, factor the quadratic equation:

$$(n-19)(n-18) = 0$$

Set each factor equal to zero and solve for n:

$$n - 19 = 0 => n = 19$$

$$n - 18 = 0 => n = 18$$

So the sum of first 18 and 19 terms of the series is 513 as 19th term will be zero.

Therefore, the answer is B, C).

23. Correct option: (B)

Given that $G_1, G_2, G_3, .. G_n$

are n geometric means between 1 and 1024.

We have to find the value of n

We know that

$$(ab)^{\frac{n}{2}} = product\ of\ G.Ms = 2^{45}$$

$$(2^{10})^{\frac{n}{2}} = 2^{5n} = 2^{45} => 5n = 45 => n = 9$$

So value of n will be 9.

Therefore, the answer is B).

24. Correct option: (B)

Given that $A_1, A_2, A_3, .. A_m$

are m arithmetic means between -2 and 1027.

We have to find the value of m.

We know that

$$\frac{n(a+b)}{2} = sum\ of\ all\ arithmetic\ means = 1025 \times 171$$

$$\frac{m(1027-2)}{2} = 1025 \times 171 => m = 171 \times 2 = 342$$

So value of m will be 342.

Therefore, the answer is B).

25. Correct option: (D)

Given that $G_1, G_2, G_3, .. G_n$

are n geometric means between 1 and 1024.

We have to find the value of n

We know that

$$(ab)^{\frac{n}{2}} = product\ of\ G.Ms = 2^{45}$$

$$(2^{10})^{\frac{n}{2}} = 2^{5n} = 2^{45} => 5n = 45 => n = 9$$

So value of n will be 9.

We can say that from the n G.M we get a G.P where first term is 1 and last term (9+2)th is 1024. So common ratio

$$r^{10} = 1024 => r = 2$$

Then sum of G.P

$$1 + 1024 + sum\ of\ G.Ms = 1(r^{11} - 1) = 2048 - 1$$
$$sum\ of\ G.Ms = 2047 - 1025 = 1022$$

Therefore, the answer is D).

26. Correct Matching is: a-R: b-P; c-S; d-Q

a) If $\frac{1}{3}, x_1, x_2, 9$ are in G.P then x_2

$$a = \frac{1}{3}, ar = x_1, ar^2 = x_2, ar^3 = 9$$

Then $r^3 = 27 => r = 3$

So value of $x_2 = \frac{1}{3} \times (3)^2 = 3$

a) -→(R)

b) The third term of a G.P. is 4 the product 6 of its five terms

We can write

$$ar^2 = 4$$

Product of first 5 terms

$$a^5 r^{1+2+3+4} = a^5 r^{10} = (ar^2)^5 = 1024$$

b) -→(P)

c)

$$\frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \cdots + \infty =$$

This is a sum of infinite G.P

$$= \frac{a}{1-r} = \frac{\left(\frac{1}{3}\right)}{\left(1 - \frac{1}{3}\right)} = \frac{1}{2}$$

c) -→(S)

d)

$$1 + 3 + 9 + 27 + .. = 364$$

Sum of G.P can be calculated as

$$= \frac{a(r^n - 1)}{r - 1} = \frac{1(3^n - 1)}{2} = 364 => 3^n - 1 = 728$$
$$729 = 3^n => n = 6$$

d) -→(Q)

27. Correct Answer is: (1)

Given that a, b, c are in A.P.

$$a \neq b$$

Then $$\frac{b-c}{a-b} = \frac{-common\ difference}{-common\ difference} = 1$$

So the value of given expression is 1.

28. Correct Answer is: (2)

Given that sum of n term of an A.P is

$$3n^2 - n, d = 6$$

Then $$S_n = \frac{n}{2}(2a + (n-1)6) = \frac{n}{2}(2a + 6n - 6) = 3n^2 - n$$
$$a + 3n - 3 = 3n - 1 => a = 2$$

So the value of first term is 2.

5. Co-ordinate Geometry

Solutions

Distance and Section Formulae

LEVEL – I

1. Correct option: (D)

Given points $(4, 2), (3, 4)$

The distance between two points can be calculated using the distance formula:

$$Distance = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Putting the values in formula

$$= \sqrt{(3-4)^2 + (4-2)^2} = \sqrt{5}$$

Then the correct option is D)

2. Correct option: (A)

Given points $A = (0, 0), B = (36, 15)$

The distance between two points can be calculated using the distance formula:

$$Distance = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Putting the values in formula

$$= \sqrt{(36)^2 + (15)^2} = \sqrt{1296 + 225} = 39$$

Then the correct option is A)

3. Correct option: (D)

The distance between two points can be calculated using the distance formula:

$$Distance = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Let's calculate the distances between these points:

1. Distance between (2, -2) and (-1, 2):

$$Distance = \sqrt{(-1-2)^2 + (2-(-2))^2}$$
$$Distance = \sqrt{9 + 16}$$
$$Distance = \sqrt{25}$$
$$Distance = 5$$

2. Distance between (2, -2) and (3, 5):

$$Distance = \sqrt{(3-2)^2 + (5-(-2))^2}$$
$$Distance = \sqrt{1 + 49}$$
$$Distance = \sqrt{50}$$

3. Distance between (-1, 2) and (3, 5):

$$Distance = \sqrt{(3-(-1))^2 + (5-2)^2}$$
$$Distance = \sqrt{16+9}$$
$$Distance = \sqrt{25}$$
$$Distance = 5$$

The distances between the points are $5, \sqrt{50} and\ 5$.

Based on these distances, we can conclude that the triangle formed by these points is an isosceles triangle, since two sides have equal lengths.

An isosceles right triangle is characterized by having two sides of equal length, while the sum of the squares of these two sides is equivalent to the square of the remaining side.

Here $5^2 + 5^2 = \sqrt{50}^2$

So the correct option is D)

4. Correct option: (C)

We have one point given as (3, 2) and the ordinate (y-coordinate) of the second point is given as -1. We also know that the distance between these two points is 5.

Given one point as (3, 2) and the y-coordinate of the second point as -1, let's plug in the values and solve for the possible x-coordinates:

Let the other point P = (x, -1)

We have: $Distance = \sqrt{(3-x)^2 + (2+1)^2}$

Simplifying further: $25 = (3-x)^2 + (3)^2 \Rightarrow 9 + (3-x)^2$

$$9 + 9 + x^2 - 6x = 25 \Rightarrow x^2 - 6x = 7$$

$$x^2 - 6x - 7 = 0$$

Solving this using quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-6 \pm \sqrt{6^2 - 4(-7)}}{2} = \frac{-6 \pm 8}{2} = 1, -7$$

Then the correct option is C)

5. Correct option: (D)

Given two opposite vertices A (1, -2) and B (4, 1) of a square, we need to calculate the length of the side of the square.

1. Calculate the distance between the two given points (diagonal length):

$$Distance\ AB = \sqrt{(4-1)^2 + \left(1-(-2)\right)^2}$$

$$Distance\ AB = \sqrt{3^2 + 3^2}$$

$$Distance\ AB = \sqrt{18}$$

$$Distance\ AB = 3\sqrt{2}$$

2. Now, in a square, the diagonal divides the square into two congruent right-angled triangles. The diagonal of the square are hypotenuses of these triangles.

We can write: $Diagonal^2 = Side^2 + Side^2$

$$\left(3\sqrt{2}\right)^2 = 2 \times Side^2$$

$$18 = 2 \times Side^2$$

$$Side^2 = \frac{18}{2}$$

$$Side^2 = 9$$

$$Side = 3$$

So, the correct answer is: D) 3

6. Correct option: (C)

Given points A (5, 3) and B (11, -5), and we are given that angle APB is 90°, meaning triangle APB is a right-angled triangle. We need to find the value of λ for the point P (12, λ).

In a right-angled triangle, the sum of the squares of the lengths of the two shorter sides (legs) is equal to the square of the length of the longest side (hypotenuse).

Let's use this property to solve for λ:

Length of side AB = $\sqrt{(11-5)^2 + (-5-3)^2}$

$$= \sqrt{36 + 64}$$

$$= \sqrt{100}$$

$$= 10$$

Length of side AP = $\sqrt{(12-5)^2 + (\lambda - 3)^2}$

$$= \sqrt{49 + (\lambda - 3)^2}$$

Length of side BP = $\sqrt{(12-11)^2 + \left(\lambda - (-5)\right)^2}$

$$= \sqrt{1 + (\lambda + 5)^2}$$

According to the property: $AP^2 + BP^2 = AB^2$

$$\left(\sqrt{49 + (\lambda - 3)^2}\right)^2 + \left(\sqrt{1 + (\lambda + 5)^2}\right)^2 = 10^2$$

$$49 + (\lambda - 3)^2 + 1 + (\lambda + 5)^2 = 100$$

$$(\lambda - 3)^2 + (\lambda + 5)^2 = 50$$

$$\lambda^2 - 6\lambda + 9 + \lambda^2 + 10\lambda + 25 = 50$$

$$2\lambda^2 + 4\lambda - 16 = 0$$

$$\lambda^2 + 2\lambda - 8 = 0$$

Factor the quadratic equation: $(\lambda + 4)(\lambda - 2) = 0$

This gives us two possible values for λ:

$$\lambda = -4 \, or \, \lambda = 2$$

So, the correct answer is: C) 2 or -4

7. Correct option: (D)

Given points: (2,5), (0,3), (2,1), (4,3)

1. Sides

Calculate the distances between consecutive points:

- Distance between (2, 5) and (0, 3): $\sqrt{(0-2)^2 + (3-5)^2} = \sqrt{8}$
- Distance between (0, 3) and (2, 1): $\sqrt{(2-0)^2 + (1-3)^2} = \sqrt{8}$
- Distance between (2, 1) and (4, 3): $\sqrt{(4-2)^2 + (3-1)^2} = \sqrt{8}$
- Distance between (4, 3) and (2, 5): $\sqrt{(2-4)^2 + (5-3)^2} = \sqrt{8}$

All sides are equal in length (√8), which satisfies the criterion for a square, rhombus.

2. Angles

- Calculate the slopes of consecutive sides:
  - Slope of AB: $\frac{3-5}{0-2} = 1$
  - Slope of BC: $\frac{1-3}{2-0} = -1$
  - Slope of CD: $\frac{3-1}{4-2} = 1$
  - Slope of DA: $\frac{5-3}{4-2} = -1$

Since the product of the slopes of AB and BC is -1, lines AB and BC are perpendicular. Similarly, the product of the slopes of CD and DA is -1, indicating that lines CD and DA are perpendicular.

This confirms that all angles are right angles (90°), which satisfies the criterion for a square.

Based on both the equal sides and the right angles, we can conclude that the points (2, 5), (0, 3), (2, 1), and (4, 3) form a square.

So, the correct answer is indeed:

D) Square

8. Correct option: (C)

Given that distance of (4,0) from (a, b) is the double the distance of point (a, b) from the x-axis.

Let's calculate the distances using the formula and then equate them as given. The distance of the point (a, b) from the x-axis can be directly determined, as the distance of a point from the x-axis is equal to its y-coordinate.

Now equation become

$$2b = \sqrt{\left(b - (4)\right)^2 + (b - 0)^2}$$

Solving further

$$4b^2 = (a-4)^2 + b^2$$
$$3b^2 = (a-4)^2$$

So the correct option is C)

9. Correct option: (B)

To find the coordinates of point R, we can use the external division formula:

For a point P(x_1, y_1) dividing the line segment joining Q$(x_2, y_2)$and R(x_3, y_3)externally in the ratio $m:n$, the coordinates of point P can be calculated as:

$$x_1 = \frac{mx_3 - nx_2}{m - n}$$
$$y_2 = \frac{my_3 - ny_2}{m - n}$$

Let point R (x, y)

Given points:

$$Q(-2,-3), R(x,y)$$

Ratio: 1:2 (P divides QR externally)

Using the formula:

$$-5 = \frac{x - 2 \times (-2)}{-1} = -(x+4) => x = 1$$
$$4 = \frac{y - 2 \times (-3)}{-1} = -(y+6) => y = -10$$

So, the coordinates of point R are $(1, -10)$.

Then the correct option is B)

10. Correct option: (C)

Given points are

$$A = (x_1, y_1), B = (x_2, y_2), C = (x_1 + t(x_2 - x_1), y_1 + t(y_2 - y_1))$$

We have to find the ration in which point C divides the line AB

Distance from A to C

$$AC = \sqrt{((x_1 + t(x_2 - x_1) - x_1)^2 + (y_1 + t(y_2 - y_1) - y_1)^2}$$
$$= t\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

1. Distance from C to B

$$CB = \sqrt{((x_1 + t(x_2 - x_1) - x_2)^2 + (y_1 + t(y_2 - y_1) - y_2)^2}$$
$$= \sqrt{\left((1-t)x_1 + (1-t)x_2\right)^2 + \left((1-t)y_1 + (1-t)y_2\right)^2}$$
$$= (1-t)\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

So the ration in which the point C divide the AB

$$k = \frac{AC}{BC} = \frac{t\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}}{(1-t)\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}} = \frac{t}{1-t}$$

Then the correct option is C)

11. Correct option: (C)

For a point P(x_1, y_1) dividing the line segment joining Q$(x_2, y_2)$and R(x_3, y_3) internally in the ratio $m:n$, the coordinates of point P can be calculated as:

$$x_1 = \frac{mx_3 + nx_2}{m + n}$$
$$y_2 = \frac{my_3 + ny_2}{m + n}$$

Value of y co-ordinate equal to zero on the x-axis

Given points

$$P = (-3,5), Q = (5,1), R = (3,2)$$

Putting the values in formulas for division

$$-3 = \frac{3m + 5n}{m + n} => \frac{3 \times \frac{m}{n} + 5}{\frac{m}{n} + 1}$$

$$-3\left(\frac{m}{n} + 1\right) = 3 \times \frac{m}{n} + 5 => -6\frac{m}{n} = 8$$

$$\frac{m}{n} = -\frac{4}{3}$$

We can say that point P divide the QR in ration of $\frac{4}{3}$ externally

Then the correct option is C)

12. Correct option: (B)

For a point P(x_1, y_1) dividing the line segment joining Q$(x_2, y_2)$and R(x_3, y_3) internally in the ratio $m:n$, the coordinates of point P can be calculated as:

$$x_1 = \frac{mx_3 + nx_2}{m + n}$$

$$y_2 = \frac{my_3 + ny_2}{m + n}$$

Given points

$$P = (x, 0), Q = (2, -3), R = (5,7)$$

Putting the values in formulas for division

$$0 = \frac{m \times 7 + n \times (-3)}{m + n} => \frac{7 \times \frac{m}{n} - 3}{\frac{m}{n} + 1}$$

$$0 = 7 \times \frac{m}{n} - 3 => \frac{m}{n} = \frac{3}{7}$$

We can say that point x axis divide the QR in ration of $\frac{3}{7}$ internally.

Then the correct option is B)

13. Correct option: (B)

For a point P(x_1, y_1) dividing the line segment joining Q$(x_2, y_2)$and R(x_3, y_3) internally in the ratio $m:n$, the coordinates of point P can be calculated as:

$$x_1 = \frac{mx_3 + nx_2}{m + n}$$

$$y_2 = \frac{my_3 + ny_2}{m + n}$$

Let the point P (x, y)

Given points

$$P = (x, y), Q = (5,7), R = (-1,3)$$

Given ration is $\frac{2}{-3} = \frac{m}{n} => m = 2, n = -3$

Putting the values in formulas for co-ordinate of P

$$x = \frac{2 * (-1) + (-3) * 5}{2 - 3} = -(-17) = 17$$

$$y = \frac{2 * 3 + (-3) * 7}{2 - 3} = -(-15) = 15$$

So the co-ordinates of the point which divide the given line segment in ratio of $\frac{2}{-3}$ are (17,15).

Then the correct option is B)

14. Correct option: (A)

Let's take the point A, B, C which divides the line PQ into four equal parts.

So B will be the midpoint of PQ , A will be the midpoint of PB and C will be the midpoint of BQ.

Given points $P = (-2,2), Q = (2,8)$

Point B

$$x = \frac{-2+2}{2} = 0$$
$$y = \frac{2+8}{2} = 5$$

So $B = (0,5)$

Point A

$$x = \frac{-2+0}{2} = -1$$
$$y = \frac{2+5}{2} = \frac{7}{2}$$

So $A = \left(-1, \frac{7}{2}\right)$

Point C

$$x = \frac{0+2}{2} = 1$$
$$y = \frac{5+8}{2} = \frac{13}{2}$$

So $C = \left(1, \frac{13}{2}\right)$

Therefore the points which divide the line segment into four equal parts are A, B, C

Then the correct option is A).

15. Correct option: (C)

Let the point P (x, y)

Given points $(-4,3), (5,7)$

Given that the point P cut line segment in $\frac{2}{3}$ and $\frac{1}{3}$ so point P cut line segment in $2:1$. Then Coordinate of the point P

$$x = \frac{5 \times 2 + (-4 \times 1)}{3} => 2$$
$$y = \frac{7 \times 2 + 3 \times 1}{3} => \frac{17}{3}$$

Then the correct option is C).

16. Correct option: (D)

For a point P(x_1, y_1) dividing the line segment joining Q$(x_2, y_2)$and R(x_3, y_3), into two equal half then coordinates of point P can be calculated as:

$$x_1 = \frac{x_3 + x_2}{2}$$
$$y_1 = \frac{y_3 + y_2}{2}$$

Let the point P (x, y)

Given points $P = \left(\frac{3}{2}, \frac{5}{2}\right), Q = (x, y+1), R = (x+1, y+2)$

Given that point P is midpoint of the line segment QR.

Midpoint formula $\frac{3}{2} = \frac{x+x+1}{2} => x = 1$

$$\frac{5}{2} = \frac{y+1+y+2}{2} => y = 1$$

We can now identify the points for which we want to find the midpoint, as they are given in the form of coordinates (x and y).

Now the points are $A = (x-1, y+1) = (0,2), B = (x+1, y-1) = (2,0)$

So midpoints of the AB line segment $x = \frac{0+2}{2} = 1, y = \frac{2+0}{2} = 1$

Then the correct option is D).

17. Correct option: (D)

For a point P(x_1, y_1) dividing the line segment joining Q$(x_2, y_2)$and R(x_3, y_3) internally in the ratio $m:n$, the coordinates of point P can be calculated as:

$$x_1 = \frac{mx_3 + nx_2}{m + n}$$
$$y_2 = \frac{my_3 + ny_2}{m + n}$$

Value of y co-ordinate equal to zero on the x-axis

Given points $P = (2,3), Q = (4,8), R = (-2,-7)$

Putting the values in formulas for division

$$2 = \frac{m \times (-2) + n \times (4)}{m + n} => \frac{-2 \times \frac{m}{n} + 4}{\frac{m}{n} + 1}$$
$$2\left(\frac{m}{n} + 1\right) = -2 \times \frac{m}{n} + 4 => 4\frac{m}{n} = 2$$
$$\frac{m}{n} = \frac{1}{2}$$

We can say that point P divide the QR in ration of $\frac{1}{2}$ internally

Then the correct option is D)

18. Correct option: (C)

Let the point P (x, y)

Given points $P = (x, y), A = (1,-2), B = (-5,6)$

For internal division

As we have to find the co-ordinate of two points which divide the QR into 3 equal sections, then the ration for the first point will be $\frac{1}{2}$ and for the second point the ratio will be $\frac{2}{1}$.

For first point P

Given ration is $\frac{1}{2} = \frac{m}{n} => m = 1, n = 2$

Putting the values in formulas for co-ordinate of P

$$x = \frac{1 \times (-5) + 2 \times 1}{1 + 2} = -\frac{3}{3} = -1$$
$$y = \frac{1 \times 6 + 2 \times (-2)}{1 + 2} = \frac{2}{3}$$

For second point Q

Given ration is $\frac{2}{1} = \frac{m}{n} => m = 2, n = 1$

Putting the values in formulas for co-ordinate of S

$$x = \frac{2 \times (-5) + 1 \times 1}{1 + 2} = \frac{-9}{3} = -3$$
$$y = \frac{2 \times 6 + 1 \times (-2)}{1 + 2} = \frac{10}{3}$$

So the co-ordinates of the points which divide the given line segment trisection is $P = \left(-1, \frac{2}{3}\right)$ and $S = (-3, \frac{10}{3})$

Then the value of PQ $\sqrt{\left(-3 - (-1)\right)^2 + \left(\frac{10}{3} - \frac{2}{3}\right)^2} = \sqrt{4 + \frac{64}{9}} = \sqrt{\frac{36+64}{9}} = \frac{10}{3}$

Then the correct option is C)

19. Correct option: (A)

For a point P(x_1, y_1) dividing the line segment joining Q$(x_2, y_2)$and R(x_3, y_3), into two equal half then coordinates of point P (mid point) can be calculated as:

$$x_1 = \frac{x_3 + x_2}{2}$$

$$y_1 = \frac{y_3 + y_2}{2}$$

We have given end points of a circle diameter as

$$A = (3, -4), B = (7,2)$$

Then, the midpoint of this diameter will be the center of the circle, as we know that the center cuts the diameter in half.

Putting the values

$$x = \frac{7+3}{2} = \frac{10}{2} = 5$$

$$y = \frac{2+(-4)}{2} = -\frac{2}{2} = -1$$

Coordinate of the center will be

$$P = (5, -1)$$

Then the correct option is A)

20. Correct option: (C)

First solve for the ratio:

$$2(m+n) = -2m + 4n$$

$$4m = 2n \Rightarrow \frac{m}{n} = \frac{1}{2}$$

The harmonic conjugate will divide the line segment in the ratio -1:2 (opposite sign because it's the conjugate).

Use the weighted average formula to find the harmonic conjugate:

$$Harmonic\ Conjugate = \left[\frac{(-2)\times(-1)+(4\times 2)}{1}, \frac{-7\times(-1)+(8)\times(2)}{1}\right]$$

Simplify: $Harmonic\ Conjugate = (10, 23)$

Then the correct option is C)

LEVEL – II

1. Correct option: (B)

Given point $P = (2,3)$

We need to calculate the distance of point P from the x-axis.

The distance of a point from the x-axis is equal to its y-coordinate:

$$Distance = P_y$$

Similarly, the distance of a point from the y-axis is equal to its x-coordinate:

$$Distance = P_x$$

Then the distance of point P will be 3.

Then the correct option is B)

2. Correct option: (B)

Given points $A = (0,6), B = (0, -2)$

Distance between two points

$$= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Putting the values in formula

$$= \sqrt{(0-0)^2 + (-2-6)^2} = \sqrt{64} = 8$$

Then the correct option is B)

3. Correct option: (C)

We know that the coordinate of origin will be $(0, 0)$

Then given points

$$P = (-6,8), O = (0,0)$$

Distance between two points

$$= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Putting the values in formula

$$= \sqrt{(0 - (-6))^2 + (0 - 8)^2} = \sqrt{36 + 64} = 10$$

Then the correct option is C)

4. Correct option: (B)

Given points

$$A = (0, 5), B = (-5, 0)$$

Distance between two points

$$= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

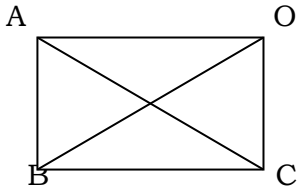
Putting the values in formula

$$= \sqrt{((-5) - 0)^2 + (0 - 5)^2} = \sqrt{50} = \sqrt{2 \times 25} = 5\sqrt{2}$$

Then the correct option is B)

5. Correct option: (A)

Given a rectangle $AOBC$ as



Given points $A = (0, 3), O = (0, 0), B = (5, 0)$

We know that the lengths of the diagonals are the same for a rectangle.

$$AC = OB$$

Distance between two points

$$= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

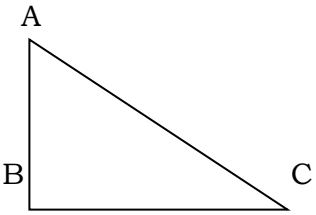
Putting the values in formula

$$OB = \sqrt{(5 - 0)^2 + (0 - 0)^2} = \sqrt{20} = 5$$

Then the correct option is A)

6. Correct option: (B)

Given a triangle as



Given points

$$A = (0, 4), B = (0, 0), C = (3, 0)$$

The perimeter of a triangle will be the sum of all its side lengths.

$$P = AB + BC + AC$$

Putting the values

$$P = \sqrt{(0 - 0)^2 + (4 - 0)^2} + \sqrt{(3 - 0)^2 + (0 - 0)^2} + \sqrt{(3 - 0)^2 + (4 - 0)^2}$$

$$P = \sqrt{16} + \sqrt{9} + \sqrt{25} = 4 + 3 + 5 = 12$$

Then the correct option is B)

7. Correct option: (C)

If point A (0, 2) is equidistant from points B (3, p) and C (p, 5), then the value of p is:

To be equidistant from two points, the distance from A to B should be equal to the distance from A to C.

Given points: $A(0,2), B(3,p), C(p,5)$

Distance from A to B:

$$\sqrt{(3-0)^2+(p-2)^2} = \sqrt{9+(p-2)^2}$$

Distance from A to C:

$$\sqrt{(p-0)^2+(5-2)^2} = \sqrt{p^2+9}$$

Setting up the equation for equidistance:

$$\sqrt{9+(p-2)^2} = \sqrt{p^2+9}$$
$$9+(p-2)^2 = p^2+9$$
$$(p-2)^2 = p^2$$
$$p^2-4p+4 = p^2$$
$$-4p+4 = 0$$
$$4p = 4$$
$$p = 1$$

So, the value of p is 1.

The correct option is C) 1.

8. Correct option: (*)

Given points $A=(-4,0), B=(4,0), C=(3,0)$

A degenerate triangle is a geometric concept referring to a triangle that has collapsed into a lower-dimensional shape. In a degenerate triangle, the sides might overlap or be of zero length, causing the triangle to become a line segment or even a single point. In essence, a degenerate triangle doesn't have the typical appearance of a triangle with distinct sides and angles.

In a non-degenerate triangle, the sum of the lengths of any two sides is always greater than the length of the third side (Triangle Inequality Theorem). In a degenerate triangle, this inequality is not satisfied.

In our case, the points (-4, 0), (4, 0), and (3, 0) are collinear and form a degenerate triangle because all three points lie on the x-axis, and therefore the triangle have zero area.

9. Correct option: (D)

To find the point that divides the line segment joining (7, -6) and (3, 4) in the ratio 1:2 internally, we can use the section formula. The section formula states that if a point divides the line segment joining two points (x_1, y_1) and (x_2, y_2) in the ratio of m:n internally, then the coordinates of the point can be calculated as:

$$x = \frac{mx_2+nx_1}{m+n}$$
$$y = \frac{my_2+ny_2}{m+n}$$

In this case, m represents the ratio 1 and n represents the ratio 2.

Given points:

$$A(7,-6)$$
$$B(3,4)$$

Let's calculate the coordinates of the point that divides the segment internally in the ratio 1:2.

$$x = \frac{3+14}{3} = \frac{17}{3}$$
$$y = \frac{4-12}{3} = -\frac{8}{3}$$

Now, let's determine in which quadrant this point lies:

- The x-coordinate is positive, so the point is not in the II or III quadrant.
- The y-coordinate is negative, so the point is in the IV quadrant.

So, the correct answer is D. IV quadrant.

10. Correct option: (B)

Let's use the slope method to find the fourth vertex D of the parallelogram ABCD.

Given vertices:

$$A(-2,3)$$
$$B(6,7)$$
$$C(8,3)$$

Let the fourth vertex D = (x, y)

Using midpoint property of diagonals of the parallelogram

$$M = \left[\frac{-2+8}{2}, \frac{3+3}{2}\right] = \left[\frac{6+x}{2}, \frac{7+y}{2}\right]$$
$$6 = 6x \Rightarrow x = 0, 7 + y = 6 \Rightarrow y = -1$$

Therefore the coordinates of the point D is

$$D = (0,-1)$$

Then the correct option is B)

11. Correct option: (B, D)

Given points

$$A = (4,2), B = (8,4), P = (2,1)$$

According the given option we have to find the values of AB, BC, AP, BP

$$AB = \sqrt{(4-2)^2 + (8-4)^2} = \sqrt{4+16} = \sqrt{20}$$
$$BP = \sqrt{(1-4)^2 + (2-8)^2} = \sqrt{9+36} = \sqrt{45}$$
$$AP = \sqrt{(1-2)^2 + (2-4)^2} = \sqrt{5}$$

Let's check the given options

A) Given $AP = \frac{1}{3}AB$

Putting the values

$$AP = \sqrt{5} \Rightarrow \frac{1}{3}AB = \frac{1}{3}\sqrt{20} \Rightarrow AP \neq \frac{1}{3}AB$$

Option A) is wrong

B) Given $BP = 3\sqrt{5}$

Putting the values

$$BP = \sqrt{45} \Rightarrow BP = \sqrt{9 \times 5} \Rightarrow BP = 3\sqrt{5}$$

Option B) is right

C) Given $PB = \frac{1}{3}AB$

Putting the values

$$BP = \sqrt{45} \Rightarrow \frac{1}{3}AB = \frac{1}{3}\sqrt{20} \Rightarrow PB \neq \frac{1}{3}AB$$

Option C) is wrong

D) Given $AP = \frac{1}{2}AB$

Putting the values

$$AP = \sqrt{5} \Rightarrow \frac{1}{2}AB = \frac{1}{2}\sqrt{4 \times 5} = \sqrt{5} \Rightarrow AP = \frac{1}{2}AB$$

Option D) is right

12. Correct option: (A, C)

To find the coordinates (x, y) where the perpendicular bisector of the line segment AB cuts the y-axis, follow these steps:

Given points:$A(1,5), B(4,6)$

1. Midpoint of AB:

$$Midpoint = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right) = \left(\frac{1+4}{2}, \frac{5+6}{2}\right) = (2.5, 5.5)$$

2. Slope of AB:

$$Slope\ of\ AB\ = \frac{y_2 - y_1}{x_2 - x_1}\ = \frac{6\ -\ 5}{4\ -\ 1}\ = \frac{1}{3}$$

3. Negative reciprocal of the slope of AB (slope of perpendicular bisector):
Slope of perpendicular bisector

$$= -\frac{1}{\frac{1}{3}} = -3$$

4. Equation of perpendicular bisector using the midpoint:
Using the point-slope form: $y - y_1 = m(x - x_1)$

$$y - 5.5 = -3(x - 2.5)$$
$$y - 5.5 = -3x + 7.5$$
$$y = -3x + 13$$

5. Finding the point of intersection with the y-axis:
When x = 0 (since the point is on the y-axis): $y = -3(0) + 13$

$$y = 13$$

So, the point of intersection with the y-axis is (x, y) = (0, 13).
Then the correct option is A, C)

13. Correct option: (A, D)

Given one end of the line segment is (2, -3) and the abscissa (x-coordinate) of the other end is 10. Let's denote the coordinates of the other end as (10, y).
The length of the line segment can be calculated using the distance formula:

$$Length\ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Given points: $A(2,-3)\ B(10,y)$
Length = 10 (given)
Using the distance formula:

$$10\ = \sqrt{(10\ -\ 2)^2 + \big(y\ -\ (-3)\big)^2}$$
$$10\ = \sqrt{64\ +\ (y\ +\ 3)^2}$$
$$100\ =\ 64\ +\ (y\ +\ 3)^2$$
$$(y\ +\ 3)^2\ =\ 100\ -\ 64$$
$$(y\ +\ 3)^2\ =\ 36$$

Taking the square root of both sides: $y + 3 = \pm 6$
Solving for y: $y = -3 \pm 6$
This gives two possible values for y: -9 and 3.
Then the correct option is A, D)

14. Correct option: (B)

The value of ordinate at x-axis will be zero. So let point P= $(x, 0)$
Now given points $P = (x, 0), A = (2,3), B = (-1,2)$
Putting the values in formulas for division

$$0\ = \frac{m \times 2 +\ n \times (3)}{m +\ n} => \frac{2 \times \frac{m}{n} + 3}{\frac{m}{n} + 1}$$
$$0 = 2 \times \frac{m}{n} + 3 => \frac{m}{n} = \frac{-3}{2}$$

We can say that x axis divide the AB in ration of $\frac{3}{2}$ externally.
Then the correct option is B)

15. Correct option: (C)

The value of abscissa at y-axis will be zero. So let point P= $(0, y)$
Now given points $P = (0, y), A = (3,2), B = (-6,5)$

Putting the values in formulas for division

$$0 = \frac{m \times (-6) + n \times 3}{m + n} => \frac{-6 \times \frac{m}{n} + 3}{\frac{m}{n} + 1}$$

$$0 = -6 \times \frac{m}{n} + 3 => \frac{m}{n} = \frac{1}{2}$$

We can say that y-axis divide the AB in ration of $\frac{1}{2}$ internally.

Then the correct option C)

16. Correct option: (D)

Let the point P (x, y)

Given points $P = (x, y), A = (1,2), R = (3,5)$

Given ration is $\frac{1}{2} = \frac{m}{n} => m = 1, n = 2$

Putting the values in formulas for co-ordinate of P

$$x = \frac{3 \times 1 + 1 \times 2}{1 + 2} = \frac{5}{3}$$

$$y = \frac{5 \times 1 + 2 \times 2}{1 + 2} = \frac{9}{3} = 3$$

So the co-ordinates of the point which divide the given line segment in ratio of $\frac{1}{2}$ are $\left(\frac{5}{3}, 3\right)$.

Then the correct option is D)

17. Correct Matching is: a-T, b-P, c-S, d-Q

a) --→ (T)

Given points are $A = (-2,5), B = (2,5)$

Applying the midpoint formula $x = \frac{2+(-2)}{2} = 0$

$$y = \frac{5 + 5}{2} = 5$$

So the midpoint of the points A and B is (0,5)

b) --→ (P)

Let the midpoint of the line PQ is R

So according the question

$$R = (2, -5), Q = (4,0)$$

As we are aware that point P lies on the Y-axis, its abscissa will be zero.

Let point P be represented as P = (0, y).

Putting the values in the midpoint formula

$$-5 = \frac{0+y}{2} => y = -10$$

Then the point P will be $(0, -10)$.

c) --→ (S)

We are given a point P on the X-axis, denoted as P(x, 0), and the distance from this point P to another point (7, -4) is $2\sqrt{5}$ units. To find the value of x, we can apply the distance formula.

Putting the given values into the distance formula will allow us to determine the exact value of x.

$$2\sqrt{5} = \sqrt{(x-7)^2 + (0-(-4))^2} = \sqrt{(x-7)^2 + 16}$$

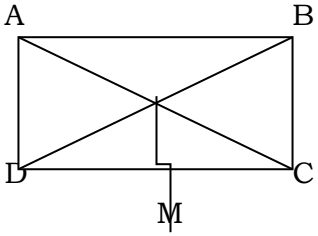
$$20 = (x-7)^2 + 16$$

$$4 = (x-7)^2 => x - 7 = \pm 2 => x = 9,5$$

So from the given option (5, 0) is matching.

d) --→ (Q)

Given that the points form a rectangle, we can leverage the concept of the midpoint of diagonals in a rectangle to readily determine the coordinates of the unknown vertex.
Let our rectangle be as follows:



The coordinates of point M can be found from points A and C. Then, by using point M, we can determine the coordinates of point D using point C.
Given points $A = (2,-2), B = (7,3), C = (11,-1)$
Then

$$M = \left(\frac{11+2}{2}, \frac{-1-2}{2}\right)$$

Let the unknown point D = (x, y)
Then

$$M = \left(\frac{7+x}{2}, \frac{3+y}{2}\right)$$

Equating the coordinates values

$$\frac{13}{2} = \frac{7}{2} + \frac{x}{2} => x = 6$$

$$-\frac{3}{2} = \frac{3}{2} + \frac{y}{2} => y = -6$$

18. Correct Answer is: 2

Using the midpoint formula:

$$(a, b) = \left(\frac{10 + k}{2}, \frac{-6 + 4}{2}\right)$$

$$(a, b) = \left(\frac{10 + k}{2}, -1\right)$$

Given that $a - 2b = 18$:
We find that b = -1, then

$$a - 2(-1) = 18$$

$$a + 2 = 18$$

$$a = 16$$

So, the coordinates of the midpoint (a, b) are (16, -1).
Solving for k:

$$16 = \frac{10 + k}{2}$$

$$32 = 10 + k$$

$$k = 32 - 10$$

$$k = 22$$

Then $\frac{k}{11}$

$$\frac{k}{11} = \frac{22}{11} = 2$$

19. Correct Answer is: 3

To find the least value of a, we can use the information provided.
The center-radius form of the equation of a circle is:

$$(x - h)^2 + (y - k)^2 = r^2$$

Where $(h, k)$the center of the circle and r is its radius
Given:
Center = $(2a, a - 7)$
Point on circle = $(11, -9)$
Diameter = $10\sqrt{2} => r = 5\sqrt{2}$
Then

$$(11 - 2a)^2 + (-9 - (a - 7))^2 = (5\sqrt{2})^2$$
$$(11 - 2a)^2 + (-9 - a + 7)^2 = 50$$
$$(11 - 2a)^2 + (-a - 2)^2 = 50$$
$$121 - 44a + 4a^2 + a^2 + 4a + 4 = 50$$
$$5a^2 - 40a + 75 = 0$$
$$a^2 - 8a + 15 = 0$$
$$(a - 5)(a - 3) = 0$$

The least value of a for the given conditions is 3.

20. Correct Answer is: 0

Given that the line segment AB, joining points A(3, 2) and B(5, 1), is divided by point P in the ratio 1:2, and it lies on the line 3x - 18y + k = 0, we can find the value of k.
Let's first find the coordinates of point P using the section formula.
Given: $A(3, 2)$

$$B(5, 1)$$
$$Ratio\ m{:}n = 1{:}2$$

Calculating the coordinates of point P:

$$x = \frac{5 + 6}{3} = \frac{11}{3}$$
$$y = \frac{1 + 4}{3} = \frac{5}{3}$$

So, the coordinates of point P are $P\left(\frac{11}{3}, \frac{5}{3}\right)$.
Since point P lies on the line $3x - 18y + k = 0$, we can substitute its coordinates into the equation: $11 - 30 + k = 0$

$$k = 19$$

Therefore, $k - 19 = 0$.

STUDENT PRACTICE QUESTIONS

1. Correct option: (B)

Given points $A = (1, 4), B = (5, 1)$

The distance between two points can be calculated using the distance formula:

$$Distance = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Putting the values in formula

$$= \sqrt{(5 - 1)^2 + (1 - 4)^2} = \sqrt{16 + 9} = \sqrt{25} = 5$$

Then the correct option is B)

2. Correct option: (*)

To determine the type of triangle formed by the given points $(\sqrt{3}\ -\ 1, \sqrt{3}\ +\ 1), (\sqrt{3}, 1), (\sqrt{3}\ +\ 1, \sqrt{3}\ -\ 1)$ We can calculate the distances between these points and see if they satisfy the conditions for any of the given types of triangles.

The distance between two points can be calculated using the distance formula:

$$Distance\ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Let's calculate the distances between these points:

1. Distance between $(\sqrt{3}\ -\ 1, \sqrt{3}\ +\ 1), (\sqrt{3}, 1)$:

$$Distance\ = \sqrt{(\sqrt{3}\ -(\sqrt{3} - 1))^2 + (\ 1 - (\sqrt{3}\ + 1))^2}$$

$$Distance\ = \sqrt{1\ +\ 3}$$

$$Distance\ = \sqrt{4}$$

$$Distance\ = 2$$

2. Distance between $(\sqrt{3}, 1), (\sqrt{3}\ +\ 1, \sqrt{3}\ -\ 1)$:

$$Distance\ = \sqrt{(\sqrt{3}\ + 1 - \sqrt{3})^2 + (\sqrt{3}\ -\ 1 - (1))^2}$$

$$Distance\ = \sqrt{1\ +\ 3 + 4 - 4\sqrt{3}}$$

$$Distance\ = \sqrt{8 - 4\sqrt{3}}$$

3. Distance between $(\sqrt{3}\ -\ 1, \sqrt{3}\ +\ 1),\ (\sqrt{3}\ +\ 1, \sqrt{3}\ -\ 1)$:

$$Distance\ = \sqrt{(\sqrt{3}\ +\ 1 - (\sqrt{3}\ -\ 1))^2 + (\sqrt{3}\ -\ 1 - (\sqrt{3}\ +\ 1))^2}$$

$$Distance\ = \sqrt{4\ +\ 4}$$

$$Distance\ = \sqrt{8}$$

The distances between the points are $2, \sqrt{8 - 4\sqrt{3}}\ and\ \sqrt{8}$.

None of the option is correct.

3. Correct option: (A)

To find the centroid of a triangle with vertices (x_1, y_1), (x_2, y_2), and (x_3, y_3), we can use the following formula: $Centroid\ = ((x_1\ +\ x_2\ +\ x_3)/3\ , (y_1\ +\ y_2\ +\ y_3)/3)$

In our case, the vertices of the triangle are (9, 3), (7, -1), and (-1, 4). Plug these coordinates into the formula:

$$Centroid\ = \left(\frac{9\ +\ 7\ -\ 1}{3}, \frac{3\ -\ 1\ +\ 4}{3}\right)$$

$$Centroid\ = \left(\frac{15}{3}, \frac{6}{3}\right)$$

$$Centroid\ = (5, 2)$$

Then the correct option is A)

4. Correct option: (B)

Given points: $(7, 1), (4, 4), (-2, -2), (1, -5)$

1. Sides

Calculate the distances between consecutive points:

- Distance between (7, 1) and (4, 4): $\sqrt{(4 - 7)^2\ +\ (4 - 1)^2}\ =\ \sqrt{18}$
- Distance between (4, 4) and (-2,-2): $\sqrt{(-2 - 4)^2\ +\ (-2 - 4)^2}\ =\ \sqrt{72}$
- Distance between (-2,-2) and (1,-5): $\sqrt{\left(1 - (-2)\right)^2 + \left(-5 - (-2)\right)^2} = \sqrt{18}$

- Distance between (1,-5) and (7, 1): $\sqrt{(7-1)^2 + \left(1-(-5)\right)^2} = \sqrt{72}$

Two sides are equal to = $\sqrt{18}$ and two other sides are = $\sqrt{72}$

Then it may be a rectangle or a Parallelogram.

2. Angles

- Calculate the slopes of consecutive sides:

Slope of AB: $\frac{4-1}{4-7} = -1$

Slope of BC: $\frac{-2-4}{-2-4} = 1$

Slope of CD: $\frac{-5-(-2)}{1-(-2)} = -1$

Slope of DA: $\frac{1-(-5)}{7-1} = 1$

Since the product of the slopes of AB and BC is -1, lines AB and BC are perpendicular. Similarly, the product of the slopes of CD and DA is -1, indicating that lines CD and DA are perpendicular.

This confirms that all angles are right angles (90°), which satisfies the criterion for a Rectangle.

Based on both the equal sides and the right angles, we can conclude that the points $(7,1), (4,4), (-2,-2), (1,-5)$ form a Rectangle.

So, the correct answer is indeed: B) Rectangle

5. Correct option: (C)

Given points: $(7,8), (1,6), (-1,0), (5,2)$

1. Sides

Calculate the distances between consecutive points:

Distance between (7, 8) and (1, 6):

$$\sqrt{(1-7)^2 + (6-8)^2} = \sqrt{36+4} = \sqrt{40}$$

Distance between (1, 6) and (-1, 0):

$$\sqrt{(-1-1)^2 + (0-6)^2} = \sqrt{4+36} = \sqrt{40}$$

Distance between (-1, 0) and (5, 2):

$$\sqrt{\left(5-(-1)\right)^2 + (2-0)^2} = \sqrt{36+4} = \sqrt{40}$$

Distance between (5, 2) and (7, 8):

$$\sqrt{(7-5)^2 + (8-2)^2} = \sqrt{36+4} = \sqrt{40}$$

All sides have equal lengths ($\sqrt{40}$), which satisfies the criterion for a square or rhombus.

2. Angles

- Calculate the slopes of consecutive sides:

Slope of (7, 8) and (1, 6):

$$\frac{6-8}{1-7} = \frac{-2}{-6} = \frac{1}{3}$$

Slope of (1, 6) and (-1, 0):

$$\frac{0-6}{-1-1} = \frac{-6}{-2} = 3$$

Slope of (-1, 0) and (5, 2):

$$\frac{2-0}{5-(-1)} = \frac{2}{6} = \frac{1}{3}$$

Slope of (5, 2) and (7, 8):

$$\frac{8-2}{7-5} = \frac{6}{2} = 3$$

Since the product of the slopes of AB and BC is 1, lines AB and BC are not perpendicular. Similarly, the product of the slopes of CD and DA is 1.
Let's find the slopes of diagonal $AC = using\ (7,8),(-1,0)\ and\ BD = using\ (1,6),(5,2)$
Slope of (7, 8) and (-1, 0):

$$\frac{0\ -\ 8}{-1\ -\ 7} = \frac{-8}{-8} = 1$$

Slope of (1, 6) and (5, 2):

$$\frac{2\ -\ 6}{5\ -\ 1} = -\frac{4}{4} = -1$$

As product of slopes of the diagonals $= -1$, we can say that diagonals are perpendicular to each other which satisfy the property of the Rhombus.
So, the correct answer is indeed: C) Rhombus

6. Correct option: (D)

For a point P(x_1, y_1) dividing the line segment joining Q$(x_2, y_2)$and R(x_3, y_3) internally in the ratio $m:n$, the coordinates of point P can be calculated as:

$$x_1 = \frac{mx_3 + nx_2}{m + n}$$
$$y_2 = \frac{my_3 + ny_2}{m + n}$$

Value of x co-ordinate equal to zero on y axis
Given points

$$P = (0, y), Q = (3,5), R = (-4,7)$$

Putting the values in formulas for division

$$0 = \frac{m \times (-4) + n \times 3}{m + n} => \frac{-4 \times \frac{m}{n} + 3}{\frac{m}{n} + 1}$$
$$0 = -4 \times \frac{m}{n} + 3 => \frac{m}{n} = \frac{3}{4}$$

We can say that point x axis divide the QR in ration of $\frac{3}{4}$ internally.
Then the correct option is D)

7. Correct option: (C)

For a point P(x_1, y_1) dividing the line segment joining Q$(x_2, y_2)$and R(x_3, y_3) internally in the ratio $m:n$, the coordinates of point P can be calculated as:

$$x_1 = \frac{mx_3 + nx_2}{m + n}$$
$$y_2 = \frac{my_3 + ny_2}{m + n}$$

Let the point P (x, y)
Given points $P = (x, y), Q = (1,7), R = (6, -3)$
For internal division
Given ration is $\frac{2}{3} = \frac{m}{n} => m = 2, n = 3$
Putting the values in formulas for co-ordinate of P

$$x = \frac{2 \times 6 + 3 \times 1}{2 + 3} = 3$$
$$y = \frac{2 \times -3 + 3 \times 7}{2 + 3} = 3$$

So the co-ordinates of the point which divide the given line segment in ratio of $2/3$ internally are $(3, 3)$.
For external division
Given ration is $-\frac{2}{3} = \frac{m}{n} => m = 2, n = -3$

Putting the values in formulas for co-ordinate of P

$$x = \frac{2 \times 6 - 3 \times 1}{2 - 3} = -9$$

$$y = \frac{2 \times (-3) - 3 \times 7}{2 - 3} = 27$$

So the co-ordinates of the point which divide the given line segment in ratio of $2/3$ externally are $(-9, 27)$.

Then the correct option is C)

8. Correct option: (B)

For a point P(x_1, y_1) dividing the line segment joining Q$(x_2, y_2)$and R(x_3, y_3) internally in the ratio $m:n$, the coordinates of point P can be calculated as:

$$x_1 = \frac{mx_3 + nx_2}{m + n}$$

$$y_2 = \frac{my_3 + ny_2}{m + n}$$

Let the point P (x, y)

Given points

$$P = (x, y), Q = (-5,2), R = (3,6)$$

For internal division

As we have to find the co-ordinate of two points which divide the QR into 3 equal sections, then the ration for the first point will be $\frac{1}{2}$ and for the second point the ratio will be $\frac{2}{1}$.

For first point

Given ration is $\frac{1}{2} = \frac{m}{n} => m = 1, n = 2$

Putting the values in formulas for co-ordinate of P

$$x = \frac{1 \times 3 + 2 \times (-5)}{1 + 2} = -\frac{7}{3}$$

$$y = \frac{1 \times 6 + 2 \times 2}{1 + 2} = \frac{10}{3}$$

For second point

Given ration is $\frac{2}{1} = \frac{m}{n} => m = 2, n = 1$

Putting the values in formulas for co-ordinate of S

$$x = \frac{2 \times 3 + 1 \times (-5)}{1 + 2} = \frac{1}{3}$$

$$y = \frac{2 \times 6 + 1 \times 2}{1 + 2} = \frac{14}{3}$$

So the co-ordinates of the points which divide the given line segment trisection is $P = \left(-\frac{7}{3}, \frac{10}{3}\right)$ and $S = \left(\frac{1}{3}, \frac{14}{3}\right)$

Then the correct option is B).

9. Correct option: (C)

For a point P(x_1, y_1) dividing the line segment joining Q$(x_2, y_2)$and R(x_3, y_3) internally in the ratio $m:n$, the coordinates of point P can be calculated as:

$$x_1 = \frac{mx_3 + nx_2}{m + n}$$

$$y_2 = \frac{my_3 + ny_2}{m + n}$$

Value of y co-ordinate equal to zero on the x-axis

Given points $P = (-3,5), Q = (5,1), R = (3,2)$

Putting the values in formulas for division

$$-3 = \frac{m \times (3) + n \times (5)}{m + n} => \frac{3 \times \frac{m}{n} + 5}{\frac{m}{n} + 1}$$

$$-3\left(\frac{m}{n} + 1\right) = 3 \times \frac{m}{n} + 5 => -6\frac{m}{n} = 8$$

$$\frac{m}{n} = -\frac{8}{6} = -\frac{4}{3}$$

We can say that point P divide the QR in ration of $\frac{4}{3}$ externally.

10. Correct option: (C)

To find the harmonic conjugate of (4, -2) with respect to the points (2, -4) and (7, 1), first, determine the ratio in which (4, -2) divides the line segment connecting (2, -4) and (7, 1).

Solve for the ratio:

$$4(m + n) = 7m + 2n$$

$$2n = 3m => \frac{m}{n} = \frac{2}{3}$$

The harmonic conjugate will divide the line segment in the ratio -2:3 (opposite sign because it's the conjugate).

Use the weighted average formula to find the harmonic conjugate:

$$Harmonic\ Conjugate = \left[\frac{7 \times (-2) + (2 \times 3)}{-2 + 3}, \frac{1 \times (-2) + (3) \times (-4)}{-2 + 3}\right]$$

Simplify: $Harmonic\ Conjugate = (-8, -14)$

Then the correct option is C)

11. Correct option: (B, D)

We have given points $P = \left(\frac{a}{3}, 4\right), Q = (-6,5), R = (-2,3)$

Then, the midpoint of these points can be calculated as

Putting the values

$$\frac{a}{3} = \frac{-2 + (-6)}{2} = \frac{-8}{2} => a = -12$$

So the correct options are B), D).

12. Correct option: (A, C)

Given points: $A(4, p) B(1, 0)$

The distance between A and B is given as 5:

$$5 = \sqrt{(1 - 4)^2 + (0 - p)^2}$$

$$25 = (3)^2 + p^2$$

$$25 = 9 + p^2$$

$$p^2 = 16$$

Taking the square root of both sides: $p = \pm 4$

So, the possible values of p are 4 and -4.

Then the correct option A, C)

13. Correct option: (A)

Given points are $A = (0,1), B = (0, -10), C = (0, \frac{1}{3})$

To accurately determine the shape formed by the points, we need to calculate the distances between them.

First of all, by observation, we can see that all three vertices lie on the x-axis. Therefore, we can easily conclude that these points are vertices of a straight line.

So the correct option will be straight line.

Then the correct option is A)

14. Correct option: (B)

Given points are $A = (a + b, a), B = (b, a + b)$

We have to find the value of AB^2

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} => AB^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$$

Putting the values $AB^2 = (b - (a + b))^2 + (a + b - (a))^2 => a^2 + b^2$

Then the correct option is B)

15. Correct Matching is: a-Q, b-T, c-RS, d-P

a) --→ (Q)

Given points are $(6,2), (0p), (4 - 7)$

Applying the centroid formula

$$-\frac{5}{3} = \frac{2 + p + (-7)}{3} => -5 = p - 5 => p = 0$$

b) --→ (T)

Distance between given point is 9

Then we can write $9^2 = (-3 - a)^2 + (-14 + 5)^2 => 9^2 = (a + 3)^2 + 9^2$

$$(a + 3)^2 = 0 => a + 3 = 0 => a = -3$$

c) --→ (R, S)

Distance between given point is 10

Then we can write $10^2 = (2 - p)^2 + (-3 - 5)^2 => 10^2 = (p - 2)^2 + 8^2$

$$(p - 2)^2 = 6^2 => p - 2 = \pm 6 => p = 8, -4$$

d) --→ (P)

Given that A, B, C are collinear

So line of equation AB is also satisfied by point C.

Find the equation of line using A and B

$$y = 2x$$

Then for C $x = 2 \times \frac{1}{2} = x = 1$

16. Correct Answer is: 4

Given points are

$$(-1, k - 1), (6, -3), (-3,6)$$

Applying the centroid formula for y coordinate

$$2 = \frac{k - 1 + (-3) + 6}{3} => 6 = k + 2 => k = 4$$

Then the value of k = 4.

17. Correct Answer is: 2

Given points

$$A(2, 9), B(a, 5), C(7, 5)$$

The sides of the triangle are AB, BC, and AC. Since B is the right angle, then the hypotenuse is AC:

$$AC^2 = AB^2 + BC^2$$

Using the distance formula to calculate the lengths of the sides:

$$(7 - 2)^2 + (5 - 9)^2 = (a - 2)^2 + (5 - 9)^2 + (7 - a)^2 + (5 - 5)^2$$

$$25 = (a - 2)^2 + (7 - a)^2 => a^2 + 4 - 4a + 49 + a^2 - 14a = 25$$

$$2a^2 - 18a + 28 = 0 => a^2 - 9a + 14 = 0$$

$$a^2 - 9a + 14 = 0 => a = 2 \text{ or } 7$$

If we consider the value a = 7, then the coordinates of point B and point C would coincide, which would result in a degenerate triangle where all three points are collinear or

coincide. In a degenerate triangle, the sides have zero length, and it's not a meaningful geometric configuration.
In the case of a = 2, the points (2, 9), (2, 5), and (7, 5) form a right-angled triangle with the right angle at vertex B. This satisfies the criteria of a non-degenerate right-angled triangle, and the value a = 2 is indeed the correct and only considerable value.

18. Correct Answer is: 2

Given that points A, B, and C are collinear, we have the following coordinates:

$$Point\ A: A = (k + 1, 2k)$$
$$Point\ B: B = (3k, 2k + 3)$$
$$Point\ C: C = (5k - 1, 5k)$$

We are given that the points are collinear, which means that the slope of AB is equal to the slope of BC.

$$slope\ of\ AB = \frac{2k + 3 - 2k}{3k - k + 1} = \frac{3}{2k - 1}$$
$$slope\ of\ BC = \frac{5k - 2k - 3}{5k - 1 - 3k} = \frac{3k - 3}{2k - 1}$$

Equating the slopes

$$\frac{3}{2k - 1} = \frac{3k - 3}{2k - 1} => k = 2$$

So, the integral value of k that results in the given points being collinear is 2.

19. Correct Answer is: 2

The points (-2, 0), (-2, -1), and (-2, 2) are 2 units away from the Y-axis towards the west direction.
To prove that the points (-2, 0), (-2, -1), and (-2, 2) are 2 units away from the Y-axis towards the west direction, we need to calculate the distance of these points from the Y-axis and demonstrate that the distance is indeed 2 units.
The distance between a point (x, y) and the Y-axis can be calculated using the formula $|x|$. The absolute value of x gives the distance of the point from the Y-axis.
Given points:

$$Point\ A: (-2, 0)$$
$$Point\ B: (-2, -1)$$
$$Point\ C: (-2, 2)$$

1. Distance of point A from Y-axis:

$$Distance_A = |x_A| = |-2| = 2\ units$$

2. Distance of point B from Y-axis:

$$Distance_B = |x_B| = |-2| = 2\ units$$

3. Distance of point C from Y-axis:

$$Distance_C = |x_C| = |-2| = 2\ units$$

All three points have a distance of 2 units from the Y-axis, and they lie on the left side of the Y-axis (west direction), as their x-coordinates are negative.
Therefore, it is proven that the points (-2, 0), (-2, -1), and (-2, 2) are 2 units away from the Y-axis towards the west direction.

20. Correct Answer is: 7

The points (-5, 7), (0, 7), (5, 7), and (7, 7) are 7 units away from the X-axis.
The distance of a point (x, y) from the X-axis is given by $|y|$.
Given points:

$$Point\ A: (-5, 7)$$
$$Point\ B: (0, 7)$$
$$Point\ C: (5, 7)$$
$$Point\ D: (7, 7)$$

1. $Distance\ of\ point\ A\ from\ X-axis: |y_A| = |7| = 7\ units$
2. $Distance\ of\ point\ B\ from\ X-axis: |y_B| = |7| = 7\ units$
3. $Distance\ of\ point\ C\ from\ X-axis: |y_C| = |7| = 7\ units$
4. $Distance\ of\ point\ D\ from\ X-axis: |y_D| = |7| = 7\ units$

All four points have a distance of 7 units from the X-axis, which is their y-coordinate. Thus, it is confirmed that the points (-5, 7), (0, 7), (5, 7), and (7, 7) are 7 units away from the X-axis.

Area Of Triangle and Quadrilaterals
LEVEL-I

1. Correct option: (C)

If $(x_1, y_1,), (x_2, y_2,), (x_3, y_3,)$ are vertices of a triangle then area of the triangle is

$$= \frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

Given vertices are $(-4, -1), (1,2), (4, -3)$

Then the area of triangle is

$$\frac{1}{2}\big|(-4)\big(2-(-3)\big) + 1\big(-3-(-1)\big) + 4\big((-1)-(2)\big)\big| = \frac{1}{2} \times 34 = 17$$

Therefore the correct option is C)

2. Correct option: (B)

The area of a triangle formed by the midpoints of the sides of another triangle is always one-fourth of the area of the original triangle.

Given that the area of triangle ABC is 20, the area of triangle PQR, which is formed by the midpoints of triangle ABC, is:

$$Area\ of\ PQR = \left(\frac{1}{4}\right) \times Area\ of\ ABC$$

$$Area\ of\ PQR = \left(\frac{1}{4}\right) \times 20$$

$$Area\ of\ PQR = 5$$

So, the area of triangle PQR is 5.

Therefore the correct option is B)

3. Correct option: (B)

The area of the triangle formed by joining the midpoints of the sides of triangle ABC is one fourth of the area of triangle ABC.

Given that the area of the triangle formed by the midpoints is 5 square units, we can set up the following equation:

$$\left(\frac{1}{4}\right) \times Area\ of\ ABC = 5$$

$$Area\ of\ ABC = 5 \times 4$$

$$Area\ of\ ABC = 20$$

So, the area of triangle ABC is 20 square units.

Therefore the correct option is B)

4. Correct option: (B)

The area of the triangle formed by joining the midpoints of the sides of triangle ABC is one fourth of the area of triangle ABC.

So area of triangle DEF can be calculated as

First find the area of triangle ABC

$$= \frac{1}{2}\big|-3(-2-6) + (-1)(6-4) + 5\big(4-(-2)\big)\big| = \frac{1}{2}|24 - 2 + 30| = 26$$

$$\left(\frac{1}{4}\right) \times Area\ of\ ABC\ =\ Area\ of\ DEF$$

$$Area\ of\ DEF\ = \frac{26}{4} = \frac{13}{2}$$

So, the area of triangle DEF is $\frac{13}{2}$ square units.

Therefore the correct option is B)

5. Correct option: (C)

The relationship between the areas of these triangles is as follows:

If Δ_1 is the area of the triangle formed by the centroid and two vertices of a triangle, and Δ_2 is the area of the triangle formed by the midpoints of the sides of the given triangle, then:

$$\Delta_1 = \frac{1}{3} \times \Delta, \Delta_2 = \frac{1}{4} \times \Delta \Rightarrow \frac{\Delta_1}{\Delta_2} = \frac{4}{3}$$

In other words, the area of the triangle formed by the centroid and two vertices is third fourth of the area of the triangle formed by the midpoints of the sides of the original triangle.

Therefore the correct option is C)

6. Correct option: (D)

Let's find the area of given triangles

For $\Delta PCB = \Delta_1$

$$= \frac{1}{2}|\alpha(5-2) + 3(2-\beta) + 4(\beta-5)| = \frac{1}{2}|3\alpha + 6 - 3\beta + 4\beta - 20|$$

$$= \frac{1}{2}|3\alpha + \beta - 14|$$

For $\Delta ABC = \Delta_2$

$$= \frac{1}{2}|6(5-2) + 3(2-3) + 4(3-5)| = \frac{1}{2}|18 - 3 - 8| = \frac{1}{2}|7|$$

Then ratio of the given areas

$$\frac{\Delta_1}{\Delta_2} = \frac{|3\alpha + \beta - 14|}{7}$$

Therefore the correct option is D)

7. Correct option: (B)

Let's find the area of given triangles

For $\Delta DCB = \Delta_1$

$$= \frac{1}{2}|x(-2-5) + 4(5-3x) + (-3)(3x+2)| = \frac{1}{2}|-28x + 14|$$

For $\Delta ABC = \Delta_2$

$$= \frac{1}{2}|6(7) + (-3)(-5) + 4(3-5)| = \frac{1}{2}|49|$$

Then given ratio of the areas

$$\frac{\Delta_1}{\Delta_2} = \frac{1}{2} \Rightarrow 2 \times (-28x + 14) = 49 \Rightarrow x = -\frac{3}{8}$$

Therefore the correct option is B)

8. Correct option: (A)

Let's find the point A using the given ratio and point PQ as

$$A = \left(\frac{3k-5}{k+1}, \frac{5k+1}{k+1}\right)$$

Area of triangle ABC

$$= \frac{1}{2}\left(\frac{3k-5}{k+1}(7) + 1\left(-2 - \frac{5k+1}{k+1}\right) + 7\left(\frac{5k+1}{k+1} - 5\right)\right)$$

$$= \frac{1}{2(k+1)}(21k - 35 - 7k - 3 - 28) = \frac{14k - 66}{2(k+1)} = \frac{7k - 33}{k+1}$$

Equating the area given as

$$\frac{7k - 33}{k+1} = \pm 2 \Rightarrow k = 7, \frac{31}{9}$$

Therefore the correct option is A)

9. Correct option: (B)

Let's find the area of given triangles

For $\triangle$

$$= \frac{1}{2}|t(5) + (-2)(1 - 2t) + 3(2t - 6)| = \frac{1}{2}|5t + 4t + 6t - 20|$$

Equating the given area

$$\frac{1}{2}|5t + 4t + 6t - 20| = 5 \Rightarrow 15t - 20 = \pm 10 \Rightarrow t = 2, \frac{2}{3}$$

Therefore the correct option is B)

10. Correct option: (D)

Let's find the area of given triangles

For $\triangle$

$$= \frac{1}{2}\left|0(0 - a^{6x}) + a^{x^2}(a^{6x} - 0) + 0(0 - 0)\right| = \frac{1}{2}|a^{7x}|$$

Equating the given area

$$\frac{1}{2}\left|a^{x^2+6x}\right| = \frac{1}{2}a^{-5} \Rightarrow x^2 + 6x + 5 = 0$$

Then

$$x = \frac{-6 \pm \sqrt{36 - 4 \times 5}}{2} = \frac{-6 \pm 4}{2} = -1, -5$$

Therefore the correct option is D)

11. Correct option: (C)

If three points are collinear we can say that area formed by the triangle made by them is 0.

Let's find the area

For $\triangle$

$$= \frac{1}{2}\left|k(4k - 6) + (-k + 1)\big(6 - 2k - (2 - 2k)\big) + (-4 - k)(2 - 2k - 2k)\right|$$

$$= \frac{1}{2}|4k^2 - 6k - 4k + 4 + 4k^2 - 8 + 14k| = \frac{1}{2}|8k^2 + 4k - 4| = 0$$

Then

$$k = \frac{-4 \pm \sqrt{144}}{16} = -1, \frac{1}{2}$$

Therefore the correct option is C)

12. Correct option: (A)

Area of the quadrilateral formed by the points given

$$\frac{1}{2}|(x_1y_2 - x_2y_1) + (x_2y_3 - x_3y_2) + (x_3y_4 - x_4y_3) + (x_4y_1 - x_1y_4)|$$

$$= \frac{1}{2}|(1 \times (-3) - 2 \times 2) + (8 - 6) + (-10) + (-5)| = \frac{1}{2}|20| = 10$$

Therefore the correct option is A)

13. Correct option: (*)

Area of the parallelogram formed by the points given

Let's find the fourth vertex using midpoint intersection of diagonals as

$$M = \left(\frac{-1 + 7}{2}, \frac{6 + 2}{2}\right) = \left(\frac{x + 4}{2}, \frac{y + 1}{2}\right) \Rightarrow x = 2, y = 7$$

Then area of parallelogram

$$\frac{1}{2}|(x_1y_2 - x_2y_1) + (x_2y_3 - x_3y_2) + (x_3y_4 - x_4y_3) + (x_4y_1 - x_1y_4)|$$

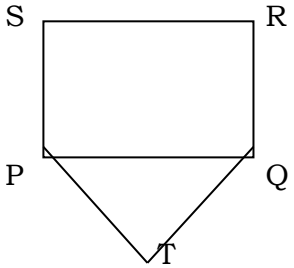
$$= \frac{1}{2}|-1 - 8 + 24 - 7 + 49 - 12 + 4 + 7| = \frac{1}{2}|56| = 28$$

14. Correct option: (B)

Total length of the string = perimeter of the square

$$12 = 4a => a = 3$$

So side of the square is 3 which will be the base of the triangle PQT as



Then are of square is $a^2 = 9$

To find the area of the given triangle we first find the height of the triangle

Given that the triangle is a isosceles triangle so the sides of the PQT will be

$$PQ = 3, PT = QT = \frac{12 - 3}{2}$$

Using the right angle triangle we can conclude that the height of the triangle

$$= \sqrt{\frac{81}{4} - \frac{9}{4}} = \frac{6}{\sqrt{2}}$$

Then the area of the triangle

$$\frac{1}{2} \times 3 \times \frac{6}{\sqrt{2}} = \frac{9}{\sqrt{2}}$$

Then the ration of area of the square to the area of the triangle is

$$\frac{9}{\frac{9}{\sqrt{2}}} = \sqrt{2} => \sqrt{2}: 1$$

Then the correct option is B)

15. Correct option: (A)

Area of the rhombus formed by the points given

$$\frac{1}{2}|(x_1y_2 - x_2y_1) + (x_2y_3 - x_3y_2) + (x_3y_4 - x_4y_3) + (x_4y_1 - x_1y_4)|$$

$$= \frac{1}{2}\big|(15 - 0) + (16— 5)\big) + (1— 8) + (0 - (-3)) = \frac{1}{2}|48| = 24$$

We can also find the area of the rhombus using the diagonals with formula as

$$= \frac{1}{2} \times d_1 \times d_2$$

Then the correct option is A)

16. Correct option: (B)

Coordinates of the point D as D is midpoint of BC

$$D = \left(\frac{7 + 1}{2}, \frac{6 + 4}{2}\right) = (4,5)$$

Then area of the triangle ABD

$$\frac{1}{2}|4(6 - 5) + 7(5 - 2) + 4(2 - 6)| = \frac{1}{2}|9| = \frac{9}{2}$$

Then the correct option is B)

17. Correct option: (D)

Area of the quadrilateral formed by the points given

$$\frac{1}{2}|(x_1y_2 - x_2y_1) + (x_2y_3 - x_3y_2) + (x_3y_4 - x_4y_3) + (x_4y_1 - x_1y_4)|$$

$$= \frac{1}{2}\big|\big(0 - (-2)\big) + (2 - 0) + \big(0 - (-6)\big) + (6 - 0)\big| = \frac{1}{2}|16| = 8$$

Then the correct option is D)

18. Correct option: (B)

Area of the quadrilateral formed by the points given

$$\frac{1}{2}|(x_1y_2 - x_2y_1) + (x_2y_3 - x_3y_2) + (x_3y_4 - x_4y_3) + (x_4y_1 - x_1y_4)|$$

$$= \frac{1}{2}|(20 - 6) + (6 - (-15)) + (9 - (-4)) + (-4 - (-12))| = \frac{1}{2}|56| = 28$$

Then the correct option is B)

19. Correct option: (C)

Slope of the line passing through the points $(x_1, y_1), (x_2, y_2)$

$$m = \frac{y_2 - y_1}{x_1 - x_2}$$

For our given points, the slope is

$$m = \frac{z^2 - y^2}{z - y} = z + y$$

Then the correct option is C)

20. Correct option: (*)

Slope of the line passing through the points $(x_1, y_1), (x_2, y_2)$

$$m = \frac{y_2 - y_1}{x_1 - x_2}$$

For our given points, the slope is given as 2 then

$$m = \frac{4x - 3 - (7)}{x - 2 - (-1)} = 3$$

$$=> 3 = \frac{4x - 10}{x - 1} => 3x - 3 = 4x - 10 => x = 7$$

LEVEL-II

1. Correct option: (B)

If $(x_1, y_1,), (x_2, y_2,), (x_3, y_3,)$ are vertices of a triangle then area of the triangle is

$$= \frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

Then the area of triangle is

$$\frac{1}{2}|1(-6 - 5) + 4(5 - 2) + 3(2 - (-6))| = \frac{1}{2} \times 25 = \frac{25}{2}$$

Therefore the correct option is B)

2. Correct option: (A)

If $(x_1, y_1,), (x_2, y_2,), (x_3, y_3,)$ are vertices of a triangle then area of the triangle is

$$= \frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

Then the area of triangle is

$$\frac{1}{2}\big|5\big(7 - (-4)\big) + 4(-4 - 2) + 7(2 - 7)\big| = \frac{1}{2} \times 4 = 2$$

Therefore the correct option is A)

3. Correct option: (B)

Coordinates of the point D as D is midpoint of BC

$$D = \left(\frac{3+5}{2}, \frac{-2+2}{2}\right) = (4, 0)$$

Area of the triangle ABD

$$\frac{1}{2}\left|4(-2-0) + 3\big(0-(-6)\big) + 4(-6-(-2))\right| = \frac{1}{2}|6| = 3$$

Area of the triangle ABC

$$\frac{1}{2}|4(-2-2) + 3(2-(-6)) + 5(-6-(-2))| = \frac{1}{2}|12| = 6$$

Then the ratio of the area of the given triangle $\triangle ABD : \triangle ABC = 1:2$

From this we can say that median of any vertex cut the triangle in two same area triangles.

Here ratio of the triangle formed $\triangle ABD : \triangle ACD = 1:1$

Then the correct option is B)

4. Correct option: (A)

Coordinates of the point D as D is midpoint of BC

$$D = \left(\frac{3+5}{2}, \frac{-2+2}{2}\right) = (4, 0)$$

We know that median of any vertex cut the triangle in two same area triangles.

Here ratio of the triangle formed $\triangle ABD : \triangle ACD = 1:1$

Then the correct option is A)

5. Correct option: (D)

Area of the quadrilateral formed by the points given

$$\frac{1}{2}|(x_1y_2 - x_2y_1) + (x_2y_3 - x_3y_2) + (x_3y_4 - x_4y_3) + (x_4y_1 - x_1y_4)|$$

$$= \frac{1}{2}|(-12-10) + (-30-28) + (-28-(30) + (-10-(12))| = \frac{1}{2}|160| = 80$$

Then the correct option is D)

6. Correct option: (A)

If $(x_1, y_1,), (x_2, y_2,), (x_3, y_3,)$ are vertices of a triangle then area of the triangle is

$$= \frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

Then the area of triangle is 0 for 3 points which are collinear

$$\frac{1}{2}|x(1-5) + 2(5+1) + 4(-2)| = \frac{1}{2} \times |-4x + 4| = 0 \Rightarrow 4x - 4 = 0$$

$$x = 1$$

Therefore the correct option is A)

7. Correct option: (B ,C)

If $(x_1, y_1,), (x_2, y_2,), (x_3, y_3,)$ are vertices of a triangle then area of the triangle is

$$= \frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

Then the area of triangle is 0 for 3 points which are collinear

$$\frac{1}{2}\left|p\big(n-(q-n)\big) + m(q-n-q) + (p-m)(q-n)\right| =$$

$$= \frac{1}{2}|pn - qn + pn + mq - nm - qm + pq - pn - mq + mn|$$

$$= \frac{1}{2}|pn - qm| = 0 \Rightarrow pn = qm \Rightarrow \frac{q}{n} = \frac{p}{m} \ (n \neq 0, m \neq 0)$$

Therefore the correct option is B, C)

8. Correct option: (D)

If $(x_1, y_1,), (x_2, y_2,), (x_3, y_3,)$ are vertices of a triangle then area of the triangle is

$$= \frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

Then the area of triangle is 0 for 3 points which are collinear

$$\frac{1}{2}|k(-2-4) + 6(4-3) + (-3)(3+2)| =$$

$$= \frac{1}{2}|-6k + 6 - 15| = 0 \Rightarrow 6k + 9 = 0 \Rightarrow k = -\frac{3}{2}$$

Therefore the correct option is D)

9. Correct option: (B)

If $(x_1, y_1,), (x_2, y_2,), (x_3, y_3,)$ are vertices of a triangle then area of the triangle is

$$= \frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

Then the area of triangle is 0 for 3 points which are collinear

$$\frac{1}{2}|7(1-2k) + 5(2k+2) + 3(-2-1)| =$$

$$= \frac{1}{2}|7 - 14k + 10k + 10 - 9| = 0 \Rightarrow 8 - 4k = 0 \Rightarrow k = 2$$

Therefore the correct option is B)

10. Correct option: (C)

If $(x_1, y_1,), (x_2, y_2,), (x_3, y_3,)$ are vertices of a triangle then area of the triangle is

$$= \frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

Then the area of triangle is 0 for 3 points which are collinear

$$\frac{1}{2}|3q(3p-1) + 0(1-0) + 1(0-3p)| =$$

$$= \frac{1}{2}|9pq - 3q - 3p| = 0 \Rightarrow 9pq - 3q - 3p = 0 \Rightarrow \frac{p+q}{pq} = 3$$

$$\frac{1}{p} + \frac{1}{q} = 3$$

Therefore the correct option is C)

11. Correct option: (A, C)

If $(x_1, y_1,), (x_2, y_2,), (x_3, y_3,)$ are vertices of a triangle then area of the triangle is

$$= \frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

Then the area of triangle is 10 for 3 points

$$\frac{1}{2}|a(6-1) + (-2)(1-2a) + 3(2a-6)| =$$

$$= \frac{1}{2}|5a - 2 + 4a + 6a - 18| = 10 \Rightarrow 15a - 20 = \pm 20 \Rightarrow a = 0, \frac{8}{3}$$

Therefore the correct option is A, C)

12. Correct option: (B, C)

If $(x_1, y_1,), (x_2, y_2,), (x_3, y_3,)$ are vertices of a triangle then area of the triangle is

$$= \frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

We are given the ratio of area of the triangle ABC to DBC = 2

Area of $\triangle ABC$

$$\frac{1}{2}|6(5+2) + (-3)(-2-3) + 4(3-5)| = \frac{49}{2}$$

Area of $\triangle DBC$

$$= \frac{1}{2}|x(5+2) + (-3)(-2-3x) + 4(3x-5)| = \frac{1}{2}|7x + 6 + 9x + 12x - 20| = \frac{1}{2}|28x - 14|$$

Then we can write as

$$\frac{49}{|28x-14|}=2 \Rightarrow \pm 49=56x-28 \Rightarrow x=\frac{77}{56},-\frac{21}{56}=\frac{11}{8},-\frac{3}{8}$$

Therefore the correct option is B, C)

13. Correct option: (D)

If three points are collinear we can say that area formed by the triangle made by them is 0.

Let's find the area

For $\triangle$

$$=\frac{1}{2}|(k+1)(2k+3-5k)+3k(5k-2k)+(5k-1)(-3)|=\frac{1}{2}|6k^2-15k+6|=0 \Rightarrow k=2,\frac{1}{2}$$

So integral value of the k is 2

Therefore the correct option is D)

14. Correct option: (A)

Area of the quadrilateral formed by the points given

$$\frac{1}{2}|(x_1y_2-x_2y_1)+(x_2y_3-x_3y_2)+(x_3y_4-x_4y_3)+(x_4y_1-x_1y_4)|$$

$$=\frac{1}{2}|(-3-7)+(14+36)+(12\times 21-14)+(7-21)|=\frac{1}{2}|264|=132$$

Then the correct option is A)

15. Correct option: (D)

Area of the quadrilateral formed by the points given

$$\frac{1}{2}|(x_1y_2-x_2y_1)+(x_2y_3-x_3y_2)+(x_3y_4-x_4y_3)+(x_4y_1-x_1y_4)|$$

$$=\frac{1}{2}|(2-12)+(18-10)+(20-9)+(6-4)|=\frac{1}{2}|11|=\frac{11}{2}$$

Then the correct option is D)

16. Correct option: (D)

Area of the quadrilateral formed by the points given

$$\frac{1}{2}|(x_1y_2-x_2y_1)+(x_2y_3-x_3y_2)+(x_3y_4-x_4y_3)+(x_4y_1-x_1y_4)|$$

$$=\frac{1}{2}|(20-6)+(6+15)+(9+4)+(-4+12)|=\frac{1}{2}|54|=27$$

Then the correct option is D)

17. Correct Matching is: a-R, b-S, c-S, d-P

Given point are $A(1,2),B(6,2),C(5,3),D(3,4)$

a) $ar(\triangle ABC)$

$$=\frac{1}{2}|(1(2-3)+6(3-2)+5(2-2)|=\frac{1}{2}|5|=\frac{5}{2}$$

Then

a) -→(R)

b) $ar(\triangle ACD)$

$$=\frac{1}{2}|(1(3-4)+5(4-2)+3(2-3)|=\frac{1}{2}|6|=3$$

Then

b) -→(S)

c) $ar(\triangle ABD)$

$$=\frac{1}{2}|(1(2-4)+6(4-2)+3(2-2)|=\frac{1}{2}|6|=3$$

Then

c) -→(S)

d) $ar(\triangle BCD)$

$$= \frac{1}{2}|(6(-1) + 5 \times 2 - 3)| = \frac{1}{2}|1| = \frac{1}{2}$$

Then

d) -→(P)

18. Correct Answer is: $(-\frac{1}{6})$

Area of the quadrilateral formed by the points given

$$\frac{1}{2}|(x_1y_2 - x_2y_1) + (x_2y_3 - x_3y_2) + (x_3y_4 - x_4y_3) + (x_4y_1 - x_1y_4)|$$

We are given that the area of the quadrilateral is 0. Then

$$= \frac{1}{2}|(6 + 10) + (20 - 23) + (-14 + 4k) + (2k + 2)| = \frac{1}{2}|6k + 1| => k = -\frac{1}{6}$$

19. Correct Answer is: (2)

If three points are collinear we can say that area formed by the triangle made by them is 0.

Let's find the area For $\triangle$

$$= \frac{1}{2}|7(1 - 2k) + 5(2k + 2) + 3(-2 - 1)| = \frac{1}{2}|7 - 14k + 10k + 10 - 9| = 0 => 8 - 4k = 0 => k = 2$$

So integral value of the k is 2

20. Correct Answer is: (0)

If $(x_1, y_1,), (x_2, y_2,), (x_3, y_3,)$ are vertices of a triangle then area of the triangle is

$$= \frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

Then the area of triangle is

$$\frac{1}{2}\left|\left(-\frac{2}{5}\right)(8 - 3) + 2(3 - 6) + (-4)(6 - 8)\right| = \frac{1}{2}|-2 - 6 + 8| = 0$$

STUDENT PRACTICE QUESTIONS

1. Correct option: (D)

Given vertices are $(a, b + c), (b, c + a), (c, a + b)$

Then the area of triangle is

$$\frac{1}{2}|a(c - b) + b(a - c) + c(b - a)| = \frac{1}{2} \times 0 = 0$$

Therefore the correct option is D)

2. Correct option: (C)

Given vertices are

$$(a + 3, a - 2), (a - 4, a + 5), (a, a)$$

Then the area of triangle is

$$\frac{1}{2}\left|(a + 3)(a + 5 - a) + (a - 4)\big(a - (a - 2)\big) + a\big(a - 2 - (a + 5)\big)\right|$$

$$= \frac{1}{2} \times |5a + 15 + 2a - 8 - 7a| = \frac{7}{2}$$

Therefore the correct option is C)

3. Correct option: (C)

Let's find the area of given triangles

For $\triangle ABD = \Delta_1$

$$= \frac{1}{2}|-3(-2 + 4) + (-1)(-4 - 4) + x(4 - (-2)| = \frac{1}{2}|2 + 6x|$$

For $\triangle ACD = \Delta_2$

$$= \frac{1}{2}|-3(6+4)+(5)(-8)+x(4-6)| = \frac{1}{2}|-2x-70|$$

Then given ratio of the areas

$$\frac{\Delta_1}{\Delta_2} = 2 => 2+6x = 2(2x+70) => x = 69$$

Therefore the correct option is C)

4. Correct option: (C)

Let's find the area of given triangles

For Δ

$$= \frac{1}{2}|1(3-4)+2(4-2)+x(2-3)| = \frac{1}{2}|3-x|$$

Equating the given area

$$\frac{1}{2}|3-x| = 40 => 3-x = \pm 80 => x = -77, 83$$

Therefore the correct option is C)

5. Correct option: (A)

If three points are collinear we can say that area formed by the triangle made by them is 0.

Let's find the area

For Δ

$$= \frac{1}{2}|t(6t-8)+2t(8-2t)+3(-4t)| = \frac{1}{2}|6t^2-8t+16t-4t^2-12t| = \frac{1}{2}|2t^2-4t| = 0 =$$
$$> 2t^2-4t =$$

Then

$$t = 0, 2$$

Therefore the correct option is A)

6. Correct option: (A)

Area of the quadrilateral formed by the points given

$$\frac{1}{2}|(x_1y_2-x_2y_1)+(x_2y_3-x_3y_2)+(x_3y_4-x_4y_3)+(x_4y_1-x_1y_4)|$$
$$= \frac{1}{2}|(-1\times(-3)-(-2\times 5))+(-2-(-15))+(39)+(39)| = \frac{1}{2}|104| = 52$$

Therefore the correct option is A)

7. Correct option: (A)

Area of the square formed by the points given

Let's find the fourth vertex using midpoint intersection of diagonals as

$$M = \left(\frac{-2+2}{2}, \frac{1+5}{2}\right) = \left(\frac{x+4}{2}, \frac{y+3}{2}\right) => x = -4, y = 3$$

Then area of parallelogram

$$\frac{1}{2}|(x_1y_2-x_2y_1)+(x_2y_3-x_3y_2)+(x_3y_4-x_4y_3)+(x_4y_1-x_1y_4)|$$
$$= \frac{1}{2}|6-4+20-(-6)+(-6)-(-20)+(-4)-6| = \frac{1}{2}|32| = 16$$

Then the correct option is A)

8. Correct option: (B)

Coordinates of the point D as D is midpoint of BC

$$D = \left(\frac{2+4}{2}, \frac{3+2}{2}\right) = \left(3, \frac{5}{2}\right)$$

Area of the triangle ABD

$$\frac{1}{2}\left|3\left(3-\frac{5}{2}\right)+2\left(\frac{5}{2}-5\right)+3(5-3)\right| = \frac{1}{2}\left|\frac{5}{2}\right| = \frac{5}{4}$$

Area of the triangle ACD

$$\frac{1}{2}\left|3\left(2-\frac{5}{2}\right)+4\left(\frac{5}{2}-5\right)+3(5-2)\right|=\frac{1}{2}\left|\frac{5}{2}\right|=\frac{5}{4}$$

Then the ratio of the area of the given triangle

$$\triangle ABD:\triangle ACD=1:1$$

Then the correct option is B)

9. Correct option: (A)

Slope of the line passing through the points $(x_1,y_1),(x_2,y_2)$

$$m=\frac{y_2-y_1}{x_1-x_2}$$

For our given points, the slope is given as 2 then

$$m=\frac{3-5}{x-2}=2 \Rightarrow -2=2(x-2) \Rightarrow x-2=-1 \Rightarrow x=1$$

Then the correct option is A)

10. Correct option: (B)

Slope of the line passing through the points $(x_1,y_1),(x_2,y_2)$

$$m=\frac{y_2-y_1}{x_1-x_2}$$

For our given points, the slope is

$$m=\frac{1-2\sqrt{3}-1}{0-2}=-\frac{2\sqrt{3}}{-2}=\sqrt{3}$$

Then the correct option is B)

11. Correct option: (B, C)

If three points are collinear we can say that area formed by the triangle made by them is 0.

Let's find the area

For $\triangle$

$$=\frac{1}{2}\left|k(4k-6)+(-k+1)\big(6-2k-(2-2k)\big)+(-4-k)(2-2k-2k)\right|$$

$$=\frac{1}{2}|4k^2-6k-4k+4+4k^2-8+14k|=\frac{1}{2}|8k^2+4k-4|=0$$

Then

$$k=\frac{-4\pm\sqrt{144}}{16}=-1,\frac{1}{2}$$

Therefore the correct option is B, C)

12. Correct option: (C)

If three points are collinear we can say that area formed by the triangle made by them is 0.

Let's find the area

For $\triangle$

$$=\frac{1}{2}\left|at_1^2(2at_2-a)+at_2^2(a-2at_1)\right|=\frac{1}{2}\left|2a^2t_1^2t_2-a^2t_1^2+a^2t_2^2-2a^2t_2^2t_1\right|$$

$$=\frac{1}{2}|2a^2t_2t_1(t_1-t_2)+a^2(t_2-t_1)(t_2+t_1)|=0$$

Then $(t_1-t_2)(2t_2t_1-t_2-t_1)=0 \Rightarrow t_2=t_1 \text{ or } 2t_2t_1=t_2+t_1$

Therefore the correct option is C)

13. Correct option: (C)

If $(x_1,y_1,),(x_2,y_2,),(x_3,y_3,)$ are vertices of a triangle then area of the triangle is

$$=\frac{1}{2}|x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2)|$$

Then the area of triangle is $\frac{1}{2}|3(8-12)+11(12-2)+8(2-8)|=$

$$= \frac{1}{2}|-12 + 110 - 48| = \frac{50}{2} => 25$$

Therefore the correct option is C)

14. Correct option: (C)

If $(x_1, y_1,), (x_2, y_2,), (x_3, y_3,)$ are vertices of a triangle then area of the triangle is

$$= \frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

Then the area of triangle is $\frac{1}{2}|3(0-4) + 7(4-0) + 8(0-0)| =$

$$= \frac{1}{2}|-12 + 28| = \frac{16}{2} => 8$$

Therefore the correct option is C)

15. Correct option: (B)

If $(x_1, y_1,), (x_2, y_2,), (x_3, y_3,)$ are vertices of a triangle then area of the triangle is

$$= \frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

Then the area of triangle is $\frac{1}{2}|6(5-(-2) + (-3)(-2-3) + 4(3-5)| =$

$$= \frac{1}{2}|42 + 15 - 8| = \frac{49}{2}$$

Therefore the correct option is B)

16. Correct Answer is: (5)

If three points are collinear we can say that area formed by the triangle made by them is 0.

Let's find the area

For $\triangle$ $= \frac{1}{2}|a(-1-4) + 1(4-1) + 11(2)| = \frac{1}{2}|-5a + 25| = 0 => 5a - 25 = 0$

$$=> a = 5$$

So integral value of the a is 5

17. Correct Answer is: (4)

If $(x_1, y_1,), (x_2, y_2,), (x_3, y_3,)$ are vertices of a triangle then area of the triangle is

$$= \frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

Then the area of triangle is $\frac{1}{2}|t(t+2-t) + (t+2)(t-t+2) + (t+3)(-4)| =$

$$= \frac{1}{2}|-8| = 4$$

18. Correct Answer is: (0)

If $(x_1, y_1,), (x_2, y_2,), (x_3, y_3,)$ are vertices of a triangle then area of the triangle is

$$= \frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

Then the area of triangle is $\frac{1}{2}|5(-1-4) + 1(4-1) + 11(1+1)| = 0$

19. Correct Answer is: (0)

If $(x_1, y_1,), (x_2, y_2,), (x_3, y_3,)$ are vertices of a triangle then area of the triangle is

$$= \frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

Then the area of triangle is $\frac{1}{2}|a(c+a-a-b) + b(a+b-b-c) + c(b+c-c-a)|$

$$= \frac{1}{2}|ac - ab + ba - bc + cb - ca| = 0$$

So the area of the given points is 0

6. Trigonometry and its Applications

Solutions

LEVEL – I

1. Correct option: (A) – 390

$$\frac{-13\pi}{6} = \frac{-13 \times 180}{6}$$
$$= -13 \times 30$$
$$= -390$$

2. Correct option: (A) $\frac{5}{14}$

Given: θ lies in first quadrant, all angles are positive
And $5\tan\theta = 4$

$$\tan\theta = \frac{4}{5}$$

So

$$\frac{5\sin\theta - 3\cos\theta}{\sin\theta + 2\cos\theta}$$

Divide by $\cos\theta$

$$\frac{5\sin\theta - 3\cos\theta}{\sin\theta + 2\cos\theta} = \frac{5\left(\frac{\sin\theta}{\cos\theta}\right) - 3\left(\frac{\cos\theta}{\cos\theta}\right)}{\left(\frac{\sin\theta}{\cos\theta}\right) + 2\left(\frac{\cos\theta}{\cos\theta}\right)}$$
$$= \frac{5\tan\theta - 3}{\tan\theta + 2}$$
$$= \frac{5\left(\frac{4}{5}\right) - 3}{\left(\frac{4}{5}\right) + 2}$$
$$= \frac{4 - 3}{\left(\frac{4 + 10}{5}\right)}$$
$$= \frac{1 \times 5}{14}$$
$$= \frac{5}{14}$$

3. Correct option: (A): $\frac{17}{31}$

Given: $\sin\theta = -\frac{7}{25}$
And θ lies in third quadrant
So, $\tan\theta$ *and* $\cot\theta$ are positive
Since

$$sin^2\theta + cos^2\theta = 1$$
$$\left(\frac{-7}{25}\right)^2 + cos^2\theta = 1$$
$$cos^2\theta = 1 - \frac{49}{625}$$
$$cos^2\theta = \frac{625 - 49}{625}$$
$$\cos\theta = \sqrt{\frac{576}{625}}$$

$$\cos\theta = \frac{24}{25}$$

Now

$$\tan\theta = \frac{\sin\theta}{\cos\theta}$$
$$= \frac{\left(\frac{-7}{25}\right)}{\left(\frac{24}{25}\right)} = \frac{7}{24}$$

And

$$\cot\theta = \frac{1}{\tan\theta} = \frac{1}{\left(\frac{7}{24}\right)} = \frac{24}{7}$$

Value of $\frac{7\cot\theta - 24\tan\theta}{7\cot\theta + 24\tan\theta}$

$$= \frac{7\left(\frac{24}{7}\right) - 24\left(\frac{7}{24}\right)}{7\left(\frac{24}{7}\right) + 24\left(\frac{7}{24}\right)}$$
$$= \frac{24-7}{24+7} = \frac{17}{31}$$

4. Correct option: (C) $\frac{1+\cos\theta}{\sin\theta}$

$$\frac{\cot\theta + \operatorname{cosec}\theta - 1}{\cot\theta - \operatorname{cosec}\theta + 1} = \frac{\cot\theta + \operatorname{cosec}\theta - (cosec^2\theta - cot^2\theta)}{\cot\theta - \operatorname{cosec}\theta + 1}$$
$$= \frac{(cot\,\theta + cosec\,\theta) - (cosec\,\theta + cot\theta)(cosec\theta - cot\theta)}{cot\,\theta - cosec\,\theta + 1}$$
$$= \frac{(cot\,\theta + cosec\,\theta)[1 - (cosec\,\theta - cot\theta)]}{cot\,\theta - cosec\,\theta + 1}$$
$$= cot\theta + cosec\theta$$
$$= \frac{\cos\theta}{\sin\theta} + \frac{1}{\sin\theta} = \frac{1+\cos\theta}{\sin\theta}$$

5. Correct option: (A) $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$

Given $x = acos^3\theta$ and $y = asin^3\theta$

$$x^{\frac{2}{3}} + y^{\frac{2}{3}} = (acos^3\theta)^{\frac{2}{3}} + (asin^3\theta)^{\frac{2}{3}}$$
$$= a^{\frac{2}{3}}(cos^3\theta)^{\frac{2}{3}} + a^{\frac{2}{3}}(sin^3\theta)^{\frac{2}{3}}$$
$$= a^{\frac{2}{3}}(cos^2\theta + sin^2\theta) = a^{\frac{2}{3}}$$

6. Correct option: (C) $\cos 2\alpha$

Given $\cos(n+1)\alpha.\cos(n-1)\alpha + \sin(n+1)\alpha\sin(n-1)\alpha$

Let $(n+1)\alpha = A$ and $(n-1)\alpha = B$

So

$$cos(n+1)\alpha.cos(n-1)\alpha + sin(n+1)\alpha\,sin(n-1)\alpha$$
$$= \cos A.\cos B + \sin A\sin B$$
$$= \cos(A-B)$$
$$= \cos[(n+1)\alpha - (n+1)\alpha]$$
$$= \cos 2\alpha$$

7. Correct option: (A) 0

$$\sin\theta + \sin(\theta + 120) - \sin(120 - \theta)$$

$$\sin(C+D) = \sin C\cos D + \cos C\sin D$$

and

$$\sin(C-D) = \sin C\cos D - \cos C\sin D$$

So
$$\sin\theta + \sin(\theta + 120) - \sin(120 - \theta)$$
$$= \sin\theta + sin\,120\,cos\,\theta + cos\,120\,sin\,\theta - (sin\,120\,cos\,\theta - cos\,120\,sin\,\theta)$$
$$= \sin\theta + sin\,120\,cos\,\theta + cos\,120\,sin\,\theta - sin\,120\,cos\,\theta + cos\,120\,sin\,\theta$$
$$= \sin\theta + 2\cos 120 \sin\theta$$

Now
$$\cos 120 = \cos(180 - 60)$$
$$= -\cos 60$$
$$= -\frac{1}{2}$$
$$\sin\theta + \sin(\theta + 120) - \sin(120 - \theta)$$
$$= \sin\theta + 2\left(-\frac{1}{2}\right)\sin\theta = \sin\theta - \sin\theta = 0$$

8. Correct option: (A) $\frac{63}{16}$

Given $\cos A = \frac{3}{5}$ and $\sin B = \frac{5}{13}$

$\cos A = \frac{3}{5}$, so
$$sin^2A + cos^2A = 1$$
$$sin^2A + \left(\frac{3}{5}\right)^2 = 1$$
$$\sin A = \sqrt{1 - \left(\frac{3}{5}\right)^2}$$
$$\sin A = \sqrt{1 - \frac{9}{25}} = \sqrt{\frac{25 - 9}{25}} = \sqrt{\frac{16}{25}} = \frac{4}{5}$$

$\sin B = \frac{5}{13}$, so
$$sin^2B + cos^2B = 1$$
$$\left(\frac{5}{13}\right)^2 + cos^2B = 1$$
$$\cos B = \sqrt{1 - \frac{25}{169}} = \sqrt{\frac{169 - 25}{169}} = \sqrt{\frac{144}{169}}$$
$$\cos B = \frac{12}{13}$$

Now
$$\tan A = \frac{sin\,A}{cos\,A} = \frac{\left(\frac{4}{5}\right)}{\left(\frac{3}{5}\right)} = \frac{4}{3}$$

And
$$\tan B = \frac{sin\,B}{cos\,B} = \frac{\left(\frac{5}{13}\right)}{\left(\frac{12}{13}\right)} = \frac{5}{12}$$

So
$$\frac{\tan A + \tan B}{1 - \tan A \tan B} = \frac{\left(\frac{4}{3}\right) + \left(\frac{5}{12}\right)}{1 - \left(\frac{4}{3}\right)\left(\frac{5}{12}\right)} = \frac{\left(\frac{16 + 5}{12}\right)}{\frac{36 - 20}{36}} = \frac{21 \times 3}{16} = \frac{63}{16}$$

9. Correct option: (B) $\frac{q-p}{pq}$

Given: $\tan A + \tan B = p$ and $\cot A + \cot B = q$

Consider
$$\cot A + \cot B = q$$
$$\frac{1}{\tan A} + \frac{1}{\tan B} = q$$

$$\frac{\tan A + \tan B}{\tan A \tan B} = q$$

$$\frac{p}{\tan A \tan B} = q$$

$$\tan A \tan B = \frac{p}{q}$$

Now $$\cot(A + B) = \frac{1}{\tan(A+B)}$$

$$= \frac{1}{\left(\frac{\tan A + \tan B}{1 - \tan A \tan B}\right)} = \frac{1 - \tan A \tan B}{\tan A + \tan B} = \frac{1 - \frac{p}{q}}{p} = \frac{q - p}{pq}$$

10. Correct option: (A) $\frac{\sqrt{3}}{2}$

$$sin^2 75° - sin^2 15° = (\sin 75° + \sin 15°)(\sin 75° - \sin 15°)$$

$$= (\sin 90°)(\sin 60°) = 1 \times \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2}$$

11. Correct option: (A) $\frac{1}{2}\cos 2\theta$

$$cos^2\left(\frac{\pi}{6} + \theta\right) - sin^2\left(\frac{\pi}{6} - \theta\right)$$

Now

$$\cos(x + y)\cos(x - y) = cos^2 x - sin^2 y$$

So

$$cos^2\left(\frac{\pi}{6} + \theta\right) - sin^2\left(\frac{\pi}{6} - \theta\right)$$

$$= \cos\left(\frac{\pi}{6} + \theta + \frac{\pi}{6} - \theta\right)\cos\left(\frac{\pi}{6} + \theta - \frac{\pi}{6} + \theta\right) = \cos\left(\frac{\pi}{3}\right)\cos 2\theta = \frac{1}{2}\cos 2\theta$$

12. Correct option: (C) $\sqrt{2}$

$$\frac{\sin\theta}{sin^2\left(\frac{\pi}{8} + \frac{\theta}{8}\right) - sin^2\left(\frac{\pi}{8} - \frac{\theta}{2}\right)}$$

$$= \frac{\sin\theta}{\left[\sin\left(\frac{\pi}{8} + \frac{\theta}{2} + \frac{\pi}{8} - \frac{\theta}{2}\right)\right]\left[\sin\left(\frac{\pi}{8} + \frac{\theta}{2} - \frac{\pi}{8} + \frac{\theta}{2}\right)\right]}$$

$$= \frac{\sin\theta}{\sin\left(\frac{\pi}{4}\right)\sin\theta} = \frac{1}{\left(\frac{1}{\sqrt{2}}\right)} = \sqrt{2}$$

13. Correct option: (C)

$$\sin 40° - \sin 80° + \sin 160°$$

$$= 2\sin\left(\frac{40 - 80}{2}\right)\cos\left(\frac{40 + 80}{2}\right) + \sin(180 - 20)$$

$$= -2\sin 20 \cos 60 + \sin 20 = -\sin 20 + \sin 20 = 0$$

14. Correct option: (D) $-\frac{1}{2}$

$$\cos 40° + \cos 80° + \cos 160° + \cos 240$$

$$= 2\cos\left(\frac{40 + 80}{2}\right)\cos\left(\frac{40 - 80}{2}\right) + \cos(180 - 20) + \cos(180 + 60)$$

$$= 2\cos 60° \cos(-20°) + (-\cos 20) + (-\cos 60)$$

As $$\text{Cos}\, C + \cos D = 2\cos\left(\frac{C+D}{2}\right).\cos\left(\frac{C-D}{2}\right)$$

$$\cos(180 + \theta) = -\theta$$

$$\cos(180 - \theta) = -\theta$$

$$\cos 40° + \cos 80° + \cos 160° + \cos 240$$

$$= 2 \times \frac{1}{2}\cos 20° - \cos 20° - \frac{1}{2} = -\frac{1}{2}$$

15. Correct option: (D) 1

$$\tan\left(\frac{\pi}{4}+\theta\right)\tan\left(\frac{\pi}{4}-\theta\right)$$

$$=\left[\frac{\tan\frac{\pi}{4}+\tan\theta}{1-\tan\frac{\pi}{4}\tan\theta}\right]\times\left[\frac{tan\frac{\pi}{4}-tan\,\theta}{1+tan\frac{\pi}{4}tan\,\theta}\right]=\frac{1+\tan\theta}{1-\tan\theta}\times\frac{1-\tan\theta}{1+\tan\theta}=1$$

16. Correct option: (D) $\frac{3}{5}$

Given: $\cot A\cot B=2,\ \cos(A+B)=\frac{3}{5}$

Now $\cos(A+B)=\frac{3}{5}$

$$cos\,A\,cos\,B-sin\,A\,sin\,B=\frac{3}{5}$$

$$\frac{cos\,A\,cos\,B-sin\,A\,sin\,B}{sin\,A\,sin\,B}=\frac{\left(\frac{3}{5}\right)}{sin\,A\,sin\,B}$$

$$\frac{cos\,A\,cos\,B}{sin\,A\,sin\,B}-1=\frac{\frac{3}{5}}{sin\,A\,sin\,B}$$

$$cot\,A\,cot\,B-1=\frac{\frac{3}{5}}{sin\,A\,sin\,B}$$

$$2-1=\frac{\frac{3}{5}}{sin\,A\,sin\,B}$$

$$1\times sin\,A\,sin\,B=\frac{3}{5}$$

$$sin\,A\,sin\,B=\frac{3}{5}$$

17. Correct option: (A) 1

Given

$$\tan 8A-\tan 5A-\tan 3A=k\tan 8A\tan 5A\tan 3A$$

$$\tan 8A-(\tan 5A+\tan 3A)=k\,tan\,8A\,tan\,5A\,tan\,3A$$

Now

$$\tan(5A+3A)=\frac{tan\,5A+tan\,3A}{1-tan\,5A\,tan\,3A}$$

$$\tan 8A\,(1-tan\,5A\,tan\,3A)=tan\,5A+tan\,3A$$

$$\tan 8A-(\tan 5A+\tan 3A)=tan\,8A-\tan 8A\,(1-tan\,5A\,tan\,3A)$$

$$=tan\,8A-tan\,8A+\tan 8A\,tan\,5A\,tan\,3A=tan\,8A\,tan\,5A\,tan\,3A$$

Hence $k=1$

18. Correct option: (C) $\sqrt{2}$

Consider

$$\cos 2x=1-2sin^2x$$

$$\cos 45°=1-2sin^2\left(\frac{45}{2}\right)$$

$$\frac{1}{\sqrt{2}}=1-2sin^2\left(\frac{45}{2}\right)$$

$$2sin^2\left(\frac{45}{2}\right)=1-\frac{1}{\sqrt{2}}$$

$$\sin\left(\frac{45}{2}\right)=\sqrt{1-\frac{1}{\sqrt{2}}}$$

Also

$$\cos 2x = 2cos^2x - 1$$

$$cos\,45° = 2cos^2\left(\frac{45}{2}\right) - 1$$

$$\cos\left(\frac{45}{2}\right) = \sqrt{1 + \frac{1}{\sqrt{2}}}$$

Now

$$\frac{\cos 22\frac{1}{2}° - \sin 22\frac{1}{2}°}{\cos 22\frac{1}{2}° + \sin 22\frac{1}{2}°} = \frac{\cos\frac{45}{2} - \sin\frac{45}{2}}{\cos\frac{45}{2} + \sin\frac{45}{2}}$$

$$= \frac{\sqrt{1 + \frac{1}{\sqrt{2}}} - \sqrt{1 - \frac{1}{\sqrt{2}}}}{\sqrt{1 + \frac{1}{\sqrt{2}}} + \sqrt{1 - \frac{1}{\sqrt{2}}}}$$

$$= \frac{\sqrt{1 + \frac{1}{\sqrt{2}}} - \sqrt{1 - \frac{1}{\sqrt{2}}}}{\sqrt{1 + \frac{1}{\sqrt{2}}} + \sqrt{1 - \frac{1}{\sqrt{2}}}} \times \frac{\sqrt{1 + \frac{1}{\sqrt{2}}} - \sqrt{1 - \frac{1}{\sqrt{2}}}}{\sqrt{1 + \frac{1}{\sqrt{2}}} - \sqrt{1 - \frac{1}{\sqrt{2}}}} = \frac{\left[\sqrt{1 + \frac{1}{\sqrt{2}}} - \sqrt{1 - \frac{1}{\sqrt{2}}}\right]^2}{1 + \frac{1}{\sqrt{2}} - 1 + \frac{1}{\sqrt{2}}}$$

$$= \frac{1 + \frac{1}{\sqrt{2}} + 1 - \frac{1}{\sqrt{2}} + 2\left(1 + \frac{1}{\sqrt{2}}\right)\left(1 - \frac{1}{\sqrt{2}}\right)}{\frac{2}{\sqrt{2}}} = \frac{2 + 2\left(1 - \frac{1}{2}\right)}{\frac{2}{\sqrt{2}}} = \frac{2\sqrt{2}}{2} = \sqrt{2}$$

19. Correct option: (B) $\alpha + \beta = \frac{\pi}{3}$

Given

$$\sin(120° - \alpha) = \sin(120 - \beta)$$

And $0 < \alpha;\ \beta < \pi$

$$120 - \alpha = 180 - (120 - \beta)$$

$$120 - \alpha = 60 + \beta$$

$$\alpha + \beta = 60°$$

$$\alpha + \beta = \frac{\pi}{3}$$

20. Correct option: (C) Acute angled

Given

In triangle,

$$\tan A + \tan B + \tan C = 6$$

And $\tan A \tan B = 2$

Now $A + B + C = 180°$

$$A + B = 180 - C$$

$$\tan(A + B) = -\tan C$$

$$\frac{\tan A + \tan B}{1 - \tan A \tan B} = -\tan C$$

$$tan\,A + tan\,B = tan\,A\,tan\,B \tan C - \tan C$$

$$tan\,A + tan\,B + \tan C = tan\,A\,tan\,B \tan C = 6$$

$$tan\,A\,tan\,B \tan C = 6$$

$$\tan C = \frac{6}{2} = 3$$

So, triangle is acute angled triangle

LEVEL – II

1. Correct option: (C) $\sqrt{3}$

$$\frac{\tan 225° - \cos 81° . \cot 69°}{\cot 261° . \tan 21°}$$

Now

$\tan 225° = 1$, $\cot 261° = \cot 81°$ and $\tan 21° = \cot 69°$

So

$$\frac{\tan 225° - \cos 81° . \cot 69°}{\cot 261° + \tan 21°} = \frac{1 - cot\, 81° . cot\, 69°}{cot\, 81° + cot\, 69°}$$
$$= -\left(\frac{cot\, 81° cot\, 69° - 1}{cot\, 81° + cot\, 69°}\right) = -\cot(81° + 69°)$$
$$= -\cot 150° = \cot 30° = \sqrt{3}$$

2. Correct option: (C) 2

Given

$$A + B = 135°$$
$$\tan(A + B) = \tan 135°$$
$$\frac{\tan A + \tan B}{1 - \tan A \tan B} = -1$$
$$tan\, A + tan\, B = tan\, A\, tan\, B - 1$$
$$tan\, A\, tan\, B - tan\, A - tan\, B = 1$$

Add 1 both sides

$$tan\, A\, tan\, B - tan\, A - tan\, B + 1 = 2$$
$$(1 - \tan B) - \tan A\,(1 - \tan B) = 2$$
$$(1 - \tan A)(1 - \tan B) = 2$$

3. Correct option: (B) $\frac{56}{33}$

Given

$0 < \alpha\beta < \frac{\pi}{4}$, $\cos(\alpha + \beta) = \frac{4}{5}$ and $\sin(\alpha - \beta) = \frac{5}{13}$

Consider

$$\cos(\alpha + \beta) = \frac{4}{5}$$
$$sin^2(\alpha + \beta) + cos^2(\alpha + \beta) = 1$$
$$\sin(\alpha + \beta) = \sqrt{1 - \left(\frac{4}{5}\right)^2} = \sqrt{\frac{25 - 16}{25}} = \frac{3}{5}$$
$$\sin(\alpha - \beta) = \frac{5}{13}$$

So

$$\cos(\alpha - \beta) = \frac{12}{13}$$

Now

$$2\alpha = \alpha + \beta + \alpha - \beta$$
$$\tan 2\alpha = \tan(\alpha + \beta + \alpha - \beta)$$
$$= \frac{\tan(\alpha + \beta) \tan(\alpha - \beta)}{1 - \tan(\alpha + \beta) \tan(\alpha - \beta)}$$

Also

$$\tan(\alpha + \beta) = \frac{\sin(\alpha + \beta)}{\cos(\alpha + \beta)} = \frac{\left(\frac{3}{5}\right)}{\left(\frac{4}{5}\right)} = \frac{3}{4}$$

And

$$\tan(\alpha-\beta)=\frac{\sin(\alpha-\beta)}{\cos(\alpha-\beta)}=\frac{\left(\frac{5}{13}\right)}{\left(\frac{12}{13}\right)}=\frac{5}{12}$$

Put the value of $\tan(\alpha+\beta)$ and $\tan(\alpha-\beta)$ in expression below

$$tan\,2\alpha=\frac{\tan(\alpha+\beta)+\tan(\alpha-\beta)}{1-\tan(\alpha+\beta)\tan(\alpha-\beta)}=\frac{\left(\frac{3}{4}\right)+\left(\frac{5}{12}\right)}{1-\left(\frac{3}{4}\right)\left(\frac{5}{12}\right)}=\frac{56}{33}$$

4. Correct option: (A) $\frac{1}{16}$

Given: $\sin x\cos y=\frac{1}{4}$

And $3\tan x=4\tan y$

$$3\frac{\sin x}{\cos x}=4\frac{\sin y}{\cos y}$$

$$3\sin x\cos y=4\sin y\cos x$$

$$sin\,y\,cos\,x=3\times\frac{1}{4}\times\frac{1}{4}$$

$$sin\,y\,cos\,x=\frac{3}{16}$$

Now $\sin(x-y)=\sin x\cos y-\sin y\cos x=\frac{1}{4}-\frac{3}{16}=\frac{4-3}{16}=\frac{1}{16}$

5. Correct option: (B) $\frac{\pi}{4}$

Given: $\sin A=\frac{1}{\sqrt{10}}$

$$A=\sin^{-1}\left(\frac{1}{\sqrt{10}}\right)$$

And $\sin B=\frac{1}{\sqrt{5}}$

$$B=\sin^{-1}\left(\frac{1}{\sqrt{5}}\right)$$

Now $A+B=\sin^{-1}\left(\frac{1}{\sqrt{10}}\right)+sin^{-1}\left(\frac{1}{\sqrt{5}}\right)$

By formula $\sin^{-1}A+\sin^{-1}B=\sin^{-1}\left(A\sqrt{1-B^2}+B\sqrt{1-A^2}\right)$

$$A+B=\sin^{-1}\left(\frac{1}{\sqrt{10}}\sqrt{1-\frac{1}{5}}+\frac{1}{\sqrt{5}}\sqrt{1-\frac{1}{10}}\right)$$

$$=\sin^{-1}\left(\frac{2}{\sqrt{50}}+\frac{3}{\sqrt{50}}\right)=\sin^{-1}\left(\frac{5}{\sqrt{50}}\right)=\sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$$

$$A+B=\frac{\pi}{4}$$

6. Correct option: (A) 1

Given $A+B+C=\frac{\pi}{2}$

$$A+B=\frac{\pi}{2}-C$$

$$\tan(A+B)=\tan\left(\frac{\pi}{2}-C\right)$$

$$\frac{\tan A+\tan B}{1-\tan A\tan B}=\cot C$$

$$\frac{tan\,A+tan\,B}{1-tan\,A\,tan\,B}=\frac{1}{\tan C}$$

$$tan\,A\,tan\,C+tan\,B\,tan\,C=1-tan\,A\,tan\,B$$

$$=tan\,A\,tan\,C+tan\,B\,tan\,C+tan\,A\,tan\,B=1$$

7. Correct option: (A) $\frac{\sqrt{3}}{4}$

$$cos^2 45° - sin^2 15$$

Now $sin^2\theta = \frac{1-\cos 2\theta}{2}$

And $\cos 45° = \frac{1}{\sqrt{2}}$

Then $cos^2 45° - sin^2 15 = \left(\frac{1}{\sqrt{2}}\right)^2 - \frac{1-\cos 30°}{2} = \frac{1}{2} - \frac{1-\frac{\sqrt{3}}{2}}{2} = \frac{\sqrt{3}}{4}$

8. Correct option: (B) -1

Given $\tan 4x + \tan 5x - \tan 9x = k \tan 4x \tan 5x \tan 9x$

Now $$\tan 9x = \tan(4x + 5x)$$

$$= \frac{\tan 4x + \tan 5x}{1 - \tan 4x \tan 5x}$$

$$\tan 9x - \tan 4x \tan 5x \tan 9x = \tan 4x + \tan 5x$$

$$tan\, 4x + tan\, 5x - tan\, 9x = -tan\, 4x\, tan\, 5x\, tan\, 9x$$

$$k = -1$$

9. Correct option: (B) $\pm\frac{\sin\theta - \cos\theta}{\sqrt{2}}$

Given $$\tan\alpha = \frac{\sin\theta - \cos\theta}{\sin\theta + \cos\theta}$$

$$\sin\alpha = \frac{\sin\theta - \cos\theta}{\sqrt{sin^2\theta + cos^2\theta + sin^2\theta + sin^2\theta}}$$

$$= \pm\frac{sin\,\theta - cos\,\theta}{\sqrt{2}}$$

10. Correct option: (D) $\frac{\sqrt{3}}{2}$

$$\sin 20° . \cos 40° + \cos 20° . \sin 40°$$

$$= \sin(20° + 40°) = \sin 60° = \frac{\sqrt{3}}{2}$$

11. Correct option: (B) $-\frac{1}{2}$

Given $\cos x + \cos y + \cos z = 0 = \sin x + \sin y + \sin z$

Now $$\cos x + \cos y = -\cos z$$

Squaring both sides $cos^2 x + 2\cos x \cos y + cos^2 y = cos^2 z \ldots$ (1)

Consider

$$\sin x + \sin y + \sin z = 0$$

$$\sin x + \sin y = -\sin z$$

Squaring both sides

$$sin^2 x + 2\sin x \sin y + sin^2 y = sin^2 z \ldots (2)$$

Adding equation (1) and (2)

$$cos^2 x + 2\cos x \cos y + cos^2 y + sin^2 x + 2\, sin\, x\, sin\, y + sin^2 y = cos^2 z + sin^2 z$$

$$2 + 2[\cos(x - y)] = 1$$

$$\cos(x - y) = -\frac{1}{2}$$

13. Correct option: (B) -1

Given $$\tan A = \frac{1}{35} \text{ and } \tan B = \frac{18}{17}$$

Now $$\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

$$= \frac{\left(\frac{1}{35}\right) - \left(\frac{18}{17}\right)}{1 + \left(\frac{1}{35}\right)\left(\frac{18}{17}\right)} = \frac{17 - 630}{595 + 18} = \frac{-613}{613} = -1$$

15. Correct options: (B) and (C)

$$\tan(A+B+C) = \tan\pi$$
$$\frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A} = 0$$
$$tan\,A + tan\,B + tan\,C - tan\,A\,tan\,B\,tan\,C = 0$$
$$tan\,A + tan\,B + tan\,C - tan\,A\,tan\,B\,tan\,C = \sin 0°$$

16. Correct option: (D) 2

Given

$$A + B = \frac{\pi}{4}$$
$$\tan(A+B) = \tan\frac{\pi}{4}$$
$$\frac{\tan A + \tan B}{1 - \tan A \tan B} = 1$$
$$tan\,A + tan\,B = 1 - tan\,A\,tan\,B$$
$$1 + tan\,A + tan\,B + tan\,A\,tan\,B = 2$$
$$(1 + \tan A)(1 + \tan B) = 2$$

17. Correct option: (B) $\tan B - tan\,A - tan\,C$

Given

$$A + C = B$$
$$\tan(A+C) = \tan B$$
$$\frac{tan\,A + tan\,C}{1 - tan\,A\,tan\,C} = \tan B$$
$$tan\,A + tan\,C = \tan B - tan\,A \tan B\, tan\,C$$
$$tan\,A \tan B\, tan\,C = \tan B - tan\,A - tan\,C$$

21. Correct Matching is: $a \to P; b \to Q; c \to R; d \to R$

a) $\cos 20° + \cos 140° + \cos 100°$

$$= (\cos 20° + \cos 140°) + \cos(90° + 10°)$$
$$= 2\cos 80° \cos 60° - \sin 10° = \cos 80° - \sin 10°$$
$$= \cos(90° - 10°) - \sin 10°$$
$$= \sin 10° - \sin 10° = 0$$

b) $(1 + \tan 20°)(1 + \tan 25°)$

$$= 2 + \tan 20° + \tan 25° + \tan 20° \tan 25°$$
$$= 2 + tan\,20° + \tan 25° - \frac{\tan 20° + \tan 25°}{\tan 45°}$$
$$= 2 + tan\,20° + tan\,25° - tan\,20° - tan\,25° = 2$$

c) $\tan 20° \tan 32° + \tan 32° \tan 38° + \tan 38° \tan 20°$

Let A = 20, B = 32, C = 38

$$A + B + C = 20 + 32 + 38 = 90$$
$$C = 90 - (A + B)$$
$$\sin C = \sin[90 - (A+B)]$$
$$\sin C = \cos(A+B)$$

And

$$\cos C = \cos[90 - (A+B)] = \sin(A+B)$$

Now

$$\tan C = \frac{\sin C}{\cos C}$$
$$= \frac{\cos(A+B)}{\sin(A+B)} = \cot(A+B)$$
$$\tan 20° \tan 32° + \tan 32° \tan 38° + \tan 38° \tan 20°$$
$$= \tan 20° \tan 32° + [\tan 32° + \tan 20°] \times \cot(20° + 32°)$$

$$= tan\,20°\,tan\,32° + [tan\,32° + tan\,20°] \times \left[\frac{1 - \tan 20° \tan 32°}{\tan 20° + \tan 32°}\right]$$
$$= tan\,20°\,tan\,32° + 1 - tan\,20°\,tan\,32°$$
$$= 1$$

d) $sin^2 10° + sin^2 110° + sin^2 130$

$$\sin 110° = sin(90° + 20°)$$
$$= \cos 20°$$
$$sin^2 110° = cos^2 20°$$

Now

$$sin^2 10° + sin^2 110° = sin^2 10° + cos^2 20° = 1$$

23. Correct Answer is: zero

$$\sum \frac{\sin(A+B)\sin(A-B)}{cos^2 A cos^2 B}$$
$$= \sum \frac{(\sin A \cos B + \cos A \sin B)(\sin A \cos B - \cos A \sin B)}{cos^2 A cos^2 B}$$
$$= \sum \frac{sin^2 A cos^2 B - cos^2 A sin^2 B}{cos^2 A cos^2 B} = \sum \frac{sin^2 A cos^2 B}{cos^2 A cos^2 B} - \sum \frac{cos^2 A cos^2 B}{cos^2 A cos^2 B}$$
$$= \sum \frac{sin^2 A}{cos^2 A} - \sum \frac{sin^2 B}{sin^2 B} = tan^2 A - tan^2 B = 0$$

24. Correct Answer is: 2

$$\frac{\tan 80° - \tan 10°}{\tan 70°} = \frac{\left(\frac{\sin 80°}{\cos 80°}\right) - \left(\frac{\sin 10°}{\cos 10°}\right)}{\left(\frac{\sin 70°}{\cos 70°}\right)}$$
$$= \frac{\frac{\sin 80° \cos 10° - \cos 80° \sin 10°}{\cos 80° \cos 10°}}{\frac{sin\,70°}{cos\,70°}}$$
$$= \frac{\sin(80° - 10°).\cos 70°}{\cos 80° \cos 10° \sin 70°} = \frac{\sin 70.cos\,70°}{cos\,80°\,cos\,10°\,sin\,70°}$$
$$= \frac{\cos 70°}{\cos 80° \cos 10°} = \frac{cos(80° - 10°)}{cos\,80°\,cos\,10°}$$
$$= \frac{\cos 80° \cos 10° + \sin 80° \sin 10°}{cos\,80°\,cos\,10°} = 1 + \tan 80° \tan 10°$$
$$= 1 + \tan(90° - 10°)\tan 10° = 1 + \cot 10° \tan 10° = 1 + 1 = 2$$

29. Correct Answer is: 1

$$\cot\left(\frac{\pi}{4} + \theta\right)\cot\left(\frac{\pi}{4} - \theta\right)$$
$$= \frac{1}{\left[tan\left(\frac{\pi}{4} + \theta\right)\right]} \times \frac{1}{\left[tan\left(\frac{\pi}{4} - \theta\right)\right]}$$
$$= \frac{1 + \tan\theta}{1 - \tan\theta} \times \frac{1 - tan\,\theta}{1 + tan\,\theta} = 1$$

30. Correct Answer is: zero

$$\cos\theta + \cos(240 + \theta) + \cos(240 - \theta)$$

Using the identity

$$\cos(A+B) + \cos(A-B) = 2\cos A \cos B$$
$$\cos\theta + \cos(240 + \theta) + \cos(240 - \theta) = \cos\theta + 2\cos 240° \cos\theta$$
$$= \cos\theta\,[1 + 2\cos 240°]$$
$$= \cos\theta\,[1 + 2\cos(270 - 30°)] = \cos\theta\,[1 + 2(-\sin 30°)]$$
$$= \cos\theta\,[1 - 1] = 0$$

STUDENT PRACTICE QUESTIONS

1. Correct option: (C) $\frac{p^2-q^2}{p^2+q^2}$

Given: $\tan\theta = \frac{p}{q}$

Value of

$$\frac{p\sin\theta - q\cos\theta}{p\sin\theta + q\cos\theta}$$

Divide by $\cos\theta$

$$\frac{p\sin\theta - q\cos\theta}{p\sin\theta + q\cos\theta} = \frac{p\tan\theta - q}{p\tan\theta + q}$$

$$= \frac{p\left(\frac{p}{q}\right) - q}{p\left(\frac{p}{q}\right) + q}$$

$$= \frac{\left(\frac{p^2 - q^2}{q}\right)}{\left(\frac{p^2 + q^2}{q}\right)} = \frac{p^2 - q^2}{p^2 + q^2}$$

2. Correct option: (A) $\sqrt{3}\sin\theta$

Given

$$\sin\theta + \cos\theta = \sqrt{3}\cos\theta$$
$$\sin\theta = \sqrt{3}\cos\theta - \cos\theta$$
$$\sin\theta = \cos\theta\left(\sqrt{3} - 1\right)$$
$$\frac{\sin\theta}{\cos\theta} = \left(\sqrt{3} - 1\right)$$
$$\frac{sin\,\theta}{cos\,\theta} = \left(\sqrt{3} - 1\right)\frac{\left(\sqrt{3} + 1\right)}{\left(\sqrt{3} + 1\right)} = \frac{3 - 1}{\sqrt{3} + 1}$$
$$\frac{sin\,\theta}{cos\,\theta} = \frac{2}{\sqrt{3} + 1}$$
$$2\cos\theta = \left(\sqrt{3} + 1\right)\sin\theta$$
$$2\cos\theta = \sqrt{3}\sin\theta + \sin\theta$$
$$2\cos\theta - \sin\theta = \sqrt{3}\sin\theta$$

3. Correct option: (B) $\frac{119}{169}$

Given B, (A+B) are acute angles and

$$\sin(A + B) = \frac{12}{13}$$
$$\sin B = \frac{5}{13}$$

Now

$$sin^2B + cos^2B = 1$$
$$\left(\frac{5}{13}\right)^2 + cos^2B = 1$$
$$\cos B = \sqrt{1 - \frac{25}{169}}$$
$$\cos B = \sqrt{\frac{169 - 25}{169}} = \sqrt{\frac{144}{169}} = \frac{12}{13}$$
$$\sin(A + B) = \sin A\cos B + \cos A\sin B$$

$$\frac{12}{13} = \sin A\left(\frac{12}{13}\right) + \cos A\left(\frac{5}{13}\right)$$
$$12 = 12\sin A + 5\cos A$$
$$12(1 - \sin A) = 5\cos A$$

Squaring both sides

$$[12(1 - \sin A)]^2 = (5\cos A)^2$$
$$144(1 - \sin A)^2 = 25cos^2A$$
$$144(1 - \sin A)^2 = 25(1 - sin^2A)$$
$$144(1 - \sin A) = 25(1 + \sin A)$$
$$144 - 144\sin A = 25 + 25\sin A$$
$$144 - 25 = 144\sin A + 25\sin A$$
$$119 = 169\sin A$$
$$\sin A = \frac{119}{169}$$

4. Correct option: (D) $-\frac{1}{2}$

Given

$$\tan 69° + \tan 66° - \tan 69°\tan 66° = 2k$$

Now, we know that

$$\tan A + \tan B = \frac{\tan A + \tan B}{1 - \tan A\tan B}$$
$$\tan 69 + \tan 66 = \frac{\tan 69 + \tan 66}{1 - \tan 69\tan 66}$$
$$\tan 135 = \frac{\tan 69 + \tan 66}{1 - \tan 69\tan 66}$$
$$(-1) = \frac{tan\,69 + tan\,66}{1 - tan\,69\,tan\,66}$$
$$tan\,69\,tan\,66 - 1 = tan\,69 + tan\,66$$
$$tan\,69 + tan\,66 - tan\,69\,tan\,66 = -1$$
$$2k = -1$$
$$k = -\frac{1}{2}$$

5. Correct option: (B) $\frac{5}{3}$

Given

$$\sec A + \tan A = 3$$
$$\frac{1}{\cos A} + \frac{\sin A}{\cos A} = 3$$
$$\frac{1 + \sin A}{\cos A} = 3$$
$$(1 + \sin A)^2 = (3\cos A)^2$$
$$1 + sin^2A + 2\sin A = 9cos^2A$$
$$1 + sin^2A + 2\sin A = 9(1 - sin^2A)$$
$$1 + sin^2A + 2\sin A = 9 - 9sin^2A$$
$$10sin^2A + 2\sin A - 8 = 0$$
$$5sin^2A + sin\,A - 4 = 0$$
$$5sin^2A + 5\sin A - 4\sin A - 4 = 0$$
$$5\sin A\,(sinA + 1) - 4(sinA + 1) = 0$$
$$(\sin A + 1)(5\sin A - 4) = 0$$
$$\sin A = -1, \sin A = \frac{4}{5}$$

Now

$$\frac{1+\sin A}{\cos A} = 3$$

$$\cos A = \frac{1+\sin A}{3}$$

$$= \frac{1+\left(\frac{4}{5}\right)}{3} = \frac{5+4}{3\times 5} = \frac{9}{15}$$

$$\cos A = \frac{3}{5}$$

$$\sec A = \frac{1}{\cos A} = \frac{1}{\left(\frac{3}{5}\right)} = \frac{5}{3}$$

6. Correct option: (A) $\sin(A+B)$

$$\cos(45°-A)\cos(45°-B) - \sin(45°-A)\sin(45°-B)$$

Let $(45°-A) = x$ and $(45°-B) = y$

$$\cos(45°-A)\cos(45°-B) - \sin(45°-A)\sin(45°-B)$$

$$= cos\, x\, cos\, y - sin\, x\, sin\, y$$

$$= \cos(x+y)$$

$$= \cos[(45°-A)+(45°-B)]$$

$$= \cos(90-A-B) = \sin(A+B)$$

7. Correct option: (B) $\cos A$

$$\sin(n+1)A\sin(n+2)A + \cos(n+1)A\cos(n+2)A$$

$$= cos(n+2)A\, cos(n+1)A + sin(n+2)A\, sin(n+1)A$$

$$= \cos[(n+2)A-(n+1)A]$$

$$= \cos(nA+2A-nA-A) = \cos A$$

8. Correct option: (B) $\frac{1}{2}$

$$\cos 28°\cos 32° - \sin 28°\sin 32°$$

Using formula

$\cos(A+B) = cos\, A\, cos\, B - sin\, A\, sin\, B$

So

$$\cos 28°\cos 32° - \sin 28°\sin 32°$$

$$= \cos(28°+32°) = \cos 60° = \frac{1}{2}$$

9. Correct option: (B) $2\sqrt{3}$

$$\tan 75° - \cot 75°$$

Now

$$\tan 75° = \tan(45°+30°)$$

$$= \frac{\tan 45° + \tan 30°}{1-\tan 45°\tan 30°}$$

$$= \frac{1+\frac{1}{\sqrt{3}}}{1-\frac{1}{\sqrt{3}}} = \frac{\sqrt{3}+1}{\sqrt{3}-1}$$

And

$$\cot 75° = \frac{1}{\tan 75°}$$

$$= \frac{1}{\left(\frac{\sqrt{3}+1}{\sqrt{3}-1}\right)} = \frac{\sqrt{3}-1}{\sqrt{3}+1}$$

So,

$$\tan 75° - \cot 75° = \frac{\sqrt{3}+1}{\sqrt{3}-1} - \frac{\sqrt{3}-1}{\sqrt{3}+1}$$
$$= \frac{\left(\sqrt{3}+1\right)^2 - \left(\sqrt{3}-1\right)^2}{\left(\sqrt{3}-1\right)\left(\sqrt{3}+1\right)}$$
$$= \frac{3+1+2\sqrt{3}-\left(3+1-2\sqrt{3}\right)}{3-1} = \frac{4\sqrt{3}}{2} = 2\sqrt{3}$$

10. Correct option: (A) $\frac{\sqrt{3}+1}{4\sqrt{2}}$

$$sin^2 52\frac{1}{2}° - sin^2 22\frac{1}{2}$$
$$= \left(\sin 52\frac{1}{2} + \sin 22\frac{1}{2}\right)\left(\sin 52\frac{1}{2} - \sin 22\frac{1}{2}\right)$$
$$= \left(\sin\frac{105}{2} + \sin\frac{45}{2}\right)\left(\sin\frac{105}{2} - \sin\frac{45}{2}\right)$$
$$= \sin 75° \sin 30° = \frac{1}{2}\sin(45° + 30°)$$
$$\sin(C + D) = \sin C \cos D + \cos C \sin D$$
$$sin^2 52\frac{1}{2}° - sin^2 22\frac{1}{2}$$
$$= \frac{1}{2}[sin\, 45°\, cos\, 30° + cos\, 45°\, sin\, 30°]$$
$$= \frac{1}{2}\left[\frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \times \frac{1}{2}\right] = \frac{1}{2}\left[\frac{\sqrt{3}+1}{2\sqrt{2}}\right] = \frac{\sqrt{3}+1}{4\sqrt{2}}$$

11. Correct option: (B)

$$\tan 100° + \tan 125° + \tan 100° \tan 125°$$

Now

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$
$$\tan(100° + 125°) = \frac{tan\, 100° + tan\, 125°}{1 - tan\, 100°\, tan\, 125°}$$
$$\tan 225° = \frac{tan\, 100° + tan\, 125°}{1 - tan\, 100°\, tan\, 125°}$$
$$1 = \frac{tan\, 100° + tan\, 125°}{1 - tan\, 100°\, tan\, 125°}$$
$$1 - tan\, 100°\, tan\, 125° = tan\, 100° + tan\, 125°$$
$$tan\, 100° + tan\, 125° + tan\, 100°\, tan\, 125° = 1$$

12. Correct option: (A)

$$\cos 35° + \cos 85° + \cos 155°$$
$$= 2\cos\left(\frac{35° + 85°}{2}\right)\cos\left(\frac{35° - 85°}{2}\right) + \cos 155°$$
$$= 2\cos 60° \cos(-25°) + \cos 155°$$
$$= 2 \times \frac{1}{2}\cos 25° + \cos 155° = \cos 25° + \cos 155°$$
$$= 2\cos\left(\frac{25° + 155°}{2}\right)\cos\left(\frac{25° - 155°}{2}\right)$$
$$= 2\cos 90° \cos 65° = 0$$

13. Correct option: (A)

$$\cos 42° + \cos 78° + \cos 162°$$
$$= cos\, 42° + cos\, 78° + cos(180° - 18°)$$
$$= cos\, 42° + cos\, 78° - cos\, 18°$$

$$= 2\,cos\left(\frac{42° + 78°}{2}\right) cos\left(\frac{78° - 42°}{2}\right) - \cos 18°$$
$$= 2\cos 60° \cos 18 - \cos 18°$$
$$= 2 \times \frac{1}{2}\cos 18 - \cos 18$$
$$= \cos 18 - \cos 18 = 0$$

14. Correct option: (B) $\frac{171}{221}$

Given: $270° < A < 360°$, $90° < B < 180°$, $\cos A = \frac{5}{13}$, $\tan B = \frac{-15}{8}$

Now

$$\cos A = \frac{5}{13}$$
$$sin^2A + cos^2A = 1$$
$$sin^2A = 1 - \left(\frac{5}{13}\right)^2$$
$$\sin A = \sqrt{1 - \frac{25}{169}} = \frac{12}{13}$$

And

$$\tan B = -\frac{15}{8}$$

Perpendicular, p = 15
Base, b = 8
Hypotenuses = h

$$h^2 = b^2 + p^2$$
$$h = \sqrt{(15)^2 + (8)^2} = \sqrt{225 + 64} = \sqrt{289}$$
$$h = 17$$

So

$$\sin B = \frac{15}{17}$$
$$\cos B = \frac{8}{17}$$

Then

$$\sin(A + B) = \sin A \cos B + \sin B \cos A$$
$$= \frac{12}{13} \times \frac{8}{17} + \frac{15}{17} \times \frac{5}{13} = \frac{96}{221} + \frac{75}{221} = \frac{171}{221}$$

15. Correct option: (A) 1

$$\tan 25° \tan 31° + \tan 31° \tan 34° + \tan 34° \tan 25°$$

Consider $a = 25,\ b = 31,\ c = 34$

$$a + b + c = 25 + 31 + 34 = 90$$

Now

$$\tan(a + b + c) = \frac{\tan a + \tan b + \tan c - \tan a \tan b \tan c}{1 - (\tan a \tan b + \tan b \tan c + \tan c \tan a)}$$

$$\tan 90° = \frac{tan\,25° + tan\,31° + tan\,34° - tan\,25°\,tan\,31°\,tan\,34°}{1 - (tan\,25°\,tan\,31° + tan\,31°\,tan\,34° + tan\,34°\,tan\,25°)}$$

$$\frac{1}{0} = \frac{tan\,25° + tan\,31° + tan\,34° - tan\,25°\,tan\,31°\,tan\,34°}{1 - (tan\,25°\,tan\,31° + tan\,31°\,tan\,34° + tan\,34°\,tan\,25°)}$$

As $\tan 90° = not\ defined$

$$1 - (tan\,25°\,tan\,31° + tan\,31°\,tan\,34° + tan\,34°\,tan\,25° = 0$$

$$tan\,25°\,tan\,31° + tan\,31°\,tan\,34° + tan\,34°\,tan\,25° = 1$$

16. Correct option: (C) $\sqrt{3}$

$$\frac{\cos 15° + \sin 15°}{\cos 15° - \sin 15°} = \frac{1 + \frac{\sin 15°}{\cos 15°}}{1 - \frac{\sin 15°}{\cos 15°}} = \frac{1 + \tan 15°}{1 - \tan 15°} = \frac{\tan 45° + \tan 15°}{1 - \tan 45° \tan 15°}$$

As $\tan 45° = 1$

$$\frac{\cos 15° + \sin 15°}{\cos 15° - \sin 15°} = \tan(45° + 15°) = \tan 60° = \sqrt{3}$$

17. Correct option: (A) $\cot(A - B)$

Given $x = \tan A - \tan B$, $y = \cot B - \cot A$

Now

$$x = \tan A - \tan B = \frac{\sin A}{\cos A} - \frac{\sin B}{\cos B}$$
$$= \frac{\sin A \cos B - \sin B \cos B}{\cos A \cos B} = \frac{\sin(A - B)}{\cos A \cos B}$$

Similarly

$$y = \cot B - \cot A$$
$$y = \frac{\sin(A - B)}{\sin A \sin B}$$

Then $\frac{1}{x} + \frac{1}{y}$

$$= \frac{1}{\left[\frac{sin(A-B)}{cos\, A \cos B}\right]} + \frac{1}{\left[\frac{sin(A-B)}{sin\, A \sin B}\right]}$$
$$= \frac{\cos A \cos B}{\sin(A - B)} + \frac{\sin A \sin B}{\sin(A - B)}$$
$$= \frac{cos\, A \cos B + sin\, A \sin B}{sin(A - B)} = \frac{cos(A - B)}{sin(A - B)} = \cot(A - B)$$

18. Correct option: (C) $a^2 + b^2$

Given $\cos(\theta - \alpha) = a$ and $\sin(\theta - \beta) = b$

$$\alpha - \beta = (\theta - \beta) - (\theta - \alpha)$$
$$\cos(\alpha - \beta) = \cos[(\theta - \beta) - (\theta - \alpha)] = \cos(\theta - \beta)\cos(\theta - \alpha) + \sin(\theta - \beta)\sin(\theta - \alpha)$$

And

$$\sin(\alpha - \beta) = sin(\theta - \beta)\, cos(\theta - \alpha) + \cos(\theta - \beta)\, sin(\theta - \alpha)$$

So

$$\cos(\alpha - \beta) = b.a + \sqrt{1 - a^2}\sqrt{1 - b^2}$$

and

$$\sin(\alpha - \beta) = a\sqrt{1 - b^2} - b\sqrt{1 - a^2}$$

Now

$$sin^2(\alpha - \beta) = \left[a\sqrt{1 - b^2} - b\sqrt{1 - a^2}\right]^2$$
$$= a^2(1 - b^2) + b^2(1 - a^2) - 2ab[\cos(\alpha - \beta) - ab]$$
$$sin^2(\alpha - \beta) + 2ab[cos(\alpha - \beta)] = a^2 - a^2b^2 + b^2 - b^2a^2 + 2a^2b^2 = a^2 + b^2$$

19. Correct option: (A)

Given: $8\tan A = -15$ and $25\sin B = -7$ and A and B are not in 4th quadrant

So, A must be in 2nd quadrant and B must be in 3rd quadrant

Now

$$\tan A = -\frac{15}{8}$$

Perpendicular, p = 15

Base, b = 8

Hypotenuse = h

$$h^2 = p^2 + b^2$$

$$h = \sqrt{(15)^2 + (8)^2}$$

$$= \sqrt{225 + 64} = \sqrt{289} = 17$$

So, $\sin A = \frac{15}{17}$ as sine is positive in 2nd quadrant and $\cos A = \frac{-8}{17}$ as cos is negative in 2nd quadrant

Also

$$sin\, B = \frac{-7}{25}$$

So

$$\cos B = \frac{-24}{25}$$

Now

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$= \frac{15}{17} \times \left(\frac{-24}{25}\right) + \left(\frac{-8}{17}\right) \times \left(\frac{17}{25}\right) = \frac{304}{425}$$

20. Correct option: (A) $\frac{c^2 - a^2}{(a^2 + b^2)}$

$$(a^2 + b^2)sin^2\theta - 2bc \sin\theta = c^2 - a^2$$

Product of equation

$$\sin\alpha \sin\beta = \frac{c^2 - a^2}{(a^2 + b^2)}$$

21. Correct option: (D) $\frac{c^2 - b^2}{a^2 + b^2}$

$$a\cos\theta + b\sin\theta = c$$

$$a\cos\theta - c = -b\sin\theta$$

Squaring both sides

$$a^2cos^2\theta + c^2 - 2ac\cos\theta = b^2sin^2\theta$$

$$a^2cos^2\theta + c^2 - 2ac\, cos\,\theta = b^2(1 - cos^2\theta)$$

$$a^2cos^2\theta + c^2 - 2ac\, cos\,\theta = b^2 - b^2cos^2\theta$$

$$(a^2 + b^2)cos^2\theta - 2ac\, sin\,\theta + c^2 - b^2 = 0$$

Product of roots $\cos\alpha \cos\beta = \frac{c^2 - b^2}{a^2 + b^2}$

22. Correct option: (C) $\frac{a^2 - b^2}{a^2 + b^2}$

$$\cos(\alpha + \beta) = \cos\alpha \cos\beta - \sin\alpha \sin\beta$$

$$= \frac{c^2 - b^2}{a^2 + b^2} - \frac{c^2 - a^2}{(a^2 + b^2)}$$

$$= \frac{a^2 - b^2}{a^2 + b^2}$$

23. Correct Matching is: $\boldsymbol{a \to Q; b \to R; c \to S; d \to R}$

a) given

$$A + B = \frac{\pi}{2}$$

$$\tan(A + B) = \tan\frac{\pi}{2}$$

$$\frac{\tan A + \tan B}{1 - \tan A \tan B} = \frac{1}{0}$$

$$1 - tan\, A\, tan\, B = 0 \Rightarrow tan\, A\, tan\, B = 1$$

$$\tan B + 2\tan(A - B) = \tan B + 2\left[\frac{\tan A - \tan B}{1 + \tan A \tan B}\right]$$

$$= \tan B + 2\left[\frac{tan\,A - tan\,B}{1+1}\right] = tan\,B + \tan A - \tan B = tan\,A$$

b) given

$$A + B = \frac{\pi}{4}$$
$$\tan(A+B) = \tan\frac{\pi}{4}$$
$$\frac{\tan A + \tan B}{1 - \tan A \tan B} = 1$$
$$tan\,A + tan\,B = 1 - tan\,A\,tan\,B$$
$$1 + tan\,A + tan\,B + tan\,A\,tan\,B = 2$$
$$(1+\tan A)(1+\tan B) = 2$$

c) Given

$$A + B = \frac{5\pi}{4}$$
$$\cot(A+B) = \cot\frac{5\pi}{4}$$
$$\frac{\cot A \cot B - 1}{\cot A + \cot B} = \cot\left(\pi + \frac{\pi}{4}\right)$$
$$\frac{cot\,A\,cot\,B - 1}{cot\,A + cot\,B} = \cot\frac{\pi}{4}$$
$$\frac{cot\,A\,cot\,B - 1}{cot\,A + cot\,B} = 1$$
$$cot\,A\,cot\,B - 1 = cot\,A + cot\,B$$

Adding $(cot\,A\,cot\,B + 1)$ both sides

$$cot\,A\,cot\,B - 1 + cot\,A\,cot\,B + 1 = cot\,A + cot\,B + cot\,A\,cot\,B + 1$$
$$2\,cot\,A\,cot\,B = \cot B\,(1+\cot A) + 1(1+\cot A)$$
$$2\,cot\,A\,cot\,B = (1+cot\,A)(1+cot\,B)$$
$$\frac{cot\,A\,cot\,B}{(1+cot\,A)(1+cot\,B)} = \frac{1}{2}$$

d) Given

$$A - B = \frac{3\pi}{4}$$
$$\tan(A-B) = \tan\frac{3\pi}{4}$$
$$\frac{\tan A - \tan B}{1 + \tan A \tan B} = -1$$
$$tan\,A + tan\,B = -\,tan\,A\,tan\,B - 1$$
$$1 - tan\,A - tan\,B + tan\,A\,tan\,B = 2$$
$$(1-\tan A)(1+\tan B) = 2$$

24. Correct Answer is: 3

Given $\cot\alpha, \cot\beta, \cot\gamma$ are in A.P.

$$2\cot\beta = \cot\alpha + \cot\gamma$$
$$= \frac{1}{\tan\alpha} + \frac{1}{\tan\gamma} = \frac{\tan\alpha + \tan\gamma}{\tan\alpha\tan\gamma} = \frac{\tan(\alpha+\gamma)(1 - tan\,\alpha\,tan\,\gamma)}{tan\,\alpha\,tan\,\gamma}$$
$$2\cot\beta = \cot\beta\left[\frac{1 - tan\,\alpha\,tan\,\gamma}{tan\,\alpha\,tan\,\gamma}\right]$$
$$2 = \frac{1 - tan\,\alpha\,tan\,\gamma}{tan\,\alpha\,tan\,\gamma}$$

Let $tan\,\alpha\,tan\,\gamma = k$

$$2 = \frac{1-k}{k}$$
$$2k = 1-k$$
$$k = \frac{1}{3}$$

So, $tan\,\alpha\,tan\,\gamma = k = \frac{1}{3}$

$$\cot\alpha\cot\beta = 3$$

25. Correct Answer is: 2

Given

$$A+B+C = \frac{\pi}{2}$$
$$A+B = \frac{\pi}{2} - C$$
$$\tan(A+B) = \tan\left(\frac{\pi}{2} - C\right)$$
$$\frac{\tan A + \tan B}{1-\tan A\tan B} = \cot C$$
$$\frac{tan\,A + tan\,B}{1-tan\,A\,tan\,B} = \frac{1}{\tan C}$$
$$\tan A\tan C + \tan B\tan C = 1-\tan A\tan B$$
$$tan\,A\,tan\,B + tan\,A\,tan\,C + tan\,B\,tan\,C = 1$$

Now

$$\sum \frac{\cos(A+B)}{\cos A\cos B} = \frac{\cos(A+B)}{\cos A\cos B} + \frac{\cos(B+C)}{\cos B\cos C} + \frac{\cos(C+A)}{\cos C\cos A}$$
$$= \frac{\cos A\cos B - \sin A\sin B}{\cos A\cos B} + \frac{cos\,B\,cos\,C - sin\,B\,sin\,C}{cos\,B\,cos\,C} + \frac{cos\,C\,cos\,A - sin\,C\,sin\,A}{cos\,C\,cos\,A}$$
$$= 1 - \frac{\sin A\sin B}{\cos A\cos B} + 1 - \frac{sin\,B\,sin\,C}{cos\,B\,cos\,C} + 1 - \frac{sin\,C\,sin\,A}{cos\,C\,cos\,A}$$
$$= 3 - \tan A\tan B - \tan B\tan C - \tan C\tan A$$
$$= 3 - (tan\,A\,tan\,B + tan\,B\,tan\,C + tan\,C\,tan\,A) = 3 - 1 = 2$$

26. Correct Answer is: 3

$cot\alpha, cot\beta, cot\gamma\ are\ in\ A.P.$

$$2cot\beta = cot\alpha + cot\gamma = \frac{tan\alpha + tan\gamma}{tan\alpha tan\gamma}$$
$$2cot\beta = \frac{tan(\alpha+\gamma)(1 - tan\alpha tan\gamma)}{tan\alpha tan\gamma}$$
$$2cot\beta = cot\beta\,\frac{(1 - tan\alpha tan\gamma)}{tan\alpha tan\gamma}$$

$Let\ tan\alpha tan\gamma = k$

$$2 = \frac{1-k}{k}$$
$$3k = 1$$
$$tan\alpha tan\gamma = k = \frac{1}{3}$$

$cot\alpha cot\gamma = 3.$

27. Correct Answer is: $\tan\frac{\pi}{4}$

Given

$$\sin A = \frac{1}{\sqrt{10}}$$
$$A = \sin^{-1}\left(\frac{1}{\sqrt{10}}\right)$$

And

$$\sin B = \frac{1}{\sqrt{5}}$$
$$B = \sin^{-1}\left(\frac{1}{\sqrt{5}}\right)$$

Now

$$A + B = \sin^{-1}\left(\frac{1}{\sqrt{10}}\right) + sin^{-1}\left(\frac{1}{\sqrt{5}}\right)$$

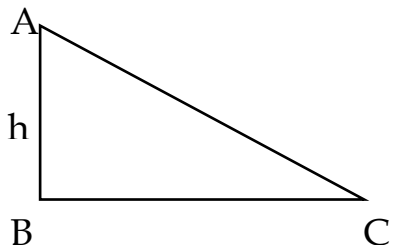
By formula

$$\sin^{-1} A + \sin^{-1} B = \sin^{-1}\left(A\sqrt{1-B^2} + B\sqrt{1-A^2}\right)$$
$$A + B = \sin^{-1}\left(\frac{1}{\sqrt{10}}\sqrt{1-\frac{1}{5}} + \frac{1}{\sqrt{5}}\sqrt{1-\frac{1}{10}}\right)$$
$$= \sin^{-1}\left(\frac{2}{\sqrt{50}} + \frac{3}{\sqrt{50}}\right)$$
$$= \sin^{-1}\left(\frac{5}{\sqrt{50}}\right) = \sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$$
$$A + B = \frac{\pi}{4}$$
$$\tan(A + B) = \tan\frac{\pi}{4}$$

Applications of Trigonometry

LEVEL – I

1. Correct option: (C) 45°



AB represents the vertical pole
BC represents the shadow on the ground

$$\tan\theta = \frac{h}{x}$$

Given that length of shadow = height of tower h = x
So

$$\tan\theta = \frac{h}{h} = 1$$
$$\tan\theta = \tan 45^\circ$$
$$\theta = 45^\circ$$

2. Correct option: (B) 30°

Let height of tower be h
So, height of shadow = $\sqrt{3}$h

$$\tan\theta = \frac{h}{\sqrt{3}h} = \frac{1}{\sqrt{3}}$$
$$\tan\theta = \tan 30^\circ$$
$$\theta = 30^\circ$$

3. Correct option: (A) 60°

Given: Height of pole = 12 m

Shadow length = $4\sqrt{3}$

$$\tan\theta = \frac{height\ of\ pole}{lenght\ of\ shadow}$$

$$= \frac{12}{4\sqrt{3}} = \frac{3}{\sqrt{3}} = \sqrt{3}$$

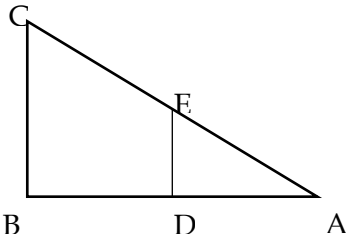
$$\tan\theta = \tan 60°$$

$$\theta = 60°$$

4. Correct option: D) 5m

Given: Length of tree, BC = 12.5 m

Length of stick, DE = 5 m



Suppose DA and BA are shadows of DE and BC respectively

Consider triangle ⊿ABC and ⊿ADE

$$\angle ABC = \angle ADE = 90°$$

$$\angle A = \angle A$$

By AA- similarity criteria

$$\Delta ABC = \Delta ADE$$

Triangles are similar, then ratio of their sides are equal

$$\frac{AB}{AD} = \frac{BC}{DE}$$

$$\frac{AB}{2} = \frac{12.5}{5}$$

$$5AB = 25$$

$$AB = 5m$$

5. Correct option: (D) 4m

Given: Foot of ladder = 2 m

Angle made by ladder = 60°

Length of wall = h

$$\cos\theta = \frac{base}{height}$$

$$\cos 60° = \frac{2}{h}$$

$$\frac{1}{2} = \frac{2}{h}$$

$$h = 4m$$

6. Correct option: (B) $10\sqrt{3}m$

Given: Foot of tower away = 30 m

Angle of elevation = 30°

$$\tan\theta = \frac{perpendicular}{base}$$

$$\tan 30° = \frac{p}{30}$$

$$p = \frac{30}{\sqrt{3}} = 10\sqrt{3}m$$

7. Correct option: (B) $150\sqrt{3}m$

Given: Height of tower = 150 m

Angle of depression = 30°

Let distance of car from tower be d

$$\tan\theta = \frac{height\ of\ tower}{distance\ of\ car}$$

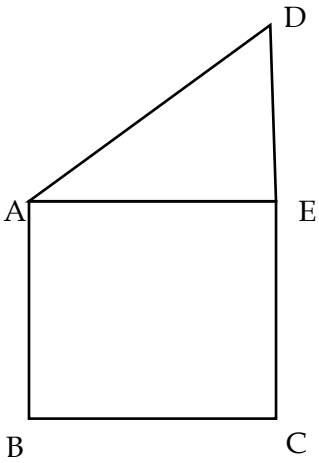
$$\tan 30° = \frac{150}{d}$$

$$\frac{1}{\sqrt{3}} = \frac{150}{d}$$

$$d = 150\sqrt{3}m$$

8. Correct option: (B) 40 m

Given: AB = 20 m



Also, CE = AB = 20 m

In triangle ABC,

$$\tan\theta = \frac{AB}{BC} = \frac{20}{BC} = \frac{20}{AE}$$

Also, in triangle ADE

$$\tan\theta = \frac{DE}{AE} = \frac{DE}{\left(\frac{20}{\tan\theta}\right)} = \frac{\text{DE}\tan\theta}{20}$$

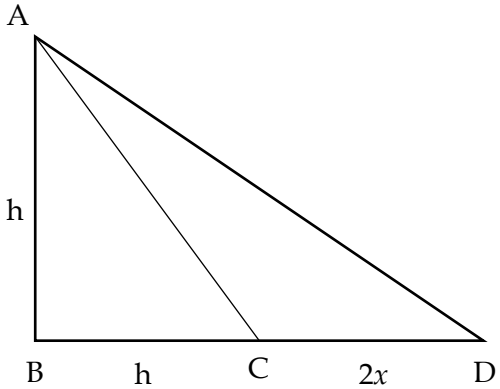
$$DE = 20m$$

Now,

$$CD = DE + CE = 20 + 20 = 40m$$

9. Correct option: (D) $x(\sqrt{3}+1)$

Let height of tower be h



Given: $\angle ACB = 45°$, and $\angle ADB = 30°$

BD is 2x than meter longer than BC

$$BD = BC + CD$$

Now
$$\tan\angle ACB = \frac{AB}{BC}$$
$$\tan 45° = \frac{h}{BC}$$
$$1 = \frac{h}{BC}$$
$$BC = h$$

So
$$BD = BC + CD = h + 2x$$

In triangle ABD

$$\tan\angle D = \frac{AB}{BD}$$
$$\tan 30° = \frac{h}{h+2x}$$
$$\frac{1}{\sqrt{3}} = \frac{h}{h+2x}$$
$$h + 2x = \sqrt{3}h$$
$$\sqrt{3}h - h = 2x$$
$$2x = h(\sqrt{3} - 1)$$
$$h = \frac{2x}{\sqrt{3}-1} = \frac{2x(\sqrt{3}+1)}{3-1} = x(\sqrt{3}+1)\,m$$

10. Correct option: (A) 30°

Given: Ratio of length of rod and shadow is $1:\sqrt{3}$

So
$$\tan\theta = \frac{1}{\sqrt{3}}$$
$$\tan\theta = \tan 30°$$
$$\theta = 30°$$

11. Correct option: (B) 6m

Height of pole be h

Shadow of pole = $2\sqrt{3}$

Angle of elevation = 60°

$$\tan\theta = \frac{height}{shadow\ length}$$
$$\tan 60° = \frac{h}{2\sqrt{3}}$$
$$\sqrt{3} \times 2\sqrt{3} = h$$
$$h = 2 \times 3$$
$$h = 6\,m$$

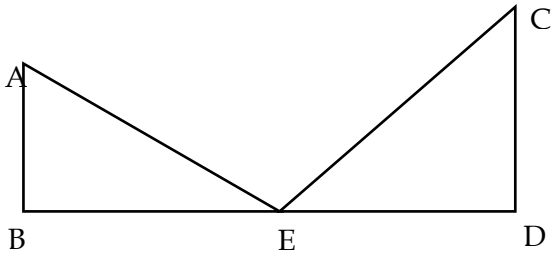
12. Correct option: (A) 30°

Given: Height of tower, AB = 20 m

Shadow of tower, BC = $20\sqrt{3}$

Now
$$\tan\theta = \frac{height}{shadow\ length} = \frac{20}{20\sqrt{3}}$$
$$\tan\theta = \frac{1}{\sqrt{3}}$$
$$\tan\theta = \tan 30°$$
$$\theta = 30°$$

13. Correct option: (C) 1:3



Given: $\angle AEB = 30°$ and $\angle CED = 60°$
Let AB = x and CD = y
Now in triangle ABE

$$\tan 30° = \frac{AB}{BE}$$
$$\frac{1}{\sqrt{3}} = \frac{x}{BE}$$
$$BE = \sqrt{3}x$$

Also, in triangle CDE

$$\tan 60° = \frac{CD}{DE}$$
$$DE = \frac{y}{\sqrt{3}}$$

Now BE = DE, as E is mid – point of BD
So

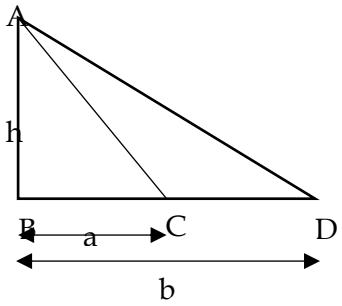
$$BE = DE = \sqrt{3}x = \frac{y}{\sqrt{3}}$$
$$3x = y$$
$$\frac{x}{y} = \frac{1}{3}$$

14. Correct option: (A) $50\sqrt{3}\,m$

Given: Strike of kite is 100m
Angle = 60°
Now

$$\sin\theta = \frac{perpendicular}{hypotenuse}$$
$$\sin 60° = \frac{h}{100}$$
$$\frac{\sqrt{3}}{2} = \frac{h}{100} \Rightarrow h = \frac{100\sqrt{3}}{2} = 50\sqrt{3}\,m$$

15. Correct option: (B) $\sqrt{ab}$



In triangle ABC

$$\angle ACB = 90° - \theta$$

So

$$\tan(90° - \theta) = \frac{AB}{BC}$$

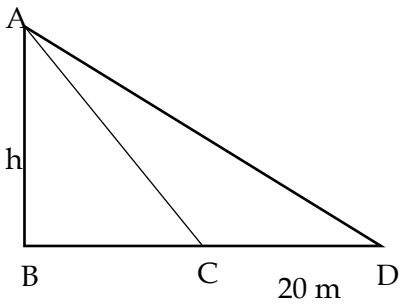
$$\cot\theta = \frac{AB}{a}$$
$$AB = a\cot\theta$$

In triangle ADC

$$\tan\theta = \frac{AB}{BD}$$
$$AB = b\tan\theta$$
$$AB^2 = ab$$
$$AB = \sqrt{ab}$$

16. Correct option: (B)$10\sqrt{3}$

Given: $\angle ADB$ = 30°, $\angle ACB$= 60°, CD = 20 m



In triangle ABC

$$\tan 60 = \frac{AB}{BC}$$
$$\sqrt{3} = \frac{h}{BC}$$
$$BC = \frac{h}{\sqrt{3}}$$

In triangle

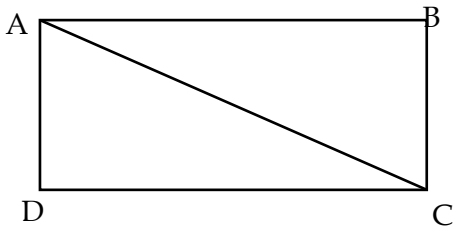
$$\tan 30° = \frac{AB}{BD}$$
$$\frac{1}{\sqrt{3}} = \frac{h}{BD}$$
$$BD = h\sqrt{3}$$

Now

$$BD - BC = 20$$
$$h\sqrt{3} - \frac{h}{\sqrt{3}} = 20$$
$$3h - h = 20\sqrt{3}$$
$$2h = 20\sqrt{3}$$
$$h = 10\sqrt{3}$$

17. Correct option: (C) $16\sqrt{3}cm^2$

Let ABCD be a rectangle



Given: Angle = 30°

Length of diagonal, AC = 8 cm

Side if rectangle

$$\cos 30° = \frac{AB}{AC}$$
$$\frac{\sqrt{3}}{2} = \frac{AB}{8}$$
$$AB = 4\sqrt{3}\ cm$$

and

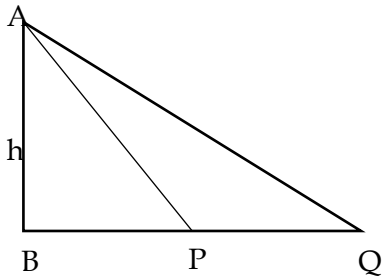
$$\sin 30° = \frac{BC}{AC}$$
$$\frac{1}{2} = \frac{BC}{8}$$
$$BC = 4cm$$

Area of rectangle = $AB \times BC$

$$= 4\sqrt{3} \times 4$$
$$= 16\sqrt{3}cm^2$$

18. Correct option: (B) $\frac{(\sqrt{3}+1)}{2} km$

Consider AB is the hill
P and Q are two consecutive stones



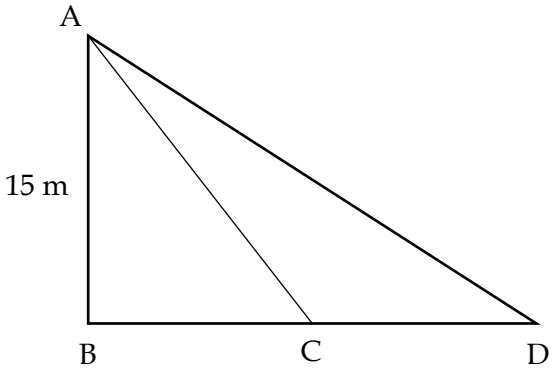
Let height of hill, AB = h m
PQ = 1 km = 1000 m
BP = x m
$\angle APB = 45°$, $\angle AQB = 30°$
In triangle ABP

$$\tan 45° = \frac{AB}{BP}$$
$$1 = \frac{h}{x}$$
$$h = x$$

In triangle ABQ
BQ = BP +PQ = x + 1000

$$\tan 30° = \frac{AB}{BQ}$$
$$\frac{1}{\sqrt{3}} = \frac{h}{x + 1000}$$
$$x + 1000 = h\sqrt{3}$$
$$\sqrt{3}h = h + 1000$$
$$h(\sqrt{3} - 1) = 1000$$
$$h = \frac{1000}{\sqrt{3} - 1} = \frac{1000}{\sqrt{3} - 1} \times \frac{\sqrt{3} + 1}{\sqrt{3} + 1} = \frac{1000(\sqrt{3} + 1)}{3 - 1}$$
$$= \frac{1000(\sqrt{3} + 1)}{2} = \frac{(\sqrt{3} + 1)}{2} km$$

19. Correct option: (C) $10\sqrt{3}$



Given: $\angle ACB = 60°, \angle ADB = 30°$

AB = 15 m

In triangle ABC

$$\tan 60° = \frac{AB}{BC}$$

$$\sqrt{3} = \frac{15}{BC}$$

$$BC = \frac{15}{\sqrt{3}} = \frac{15\sqrt{3}}{\sqrt{3} \times \sqrt{3}} = 5\sqrt{3}$$

In triangle ABD

$$\tan 30° = \frac{AB}{BD}$$

$$\frac{1}{\sqrt{3}} = \frac{15}{BD}$$

$$BD = 15\sqrt{3}$$

Now, difference between the length of the shadow

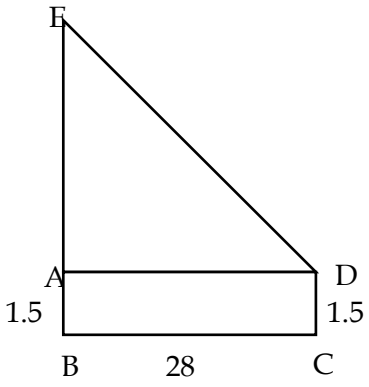
$$= BD - BC = 15\sqrt{3} - 5\sqrt{3} = 10\sqrt{3}$$

20. Correct option: (B) 30 m

Given: Height of observer, DC = AB = 1.5 m

BC = AD =28.5 m

AE = $h - 1.5$



In right triangle EAD

$\angle ADE = 45°$,

$$\tan 45° = \frac{AE}{AD}$$

$$1 = \frac{h - 1.5}{28.5}$$

$$h = 28.5 + 1.5 = 30\ m$$

LEVEL – II

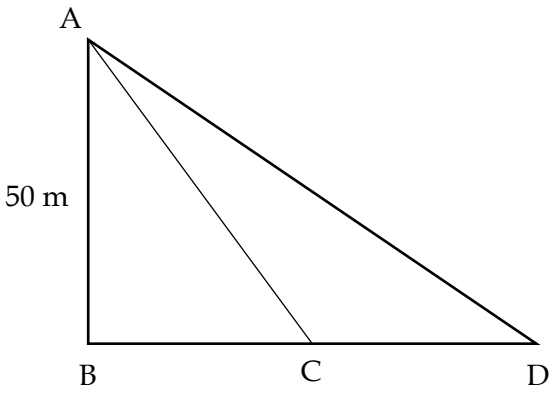
1. Correct options: (B) and (C)

Given Height of tower, h = 50 m

$\angle ACB = 45°$, $\angle ADB = 30°$

$BD - BC = x$, so

CD = x meter



In right triangle ABD

$\angle ADB = 30°$,

$$\tan 30° = \frac{AB}{BD}$$
$$\frac{1}{\sqrt{3}} = \frac{50}{BD}$$
$$BD = 50\sqrt{3}\, m$$

In triangle ABC

$$\tan 45° = \frac{AB}{BC}$$
$$1 = \frac{50}{BC}$$
$$BC = 50\, m$$

Now

$$BD = BC + CD$$
$$CD = BD - BC$$
$$= 50\sqrt{3} - 50$$
$$= 50\left(\sqrt{3} - 1\right)$$
$$= 50(1.732 - 1)$$
$$= 50 \times 0.732$$
$$CD = 36.6\, m$$

2. Correct options: (A) and (C)

Given

Height = h meters

Angle of elevation = 60°

Distance = 25 m

$$\tan 60° = \frac{height}{distance}$$
$$\sqrt{3} = \frac{h}{25}$$
$$h = 25\sqrt{3}$$
$$h = 25 \times 1.732 = 43.3\, m$$

3. Correct options: (B) and (C)

Given: Height of ladder = 15 m

Angle of elevation = 60°

Wall height = h

Now

$$\cos 60° = \frac{h}{15}$$

$$\frac{1}{2} = \frac{h}{15}$$

$$h = \frac{15}{2} \Longrightarrow h = 7.5\ m$$

4. Correct option: (C) 60°

Given: Height of tower = 30 m

Shadow length = $10\sqrt{3}\ m$

Now

$$\tan\theta = \frac{30}{10\sqrt{3}} = \frac{3}{\sqrt{3}}$$

$$\tan\theta = \sqrt{3}$$

$$\tan\theta = \tan 60°$$

$$\theta = 60°$$

5. Correct options: (B) and (D)

Let AB be the hill

CD be tower

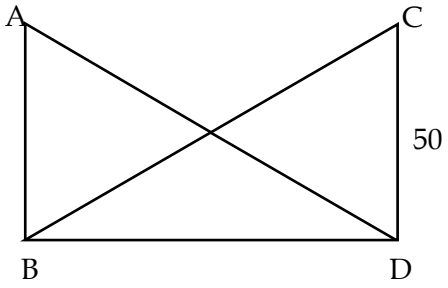
Angle of elevation at foot of tower,

$$\angle ADB = 60°$$

Angle of depression from top of tower,

$$\angle CBD = 30°$$

CD = 50 m



Now in right triangle CDB

$$\tan 30° = \frac{CD}{BD}$$

$$\frac{1}{\sqrt{3}} = \frac{50}{BD}$$

$$BD = 50\sqrt{3}$$

Also, in right triangle ABD

$$\tan 60° = \frac{AB}{BD}$$

$$\sqrt{3} = \frac{AB}{50\sqrt{3}}$$

$$AB = 50 \times 3 = 150\ m = \frac{300}{2}\ m$$

6. Correct option: (D) 80 m

Now in right triangle ABC $\tan 60° = \frac{AB}{BC}$

$$\sqrt{3} = \frac{60}{BC} \Rightarrow BC = \frac{60}{\sqrt{3}}$$

$$BC = \frac{60}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = 20\sqrt{3} = AD$$

Also, in right triangle ACE $\tan 30° = \frac{ED}{AD}$

$$\frac{1}{\sqrt{3}} = \frac{ED}{20\sqrt{3}}$$

$$ED = 20\,m$$

Now $EC = ED + DC = ED + AB = 60 + 20 = 80\,m$

7. Correct option: (A) 20 m

Height of building = 60 m

Height of tower =80 m

Difference between height and tower $80 - 60 = 20\,m$

8. Correct option: (B) $20\sqrt{3}$

Distance between tower is AD = $20\sqrt{3}$

9. Correct option: (D) $34.64\,m$

In right triangle ABC $\tan 60° = \frac{AB}{AC}$

$$\sqrt{3} = \frac{60}{AC}$$

$$AC = \frac{60}{\sqrt{3}}$$

$$= \frac{60}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$$

$$= 20\sqrt{3}$$

$$= 20 \times 1.732$$

$$= 34.64\,m$$

$$= DE$$

10. Correct option: (A) 40 m

In right triangle BDE $\tan 30° = \frac{BE}{DE}$

$$\frac{1}{\sqrt{3}} = \frac{(60 - h)\sqrt{3}}{60}$$

$$60 = 3(180 - h)$$

$$60 = 180 - 3h$$

$$3h = 120$$

$$h = 40\,m$$

11. Correct option: (B) 20 m

$$BE = 60 - h$$

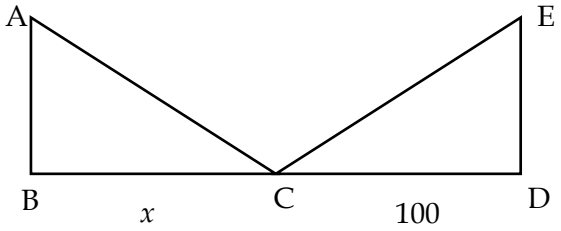
$$BE = 60 - 40$$

$$BE = 20\,m$$

12. Correct Matching is: $a \to T; b \to R; c \to P; d \to S$

Let AB and ED be two pillars of height h meters

C be a point on road such that BC = x



$\angle ACB = 60°$, $\angle ECD = 30°$

a) In triangle ABC

$$\tan 60° = \frac{AB}{BC}$$

$$\sqrt{3} = \frac{h}{x}$$

$$h = \sqrt{3}x$$

Also, in triangle ECD

$$\tan 30° = \frac{ED}{CD}$$

$$\frac{1}{\sqrt{3}} = \frac{h}{100 - x}$$

$$\sqrt{3}h = (100 - x)$$

$$\sqrt{3}(\sqrt{3}x) = 100 - x$$

$$3x = 100 - x$$

$$4x = 100$$

$$x = 25\ m$$

Height is

$$h = \sqrt{3}x = \sqrt{3} \times 25 = 25 \times 1.732 = 43.3\ m$$

b) Sum of heights of pillars

$$= 25\sqrt{3} + 25\sqrt{3} = 50\sqrt{3}\ m$$

c) Position of point from first pillar

$$= x = 25\ m$$

d) Position of point from second pillar

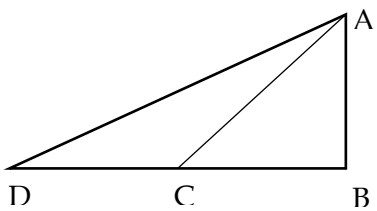
$$= 100 - x = 100 - 25 = 75\ m$$

13. Correct Answer is: 5

Let height of tower, AB = h

Now, BC = 5 m and BD = 20 m

$\angle ACB = \theta$, $\angle ADB = 90° - \theta$



In triangle ABC

$$\tan\theta = \frac{AB}{BC}$$

$$= \frac{h}{5}$$

In triangle ABD $\cot(90 - \theta) = \frac{BD}{AB}$

$$\tan\theta = \frac{20}{h}$$

Now $\frac{20}{h} = \frac{h}{5}$

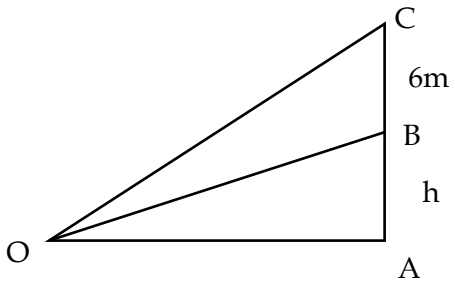
$$(h)^2 = 100$$

$$h = \sqrt{100}$$

$$h = 10\ m$$

Then $\frac{h}{2} = \frac{10}{2} = 5\ m$

14. Correct Answer is: 3



Let AB be height of tower and BC be the flagstaff
BC = 6 m, $\angle AOB = 30°$ and $\angle AOC = 60°$
AB = h m
Now in triangle AOB

$$\tan 30° = \frac{AB}{OA}$$

$$\frac{1}{\sqrt{3}} = \frac{h}{OA} \Rightarrow OA = h\sqrt{3}$$

Also, in triangle AOC

$$\tan 60° = \frac{AC}{OA}$$

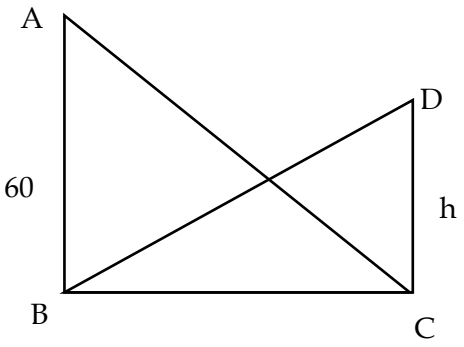
$$\sqrt{3} = \frac{AB + BC}{h\sqrt{3}}$$

$$3h = h + 6$$

$$2h = 6$$

$$h = 3\ m$$

15. Correct Answer is: 20



Given
$\angle DBC = 30°$ and $\angle ACB = 60°$
Height of tower = 60 m

In triangle BCD

$$\tan 30° = \frac{CD}{BC}$$
$$\frac{1}{\sqrt{3}} = \frac{h}{x}$$
$$x = \sqrt{3}h$$

In triangle ABC

$$\tan 60° = \frac{AB}{BC}$$
$$\sqrt{3} = \frac{60}{x}$$
$$\sqrt{3} = \frac{60}{\sqrt{3}h}$$
$$3h = 60$$
$$h = 20\ m$$

16. Correct Answer is: 5

Given: Ratio of length of pole and its shadow is $\sqrt{3}:1$

Now

$$\tan x = \frac{\sqrt{3}}{1}$$
$$\tan x = \tan 60°$$
$$x = 60°$$

Then

$$\frac{x°}{12} = \frac{60}{12} = 5°$$

STUDENT PRACTICE QUESTIONS

2. Correct option: (A) 30°

Let height of tower be h

So, height of shadow = $\sqrt{3}$h

$$\tan\theta = \frac{h}{\sqrt{3}h} = \frac{1}{\sqrt{3}}$$
$$\tan\theta = \tan 30°$$
$$\theta = 30°$$

7. Correct option: (C) $\frac{15}{2}m$

Given: length of ladder = 15 m

Angle made by ladder = 60°

Height = h

$$\cos\theta = \frac{base}{height}$$
$$\cos 60° = \frac{h}{15}$$
$$\frac{1}{2} = \frac{h}{15} \Rightarrow h = \frac{15}{2}m$$

10. Correct option: (B) 30°

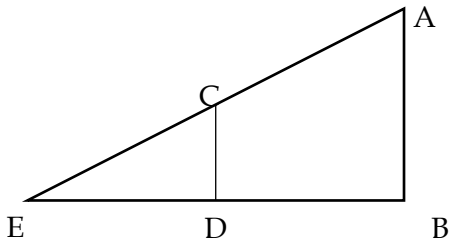
Given: Height of kite from ground = 30 m

Length of string = 60 m

Angle of elevation

$$\sin\theta = \frac{30}{60} = \frac{1}{2}$$
$$\sin\theta = \sin 30° \Rightarrow \theta = 30°$$

12. Correct option: (C) 2.5 m



Let AB be the lamp post
CD be girl = 1.5 m
Shadow of girl, DE = 4.5 m and BD = 3m
Now in triangle CDE

$$\tan\theta = \frac{CD}{DE} = \frac{1.5}{4.5} = \frac{1}{3}$$

Also, in triangle ABE

$$\tan\theta = \frac{AB}{BE}$$
$$\frac{1}{3} = \frac{AB}{4.5+3}$$
$$3AB = 7.5$$
$$AB = \frac{7.5}{3} = 2.5m$$

18. Correct option: (B) $10\sqrt{3}$

Given: Angle of elevation =30°
Distance between foot of tower = 30 m
Height of tower be h
Now,

$$\tan\theta = \frac{heigth}{distane}$$
$$\tan 30° = \frac{h}{30}$$
$$\frac{1}{\sqrt{3}} = \frac{h}{30} \Rightarrow h = \frac{30}{\sqrt{3}} = 10\sqrt{3}$$

7. Surface Areas and Volumes
Solutions

CYLINDER, CONE AND SPHERE
LEVEL-I

1. Correct option: (B)

The curved surface area of a right circular cylinder of the base radius 'r' and height 'h' is 2πrh

The curved surface area of the pillar = ?

Radius of the cylinder, r = 0.7 m

The Curved surface area of the pillar $= 2\pi rh$

Curved surface area of the pillar $= 2 \times \frac{22}{7} \times 0.7\ m \times (1) = 2 \times 22 \times 0.1$

Curved surface area of the pillar $= 44 \times 0.1 = 4.4\ m^2$.

Therefore, the Curved surface area of the pillar is $4.4\ m^2$.

2. Correct option: (A)

Curved surface Area of cylindrical pillar $= 2\pi rh$

Curved surface Area of cylindrical pillar $= 2 \times \frac{22}{7} \times 1.75 \times 15$

Curved surface Area of cylindrical pillar $= 2 \times 22 \times 0.25 \times 15$

Curved surface Area of cylindrical pillar $= 165$ m²

Now, Cost of painting 1 m² curved surface of the pillar $= Rs.25$

Cost of painting 165 m² curved surface of the pillar $= Rs.25 \times 165 = Rs.4125$.

3. Correct option: (C)

Given, curved surface area =176 cm^2 and and base area 38.5 cm^2.

$\Rightarrow \pi r^2 = 38.5$

$\Rightarrow r^2 = 38.5 \times \frac{22}{7} = 12.25$

$\Rightarrow r = 3.5\ m.$

$Again, curved\ surface\ area = 176\ cm^2$

$\Rightarrow 2\pi rh = 176$

$\Rightarrow 2 \times \frac{22}{7} \times 3.5 \times h = 176$

$\Rightarrow h = 8.$

Volume $= \pi r^2 h = \frac{22}{7} \times 3.52 \times 8 = 308\ cm^3$

Therefore, Volume is 308 cm^3.

4. Correct option: (A)

Length of the rectangular paper will be height of the cylinder and width of the rectangular paper is circumference of the cylinder

$\therefore 2\pi r = 22cm$

$r = \frac{7}{2}$

$and\ h = 10\ cm$

volume of cylinder(V) $= \pi r^2 h$

$= \frac{22}{7} \times \left(\frac{7}{2}\right)^2 \times 10$

$= \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times 10$

$= 385\ cm^3$

Therefore, the volume of cylinder is 385 cm^3.

5. Correct option: (C)

⇒ Dimension of cuboid = $44cm \times 30cm \times 15cm$

⇒ Height of a cylinder (H)=28cm

According to the question,

⇒ Volume of a cuboid = Volume of cylinder.

⇒ $l \times b \times h = \pi \times r^2 \times H$

⇒ $44 \times 30 \times 15 = 722 \times (r^2) \times 28$

⇒ $44 \times 30 \times 15 = 22 \times 4 \times r^2$

⇒ $r^2 = \dfrac{44 \times 30 \times 15}{22 \times 4}$

⇒ $r^2 = 225\ cm^2$

∴ $r = 15\ cm.$

Therefore, radius is 15 cm.

6. Correct option: (C)

Let the height of X and Y be `h', and their radii be $r\ and\ 2r$ respectively.

Then, volume of $X\ =\ \pi r^2 h$ and

volume of Y = $\pi\ (2r)^2$, h = $\ 4\ \pi r^2 h$

$New\ height\ of\ X\ =\ 2h,$

$So, new\ volume\ of\ X\ = \pi r^2 (2h) =\ 2\pi r^2 h$

$= \frac{1}{2}(4\pi r^2 h) = \frac{1}{2} volume\ of\ Y.$

Therefore, Volume of X becomes $Half\ volume\ of\ Y$.

7. Correct option: (D)

Wire is in the shape of a cylinder.

Let the radius of the original cylinder =r cm and height =h cm

Original volume = $\pi r^2 h$

New Radius after decrease = $3r$

Let the new height be H.

Given, $\pi r^2 h = \pi \left(\frac{r}{2}\right)^2 H$

$H = 9h$

Hence, the height becomes 9 times.

8. Correct option: (B)

Volume of earth dug out

$= \frac{22}{7} \times 7 \times 7 \times 10\ m^3 =\ 1540\ m^3$

Area of embankment

$= \dfrac{22}{7} \times ((28)^2 - (7)^2) m^2$

$= \dfrac{22}{7} \times ((28 + 7)(28 - 7)) m^2$

$= 2310\ m^2$

Height of embankment = $\frac{Volume}{Area}$

Therefore, Height of embankment = $\frac{1540}{2310}\ m = \frac{2}{3}\ m.$

9. Correct option: (C)

Circumference=12m and height (h)=πm Therefore

$2\pi r = 12\ or$

$r = \dfrac{12}{2\pi} = \dfrac{6}{\pi}$

$Now\ V = \pi r^2 h$

$= \pi \times \left(\frac{6}{\pi}\right)^2 \times \pi$

$= \frac{\pi \times 36 \times \pi}{\pi \times \pi} = 36m^3.$

Therefore, Volume is $36m^3$.

10. Correct option: (C).

Given, Diameter $d = 14cm$

$\therefore r = 214 = 7cm, l = 10cm$

Curved surface area of a cone= $\pi rl = \frac{22}{7} \times 7cm \times 10cm = 220cm^2$.

Therefore, Curved surface area of a cone is $220cm^2$.

11. Correct option: (B)

Curved surface area of cone painted $= \frac{35.20}{0.05} = 704\ cm^2$.

$\Rightarrow \boldsymbol{\pi r(r + l)} = 704\ cm^2.$

$\Rightarrow 704 = \pi r(r + l)$

$\Rightarrow 704 = \frac{22r(r + 25)}{7}$

$\Rightarrow \frac{704 \times 7}{22} = r^2 + 25r$

$\Rightarrow 224 = r^2 + 25r$

$\Rightarrow 0 = r^2 + 25r - 224$

$\Rightarrow 0 = r^2 + 32r - 7r - 224$

$\Rightarrow 0 = r(r + 32) - 7(r + 32)$

$\Rightarrow 0 = (r - 7)(r + 32)$

$\Rightarrow 0 = r - 7\ or\ 0 = r + 32$

$\Rightarrow r = 7\ or\ -32.$

Negative value can't be radius.

⇒ radius of cone = 7 cm

By Pythagoras Theorem,

$height\ of\ cone = (l^2 - r^2)^{\frac{1}{2}} = (25^2 - 7^2)^{\frac{1}{2}} = (625 - 49)^{\frac{1}{2}} = (576)^{\frac{1}{2}} = 24\ cm$

Volume of cone $= \frac{\pi r^2 h}{3} = \frac{22 \times 7 \times 7 \times 24}{3 \times 7} = 1232\ cubic\ cm$

Therefore, volume of cone is 1232 *cubic cm.*

12. Correct option: (C)

for cone

$r = 3cm$

$h = 5cm$

$v_1 = \frac{\pi r^2 h}{3}$

$v_1 = \frac{\pi \times (3)^2 \times (5)}{3}$

$v_1 = \frac{\pi \times 9 \times (5)}{3}$

$v_1 = \frac{\pi \times 45}{3}$

$v_1 = 15\pi$

for cylinder

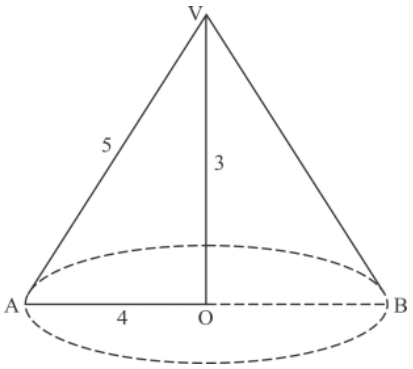
$r = 2cm$

$h = 4cm$

$v_2 = \pi r^2 h$

$v_2 = \pi \times (2)^2 \times (4)$
$v_2 = \pi \times 4 \times 4$
$v_2 = 16\pi$
$\frac{v_1}{v_2} = \frac{15\pi}{16\pi} = \frac{15}{16}$
$v_1 : v_2 = 15 : 16.$
Therefore, $v_1 : v_2 = 15 : 16.$

13. Correct option: (C).
Radius of cone VAOB, r = 4 cm
Height of cone VAOB, h = 3 cm



The volume of cone VAOB $= \frac{\pi \times (4)^2 \times (3)}{3} = \frac{\pi \times 16 \times 3}{3} = 16\pi\ c\ m^3$.
Therefore, volume of cone VAOB is $16\pi\ c\ m^3$.

14. Correct option: (D).
Let V_1 and V_2 are the volumes of cylinder and cone respectively.
Cylinder is melted to form a cone,
$\therefore V_1 = V_2$
$\Rightarrow \pi \times (8)^2 \times 2 = \frac{1}{3} \times \pi \times r^2 \times 6$ (where r is the radius of cone)
$\Rightarrow r^2 = (8)^2$
$\Rightarrow r = 8cm.$
Therefore, radius is 8 cm.

15. Correct option: (B)
Radius of a sphere $(r) = 14\ cm$
$\therefore$ Surface area of $sphere = 4\pi r^2 = 4 \times \frac{22}{7} \times 14 \times 14\ cm^2$
Surface area of $sphere = 2464\ cm^2$
Therefore, Surface area of $sphere\ is\ 2464\ cm^2$.

16. Correct option: (D)
Let the radius of the sphere before increase= $\boldsymbol{r}$.
Given, increased surface area = $352cm^2$.
Also, the surface area of the sphere before the increase is A= $4\pi r^2$
After increasing radius by 2 cm, the area is increased by $352cm^2$.
Applying the above conditions:
$\therefore 4\pi r^2 + 352 = 4\pi(r+2)2$
$\Rightarrow 352 = 16\pi + 16\pi r$
$\Rightarrow 1 + r = \frac{352}{16\pi}$
$\Rightarrow 1 + r = 7$
$\Rightarrow r = 6cm.$
Therefore, Radius is 6 cm.

17. Correct option: (D)

Surface area of the sphere = $4\pi r^2$

The ratio of the surface area of A to the surface area of B is $\frac{A}{B} = \frac{(40)^2}{(10)^2} = \frac{1600}{100} = \frac{16}{1}$

Therefore, ratio of the surface area of A to the surface area of B is 16:1.

18. Correct option: (B)

Volume of sphere of radius r = $\frac{4}{3} \times \pi \times (r)^3$

Volume of sphere of radius r = $\frac{r}{3} \times 4\pi r^2$.

We know that,

Surface area of the sphere = $4\pi r^2$

Volume of sphere of radius r = $\frac{r}{3}$ ×surface area of sphere.

Hence, the volume of sphere of radius 'r' is obtained by multiplying its surface area by $\frac{r}{3}$.

19. Correct option: (C)

The sphere will have the diameter equal to the side of the cube as they are of the same height.

If the length of the side of the cube is a, then the diameter of the sphere =a unit

∴ Radius of the sphere = $\frac{a}{2}$

Volume of a sphere of radius 'r' = $\frac{4}{3} \times \pi \times (r)^3$

Hence, volume of this sphere = $\frac{4}{3} \times \pi \times \left(\frac{a}{2}\right)^3 = \frac{\pi a^3}{6}$

Volume of a cube of edge $a = a^3$

Ratio of their volumes = $\frac{\pi a^3}{6} : \ a^3$

= $\frac{\pi}{6} : \ 1 = \frac{\frac{22}{7}}{6} : \ 1 = \frac{22}{7 \times 6} : \ 1 = 11:21.$

1. Therefore, Ratio of their volumes is 11: 21.

20. Correct option: (A)

Given.

A cone, a hemisphere, and a cylinder stand on equal bases.

Hence radius of base of cone = radius of hemisphere = radius of cylinder=r

And,

They also have same height =h

Height of cone =h

Height of cylinder=h

Height of Hemisphere r=h (r is the radius of sphere)

Now,

Volume of cone $V_C = \frac{1}{3}\pi r^2 h$.....(1)

Volume of hemisphere $V_H = \frac{2}{3}\pi r^2 h$ .....(2)

Volume of cylinder $V_c = \pi r^2 h$.....(3)

From equation 1,2 and 3

The ratio of their volume

$V_C : V_H : V_c = \frac{1}{3}\pi r^2 h : \frac{2}{3}\pi r^2 h : \pi r^2 h$

$V_C : V_H : V_c = \frac{1}{3} : \frac{2}{3} : 1.$

$V_C : V_H : V_c = \ 1 : 2 : \ 3$

Therefore, the ratio of their volume $V_C : V_H : V_c = \ 1 : 2 : \ 3.$

LEVEL-II

1. Correct option: (B).

Let the length of the wire be h

Then,

$\pi \times \frac{3}{20} \times \frac{3}{20} \times h = \frac{4}{3}\pi \times 4 \times 4 \times 4$

$\Rightarrow h = \frac{4 \times 4 \times 4 \times 4 \times 20 \times 20}{3 \times 3 \times 3} cm$

$\Rightarrow h = \frac{102400}{27} cm$

$\Rightarrow h = 3792.5cm \qquad (1\ m = \ 100\ cm\ and\ 1\ cm\ = \ 0.01\ m)$

$\Rightarrow h = 37.9m.$

2. Correct option: (C)

Increase in volume of cylinder = volume of sphere

Let increase in the volume of cylinder be $\pi r^2 h$, where r is its radius and h is its height.

So, $\pi r^2 h = \frac{4}{3}\pi r^3$

$\Rightarrow 16\ \pi rh = \frac{4}{3}\pi r^3$

where radius of the cylinder=4cm and radius of the sphere=3cm

$\Rightarrow h = \frac{4}{3} \times \frac{27}{16} = \frac{36}{16} = \frac{9}{4}$

$\Rightarrow h = \frac{9}{4}$

Therefore, water rises by $\frac{9}{4}$cm in the vessel.

3. Correct option: (B)

Given : A Right circular cylinder

Height =14cm

curved surface area (CSA)=704sq.cm

Volume = ?

We know, Curved surface area of cylinder = $\boldsymbol{2\pi rh}$

$704 = 2\pi r \times 14$

$r = \frac{704}{2\pi r \times 14} = \frac{352}{14\pi}$

$r = \frac{176}{7\pi} \qquad \left(\pi = \frac{22}{7}\right)$

$r = \frac{176 \times 7}{7 \times 22} = 8$

$r = 8\ cm.$

Volume of a Right circle cylinder (v) = $\pi r^2 h$

$= \frac{22}{7} \times 8 \times 8 \times 14 cm^3 = 2816\ cm^3.$

Therefore**,** Volume of a Right circle cylinder is $2816\ cm^3$.

4. Correct option: (C)

Given: Circumference of the base= $\boldsymbol{2\pi r = 66\ cm}$

$\Rightarrow 2 \times \frac{22}{7} \times r = 66$

$\Rightarrow r = \frac{66 \times 7}{2 \times 22} = \frac{21}{2} = 10 \cdot 5$

$\Rightarrow r = 10 \cdot 5cm.$

Curved Surface area of a cylinder= $\boldsymbol{2\pi rh = 2640\ sq.cm}$

$\Rightarrow h = \frac{2640}{2\pi \times 10.5} = \frac{2640 \times 7}{2 \times 22 \times 10.5} = 40\ cm$

$\Rightarrow h = 40\ cm$

Volume of cylinder $= \pi r^2 h$

Volume of cylinder $= \frac{22}{7} \times 10.5 \times 10.5 \times 40 = 13860 cm^3$.

Therefore, Volume of cylinder is 13860 cm^3.

5. Correct option: (D)

It is given that

Diameter = 3mm = 0.3cm

Radius = $\frac{0{\cdot}3}{2}$ cm

We know that Number of lead shots = $\frac{\text{volume of cuboid}}{\text{volume of 1 lead shot}}$.

By substituting the values Number of lead shots = $\frac{12 \times 11 \times 9}{\frac{4}{3} \times \frac{22}{7} \times \left(\frac{0.3}{2}\right)^3}$.

On further calculation

Number of lead shots = $\frac{12 \times 11 \times 9}{\frac{4}{3} \times \frac{22}{7} \times \frac{0.0027}{8}}$

So we get Number of lead shots = 84000

Therefore, the number of lead shots are 84000.

6. Correct option: (C)

Volume of sphere $= \frac{4}{3}\pi r^3$

⇒r=7cm

vol. of sphere $= \frac{4}{3}\pi(7)^3 = \frac{1372}{3}\pi cm^3$__(1)

vol of cylinder $\pi R^2 h$__(2)

from eq (1) and (2)

$\frac{4}{3}\pi(7)(7)(7) = \pi(R)(R)\frac{7}{3}$

$R = 14cm$

Diameter of cylinder = 2 R = 2 $\times 14$ = 28cm

Therefore, Diameter of cylinder is 28cm.

7. Correct option: (A)

T.S.A. of cylinder $= 2\pi rh + 2\pi r^2 = 2\pi \times \left(\frac{35}{100}\right) \times 1.25 + 2\pi \left(\frac{35}{100}\right)^2$

$\Rightarrow T.S.A = 3.52\, m^2$

cost of $1m^2$ is Rs 80

$\Rightarrow Total\ cost = 3.52 \times 80 = Rs.281.6$

Therefore, total cost is $Rs.281.6$.

8. Correct option: (D)

Diameter of circular well= $2n$

∴ Radius= $\frac{2}{2} = 1\, m$

Depth $(h) = 14\, m$

Volume of earth dug out $= \pi r^2 h$

Volume of earth dug out $= \frac{22}{7} \times 1 \times 1 \times 14 = 44\, m^3$

∴ Volume of earth dug out is $44\, m^3$.

9. Correct option: ()

C.S.A. of the cone $= \pi rl$

$119\frac{5}{7} cm^2 = \frac{838}{7} cm^2$

Base area= πr^2

$254\frac{4}{7} cm^2 = \frac{1782}{7} cm^2$

Now,

$\pi \times r^2 = \frac{1782}{7} cm^2$

$\frac{22}{7} \times r^2 = \frac{1782}{7}\ cm^2$

$r^2 = \frac{1782 \times 7}{7 \times 22}\ cm^2$

$r^2 = \frac{1782}{22} = 81\ cm^2$

$r = 9\ cm$

Therefore, radius is 9 cm.

Now, $\pi rl = \frac{838}{7} cm^2$

$\frac{22}{7} \times 9 \times l = \frac{838}{7} cm^2$

$l = \frac{838 \times 7}{7 \times 22 \times 9} cm^2$

$l = \frac{838}{198} cm^2$

$l = 4.232\ cm$

$h = \sqrt{r^2 + l^2}$

$h = \sqrt{(9)^2 + (4.232)^2} = \sqrt{81 + 18} = \sqrt{99} = 9.94\ = 10$

$h = \ 10\ cm$

Volume of the cone $= \frac{1}{3}\pi r^2 h$

$\frac{1}{3}\pi r^2 h = \frac{1}{3} \times \frac{22}{7} \times 9 \times 9 \times 10 = \ 848.57$

Therefore, Volume of the cone is 848.57 cm^3.

10. Correct option: (B)

Given, $r\ +\ h\ =\ 37\ cm$

Total surface area of the cylinder = 1628 cm2 (Given)

$\therefore 2\pi r(r + h) = 1628$

$\Rightarrow 2\ \pi r\ \times\ 37\ =\ 1628$

$\Rightarrow\ r\ =\ 7\ cm$

Now, $r\ +\ h\ =\ 37\ cm$

$\Rightarrow 7\ +\ h\ =\ 37$

$\Rightarrow\ h\ =\ 37 - 7\ =\ 30\ cm$

Volume of the cylinder = $\pi r^2 h = \frac{22}{7} \times 7 \times 7 \times 30 = \ 4620\ cm^3$.

Therefore, Volume of the cylinder is $\ 4620\ cm^3$.

11. Correct option: (B)

Let the radius of the cylinder is r and height is h

$\therefore$ Volume of the cylinder = $\pi r^2 h$

According to the question new radius is half of the initial radius

$\therefore$New radius=$\frac{r}{2}$

Height is h.

$\therefore$New volume of the cylinder $\pi \times \left(\frac{r}{2}\right)^2 \times h = \pi \times \frac{r^2}{4} \times h$

$\therefore$ The ratio of the new volume of the cylinder and initial volume of the original cylinder

$$= \frac{\pi \times \frac{r^2}{4} \times h}{\pi r^2 h} = \frac{1}{4}$$

$Therefore,\ Ratio\ = 1:4.$

12. Correct option: ()

Volume of spherical ball $V_s = \frac{4}{3}\pi(R^3 - r^3)$.....(1)

Radius = $\frac{d}{2}$

Radius $r = \frac{6}{2} = 3\ cm$

Radius $R = \frac{8}{2} = 4\ cm$

Volume of spherical ball $V_s = \frac{4}{3}\pi\ (4^3 - 3^3) = \frac{4}{3}\pi\ (64 - 27) = \frac{4}{3}\pi\ (37) = \frac{148}{3}\pi\ cm^3$

Volume of spherical ball $V_s = \frac{148}{3}\pi\ cm^3$

Volume of cone $V_c = \frac{1}{3}\pi r^2 h$ .....(2)

Radius = $\frac{d}{2}$

Radius $= \frac{8}{2} = 4\ cm$

$V_c = \frac{1}{3}\pi(4)^2 h = \frac{16}{3}\pi h = \frac{16}{3}\pi h\ cm^3.$

$V_c = \frac{16}{3}\pi h\ cm^3.$

Now,

$V_s = V_c$

$\frac{148}{3}\pi = \frac{16}{3}\pi h$

$\frac{148}{16} = h$

$h = \frac{74}{8} = 9.25\ cm$

Therefore, the height of cone is 9.25 cm.

13. Correct option: (D)

Given the radius of the cylinder, r = 12 cm

It is also given that a spherical iron ball is dropped into the cylinder and the water level raised by 6.75 cm

Hence volume of water displaced = volume of the iron ball

Height of the raised water level, h = 6.75 m

Volume of water displaced = $\pi r^2 h$

$= \pi \times 12 \times 12 \times 6.75\ cm^3$

$\Rightarrow$ Volume of iron $ball = \pi \times 12 \times 12 \times 6.75\ cm^3 \rightarrow$ (1)

But, volume of iron ball = $\frac{4}{3}\pi r^3$---(2)

From (1) and (2) we get

$\frac{4}{3}\pi r^3 = \pi \times 12 \times 12 \times 6.75$

$r^3 = \pi \times 12 \times 12 \times 6.75 \times \frac{3}{4}$

$r^3 = 729$

$r^3 = 3^3$

r = 9

Thus the radius of the iron ball is 9 cm.

14. Correct option: (A)

Diameter of the vessel =4 cm

Radius of vessel =2 cm

Rise in water level =0.8 cm

Volume of the increased water level = $\pi r^2 h = \frac{22}{7} \times (2)^2 \times 0.8 = 10.06\ cu.cm$

$Volume\ of\ 300\ balls = 300 \times \frac{4}{3} \times \frac{22}{7} \times r^3$

$\Rightarrow r^3 = 10.06 \times \frac{1}{300} \times \frac{4}{3} \times \frac{22}{7}$

$\Rightarrow r^3 = 0.008$

$\Rightarrow r = 0.2$

So, the radius of each ball is 0.2 cm.

Diameter of each ball is $2 \times 0.2 = 0.4\ cm.$

Therefore, radius of each lead ball is 0.2 cm.

15. Correct option: (B)

Diameter of the vessel =4 cm

Radius of vessel =2 cm

Rise in water level =0.8 cm

Volume of the increased water level = $\pi r^2 h = \frac{22}{7} \times (2)^2 \times 0.8 = 10.06\ cu.cm$

Therefore, Volume of the increased water level is 10.06 cu.cm.

16. Correct option: (D)

Diameter of the vessel =4 cm

Radius of vessel =2 cm

Rise in water level =0.8 cm

Volume of the increased water level = $\pi r^2 h = \frac{22}{7} \times (2)^2 \times 0.8 = 10.06\ cu.cm$

$Volume\ of\ 300\ balls = 300 \times \frac{4}{3} \times \frac{22}{7} \times r^3$

$\Rightarrow r^3 = 10.06 \times \frac{1}{300} \times \frac{4}{3} \times \frac{22}{7}$

$\Rightarrow r^3 = 0.008$

$\Rightarrow r = 0.2$

So, the radius of each ball is 0.2 cm.

$Volume\ of\ 300\ balls = 300 \times \frac{4}{3} \times \frac{22}{7} \times r^3$

$Volume\ of\ 300\ balls = 300 \times \frac{4}{3} \times \frac{22}{7} \times (0.2)^3$

$= 300 \times \frac{4}{3} \times \frac{22}{7} \times 0.008$

$= \frac{211.2}{21}$

$Volume\ of\ 300\ balls = 10.05\ cu.cm$

Therefore $Volume\ of\ 300\ balls\ is\ 10.05\ cu.cm.$

17. Correct Matching is: $a \rightarrow R, b \rightarrow P, c \rightarrow T, d \rightarrow S$

a) If the base radii of two cylinders are equal, then the ratio of their curved surface area is r: R.

If the base radii of two cylinders are equal, then the ratio of their curved surface areas is directly proportional to the ratio of their heights. Therefore, the correct ratio would be r:R.

b) If the curved surface areas are equal, then the ratio of their volumes is $is\ r^2 + R^2$.

If the curved surface areas of two objects are equal, then the ratio of their volumes is given by $r^2 + R^2$. This is applicable for objects that have the same shape and only differ in size. The ratio of their volumes will be equal to the square of the ratio of their respective radii (or dimensions).

c) If the volumes of two objects are equal, then the ratio of their curved surface areas is h : H. This is known as the isoperimetric inequality. The isoperimetric inequality states

that among all objects with a given volume, a sphere has the minimum surface area. Therefore, if two objects have the same volume, their curved surface areas will have a similar ratio of h : H, where h represents the curved surface area of the object with a smaller surface area, and H represents the curved surface area of the object with a larger surface area.

d) If the heights are equal, then the ratio of their volumes is R : r.

If the heights of two cylinders are equal, then the ratio of their volumes is determined by the ratio of their base radii.

Therefore, $a \to R, b \to P, c \to T, d \to S$.

18. Correct Answer is: 12 cm

Radius of the sector, 'r' = 15 cm

Angle of the sector, 'x' = $216°$

$\therefore$ Length of the arc, $l = \frac{x}{360} \times 2\pi r = \frac{216}{360} \times 2\pi r = \frac{3}{5} \times 2\pi r$

Perimeter of the base of the cone = Length of the arc

$2\pi r$ of cone = $\frac{6}{5}\pi r$ of the circle

Radius of the cone 'r' $= \frac{3}{5} \times 15 = 9$

Radius 'r' of the circle = slant height l of the cone = 9 cm

$height\ of\ cone = (l^2 - r^2)^{\frac{1}{2}} = (15^2 - 9^2)^{\frac{1}{2}}$

$= (225 - 81)^{\frac{1}{2}} = (144)^{\frac{1}{2}}$

$= (12^2)^{\frac{1}{2}} = 12\ cm$

Height of cone is 12 cm.

19. Correct Answer is: 24 cm.

$Radius\ of\ cone = (l^2 - h^2)^{\frac{1}{2}} = (25^2 - 7^2)^{\frac{1}{2}}$

$= (625 - 49)^{\frac{1}{2}} = (576)^{\frac{1}{2}}$

$= (24^2)^{\frac{1}{2}} = 24\ cm.$

Therefore, radius is 24 cm.

20. Correct Answer is: 5 cm

Given diameter of metallic cylinder=5 cm

So radius of cylinder=2.5 cm

And its height $\mathbf{3\frac{1}{3} = \frac{10}{3}\ cm}$

Then volume of cylinder = $\pi r^2 h = \pi \times (2 \cdot 5)^2 \times \frac{10}{3}\ cu.cm$

Let radius of sphere is r

Then volume of sphere =$\frac{4}{3}\pi r^3$

Then volume of cylinder =volume of sphere

$\therefore \frac{4}{3}\pi r^3 = \pi \times (2 \cdot 5)^2 \times \frac{10}{3}$

$\Rightarrow r^3 = \frac{2.5}{4} \times 2.5 \times 10 = \frac{62.5}{4} = 15.625 \Rightarrow r = 2.5cm$

$So\ diameter\ of\ sphere = 2 \times 2.5 = 5cm$

Therefore, diameter is 5 cm.

STUDENT PRACTICE QUESTIONS

1. Correct option: (D)

The area of the metal sheet required = The total surface area of the cylinder = $2\pi(r + h)$

$2\pi(r + h) = 2 \times \frac{22}{7} \times 7(7 + 10) = 44(17) = 748\ cm^2.$

Therefore, the area of metal sheet required is $748\ cm^2$.

2. Correct option: (B)

Given, height of cylinder = 6.5 cm and total surface area of cylinder = 220 cm^2.

$2\pi r(r + h) = 220\ cm^2$.

$r(r + 6.5) = 220 \times \frac{7}{2} \times 22$

$r^2 + 6.5r - 35 = 0$

$r^2 - 3.5r + 10r - 35 = 0$

$(r - 3.5)(r + 10) = 0$

$r = 3.5, -10$

Radius can not be negative, so take ,positive value of r ,

r = 3.5 cm

volume of cylinder(V) = $\pi r^2 h$

$= \frac{22}{7} \times (3.5)^2 \times 6.5$

$= \frac{1751.75}{7}$

$= 250.25\ cm^3$

Therefore, the volume of cylinder is 250.25 cm^3.

3. Correct option: (B)

We know, the base of a cone is circular.

Then, Area of a circle = πr^2=616.

$\Rightarrow r^2 = \frac{616 \times 7}{22} = 196$

$\Rightarrow r^2 = 196$

$=> r = 14cm.$

Also, total surface area of a cone = $\boldsymbol{\pi r(r + l)}$ where, r is the radius of the cone and l is the slant height.

For a cone, slant height $l = \sqrt{r^2 + h^2}$ where, h is the perpendicular height.

Hence, $l = \sqrt{(48)^2 + (14)^2} = \sqrt{2500}$

$\Longrightarrow l = 50\ cm.$

Hence, Total surface area of this cone, $= \frac{22}{7} \times 14 \times (14 + 50) = 2816\ cm^2$.

Therefore, Total surface area of this cone is 2816 cm^2.

4. Correct option: (D)

The amount of canvas used to make the tent = Curved surface area of cylindrical part + Curved surface area of the conical part.

Curved Surface Area of a Cylinder of $Radius\ "R"\ and\ height\ "h"\ = 2\pi Rh$

Radius of the cylindrical part = $\frac{\text{Diameter}}{2} = \frac{9}{2}\ m$

Curved surface area of a $cone\ = \pi rl,$ where r is the radius of the cone and l is the slant height.

Radius of the conical part = $\frac{\text{Diameter}}{2} = \frac{9}{2}$ m

Height of the conical part $= 10.8 - 4.8 = 6\ m$

For a cone, $l = \sqrt{r^2 + h^2}$, where h is the height.

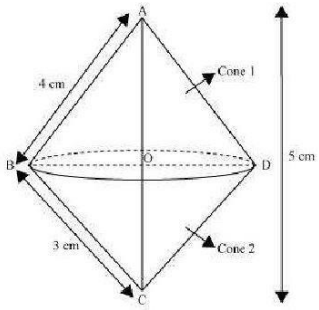
Hence, $l = \sqrt{(6)^2 + (4.5)^2}$

$\therefore l = 7.5\ m$

Hence, area of canvas $= 2 \times \pi \times 4.5 \times 4.8 + \pi \times 4.5 \times 7.5 = 76.95\pi = 241.84\ m^2$

Hence, area of canvas is 241.84 m^2.

5. Correct option: (B)



The double cone so formed by revolving this right-angled triangle ABC about its hypotenuse is shown in the figure.

Hypotenuse $AC = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$

$AC = 5\ cm$

$$\text{Area of } \Delta ABC = \frac{1}{2} \times AB \times AC$$

$\frac{1}{2} \times AC \times OB = \frac{1}{2} \times 4 \times 3$

$= \frac{1}{2} \times 5 \times OB = 6$

$OB = \frac{12}{5} = 2.4\ cm.$

$Volume\ of\ double\ cone\ = Volume\ of\ cone\ 1\ +\ Volume\ of\ cone\ 2$

$= \frac{1}{3} \times \pi r^2 h_1 + \frac{1}{3} \times \pi r^2 h_2 = \frac{1}{3} \times \pi r^2 (h_1 + h_2)$

$= \frac{1}{3} \times \pi r^2 (OA + OC)$

$= \frac{1}{3} \times 3.14 \times (2.4)^4 \times (5) = 30.14 cm^3.$

Therefore, $Volume\ of\ double\ cone\ is\ 30.14 cm^3$.

6. Correct option: (D)

let the Height of the $\boldsymbol{cone = h\ cm}$

Radius of the $\boldsymbol{cone = r\ \ cm}$

Then volume of the old cone= $\frac{1}{3} \times \pi \times r^2 \times h$

If the height and radius are double then

$Height = 2h$

$Radius = 2r$

Then new volume= $\frac{1}{3} \times \pi \times (2r)^2 \times 2h.$

⇒new volume $= \frac{1}{3} \times \pi \times 4r^2 \times 2h$

new volume $= 8 \times \frac{1}{3} \times \pi \times r^2 \times h$

Hence it is 8 times of the old volume.

7. Correct option: (B)

Let the radii of Cone1 and Cone2 are '$2r$ ' and 'r ' respectively.

⇒ Height of Cone1 and Cone2 are H_1 and H_2 respectively.

$\frac{1}{3} \times \pi \times (2r)^2 \times H_1 = \frac{1}{3} \times \pi \times (r)^2 \times H_2$

$\Rightarrow 4 \times (r)^2 \times H_1 = (r)^2 \times H_2$

$\Rightarrow 4 \times H_1 = H_2$

$\Rightarrow \frac{H_1}{H_2} = \frac{1}{4}$

∴ The ratio of the heights of two cones are 1 : 4.

8. Correct option: (A)

Let the radius of the globe be r.

The length of the equator of the globe means the circumference of the globe.

Circumference of the globe = 88cm

$2\pi r = 88$

$2 \times \frac{22}{7} \times r = 88$

$r = \frac{88 \times 7}{44} = 2 \times 7 = 14\ cm$

$r = 14\ cm$

The surface area of the globe is computed as,

∴Surface area = $4\pi r^2$

$4 \times \frac{22}{7} \times 14 \times 14 = 2464 cm^2$

Hence, the surface area is $2464 cm^2$.

9. Correct option: (A)

Diameter of spherical ball of lead = 3 cm

Radius(r)= $\frac{3}{2}$cm

Volume=$\frac{4}{3} \times \pi \times (r)^3 = \frac{4}{3} \times \pi \times \left(\frac{3}{2}\right)^3 = \frac{9}{2} \times \pi$

∴Volume=$\frac{9}{2} \times \pi\ cm^3$.

Diameter of spherical ball of lead = $\frac{3}{2}$ cm

Radius(r_1)= $\frac{3}{2}$cm

Volume=$\frac{4}{3} \times \pi \times (r)^3 = \frac{4}{3} \times \pi \times \left(\frac{3}{2}\right)^3 = \frac{9}{16} \times \pi$

∴Volume=$\frac{9}{16} \times \pi\ cm^3$

Diameter of second smaller ball = 2 cm

and radius (r_2)= $\frac{2}{2} = 1$cm

∴Volume= $\frac{4}{3} \times \pi \times (r)^3 = \frac{4}{3} \times \pi \times (1)^3 = \frac{4}{3} \times \pi$

∴Volume= $\frac{4}{3} \times \pi\ cm^3$

Volume of third smaller ball

$= \frac{9}{2}\pi - \left(\frac{9}{16}\pi + \frac{4}{3}\pi\right)$

$= \left(\frac{9}{2} - \frac{9}{16} + \frac{4}{3}\right)\pi\ \ cm^3.$

$= \left(\frac{216-27-64}{48}\right)\pi\ \ cm^3.$

$= \left(\frac{125}{48}\right)\pi\ \ cm^3.$

∴ Radius of the third ball = $\left(\frac{volume \times 3}{4\pi}\right)^{\frac{1}{3}}$

$= \left(\frac{125 \times 3}{48 \times 4\pi}\right)^{\frac{1}{3}} = \left(\frac{125}{64}\right)^{\frac{1}{3}} = \left(\left(\frac{5}{4}\right)^3\right)^{\frac{1}{3}} = \frac{5}{4}$

$= \frac{5}{4} = 1.25\ cm.$

$\boldsymbol{Diameter\ of\ third\ ball} = 2 \times radius$

$= 2 \times 1.25 = 2.5 cm$

∴ ***Diameter of third ball is* 2. 5 *cm*.**

10. Correct option: (B)

Let r be the radius of single sphere.

Now, The volume of single sphere = sum of volume of three spheres

$\frac{4}{3}\times\pi\times(r)^3=\frac{4}{3}\times\pi\times(61)^3+\frac{4}{3}\times\pi\times(8)^3+\frac{4}{3}\times\pi\times(10)^3$

$\frac{4}{3}\times\pi\times(r)^3=\frac{4}{3}\times\pi\times(216+512+1000)$

$r^3=1728$

$r=12\,cm.$

Diameter $=20\times r=24\,cm.$

Therefore, the diameter $is\ 24\,cm.$

11. Correct option: (B)

We know the total surface area of a hemisphere = $3\pi r^2$

Here, r=14cm

Area of hemisphere = $3\pi r^2$

$\Rightarrow 3\times\frac{22}{7}\times 14$

$\Rightarrow 462\ cm^2$

Hence, the total surface area of a hemisphere of radius 14cm is 462 cm^2

12. Correct option: (A)

Given,

Radius of the hemisphere = Radius of the cone.

Now, Volume of the hemisphere $=\frac{2}{3}\pi R^3$ and

Volume of the cone $=\frac{1}{3}\pi R^2H$

Volume of the hemisphere = volume of the cone

$\frac{2}{3}\pi R^3=\frac{1}{3}\pi R^2$

$2R=H$

$H=2R$

Therefore, If 'H' is the height of the cone, then $H=2R.$

13. Correct option: (A)

Radius of the sector, 'r' = 15 cm

Angle of the sector, 'x' = 216°

$\therefore$ Length of the arc, $l=\frac{x}{360}\times 2\pi r=\frac{216}{360}\times 2\pi r=\frac{3}{5}\times 2\pi r$

Perimeter of the base of the cone = Length of the arc

2πr of cone = $\frac{6}{5}$ πr of the circle

Radius of the cone 'r' $=\frac{3}{5}\times 15=9$

Radius 'r' of the circle = slant height l of the cone = 9 cm

$height\ of\ cone=(l^2-r^2)^{\frac{1}{2}}=(15^2-9^2)^{\frac{1}{2}}$

$=(225-81)^{\frac{1}{2}}=(144)^{\frac{1}{2}}$

$=(12^2)^{\frac{1}{2}}=12\ cm$

Height of cone is 12 cm.

Volume of the cone $=\frac{1}{3}\pi r^2h$

Volume of the cone $=\frac{1}{3}\times\frac{22}{7}\times(9)^2\times 12$

Volume of the cone = $1018.28\,cu.cm$

Therefore, Volume of the cone = $1018.28\,cu.cm.$

14. Correct option: (D)

The total surface area of a cone = $\boldsymbol{\pi r(r+l)}$ where, r is the radius of the cone and l is the slant height.

The total surface area of a cone = $\frac{22}{7} \times \mathbf{9(9+15)} = \frac{22}{7} \times 9(24) = \frac{4752}{7} = 678.58$

Therefore, the total surface area of a cone is 678.58 cm^2.

15. Correct option: (B)

Radius of the sector, 'r' = 15 cm

Angle of the sector, 'x' = 216°

∴ Length of the arc, $l = \frac{x}{360} \times 2\pi r = \frac{216}{360} \times 2\pi r = \frac{3}{5} \times 2\pi r$

Perimeter of the base of the cone = Length of the arc

2πr of cone = $\frac{6}{5}$πr of the circle

Radius of the cone 'r' $= \frac{3}{5} \times 15 = 9$

Radius 'r' of the circle = slant height l of the cone = 9 cm

$height\ of\ cone = (l^2 - r^2)^{\frac{1}{2}} = (15^2 - 9^2)^{\frac{1}{2}}$

$= (225 - 81)^{\frac{1}{2}} = (144)^{\frac{1}{2}}$

$= (12^2)^{\frac{1}{2}} = 12\ cm$

Height of cone is 12 cm.

For a cone, slant height $l = \sqrt{r^2 + h^2}$ where, h is the perpendicular height.

Hence, $l = \sqrt{(9)^2 + (12)^2} = \sqrt{81 + 144} = \sqrt{225}$

$\Rightarrow l = 15\ cm.$

Therefore, slant height of a cone is 15 cm.

16. Correct Matching is: $\boldsymbol{a \to Q, b \to P, c \to T, d \to S, R}$

a) The curved surface area is defined as the area of only curved surface, leaving the circular top and base.

Consider a cylinder having a height 'h' and base radius 'r'.

The curved surface area of a cylinder is given as **C.S.A.= $\boldsymbol{2\pi rh}$**.

b) The volume of a cylinder is the number of unit cubes (cubes of unit length) that can be fit into it. It is the space occupied by the cylinder as the volume of any three-dimensional shape is the space occupied by it.

Therefore, **volume of Cylinder = $\boldsymbol{\pi r^2 h}$**.

c) A cone's volume is defined as the amount of space or capacity it fills.

Volume of the cone = $\boldsymbol{\frac{1}{3}\pi r^2 h}$.

d) A sphere is the 3- D shape where the curved surface area equals to the total surface area of the figure. The curved surface area is the area in which only the area of the curved part is covered. The formula does not take into account the circular base.

The total surface area, on the other hand, is a combination of the curved area along with the area of the base The formula for the total surface area and the curved surface area of the sphere is **Surface area (TSA) = CSA = 4πr² square units.**

Therefore, : $a \to Q, b \to P, c \to T, d \to S, R.$

17. Correct Answer is: 7 cm.

Let radius of cylinder be r cm

Circumference of base of cylinder = Length of rectangle

$2\pi r = 44$

$r = \frac{44 \times 7 =}{2 \times 22} 7$

∴ Radius of the cylinder is 7 cm.

18. Correct Answer is: 1232 cubic cm.

Curved surface area of cone painted $= \frac{35.20}{0.05} = 704\ cm^2$.

$\Rightarrow \boldsymbol{\pi r(r+l)} = 704\ cm^2$.

$\Rightarrow 704 = \pi r(r+l)$

$\Rightarrow 704 = \frac{22r(r+25)}{7}$

$\Rightarrow \frac{704 \times 7}{22} = r^2 + 25r$

$\Rightarrow 224 = r^2 + 25r$

$\Rightarrow 0 = r^2 + 25r - 224$

$\Rightarrow 0 = r^2 + 32r - 7r - 224$

$\Rightarrow 0 = r(r+32) - 7(r+32)$

$\Rightarrow 0 = (r-7)(r+32)$

$\Rightarrow 0 = r - 7\ or\ 0 = r + 32$

$\Rightarrow r = 7\ or\ -32.$

Negative value can't be radius.

⇒ radius of cone = 7 cm

By Pythagoras Theorem,

$height\ of\ cone = (l^2 - r^2)^{\frac{1}{2}} = (25^2 - 7^2)^{\frac{1}{2}} = (625 - 49)^{\frac{1}{2}} = (576)^{\frac{1}{2}} = 24\ cm$

Volume of cone $= \frac{\pi r^2 h}{3} = \frac{22 \times 7 \times 7 \times 24}{3 \times 7} = 1232\ cubic\ cm$

Therefore, volume of cone is $1232\ cubic\ cm.$

19. Correct Answer is: 9

To find the sum of the digits in the length of the wire, we can use the formula for the volume of a sphere and the formula for the volume of a cylinder.

The volume of a sphere is given by the formula: Volume of sphere = $\frac{4}{3}\pi r^3$

where r is the radius of the sphere.

Given that the diameter of the copper sphere is 6 cm, the radius (r) is half of the diameter, so r = $\frac{6}{2}$= 3 cm.

Now, let's calculate the volume of the sphere: Volume sphere = $\frac{4}{3}\pi(3)^3 = \frac{4}{3}\pi(27) = 36\pi\ cm^3$

The volume of the sphere is equal to the volume of the wire, since the copper sphere is beaten and drawn into a wire.

The volume of a cylinder is given by the formula:

Volume of cylinder = $\pi r^2 h$, where r is the radius of the cylinder and h is the height (length) of the cylinder (the wire).

We are given that the diameter of the wire is 0.2 cm, so the radius (r) is 0.1 cm.

Let's substitute the values into the formula to find the length of the wire: $36\pi = \pi r^2 h$

$36 = 0.01 \times h$

$h = \frac{36}{0.01} = 3600\ cm$

The length of the wire is 3600 cm.

To find the sum of the digits in the length of the wire, we add up the individual digits: $3 + 6 + 0 + 0 = 9$.

Therefore, the sum of the digits in the length of the wire is 9.

20. Correct Answer is: 6 cm.

Volume of cone $= \frac{\pi r^2 h}{3}$.

Given base area of cone= $\pi r^2 = 38.5\ cm^3$.

Volume of cone = $77\ cm^3$.

$\frac{(38.5)h}{3} = 77$

$h = \frac{231}{38.5} cm.$

$h = 6\ cm$

$Therefore,$ height is 6 cm.

8. Statistics and Probability

Solutions

LEVEL-I

1. Correct option: (B)

Particular item occurs multiple times in a class interval is called as Frequency

The correct Answer is 'B'

2. Correct option: (B)

The groups into which large data is condensed are called classes.

The correct Answer is 'B'

3. Correct option: (A)

Range of data = Maximum value -Minimum value

$\therefore 98.3-89=9.3$

The correct Answer is 'A'

4. Correct option: (C)

From the bar graph it is evident that the number of students born in the months of October and November is '4' respectively

And hence the difference between two numbers is zero

Also from the same bar graph it is evident that the number of students born in the months of March and April is '3' respectively

And hence the difference between two numbers is zero

The correct Answer is 'C'

5. Correct option: (C)

$2,3,5,7,11,13,17,19,23,29$

$n = 10$

$\frac{n}{2} = 5^{th} term = 11$

$\frac{n}{2} + 1 = 6^{th} term = 13$

$Median = \frac{11+13}{2} = 12$

The correct Answer is 'C'

6. Correct option: (C)

A frequency polygon, is drawn by joining the midpoints of all the top edges of the rectangles in a histogram by straight lines

The correct Answer is 'C'

7. Correct option: (B)

The given class intervals are 10-20, 20-30. In these class intervals the value 20 is lies in the class interval 20-30.

The correct Answer is 'B'

8. Correct option: (A)

11, 13, 15, 19, p + 2, p + 4, 30, 35, 39, 46

The median of given observations arranged in ascending order is 25.

$$Median = \frac{(p+2)+(p+4)}{2} = 25$$

$$\Rightarrow p + 3 = 25$$

$$\Rightarrow p = 22$$

The correct Answer is 'A'

9. Correct option: (B)

Range is not a measure of central tendency

The correct Answer is 'B'

10. Correct option: (C)

The given frequency range varies from 14 to 112

The class intervals for the given set of data are

$$13-22,23-32,33-42,43-52,53-62$$
$$63-72,73-82,83-92,93-102,103-112$$

Hence, number of class interval = 10

The correct Answer is 'C'

11. Correct option: (B)

Width of each class = 2.5

No. of classes = 9

Lower class boundary of the lowest class = 10.6

∴ Upper class limit of highest class= 10.6 + 9×2.5

= 10.6 + 22.5 = 33.1

The correct Answer is 'B'

12. Correct option: (C)

Ogive is the graphical representation of cumulative frequency.

The correct Answer is 'C'

13. Correct option: (A)

Let the 'n' observations be $X_1, X_2 .....X_n$

Let $\overline{X}$ be the mean of n observations, then we have

$$\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$
$$\Rightarrow \sum_{i=1}^{n} X_i = n\overline{X}$$

Multiply a constant k to each of the observations. Then the observations become $kX_{i,} .....kX_n$

If $\overline{Y}$ be the mean of n observations, then we have

$$\overline{Y} = \frac{1}{n}\sum_{i=1}^{n} kX_i$$
$$= \frac{k}{n}\sum_{i=1}^{n} X_i$$
$$= k.\frac{1}{n}\sum_{i=1}^{n} X_i$$
$$= k\overline{X}$$

The correct Answer is 'A'

14. Correct option: (A,B)

In set 2, 2, 2, 2, 4

Mean = $\frac{2+2+2+2+4}{5} = 2.4 \approx 2$

Median is the 3rd term which is equal to 2

Since 2 occurs maximum number of times mode =2

In the set 1, 3, 3, 3, 5

Mean = $\frac{1+3+3+3+5}{5} = 3$

Median is the 3rd term which is equal to 3

Since 3 occurs maximum number of times mode =3

The correct Answer is 'A' & 'B'

15. Correct option: (B)

The given set is 2, 2, 4, 5, and 12

Mean = $\frac{2+2+4+5+12}{5} = 5$

Median is the 3rd term which is equal to 4

Since 2 occurs maximum number of times mode =2

Mean > Mode

The correct Answer is 'B'

16. Correct option: (A)

Mean – Mode = 3 (Mean - Median)

If we solve the known relationship between the mean, median and mode, we get,

Mean – Mode = 3 Mean – 3 Median

On transposing Mean from the LHS to the RHS, we get,

– Mode = 3 Mean – Mean – 3 Median

– Mode = 2 Mean – 3 Median

On transposing -3 Median from RHS to the LHS, we get,

3 Median – Mode = 2 Mean

It can be also written as, 2 Mean = 3 Median – Mode

The correct Answer is 'A'

17. Correct option: (A)

The algebraic sum of deviation of a set of n values from A.M. is zero. This is one of the mathematical properties of arithmetic mean.

The correct Answer is 'A'

18. Correct option: (D)

$$\frac{x_1 + x_2 + ... + x_n}{n} = a$$

$$\Rightarrow x_1 + x_2 + ... + x_n = a.n$$

$$\Rightarrow \frac{x_1 + \alpha + x_2 + \alpha .... + x_n + \alpha}{n}$$

$$= \frac{x_1 + x_2 + ... + x_n + n\alpha}{n}$$

$$= \frac{na + n\alpha}{n}$$

$$= a + \alpha$$

The correct Answer is 'D'

19. Correct option: (A)

Arranging numbers in ascending order 17, 26, 29, 32, 33, 34, 45, 56, 60

Here n = 9 which is odd.

So Median =((n+1)/2) term

Median = ((9+1)/2) term

Median = (10/2) term

Median = 5th term

Median = 33

Hence median is 33.

If 26 is replaced by 62, new set of numbers in ascending order is shown below.

17, 29, 32, 33, 34, 45, 56, 60, 62

Here n = 9 which is odd.

So Median =((n+1)/2) term

Median = ((9+1)/2) term
Median = (10/2) term
Median = 5th term
Median = 34
Hence median is 34.
The correct Answer is 'A'

20. Correct option: (A)

$$mean = \frac{Sum of all observations}{total no of observations}$$
$$= \frac{1+4+3+3+4+2+2+3+3+5}{10}$$
$$= \frac{30}{10}$$
$$= 3$$

The correct Answer is 'A'

LEVEL-II

1. Correct option: (A)

$$Class\ mark = \frac{Lower\ limit + Upper\ limit}{2}$$
$$\Rightarrow 25 = \frac{Lower\ limit + 40}{2}$$
$$\Rightarrow Lower\ limit = 10$$

The correct Answer is 'A'

2. Correct option: (A)

Lower boundary means the lower limit of the class interval.
The lower limit of 20−30 is 20.
The correct Answer is 'A'

3. Correct option: (C)

The first 'n' natural numbers are 1, 2, 3, 4… n

Sum of first n natural numbers $= \frac{n(n+1)}{2}$

Arithmetic Mean

$$= \frac{n(n+1)}{\frac{2}{n}}$$
$$= \frac{n(n+1)}{2n}$$
$$= \frac{(n+1)}{2}$$

The correct Answer is 'C'

4. Correct option: (A)

The mean weight of children is $\text{mean weight of the children} = \frac{\text{TotalWeight (in kg)}}{\text{No. of children}}$

$$\Rightarrow \text{mean weight of the children} = \frac{(40).(2)+(41).(3)+(42).(6)+(43).(2)+44.(5)+45.(5)}{2+3+6+2+5+5}$$
$$= \frac{986}{23}$$
$$= 42.86kg$$

The correct Answer is 'A'

5. Correct option: (A)

$$\text{mean of data} = \frac{\sum F.X}{\sum F}$$

$$\Rightarrow \text{mean weight of the children} = \frac{(4).(11)+(8).(2)+(6).(3)+(7).(y)}{11+2+3+y}$$

$$\Rightarrow 5.3 = \frac{78+7y}{16+y}$$

$$= 4$$

The correct Answer is 'A'

6. Correct option: (B)

$$Mode = l + \left[\frac{h \times (f - f_1)}{(2f - f_1 - f_2)}\right]$$

$l = 10.5,\ f = 10, f_1 = 4, f_2 = 6, h = 15 - 11 = 4$

$$\therefore Mode = 10.5 + \left[\frac{4 \times (10-4)}{(2 \times 10 - 4 - 6)}\right] = 12.9 \approx 13$$

The correct Answer is 'B'

7. Correct option: (B)

Class Interval	Mean of class interval	Frequency	Deviation	fd
100-104	102	15	110-102=8	120
104-108	106	20	4	80
108-112	110	32	0	0
112-116	114	18	-4	-72
116-120	118	15	-8	-20

$$\text{Mean} = \frac{108+112}{2} = 110$$

The correct Answer is 'B'

8. Correct option: (C)

Class interval having more frequency 32 is (108-112)

And hence class width is 112-108=4

The correct Answer is 'C'

9. Correct option: (B)

Class Interval	Mean of class interval	Frequency	Deviation	fd
100-104	102	15	110-102=8	120
104-108	106	20	4	80
108-112	110	32	0	0
112-116	114	18	-4	-72
116-120	118	15	-8	-20

$\therefore$ *Assumedmean*, $a = 110$,

Class width, $h = 4$

and total observation, $N = 100$

By assumed mean method,

$$\text{Mean} = a + \frac{\sum fd}{N}$$

$$= 110 + \frac{-8}{100} = 109.9 \approx 109$$

The correct Answer is 'B'

10. Correct Matching is: a - s, b – r, c – q, d – p,

a. The most frequent value in data is known as Mode

The correct Answer is 'S'

b. The mean cannot be calculated from the graph.

The correct Answer is 'R'

c. Ogive is used to determine median of a continuous data.

The correct Answer is 'Q'

d. Standard deviation is not a measure of central tendency

The correct Answer is 'P'

11. Correct Answer is: 6

$$median = x$$
$$mode = 3.median - 2mean$$
$$\Rightarrow 55.2 = 3x - 2(53.4)$$
$$\Rightarrow 3x = 55.2 + 2(53.4)$$
$$\Rightarrow x = 54 \Rightarrow \frac{x}{9} = 6$$

12. Correct Answer is: 10

If M is the mid-value of class interval 20 − 30 then

$$M = \frac{20 + 30}{2} = 25$$
$$\therefore \frac{M}{2.5} = \frac{25}{2.5} = 10$$

13. Correct Answer is: 6

If the difference between the mode and median is 2, then the difference between the median and mean is _____

$$mode - median = 2$$
$$mode = 3.median - 2mean$$
$$\Rightarrow 55.2 = 3x - 2(53.4)$$
$$\Rightarrow 3x = 55.2 + 2(53.4)$$
$$\Rightarrow x = 54 \Rightarrow \frac{x}{9} = 6$$

STUDENT PRACTICE QUESTIONS

1. Correct option: (D)

The first 5 whole numbers are

$$0, 1, 2, 3, 4$$
$$\frac{0+1+2+3+4}{5} = \frac{10}{2} = 5$$

The correct Answer is 'D'

2. Correct option: (A)

From above data, we see that modal shirt size is 39 as it has the maximum frequency.
The correct Answer is 'A'

3. Correct option: (B)

Class width =20-15=5
Here a=20, h=5

Lower limit of the class $20-\frac{5}{2}=17.5$

Upper limit of the class $20+\frac{5}{2}=22.5$

The correct Answer is 'B'

4. Correct option: (B)

$Number\ of\ boys\ in\ the\ school\ =\ 90$

$Mean\ marks\ of\ boys\ =45\%$

$Total\ marks\ of\ 90\ boys\ =90\times45=4050$

$Number\ of\ girls\ in\ the\ school\ =\ 30$

$Mean\ marks\ of\ girls\ =70\%$

$\therefore Total\ marks\ of\ 30\ girls\ =30\times70=2100$

$Total\ marks\ secured\ by\ 90\ boys\ and\ 30\ girls=\ 4050\ +\ 2100\ =\ 6150$

$Number\ of\ boys\ and\ girls\ who\ have\ appeared\ for\ exam\ from\ the\ school=\ 90\ +\ 30\ =\ 120$

$\therefore\ Average\ marks\ \%\ of\ school=\frac{6150}{120}=51.25\%$

The correct Answer is 'B'

5. Correct option: (C)

In statistics, a central tendency (or measure of central tendency) is a central or typical value for a probability distribution. Colloquially, measures of central tendency are often called averages.
The correct Answer is 'C'

6. Correct option: (C)

The arithmetic mean of 30, 36, 39, 23 and 27.

$$\frac{30+36+39+23+27}{5}$$
$$=\frac{155}{5}$$
$$=31$$

The correct Answer is 'C'

7. Correct option: (C)

$$\frac{a_1+a_2+a_3+a_4+a_5+a_6+a_7+a_8}{8}=18$$
$$\frac{a_9+a_{10}+a_{11}+a_{12}+a_{13}+a_{14}+a_{15}+a_8}{8}=20$$
$$\frac{a_1+a_2+a_3+a_4+a_5+a_6+a_7+a_8+\ldots.a_{15}}{15}=19$$
$$\therefore a_1+a_2+a_3+a_4+a_5+a_6+a_7+a_8+\ldots.a_{15}=(18+20)\times8$$
$$\Rightarrow 19\times15+a_8=38\times8$$
$$\therefore a_8=19$$

The correct Answer is 'C'

8. Correct option: (B)

Let x and y be the lower and upper class limit of a continuous frequency distribution
Now, mid - point of a class

$$\Rightarrow \frac{x+y}{2} = m$$
$$x = lower\ class\ limit$$
$$y = l, upper\ class\ limit$$
$$\therefore \frac{x+l}{2} = m \Rightarrow x = 2m - l$$
$$\Rightarrow x = 2m - l$$

The correct Answer is 'B'

9. Correct option: (B)

The mid value of the class =10
Width of the interval=6

$$Lower\ \lim it = \text{midvalue} - \frac{width}{2}$$

$$= 10 - \frac{6}{2} = 7$$

The correct Answer is 'B'

10. Correct option: (D)

A frequency polygon is constructed by plotting frequency of the class interval and mid value of the class.
The correct Answer is 'D'

11. Correct option: (A)

Let the total wages of laborers working in shift 1 is x
Let the total wages of laborers working in shift 2 is y
Let the total wages of laborers working in shift 3 is z

Mean wages of 150 laborers $= \frac{x+y+z}{a+b+c}$

Where a, b and c is the number of laborers working is shift 1, 2 and 3 respectively
Mean wages =114 Rs

$$\therefore \frac{x+y+z}{a+b+c} \Rightarrow$$
$$\frac{x+y+z}{150} = 114 \Rightarrow x+y+z = 17100$$

Mean wages of laborers in shift 1 is $\frac{x}{a} = 121.5 \Rightarrow x = 121.5 \times 60 = 7290$

Mean wages of laborers in shift 2 is $\frac{y}{b} = 107.5 \Rightarrow x = 107.5 \times 40 = 4310$

$$x+y+z = 17100$$
$$\Rightarrow z = 17100 - 7290 - 4310 = 5500$$

Let the mean of laborers in shift 3=c

$$\therefore m = \frac{z}{c} = \frac{5500}{50} = 110$$

Then mean wage of laborer working in third shift=Rs 110
The correct Answer is 'A'

12. Correct option: (A)

$x_1, x_2, ...x_{75}$

$$mean = \frac{x_1 + x_2 + ... + x_{75}}{75} = 25$$

$$\Rightarrow x_1 + x_2 + ... + x_{75} = 1875$$

$$Mean\ of\ \ new\ numbers = \frac{\frac{x_1}{5} + \frac{x_2}{5} + ... + \frac{x_{75}}{5}}{75}$$

$$= \frac{x_1 + x_2 + ... + x_{75}}{5 \times 75}$$

$$= \frac{25 \times 75}{5 \times 75} = 5$$

The correct Answer is 'A'

13. Correct option: (B)

The even numbers between 50 and 60 are 52, 54, 56 and 58.

The odd numbers between 50 and 60 are 51, 53, 55, 57 and 59.

The AM of even numbers implies $\frac{52+54+56+58}{4} = 55$

The AM of odd numbers implies $\frac{51+53+55+57+59}{5} = 55$

The difference between two means is zero

The correct Answer is 'B'

14. Correct option: (D)

Here, total score in 10 innings= 60 × 10 = 600

After 11th innings, mean score raised to 62.

So, 62 × 11 = 682

Score of 11th innings = 682 – 600 = 82

The correct Answer is 'D'

15. Correct option: (D)

Arranging in ascending order, we get -3,-3,-1,0,2,2,2,5,5,5,5,6,6,6

Here n = 14 (which is even)

$$\therefore Median = \frac{1}{2}\left[\frac{n}{2}term + (\frac{n}{2} + 1)term\right]$$

$$= \frac{1}{2}\left[7^{th}term + 8^{th}term\right]$$

$$= \frac{1}{2}[2 + 5] = 3.5$$

The correct Answer is 'D'

16. Correct option: (D)

The mean of 20 numbers is 40

$$Mean = \frac{n_1 + n_2 + \ldots + n_{20}}{20}$$

$$\Rightarrow \frac{n_1 + n_2 + \ldots + n_{20}}{20} = 40$$

$$\Rightarrow n_1 + n_2 + \ldots + n_{20} = 40 \times 20$$

$$\therefore mean = \frac{(n_1 - 5) + (n_2 - 5) + \ldots + (n_{20} - 5)}{20}$$

$$= \frac{(n_1 + n_2 + \ldots + n_{20}) - 100}{20}$$

$$= \frac{800 - 100}{20}$$

$$= 35$$

The correct Answer is 'D'

17. Correct option: (D)

Given that average of a, b, c, d, e is 28

Mean of a, c, e is 24

$$\frac{a+b+c+d+e}{5}=28$$

$$\Rightarrow a+b+c+d+e=140$$

$$\frac{a+c+e}{3}=24$$

$$\Rightarrow a+c+e=72$$

$$\therefore b+d+72=140 \Rightarrow b+d=140-72=68$$

$$\therefore \frac{b+d}{2}=34$$

The correct Answer is 'D'

18. Correct option: (D)

$$n=50, \overline{X}=250$$

$$\overline{X}=\frac{1}{n}\sum X_i$$

$$\Rightarrow \sum X_i = 50\times 250 = 12500$$

$$Therefore,\ incorrect\ value\ of \sum X_i = 12500$$

$$correct\ value\ of \sum X_i = 12500-115+165=12550$$

$$correct\ \text{mean}=\frac{12550}{50}=251$$

The correct Answer is 'D'

19. Correct option: (A)

$$\therefore \frac{p+p+2+p+4+p+6+p+8p-2+p-4+p-6+p-8}{9}=10$$

$$\Rightarrow \frac{9p}{9}=10$$

$$\Rightarrow p=10$$

$$The\ mean\ of\ least\ 5\ terms\ are$$

$$\frac{p+p-2+p-4+p-6+p-8}{5}=\frac{5p-20}{5}$$

$$\Rightarrow \frac{5(10)-20}{5}=6$$

The correct Answer is 'A'

20. Correct option: (A)

Mean of 6 numbers=20

Sum of 6 numbers= 120.

New mean=15.

Sum of 5 numbers =75.

Deleted number =120−75 = 45.

The correct Answer is 'A'

Probability

LEVEL-I

1. Correct option: (B)

When a die is rolled total number of events -6 {1, 2, 3, 4, 5, 6}

The favorable cases for even number -3 {2, 4, 6}

$$probability = \frac{no.offavoriblecases}{totaln\,o\,ofcases} = \frac{3}{6} = \frac{1}{2}$$

The correct Answer is 'B'

2. Correct option: (C)

When two die are rolled total number of events $=6^2=36$

The favorable case is getting a single score of '7'

Total number of favorable cases is {(1, 6), (2, 5), (3, 4) (4, 3) (5, 2) (6, 1)}

$$probability = \frac{no.offavoriblecases}{totaln\,o\,ofcases} = \frac{6}{36} = \frac{1}{6}$$

The correct Answer is 'C'

3. Correct option: (A)

When two die are rolled total number of events $=6^2=36$

The favorable case is getting equal numbers

Total number of favorable cases is {(1, 1), (2, 2), (3, 3) (4, 4) (5, 5) (6, 6)}

$$probability = \frac{no.offavoriblecases}{totaln\,o\,ofcases} = \frac{6}{36} = \frac{1}{6}$$

The correct Answer is 'A'

4. Correct option: (A)

When a die is rolled twice total number of events $=6^2=36$

The favorable case is getting doublet

Total number of favorable cases is {(1, 1), (2, 2), (3, 3) (4, 4) (5, 5) (6, 6)}

$$probability = \frac{no.offavoriblecases}{totaln\,oofcases} = \frac{6}{36} = \frac{1}{6}$$

The correct Answer is 'A'

5. Correct option: (B)

When a die is rolled twice total number of events $=6^2=36$

The favorable case is getting both are even numbers

Total number of favorable cases is {(2, 2), (2, 4) (2, 6) (4, 2) (4, 4) (4, 6) (6, 2) (6, 4) (6, 6)}

$$probability = \frac{no.offavoriblecases}{totaln\,oofcases} = \frac{9}{36} = \frac{1}{4}$$

The correct Answer is 'B'

6. Correct option: (A)

The sum on the uppermost faces of the dice is an odd number. Hence exactly one of the two numbers on the uppermost face is 2.

So the different possible combinations

are (2,3),(2,5),(2,7),(2,11),(2,13),(3,2),(5,2),(7,2),(11,2) and (13,2)

Total required combinations =10

$$probability = \frac{no.offavoriblecases}{totaln\,oofcases} = \frac{10}{36} = \frac{5}{18}$$

The correct Answer is 'A'

7. Correct option: (A)

When a two dies is rolled total number of events $=6^2=36$

The favorable case is getting the dots on the second die is less than that on the first die is

Total number of favorable cases is {(2, 1), (3, 1) (3, 2) (4, 1) (4, 2) (4, 3) (5, 1) (5,2) (5, 3), (5, 4) , (6,1), (6, 2), (6, 3), (6, 4) (6, 5}

$$probability = \frac{no.offavoriblecases}{tota\ln oofcases} = \frac{15}{36} = \frac{5}{12}$$

The correct Answer is 'A'

8. Correct option: (B)

When three dies is rolled total number of events $=6^3=216$

The favorable case is the sum of the numbers on them is 6

Total number of favorable cases is {(1, 1, 4)(1, 4, 1)(1,2,3),(1, 3, 2) (2, 3, 1)(2, 1, 3) (2, 2, 2) (3, 2, 1) (3, 1, 2) (4, 1,1)}

$$probability = \frac{no.offavoriblecases}{tota\ln oofcases} = \frac{10}{216} = \frac{5}{108}$$

The correct Answer is 'B'

9. Correct option: (B)

When three dies is rolled total number of events $=6^3=216$

The favorable case is getting triplet

Total number of favorable cases is {(1,1,1) (2,2,2) (3,3,3) (4,4,4) (5,5,5) (6,6,6)}

$$probability = \frac{no.offavoriblecases}{tota\ln oofcases} = \frac{6}{216} = \frac{1}{36}$$

The correct Answer is 'B'

10. Correct option: (A)

The total no .of cards is 52

The favorable case is getting king

Total number of favorable cases is 4

$$probability = \frac{no.offavoriblecases}{tota\ln oofcases} = \frac{4}{52} = \frac{1}{13}$$

The correct Answer is 'A'

11. Correct option: (B)

A leap year has 366 days or 52 weeks and 2 odd days. The two odd days can be {Sunday,Monday},{Monday,Tuesday},{Tuesday,Wednesday},{Wednesday,Thursday},{Thursday,Friday},{Friday,Saturday},{Saturday,Sunday}. So there are 7possibilities out of which 2 have a Sunday.

So the probability of 53 Sundays is 2/7

The correct Answer is 'B'

12. Correct option: (A)

There are 365 days in a non-leap year.365 days =52 weeks +1 day

Which means, every day of the week will come at least 52 times in a year. To have 53 Tuesdays in a year, the remaining one day should be Tuesday.

Since, the remaining one day can be Sunday, Monday, Tuesday, Wednesday, Thursday, Friday or Saturday.

So the probability of 53 Tuesdays is 1/7

The correct Answer is 'A'

13. Correct option: (D)

Probability of selecting a month = $\frac{1}{12}$

Probability of getting Friday is = $\frac{1}{7}$

Hence required probability $\frac{1}{12} \times \frac{1}{7} = \frac{1}{84}$

The correct Answer is 'D'

14. Correct option: (C)

When a Card is drawn at random from a pocket of 100 cards numbered 1 to 100 then the number of events will be 100
The favorable case is drawing a number which is a square is
Therefore the number of squares between 1to 100 is 10

$$probability = \frac{no.offavoriblecases}{tota\ln oofcases} = \frac{10}{100} = \frac{1}{10}$$

The correct Answer is 'C'

15. Correct option: (D)

S is a sample space $S = x \in N : 1 \le x \le 100$ this implies that in Sample space there are 99 events are there

$$E = \{x : (x+1)(x-1) \in S\}$$
$$= x^2 - 1$$
$$\therefore xcanbe 2,3,4,5,6,7,8,9,10$$

So that only x will be in 'S'
So the favorable cases are '9'

$$probability = \frac{no.offavoriblecases}{tota\ln oofcases} = \frac{9}{99} = \frac{1}{11}$$

The correct Answer is 'D'

16. Correct option: (A)

A number n is chosen at random from 1,2,3,4... 1000 therefore there are 1000 events
Favorable case is that n is a number that leaves remainder 1 when divided by 7 is
There are 142 multiples for 7 less than 1000; therefore number of favorable cases is 142

$$probability = \frac{no.offavoriblecases}{tota\ln oofcases} = \frac{142}{1000} = \frac{71}{500}$$

The correct Answer is 'A'

17. Correct option: (B)

Since x and y each can take values from 0 to 10, so the total number of ways of selecting x and y is 11×11=121.
Now, |x-y| >5 ⇒ x - y < - 5 or x-y > 5,
when x - y > 5, the favorable cases are {(6,0), (7,0) (7,1) (8,0) (8,1) (8,2) (9,0), (9,1), (9,2), (9,3),(10,0), (10,1), (10,2), (10,3), (10,4)} total 15
The reverse of the above sets are favorable cases for x - y <- 5 and hence another 15 are there in favorable cases
Therefore total favorable cases are 30

$$probability = \frac{no.offavoriblecases}{tota\ln oofcases} = \frac{30}{121}$$

The correct Answer is 'B'

18. Correct option: (C)

19. Correct option: (A)

$$P(A) = \frac{3}{4}, P(B) = \frac{2}{5}, P(A \cap B) = \frac{1}{4}$$
$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$
$$= \frac{3}{4} + \frac{2}{5} - \frac{1}{4}$$
$$= \frac{9}{10}$$
$$= 0.9$$

The correct Answer is 'A'

20. Correct option: (C)

$P(A\cup B)=0.65, P(A\cap B)=0.15$

$P(\overline{A})+P(\overline{B})=1-P(A)+1-P(B)$

$=2-(P(A)+P(B))$

$P(A\cup B)=P(A)+P(B)-P(A\cap B)$

$P(A)+P(B)=P(A\cup B)+P(A\cap B)$

$\therefore 2-(P(A)+P(B))=2-P(A\cup B)-P(A\cap B)$

$=2-0.65-0.15$

$=1.2$

The correct Answer is 'C'

LEVEL-II

1. Correct option: (A)

$P(A\cup B)=0.8, P(A\cap B)=0.3$

$P(\overline{A})+P(\overline{B})=1-P(A)+1-P(B)$

$=2-(P(A)+P(B))$

$P(A\cup B)=P(A)+P(B)-P(A\cap B)$

$P(A)+P(B)=P(A\cup B)+P(A\cap B)$

$\therefore 2-(P(A)+P(B))=2-P(A\cup B)-P(A\cap B)$

$=2-0.8-0.3$

$=0.9$

The correct Answer is 'A'

2. Correct option: (C)

$P(A\cup B)=0.7, P(A\cap B)=0.2$

$P(\overline{A})+P(\overline{B})=1-P(A)+1-P(B)$

$=2-(P(A)+P(B))$

$P(A\cup B)=P(A)+P(B)-P(A\cap B)$

$P(A)+P(B)=P(A\cup B)+P(A\cap B)$

$\therefore 2-(P(A)+P(B))=2-P(A\cup B)-P(A\cap B)$

$=2-0.7-0.2$

$=1.1$

The correct Answer is 'C'

3. Correct option: (B)

$P(A\cup B)=1-P(\overline{A}\cap\overline{B})$

$\therefore P(\overline{A}\cap\overline{B})=1-P(A\cup B)$

$=1-0.14$

$=0.86$

4. Correct option: (B)

$P(A\cap B)=\frac{1}{2}, P(\overline{A}\cap\overline{B})=\frac{1}{3}, P(A)=p, P(B)=2p$

$P(A\cup B)=P(A)+P(B)-P(A\cap B)$

$P(A\cup B)=1-P(\overline{A}\cap\overline{B})=1-\frac{1}{3}=\frac{2}{3}$

$\Rightarrow \frac{2}{3}=3p-\frac{1}{2}$

$\Rightarrow 3p=\frac{7}{6}\Rightarrow p=\frac{7}{18}$

The correct Answer is 'B'

5. Correct option: (A)

If A, B are two mutually exclusive events such that $P(A\cap B)=0$

$$P(A\cup B)=P(A)+P(B)-P(A\cap B)$$
$$\therefore P(A\cup B)=P(A)+P(B)$$
$$=0.5+0.3$$
$$=0.8$$
$$\therefore P(A\cup B)=1-P(\overline{A}\cap\overline{B})$$
$$\therefore P(\overline{A}\cap\overline{B})=1-P(A\cup B)$$
$$=1-0.8$$
$$=0.2$$

The correct Answer is 'A'

6. Correct option: (B)

Let A be the event that the card drawn is spade,
B is the event that card drawn is king and S is the sample space,
When a card is drawn from a pack of cards.

$n(S)={}^{52}C_1=52, n(A)={}^{13}C_1=13, n(B)={}^{4}C_1=4$

$n(A\cap B)={}^{1}C_1=1$

$$\therefore P(A)=\frac{n(A)}{n(S)}=\frac{13}{52}=\frac{1}{4}$$

$$\therefore P(B)=\frac{n(B)}{n(S)}=\frac{4}{52}=\frac{1}{13}$$

$$\therefore P(A\cap B)=\frac{n(A\cap B)}{n(S)}=\frac{1}{52}$$

$$\therefore P(A\cup B)=\frac{13}{52}+\frac{4}{52}-\frac{1}{52}=\frac{16}{52}=\frac{4}{13}$$

The correct Answer is 'B'

7. Correct option: (C)

When two die are rolled total number of events $=6^2=36$

The favorable case is getting sum of the numbers at most 10

Total number of un favorable cases is {(5, 6), (6, 5), (6, 6)}

Therefore total number of favorable cases will be 36-3=33

$$probability=\frac{no.offavoriblecases}{total\,no\,ofcases}=\frac{33}{36}=\frac{11}{12}$$

The correct Answer is 'C'

8. Correct option: (A)

When two die are rolled total number of events $=6^2=36$
The favorable case is getting sum as a prime number
Total number of favorable cases is {(1,1), (1, 2), (1,4) (1, 6) (2,1) (2,3) (2,5) (3,2) (3, 4)(4,1) (4,3) (5, 2) (5,6) (6,1) (6,5)}

$$probability=\frac{no.of favoriblecases}{total\,no\,of cases}=\frac{15}{36}=\frac{5}{12}$$

The correct Answer is 'A'

9. Correct option: (A)

A throws 16

B throws more than 16, means B throws, 17 or 18.
Total number of outcomes on throwing 3 dices = 6×6×6=216
Favorable outcomes = {(5,6 ,6), (6,5,6), (6,6,5) (6,6,6)}
No. of favorable outcomes = 4
Probability of B throwing a number greater than 16=4/216=1/54
The correct Answer is 'A'

10. Correct option: (B)

If 3 dices are thrown simultaneously then total outcomes will be 216
Favorable outcomes = 20
Probability of event =20/216=5/54
The correct Answer is 'B'

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. Correct option: (A,B,C,D)

A.

$$P(E_1) = 1 - P(RRR)$$
$$= 1 - \left[\frac{1}{2}\frac{2}{4}\frac{3}{5}\right] = 0.9$$

B.

$$P(E_2) = 3P(BRR)$$
$$= 3\left[\frac{2}{3}\frac{1}{4}\frac{2}{5}\right] = 0.5$$

C.

$$P(E_3) = \frac{P(RRR)}{P(RRR) + P(BBB)}$$
$$= \frac{0.1}{0.1 + 0.4}$$
$$= 0.2$$

D.

$$P(E_4) = 1 - P(BBB)$$
$$= 1 - 0.4$$
$$= 0.6$$

12. Correct option: (A, B)

13.Correct option: (A, C, D)

$P(\mathrm{E}_1) = \frac{1}{4}, P\left(\frac{\mathrm{E}_2}{\mathrm{E}_1}\right) = \frac{1}{2}, P\left(\frac{\mathrm{E}_1}{\mathrm{E}_2}\right) = \frac{1}{4}$

$P\left(\frac{\mathrm{E}_2}{\mathrm{E}_1}\right) = \frac{1}{2} = \frac{P(\mathrm{E}_2 \cap \mathrm{E}_1)}{P(\mathrm{E}_1)}$

$\Rightarrow P(\mathrm{E}_2 \cap \mathrm{E}_1) = \frac{1}{8}$

$P\left(\frac{\mathrm{E}_1}{\mathrm{E}_2}\right) = \frac{1}{4} = \frac{P(\mathrm{E}_1 \cap \mathrm{E}_2)}{P(\mathrm{E}_2)} \Rightarrow P(\mathrm{E}_2) = \frac{1}{2}$

$\therefore P(\mathrm{E}_1 \cap \mathrm{E}_2) = P(\mathrm{E}_2)P(\mathrm{E}_1) = \frac{1}{8}$

Therefore E_1 and E_2 are Independent events.

$\therefore P(\mathrm{E}_2) = 2 \times P(\mathrm{E}_1)$

also $P(\mathrm{E}_1)^2 = P(\mathrm{E}_2) \times P(\mathrm{E}_1 \cap \mathrm{E}_2)$

$\therefore P(\mathrm{E}_1 \cap \mathrm{E}_2), P(\mathrm{E}_2), P(\mathrm{E}_1)$ *are in G.P.*

The correct Answer is 'A' 'C' and 'D'

14. Correct option: (B)

P (both of them are selected) = P (A ∩ B) = P (A) × P (B)

$$P(\mathrm{A}) = \frac{1}{3}, P(\mathrm{B}) = \frac{1}{4}$$

$$\therefore P(\mathrm{A} \cap \mathrm{B}) = \mathrm{P}(\mathrm{A})\,\mathrm{P}(\mathrm{B}) = \frac{1}{12}$$

The correct Answer is 'B'

15. **Correct option: (B)**

P (none of them are selected) = P (A' ∩ B') = P (A') × P (B')

$$P(\mathrm{A}') = 1 - P(\mathrm{A}) = \frac{2}{3}, P(\mathrm{B}') = 1 - P(\mathrm{B}) = \frac{3}{4}$$

$$\therefore P(\mathrm{A}' \cap \mathrm{B}') = \mathrm{P}(\mathrm{A}')\,\mathrm{P}(\mathrm{B}') = \frac{1}{2}$$

The correct Answer is 'B'

16. **Correct option: ()**

P (one of them are selected) = P (A ∩ B') + P (B ∩ A')

$$P(A)\,P(B') + P(B)\,P(A')$$

$$= \frac{1}{3}.\frac{3}{4} + \frac{1}{4}\frac{2}{3}$$

$$= \frac{1}{6} + \frac{1}{4}$$

$= \frac{5}{12}$

17. Correct Matching is: a-S; b-P; c-Q; d-R

$$P(\mathrm{A}) = \frac{3}{4}, P(\mathrm{B}) = \frac{2}{5} P(\mathrm{A} \cap \mathrm{B}) = \frac{1}{4}$$

$$P(\mathrm{A} \cup \mathrm{B}) = P(\mathrm{A}) + P(\mathrm{B}) - P(\mathrm{A} \cap \mathrm{B})$$

$$= \frac{3}{4} + \frac{2}{5} - \frac{1}{4}$$

$$= \frac{9}{10}$$

The Correct Match is 'S'

$$P(\mathrm{A}) = \frac{1}{4}, P(\mathrm{B}) = \frac{1}{2}, P(\mathrm{A} \cup \mathrm{B}) = \frac{5}{8} P(\mathrm{A} \cap \mathrm{B}) = \frac{1}{4}$$

$$P(\mathrm{A} \cup \mathrm{B}) = P(\mathrm{A}) + P(\mathrm{B}) - P(\mathrm{A} \cap \mathrm{B})$$

$$\frac{5}{8} = \frac{1}{4} + \frac{1}{2} - P(\mathrm{A} \cap \mathrm{B})$$

$$\therefore P(\mathrm{A} \cap \mathrm{B}) = \frac{3}{4} - \frac{5}{8} = \frac{1}{8}$$

The Correct Match is 'P'

$$P(\mathrm{A}) = \frac{1}{4}, P(\mathrm{B}) = \frac{1}{2}, P(\mathrm{A} \cap \mathrm{B}) = \frac{1}{5}$$

$$P(\overline{A} \cap \overline{B}) = 1 - P(\mathrm{A} \cup \mathrm{B})$$

$$= 1 - \frac{1}{4} - \frac{1}{2} - \frac{1}{5}$$

$$= \frac{9}{20}$$

The Correct Match is 'Q'

$$P(A\cup B)=\frac{3}{4}, P(A\cap B)=\frac{1}{5}$$
$$P(\overline{A})+P(\overline{B})=1-P(A)+1-P(B)$$
$$=2-\{P(A)+P(B)\}$$
$$=2-\{P(A\cup B)+P(A\cap B)\}$$
$$=2-\{\frac{3}{4}+\frac{1}{5}\}$$
$$=\frac{21}{20}$$

The Correct Match is 'R'

18. Correct Answer is:

$$P(A\cap B')=P(A\cup B)-P(B)=\frac{1}{30}$$
$$P(A\cup B)=\frac{1}{6}$$
$$\Rightarrow P(B)=\frac{1}{6}-\frac{1}{30}=\frac{4}{30}$$
$$\therefore 10.P(B)=\frac{4}{3}$$

19. Correct Answer is: 7

A and B throws a die

Ways that B's throw always greater than A's is

If A=1 then B=1, 2, 3, 4, 5, 6 $P_1=\frac{1}{6}\times\frac{6}{6}$ $P(\text{cluborheart})=\frac{13}{52}+\frac{13}{52}-\frac{6}{52}=\frac{20}{52}=\frac{5}{13}$

If A=2 then B=2, 3, 4, 5, 6 $P_2=\frac{1}{6}\times\frac{5}{6}$

If A=3 then B=3, 4, 5, 6 $P_3=\frac{1}{6}\times\frac{4}{6}$

If A=4 then B=4, 5, 6 $P_4=\frac{1}{6}\times\frac{3}{6}$

If A=5 then B=5, 6 $P_5=\frac{1}{6}\times\frac{2}{6}$

If A=6 then B=6 $P_6=\frac{1}{6}\times\frac{1}{6}$

Total probability= $\frac{1}{36}(6+5+4+3+2+1)=\frac{7}{12}$

$12p=12\times\frac{7}{12}=7$

20. Correct Answer is: 3

The number of determinant of the second order that can be formed with 0, 1=2×2×2×2=16

Since each of the four places of elements can be filled in 2 ways: by 0 or 1

∴n(S) =16

Let E= the events of getting non-negative determinant.

Then E′= the event of getting negative determinant.

Clearly negative determinant of second order that can be made with 0,1 are

$$\begin{vmatrix}0 & 1\\ 1 & 0\end{vmatrix}\begin{vmatrix}1 & 1\\ 1 & 1\end{vmatrix}\begin{vmatrix}1 & 1\\ 1 & 0\end{vmatrix}$$

$$\therefore P(E') = \frac{n(E')}{n(S)} = \frac{3}{16}$$

$$\therefore \frac{p}{16} = \frac{3}{16} \Rightarrow p = 3$$

STUDENT PRACTICE QUESTIONS

1. Correct option: (B)

When two die are rolled total number of events $=6^2=36$

The favorable case is getting ten in a single throw

Total number of favorable cases is {(4, 6), (5, 5), (6, 4)}

$$probability = \frac{no.offavoriblecases}{total\,no.ofcases} = \frac{3}{36} = \frac{1}{12}$$

The correct Answer is 'B'

2. Correct option: (A)

When a die is rolled twice total number of events $=6^2=36$

The favorable case is the numbers on them are different i.e. they must not be equal numbers

Total number of not favorable cases is {(1, 1), (2, 2), (3, 3) (4, 4) (5, 5) (6, 6)}

$$\text{Non favorable } probability = \frac{no.ofnonfavoriblecases}{total\,no.ofcases} = \frac{6}{36} = \frac{1}{6}$$

$$\therefore favorableprobablity = 1 - nonfavorableprobablity$$

$$\Rightarrow 1 - \frac{1}{6} = \frac{5}{6}$$

The correct Answer is 'A'

3. Correct option: (C)

When a die is rolled twice total number of events $=6^2=36$

The favorable case is getting an even number and an odd number

Total number of favorable cases is {(1, 2), (1, 4) (1, 6) (2, 1) (2, 3) (2, 5) (3, 2) (3, 4) (3, 6) (4,1) (4,3), (4,5) (5,2) (5,4), (5,6) , (6,1), (6,3), (6,5)}

$$probability = \frac{no.offavoriblecases}{total\,no.ofcases} = \frac{18}{36} = \frac{1}{2}$$

The correct Answer is 'C'

4. Correct option: (A)

When three dies is rolled total number of events $=6^3=216$

The favorable case is the sum of the numbers on them is 16

Total number of favorable cases is {(5,5,6) (5,6,5) (6,5,5) (6,6,4) (4,6,6) (6,4,6)}

$$probability = \frac{no.offavoriblecases}{total\,no.ofcases} = \frac{6}{216} = \frac{1}{36}$$

The correct Answer is 'A'

5. Correct option: (B)

The total number of cards = 52. Symbols that cards contain are 'Heart', 'Diamond', 'Spade', 'Club'. Every symbol has 13 cards.

Number of favorable outcomes i.e. 'a card of heart' is 13.

$$probability = \frac{no.offavoriblecases}{total\,no.ofcases} = \frac{13}{52} = \frac{1}{4}$$

The correct Answer is 'B'

6. Correct option: (D)

The 2 odd days may be the combination of Sunday & Monday, Monday & Tuesday, Tuesday & Wednesday, Wednesday & Thursday, Thursday & Friday, Friday & Saturday or Saturday & Sunday.

Therefore, the total number of possible outcomes or elements of a sample space is 7.

probability of 2 Odd days not to be Monday & Tuesday, Tuesday & Wednesday, Wednesday & Thursday, Thursday & Friday or Friday & Saturday

Expected events of A = {Sun & Mon}, {Wed & Thu}, {Thu & Fri}, {Fri & Sat}, {Sat & Sun} = 5

So the probability of 52 Tuesdays is 5/7

The correct Answer is 'D'

7. Correct option: (D)

Probability of 1 Odd day not to be Monday

The sample space S = {Sun, Mon, Tue, Wed, Thu, Fri, Sat}

Expected events of A = {Sun, Tue, Wed, Thu, Fri, Sat}

P (A) = {Sun, Tue, Wed, Thu, Fri, Sat}/{Sun, Mon, Tue, Wed, Thu, Fri, Sat}

P(A)=6/7=0.86

6/7 or 0.86 is probability for 52 Mondays in a non-leap year.

The correct Answer is 'D'

8. Correct option: (D)

When a number is picked at random from 1 to 900 then the number of events will be 900

The favorable case is drawing a number which is a square is

Therefore the number of squares between 1to 900 is 30

$$probability = \frac{no.offavoriblecases}{tota\ln oofcases} = \frac{30}{900} = \frac{1}{30}$$

The correct Answer is 'D'

9. Correct option: (B)

$$P(\mathrm{A}) = \frac{2}{7}, P(\mathrm{B}) = \frac{3}{5}, P(\mathrm{A} \cap \mathrm{B}) = 0$$

$$P(\mathrm{A} \cup \mathrm{B}) = P(\mathrm{A}) + P(\mathrm{B})$$

$$= \frac{2}{7} + \frac{3}{5}$$

$$= \frac{31}{35}$$

The correct Answer is 'B'

10. Correct option: (B)

$$P(\mathrm{A}) = \frac{1}{4}, P(\mathrm{B}) = \frac{1}{2}, P(\mathrm{A} \cup \mathrm{B}) = \frac{5}{8}$$

$$\frac{5}{8} = \frac{1}{4} + \frac{1}{2} - P(\mathrm{A} \cap \mathrm{B})$$

$$P(\mathrm{A} \cap \mathrm{B}) = -\frac{5}{8} + \frac{3}{4}$$

$$= \frac{1}{8}$$

The correct Answer is 'B'

11. Correct option: (C, D)

$$P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)\,P(B)}{P(B)} = P(B)$$

$$P((A \cup B)') = P(A' \cap B') = P(A')\,P(B')$$

The correct Answer is 'C' & 'D'

12. Correct option: (B, C, D)

If M *and* N *are independent then*

$$P(M \cap N) = P(M)\,P(N)$$

$$P(M \cap \overline{N}) = P(M) - P(M \cap N)$$

$$= P(M) - P(M)\,P(N)$$

$$= P(M)\,P(\overline{N})$$

$\therefore$ M, $\overline{N}$ are independent

$$P(\overline{M} \cap \overline{N}) = 1 - P(M \cup N)$$

$$= 1 - P(M) - P(N) - P(M \cap N)$$

$$= P(\overline{M})P(\overline{N})$$

$\therefore \overline{M}, \overline{N}$ are independent

The correct Answer is 'B' 'C' and 'D'

13. Correct option: (A)

If 2 dices are thrown simultaneously then total outcomes will be 36

Favorable outcomes = (1,5) (2,4)(3,3)(3,5)(4,2)(4,4)(5,1)(5,3)(6,2)

Probability of event =10/36=5/18

The correct Answer is 'A'

14. Correct option: (A)

Probability of getting a total of 10:

To get a total of 10 with two dice, you can have the following combinations: (4, 6), (5, 5), and (6, 4). There are three favorable outcomes.

Probability of getting a total of an even number:

An even total can be obtained in multiple ways: (2, 2), (2, 4), (2, 6), (4, 2), (4, 4), (4, 6), (6, 2), (6, 4), and (6, 6). There are nine favorable outcomes.

$$\frac{3}{36} + \frac{9}{36} = \frac{1}{3}$$

The correct Answer is 'A'

15. Correct Matching is: a-R; b-Q; c-P; d-S

A

Let A: event of drawing a king $P(A) = \frac{4}{52} = \frac{1}{13}$

B: event of drawing a spade = $P(B) = \frac{13}{52} = \frac{1}{4}$

$P(A \cap B) = \frac{1}{52}$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{1}{13} + \frac{1}{4} - \frac{1}{52}$$

$$= \frac{4}{13}$$

The Correct Match is 'R'

B

The number of cards in a pack = 52

The number of the kings in a pack = 4

If we draw one card at random from the 52 cards,

Then the probability of getting a king = number of kings/number of all the cards i.e. probability of getting a king = $\frac{4}{52}$

The number of the queens in a pack = 4

If we draw one card at random from the 52 cards,

Then the probability of getting a queen = number of queens/number of all the cards

i.e. probability of getting a queen = $\frac{4}{52}$

The probability of getting a king or a queen out of all the cards = probability of getting a king out of all the cards + probability of getting a queen out of all the cards $\frac{4}{52}+\frac{4}{52}=\frac{2}{13}$

Hence the probability of getting a king or a queen out of 52 cards is $\frac{2}{13}$

The Correct Match is 'Q'

C

A=the drawn card is a spade; B= the drawn card is a heart.

$P(\text{A})=\frac{13}{52}$ (As there are 13 spade cards in a fair deck of 52 cards)

$P(\text{B})=\frac{13}{52}$ (As there are 13 heart cards in a fair deck of 52 cards)

Events A and B are *mutually exclusive events.

As by addition theorem of probability, P (A or B) = P (A) + P (B) for mutually exclusive events.

P (spade or heart) = P (A U B) = $\frac{13}{52}+\frac{13}{52}=\frac{1}{2}$

The Correct Match is 'P'

D

There are 13 cards in each suit, so there are 26 cards that are either hearts or clubs.

However, we have double counted the cards that are both a heart and a club (i.e., the Ace, King, Queen, Jack, and 10 of hearts and clubs).

The probability of drawing a heart or a club is then:

P(heart or club) = $=P(\text{cluborheart})=\frac{13}{52}+\frac{13}{52}-\frac{6}{52}=\frac{20}{52}=\frac{5}{13}$

The Correct Match is 'S'

16. Correct Answer is:

$(x^4-y^4)=(x^2+y^2)\,(x+y)\,(x-y)$

Let us consider five sets $(5k, 5k+1, 5k+2, 5k+3, 5k+4)$.

Case I: $(x-y)$ *is divisible by* 5.

Elements are to be selected from the same set, 5. ${}^{n}C_{2}$

Case II: $x+y$ *is divisible by* 5 *but* $x-y$ *is not divisible by* 5

It can be done from $(5k+1, 5k+4)$ *or* $(5k+2, 5k+3)$.

$Total\ ways\ = 2n^2 \quad \ldots.(2)$

$Case\ III\colon\ x^2 + y^2\ is\ divisible\ by\ 5\ but\ x - y\ or\ x + y\ is\ not\ divisible\ by\ 5 \quad \ldots(3)$

$We\ can\ select\ one\ number\ either\ from (5k + 1, 5k + 4)\ and\ the\ second\ number\ from\ (5k + 2, 5k + 3).$

$Hence, the\ number\ of\ cases\ = 2n \times 2n = 4n^2 \ \ldots\ldots\ldots\ldots\ldots(3)$

$Total\ number\ of\ \ favourable\ cases\ = 5\dfrac{n(n-1)}{2} + 6n^2 = \dfrac{17n^2 - 5n}{2}$

$Total\ number\ of\ cases\ = 5\dfrac{n(n-1)}{2}$

$Pr\ o\ bailty = \dfrac{17n - 5}{25n - 5}$

$here, n = 3 \Rightarrow Pr\ o\ bailty(p) = \dfrac{46}{70} = \dfrac{23}{35}$

$\therefore \dfrac{p}{23} = \dfrac{1}{35}$

17. Correct Answer is:1

$P(A \cap B) = \frac{1}{12}, P(\overline{A} \cap \overline{B}) = \frac{1}{2}, P(A) = \frac{1}{m}$

$$P(\overline{A} \cap \overline{B}) = P(\overline{A}).P(\overline{B}) = \frac{1}{2}$$

$$\Rightarrow [1 - P(A)].[1 - P(B)] = \frac{1}{2}$$

$$\Rightarrow 1 - P(A) - P(B) + P(A).P(B) = \frac{1}{2}$$

$$P(A \cap B) = P(A)\,P(B) = \frac{1}{m}P(B) = \frac{1}{12}$$

$$\therefore P(B) = \frac{m}{12}$$

$$\Rightarrow 1 - \frac{1}{m} - \frac{m}{12} + \frac{1}{12} = \frac{1}{2}$$

$$\Rightarrow 1 + \frac{1}{12} - \frac{1}{2} = \frac{1}{m} + \frac{m}{12}$$

$$\Rightarrow \frac{7}{12} = \frac{m^2 + 12}{12m}$$

$$\Rightarrow 12m^2 + 144 - 24m = 0$$

$$\Rightarrow m = 1 \pm \sqrt{-11}$$

$$\therefore [m] = 1$$

$$\Rightarrow \left[\frac{m}{3}\right] = \frac{1}{3}$$